



**Mapping the Fashion Entrepreneurship Ecosystem in Saudi Arabia:
Impacts on Entrepreneurial Orientation and Business Performance of
Micro, Small, and Medium Enterprises**

A thesis submitted in fulfilment of the requirements
for the degree of Doctor of Philosophy

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DECLARATION

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ACKNOWLEDGEMENTS

All praise and thanks are due to Allah, the Most Gracious, the Most Merciful, for His countless blessings and divine guidance throughout my academic journey.

To my cherished parents, **Dakhailallah** and **Khazna**, your prayers and unconditional love have been the foundation of my strength and determination. Your unwavering faith in me has been a constant source of motivation, guiding me through every challenge.

To my beloved husband **Abdulrahman**, who stood by my side through the most difficult times, taking on countless responsibilities with patience and love. Your steadfast support has been my anchor, and for that, I am forever thankful.

To my dear son, **Mutaz**, who grew alongside me throughout this journey, filling my days with love and encouragement while affectionately calling me “Super Mommy”. You are and will always remain my greatest source of inspiration. And to my daughter, **Laila**, whose arrival brought immense joy and beautifully marked the conclusion of this journey you are the brightest light in my life.

To my siblings, whose love and support have been invaluable. A special thank you to my sisters **Laila** and **Sarah** for their constant kindness and care, and to **Sumaya**, **Kholoud** and **Sarah** for their unwavering encouragement and support.

I am deeply grateful to my supervisors, **Dr. Caroline Tan** and **Dr. Carolina Quintero Rodriguez**, for their invaluable guidance, insightful advice, and continuous encouragement.

I would also like to extend my sincere thanks to the editor and proofreader, **David Langdon** and **Rob Sheehan**, for their meticulous work in refining this thesis.

Finally, I wish to express my heartfelt gratitude to the **Saudi Arabian Cultural Mission** in Australia for their generous financial support and continued guidance. This

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ABBREVIATIONS

Abbreviation	Terminology
AVE	Average Variance Extracted
BP	Business Performance
CMB	Common Method Bias
CR	Composite Reliability
EE	Entrepreneurship Ecosystem
EO	Entrepreneurial Orientation
FEE	Fashion Entrepreneurship Ecosystem
G20	The Group of Twenty
GCC	Gulf Cooperation Council
GDP	Gross Domestic Product
GEM	Global Entrepreneurship Monitor
HTMT	Heterotrait-Monotrait Ratio
KSA	Kingdom of Saudi Arabia
MENA	Middle East and North Africa
MSME	Micro, Small and Medium-Sized Enterprises
NECI	National Entrepreneurship Context Index
OPEC	Organization of the Petroleum Exporting Countries
PLS- SEM	Partial Least Square Structural Equation Modelling
R&D	Research and Development
SAR	Saudi Arabia Riyal
SPSS	Statistical Package for the Social Sciences

RESEARCH SUMMARY

This study investigates the influence of the fashion entrepreneurship ecosystem (FEE) on entrepreneurial orientation (EO) and business performance (BP) within fashion micro, small, and medium enterprises (MSMEs) in Saudi Arabia. Drawing on the Resource-Based View (RBV) theory (Barney, 1991), EO, defined through innovativeness, proactiveness, and risk-taking (Lumpkin & Dess, 2015), is treated as a core intangible resource that mediates the relationship between external ecosystem elements and firm success (Sarsiti, 2024). The FEE is conceptualised as a dynamic interplay of domains, including finance, education, policy, market, culture, and support (Isenberg, 2011), each contributing to entrepreneurial development in line with Saudi Arabia's Vision 2030 (Nurunnabi, 2017). While prior studies have explored these domains separately or in different sectors (Spigel, 2017), no research has integrated them in the specific context of Saudi fashion MSMEs.

Saudi fashion enterprises represent a growing but under-researched segment within the creative economy, with the sector projected to grow by 11.6% annually and reach a value of USD 7.57 billion by 2029 (Statista, 2024). These businesses are often led by women and face unique challenges, including limited access to finance, weak entrepreneurial education, and global market barriers (Alshibani et al., 2024; Maghrabi & Abalkail, 2024). This study addresses this critical gap by offering a context-specific analysis of how FEE factors impact EO and BP, thereby supporting policy efforts to diversify the economy and strengthen the creative sector (Almuzel & Anderson, 2020).

This thesis comprises two sequentially linked studies using a mixed-methods explanatory design (Creswell, 2017). Study 1 employed quantitative analysis (PLS-SEM) to test the direct and mediating effects of FEE on EO and BP. Study 2 involved thematic qualitative interviews with fashion entrepreneurs and ecosystem stakeholders to explain and contextualise these quantitative findings. This integration enables a comprehensive understanding of how systemic factors influence entrepreneurial success, particularly when outcomes diverge from expectations (Dionysus & Arifin, 2020). The findings offer theoretical contributions to the entrepreneurship literature and practical insights into policy reform, ecosystem development, and educational innovation in Saudi Arabia.

Study 1

Participants

The participants in this study included 223 fashion MSMEs operating in Saudi Arabia. The sample primarily consisted of female participants (91%), with 67.3% holding a bachelor's degree. Most participants (50.7%) studied fashion, while 36.3% attended business courses. Approximately 44.4% of businesses were established between 2017 and 2020. Most businesses (94.2%) were microenterprises in size.

Methods

The survey instrument measured FEE, EO, and BP using validated frameworks. FEE was assessed using the Global Entrepreneurship Monitor (GEM) National Expert Survey (NES), which evaluates core ecosystem factors: finance, policy, market, culture, social norms, entrepreneurial education, talent, and support. EO was measured using Covin and Slevin's (1989) three core components: innovativeness, proactiveness, and risk-taking. BP was assessed using five key indicators: ROI, sales increase, customer satisfaction, employee satisfaction, and overall firm performance. This approach aligns with prior research on performance evaluations (Murphy et al., 1996; Hult et al., 2008). Financial and non-financial metrics provide a comprehensive view of business success (Harter et al., 2002; Wong & Low, 2013).

Participants were contacted via email and invited to participate in the study. After providing consent, participants accessed the online survey through Qualtrics. The survey questions were compulsory, ensuring that no data points were lost. Data analysis involved several key steps using IBM SPSS and SmartPLS software. Measurement reliability and validity were assessed using Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE) values. The measurement model was evaluated for internal consistency (Cronbach's alpha > 0.70), composite reliability (CR > 0.70), and discriminant validity using the Fornell-Larcker criterion and the heterotrait-monotrait (HTMT) ratio. Structural equation modelling was used to assess the direct and indirect effects. Path coefficients, R-squared (R^2) values, and bootstrapping methods were used to confirm the significance and model fit.

Results

Direct Impact of FEE on EO: Path analysis revealed a significant positive effect of FEE on EO ($\beta = 0.308$, $t = 4.145$, $p < 0.001$). The R^2 value for EO was 0.095, indicating that FEE explained 9.5% of the variance in EO. This finding suggests that a supportive entrepreneurial ecosystem positively influences fashion MSMEs' innovativeness, proactiveness, and risk-taking.

Direct Impact of FEE on BP: A significant but smaller positive effect of FEE on BP was observed ($\beta = 0.149$, $t = 2.113$, $p = 0.035$). The R^2 value for BP was 0.286, indicating that FEE explained 28.6% of the variance in BP. This suggests that the entrepreneurial ecosystem supports business growth and performance, although other factors may also contribute.

Direct Impact of EO on BP: EO had a strong positive influence on BP ($\beta = 0.469$, $t = 8.078$, $p < 0.001$). This indicates that higher levels of innovativeness, proactiveness, and risk-taking directly enhance the business performance. This strong relationship confirms that EO is a strategic driver of business success.

Mediating Role of EO: EO partially mediated the relationship between FEE and BP. The direct effect of FEE on BP was $\beta = 0.149$ ($p < 0.05$), while the indirect effect through EO was $\beta = 0.145$ ($p < 0.001$). The total effect of FEE on BP was $\beta = 0.294$ ($p < 0.001$). This suggests that FEE influences business performance directly and indirectly through EO.

Model Fit and Predictive Relevance: The model demonstrated moderate predictive relevance for BP ($Q^2 = 0.176$) and weak predictive relevance for EO ($Q^2 = 0.045$). All constructs demonstrated high factor loadings (>0.70), satisfactory composite reliability (>0.70), and discriminant validity. Path analysis confirmed that the model fit the data well.

Study 2

This qualitative study is based on 15 interviews with key stakeholders in the Saudi fashion industry. It focuses on mapping and exploring the current landscape of the Saudi FEE and how each domain influences EO and BP. Based on inductive thematic analysis, Study 2 revealed several influential ecosystem factors shaping EO and BP: most fashion entrepreneurs relied on informal sources, such as personal savings or family loans, due to limited access to venture capital or incubator funding. The Social Development Bank was a key source of accessible loans, but procedural complexities, limited financial literacy, and cultural/institutional biases especially against women restricted access to broader financial support. Access to finance directly enhances innovation, sustainability practices, and risk-taking (Alshibani et al., 2024; Alsaiari, 2022; Adam, 2024).

While entrepreneurship is culturally encouraged in Saudi Arabia, fashion as a profession faces lingering stigma, particularly tailoring, which is viewed as a low-status profession. Regional and religious values shape design preferences and business models, and fear of failure discourages risk-taking, especially among women, due to societal pressure and a preference for government jobs (Alzahrani et al., 2015, Fridhi, 2020, Elshaer & Sobaih, 2022). The study found a significant mismatch between the available incubators and the specific needs of fashion startups. Most support structures are designed for tech startups and offer irrelevant services, such as software mentorship. Consultants are often competitors, creating trust issues. Entrepreneurs in rural areas also face geographic barriers to support access (Braun & Suoranta, 2024; Kazhenov, 2023; Nana et al., 2021).

While the government has expanded fashion programs and scholarships, participants reported limited institutional support for starting businesses. Overseas universities offer stronger entrepreneurship support. Local institutions lack practical, fashion-specific business training, structured entrepreneurial curricula, and access to industry professionals (Alshebami, 2022; Raju & Singh, 2023; Lang & Liu, 2019). The evidence from this study supports the idea that variations in entrepreneurial performance in the Saudi fashion sector depend not only on access to resources but also on the entrepreneur's ability to transform these assets into a competitive edge. This transformation aligns with the RBV's focus on resource deployment as strategic behaviour (Wernerfelt, 1984) and is supported by research highlighting EO as a critical mediator of resource effectiveness (Dionysus & Arifin, 2020; Ferreira et al., 2011).

CHAPTER 1: INTRODUCTION

1.1 Chapter Overview

This chapter introduces the research setting and social context of Saudi Arabia, focusing on its fashion industry. It highlights key concepts, such as fashion entrepreneurship, the fashion entrepreneurship ecosystem (FEE), and the role of micro, small, and medium enterprises (MSMEs) in the Saudi fashion sector. This chapter explores the geographical, cultural and economic landscape of Saudi Arabia to establish a foundation for investigating the local fashion industry. Additionally, it outlines the research rationale, aims and questions, and contributions. The chapter concludes with a summary and overview of the structure of the thesis.

1.2 Research Background

Entrepreneurship plays a central role in driving the growth and success of industries, including creative industries (Loots et al., 2021; Nambisan, 2017; Schulte-Holthaus, 2018). These industries, encompassing sectors such as fashion, film, music, and design, depend on entrepreneurial activities to transform creative ideas into economic value (Potts, 2011; Wardani et al., 2015). In these contexts, the entrepreneurial process facilitates the commercialisation of creative outputs and enhances the overall dynamism and competitiveness of the economy (Daneshjoovash et al., 2020; Rostami et al., 2019).

Creative industries contribute significantly to the global economy by creating jobs and professions, generating revenue, and increasing competitiveness (Issakhova & Jumasseitova, 2024; Riabov & Riabova, 2021; Sung, 2015). Within these industries, the fashion industry is a key contributor to global economies, valued at nearly USD 1.8 trillion in 2023, and contributing 2% to the global Gross Domestic Product (GDP) (Euromonitor International, 2024; Peters et al., 2021). Consequently, scholars are increasingly focusing on the unique characteristics and dynamics of entrepreneurial activity in these sectors (Lee et al., 2014; Purbasari & Wijaya, 2019).

The growing interest in fashion entrepreneurship has led scholars to examine the factors that contribute to success in this industry (Zeng, 2020; Zeng et al., 2024). Studies have utilised the entrepreneurial ecosystem framework to explore the influence of culture, financial support, education, market conditions, human resources, and policy on the success and performance of fashion enterprises (Brydges & Pugh, 2021; Salonoja, 2013). Additionally, recent research emphasises fashion MSMEs, where entrepreneurial orientation (EO) is crucial for success in a

dynamic, trend-driven industry, where adaptability and innovation are critical (Miller, 1983; Mulyana & Sutapa, 2016). High EO levels enable fashion MSMEs to maintain their competitiveness by anticipating trends and responding swiftly to evolving market demands (Juliani & Nuvriasari, 2024).

1.3 The Saudi Context

1.3.1 Country Profile: The Kingdom of Saudi Arabia

Saudi Arabia is the largest and most influential country on the Arabian Peninsula and occupies a strategic position in the Middle East (Central Intelligence Agency, 2024). Figure 1.1 shows the geographical boundaries of the country, with Jordan and Iraq situated to the north, Kuwait to the northeast, and Qatar, Bahrain, and the United Arab Emirates along its eastern border. The Red Sea lies to the west, and the Arabian Gulf is located to the east, with Oman and Yemen to the south (World Atlas, 2022). This geographical positioning provides Saudi Arabia with significant access to major waterways and trade routes, thereby enhancing its geopolitical importance in regional and global contexts. The Kingdom of Saudi Arabia comprises 13 administrative regions, with the capital city of Riyadh situated near the country's geographical centre.



Figure 1.1: Geographic Location of Saudi Arabia and Surrounding Features (World Atlas, 2022).

Saudi Arabia is recognised as the birthplace of Islam, hosting Mecca and Medina, which are two of the holiest cities in the Muslim religion. This religious significance attracts millions of Muslim pilgrims annually, both domestically and internationally, contributing to the country's economy and enhancing its cultural heritage, and is a UNESCO World Heritage Site. Thus, the annual Hajj pilgrimage underscores Saudi Arabia's central role in the Islamic faith and its commitment to facilitating religious practices (Bowen, 2024; Czornik, 2020).

Since its unification in 1932, Saudi Arabia has been an absolute monarchy. King Salman bin Abdulaziz Al Saud has ruled since 2015, and under the Saudi Vision 2030 blueprint for development, Crown Prince Mohammed bin Salman is leading reforms aimed at modernising the kingdom. Additionally, Saudi Arabia plays a significant role in global diplomacy as a member of international organisations such as the Gulf Cooperation Council, the United Nations, and the Organization of Islamic Cooperation (Bowen, 2024).

1.3.2 Demographic and Policy Implications

Saudi Arabia's population in 2024 is estimated to be approximately 36.5 million, with notable shifts in the composition of residents, particularly among foreign nationals (Central Intelligence Agency, 2024). Historically, the proportion of foreign residents has fluctuated between 36% and 39%. However, this figure has declined due to the Saudization policy and the impact of the COVID-19 pandemic (General Authority for Statistics, 2023). Saudization (Nitaqat) is a labour strategy aimed at increasing Saudi employment in the private sector by mandating quotas for hiring nationals (Al-Asfour & Khan, 2014; Potnuru & Arora, 2022). Nevertheless, recent developments emerging from the surge in megacity construction projects under Vision 2030 have contributed to an increase in the foreign resident population (Stiftung, 2020). Although Saudi citizens are 100% Muslim, foreign residents have diverse religious backgrounds, including Eastern Orthodox, Buddhist, Protestant, Jewish, Hindu, Roman Catholic, and Sikh (Central Intelligence Agency, 2024; Stiftung, 2020).

According to a demographic survey conducted by the General Authority for Statistics (2023), Saudi Arabia has a predominantly young demographic profile. As shown in Figure 1.2, the working-age population (ages 15 to 64) comprises approximately 72.4% of the total population, translating to approximately 23.7 million individuals in 2023. Projections suggest that this figure could rise to 26.8 million by 2030, underscoring a demographic shift towards a younger, economically active populace (Rizvi & Hussain, 2022). This trend presents significant economic growth and innovation opportunities, contingent on adequate investment in education and employment. However, this trend also poses challenges that require careful

policy consideration, particularly in terms of effectively integrating this burgeoning workforce into the economy (Rizvi & Hussain, 2022).

Approximately 84.7% of the population lives in urban areas. Riyadh, the capital, is the largest city, followed by other major urban centres such as Jeddah and Dammam. Extensive urbanisation is supported by the comprehensive reach of state administration, which embeds access to essential utilities, even in the most remote regions. The government’s commitment to infrastructure development and service provision is crucial for supporting the growing urban population and fostering a conducive environment for maintaining the labour force necessary for economic growth and development (Rizvi & Hussain, 2022). Achieving Vision 2030 requires balancing demographic trends and policy reform. As Saudi Arabia adapts to its native and foreign residents, addressing societal needs such as civic participation and inclusivity is essential. The rapidly evolving socioeconomic landscape highlights the importance of managing diversity in workforce integration, including cultural differences and varying education and skill levels (MarketLine, 2024; Stiftung, 2020).

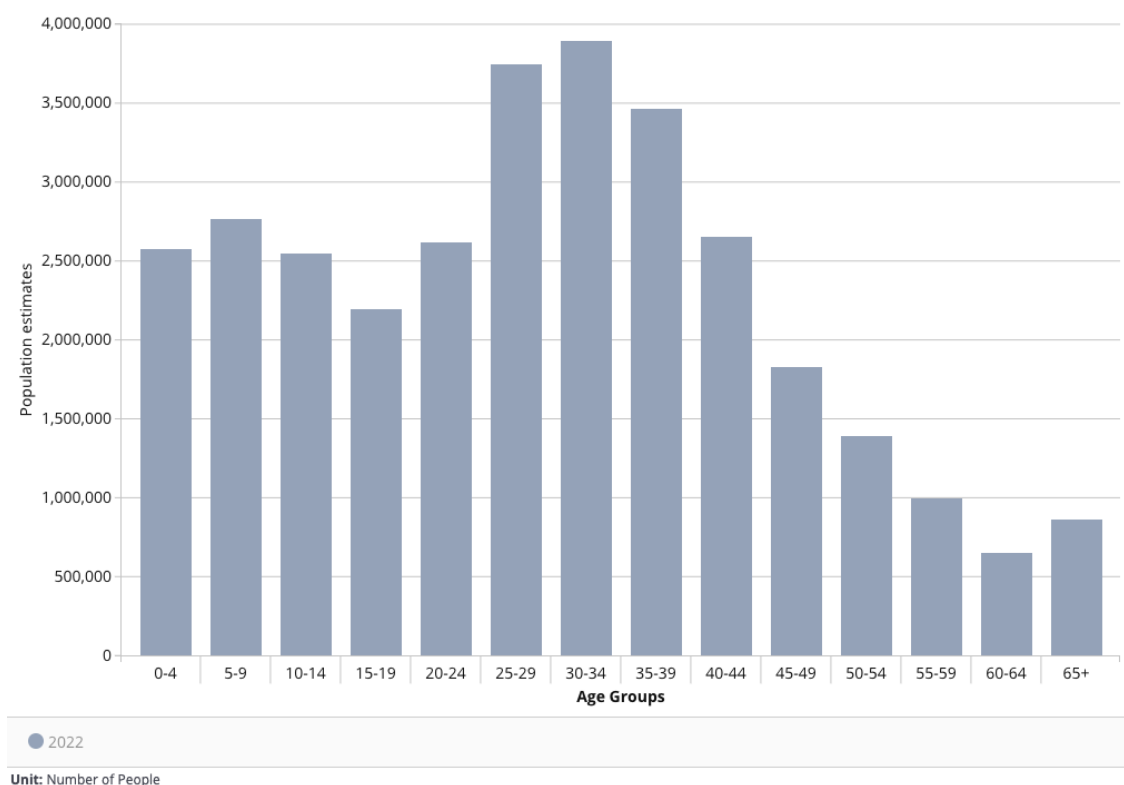


Figure 1.2: Total Saudi and Non-Saudi Population, by Age Group, 2022 (General Authority for Statistics, 2023).

1.3.3 Economic Environment and Vision 2030

Saudi Arabia plays a crucial role in the global economy as one of the world's largest oil exporters and a key member of international organisations such as the Group of Twenty (G20), Organisation of the Petroleum Exporting Countries (OPEC), and Gulf Cooperation Council (GCC) (Zhukov et al., 2021). The Kingdom's substantial petroleum reserves have provided it with considerable economic influence, particularly within the oil and gas sector which remains the dominant source of exports and government revenue (European Parliamentary Research Service (EPRS), 2023). Additionally, Saudi Arabia is endowed with other natural resources, such as natural gas, gold, copper, and iron, which contribute to its broader economic structure (Central Intelligence Agency, 2024).

In 2014, Saudi Arabia's GDP reached USD 753.8 billion, with merchandise exports totalling USD 341 billion, primarily from oil and gas (International Monetary Fund, 2022). However, this dependence on oil, which accounts for approximately 85% of the government's revenue, has raised questions regarding the sustainability of Saudi Arabia's economic model. The volatility of global oil prices poses significant risks to the kingdom's economic stability and long-term growth (OPEC, 2017). With oil revenues currently contributing to the majority of government income, the Saudi economy remains highly susceptible to global economic downturns and shifts in energy consumption patterns (World Bank, 2024). In response to these challenges, Saudi Arabia launched Vision 2030 in 2016 as a comprehensive strategy to restructure the economy by reducing its dependence on oil revenues and fostering growth in non-oil sectors such as tourism, manufacturing, renewable energy, and cultural industries (Government of Saudi Arabia, 2024; Moshashai et al., 2020; Nurunnabi, 2017).

A core component of Vision 2030 is support for MSMEs, which play a crucial role in driving innovation and economic growth. The establishment of the General Authority for Small and Medium Enterprises (Monsha'at) has significantly boosted entrepreneurial activity, with microenterprises accounting for 87% of all businesses in the country (Government of Saudi Arabia, 2024). MSMEs in Saudi Arabia are classified based on employee numbers and annual income. Micro enterprises have 1-5 employees and earn less than SAR3 million annually. Small enterprises employ 6-49 staff with annual revenues between SAR3 and SAR50 million. Medium-sized enterprises have 50-249 employees, with annual incomes from SAR50 million to SAR200 million (Monsha'at, 2023). These enterprises contribute substantially to job creation and economic diversification, aligning with the broader goals of Vision 2030. Recent data from General Authority for Statistics (2024a) show a notable shift in Saudi Arabia's economy, with a 4.9% growth in the non-oil sector from 2023 to 2024.

In contrast, oil-related activities decreased by 8.9% during the same period. This trend underscores diversification efforts within the economy, which is central to this study’s focus on the evolving business landscape. Figure 1.3 illustrates the annual Gross Domestic Product (GDP) in billions from 2010 to 2019, segmented by the oil, non-oil, and government and private sectors, revealing sector-specific growth patterns and economic diversification over the decade. Understanding these dynamics is crucial for assessing their impact on local entrepreneurial ecosystems.

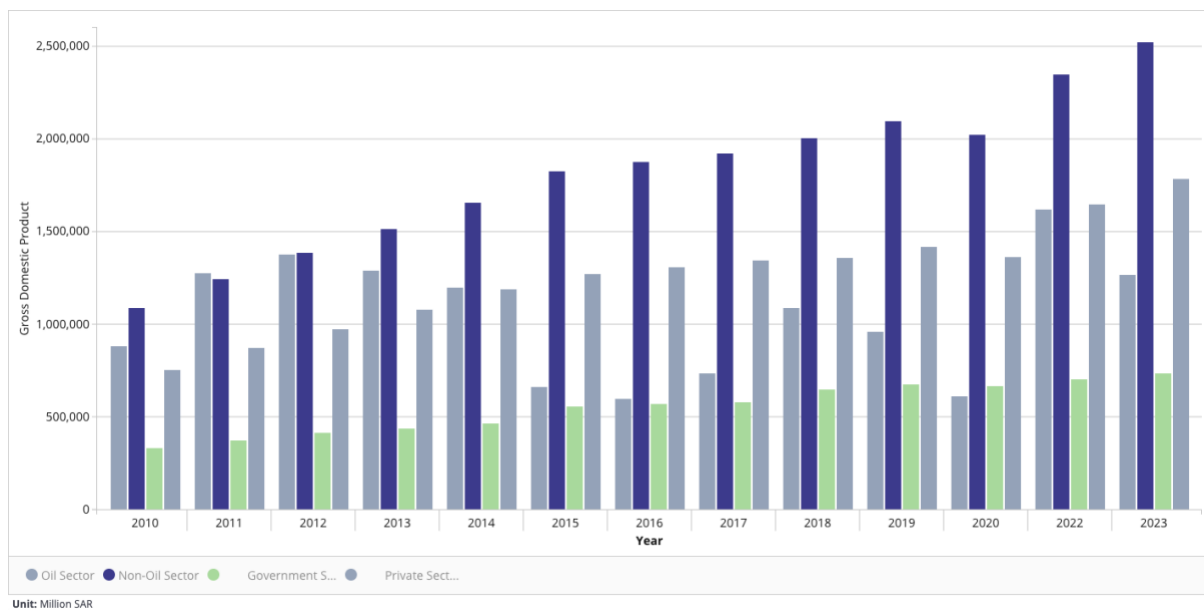


Figure 1.3: Saudi GDP at Current Price by Institutional Sectors, 2023 (General Authority for Statistics, 2024b).

1.3.4 Vision 2030 and the Entrepreneurship Ecosystem

This section provides an overview of Saudi Arabia’s current entrepreneurship ecosystem in relation to Vision 2030. It discusses the role of Vision 2030 in shaping EE for fashion MSMEs by focusing on several key domains: entrepreneurship education, talent development, support structures, finance, government policies, culture, and market access, including both advancements and challenges.

Vision 2030 has three primary themes: an ambitious nation, a thriving economy, and a vibrant society. Each theme encompasses numerous initiatives designed to enhance citizens’ quality of life, stimulate economic growth, and improve government efficiency and transparency. According to the Vision 2030 framework, “Saudi Arabia is determined to become a global investment powerhouse” and aims to establish “a strong foundation for

economic diversification and entrepreneurship” (Government of Saudi Arabia, 2024). Central to this vision is fostering entrepreneurship, which is described as ‘a critical driver of economic diversification and innovation in Saudi Arabia’ (Alharthi et al., 2019; Government of Saudi Arabia, 2024; Nurunnabi, 2017).

To measure progress in fostering a supportive entrepreneurial environment, the National Entrepreneurship Context Index (NECI), developed by the Global Entrepreneurship Monitor (GEM), offers a globally recognised and comprehensive framework. The NECI consolidates expert evaluations into a single numerical score out of 10, reflecting the overall conditions that support entrepreneurship (Akulava et al., 2020; Global Entrepreneurship Monitor, 2024b; Hill et al., 2023; Rietveld & Patel, 2022). This score is based on assessments of 13 pillars grouped into nine key areas:

1. Financing for Entrepreneurs
2. Government Policies
3. Government Programs
4. Education and Training
5. Research and Development (R&D) Transfer
6. Commercial and Legal Infrastructure
7. Internal Market Dynamics
8. Physical Infrastructure
9. Cultural and Social Norms

According to the Global Entrepreneurship Monitor (2024a), Saudi Arabia’s entrepreneurship ecosystem has undergone significant advancements, particularly in 2023. These developments are underpinned by evolving government initiatives which encourage societal perceptions that increasingly recognise entrepreneurship as a desirable career pathway and a critical driver of economic growth. Their findings indicate that 90% of Saudi adults report knowing someone who has recently started a business, the highest proportion among GEM-participating economies. Another 93% of respondents perceived good opportunities to start a business in their local area. Furthermore, 91% express confidence in their ability to establish a business, reflecting a supportive societal environment for entrepreneurial activity.

In 2023, Saudi Arabia’s NECI score reached 6.3, placing the country among the top three globally in the GEM rankings. This marks a notable increase from a score of 4.5 in 2016, reflecting substantial advancements across the critical domains of the EE in the country. Saudi

Arabia’s ranking second among the 15 high-income countries underscores the robustness of its entrepreneurial environment and the effectiveness of policy interventions under Vision 2030 (Table 1.1).

Table 1.1: NECI Ranking of 16 High-Income Economies (Global Entrepreneurship Monitor, 2023)

1	United Arab Emirates	7.68
2	Saudi Arabia	6.32
3	Qatar	6.03
4	Netherlands	5.94
5	Republic of Korea	5.85
6	Switzerland	5.52
7	France	4.91
8	Canada	4.85
9	Slovenia	4.83
10	United States	4.82
11	Germany	4.81
12	Norway	4.81
13	Sweden	4.73
14	Luxembourg	4.64
15	United Kingdom	4.64
16	Italy	4.5

As shown in Figure 1.4, Saudi Arabia’s highest indicators are sufficient financing for entrepreneurs and physical infrastructure, where the scores align closely with or surpass the average for high-income economies (Global Entrepreneurship Monitor, 2024b). This suggests that Saudi Arabia has made notable progress in establishing financial support mechanisms and developing robust infrastructure to facilitate entrepreneurial activities. Strong financing performance reflects the increased availability of capital and investment opportunities, which are critical enablers of new ventures.

In contrast, the lowest indicators were observed for entrepreneurial education at the school stage, entrepreneurial education at the post-school stage, and internal market dynamics. Low scores for entrepreneurial education highlight significant gaps in the educational system’s capacity to equip individuals with the necessary entrepreneurial skills and mindsets at all stages. Despite these improvements, this weakness suggests that current programs remain largely theoretical and lack practical applications and alignment with market demand. Furthermore, poor performance in internal market dynamics indicates limited opportunities and competitiveness in domestic markets, which can hinder entrepreneurial growth. Appendix A, p.329 provides additional information, including the ratings for the individual NECI pillars across 16 high-income economies. The following section examines each domain of the

entrepreneurship ecosystem in detail.



Figure 1.4: Ratings for Individual NECI Pillars for Saudi Arabia, and Average for High-Income Economies in 2023 (Global Entrepreneurship Monitor, 2024a)

However, while these metrics suggest broad ecosystem progress, they may not fully reflect persisting foundational weaknesses. Despite Saudi Arabia's strong performance in the NECI rankings, this high score conceals several structural challenges that continue to affect the effectiveness of the EE (Coduras et al., 2023), particularly in areas such as entrepreneurial education and internal market dynamics. Studies indicate entrepreneurship education remains underdeveloped, with university curricula and teaching methods often misaligned with labour market demands, limiting the practical application of entrepreneurship learning outcomes (Abdullateef et al., 2023). Additional studies confirm that there is a disconnect between higher education outcomes and job market needs, especially in preparing graduates for innovation-driven entrepreneurship (Elmonshid & Sayed, 2024; Khoirunnisa & Nurhaliza, 2024). Moreover, market dynamics in Saudi Arabia continue to present constraints, particularly for SMEs, due to barriers like limited internal competition and underdeveloped support infrastructure for business growth (Bateh, 2020). These issues persist despite government efforts to streamline regulatory environments and stimulate entrepreneurial activity. Nonetheless, the country's overall NECI performance reflects significant improvements in

other ecosystem pillars such as access to funding and regulatory efficiency brought about through Vision 2030 initiatives. These advancements demonstrate that while progress is being made, a comprehensive and targeted effort is still needed to reconcile high-level competitiveness scores with the foundational realities of entrepreneurial practice (Meeralam & Adeinat, 2021; Elmonshid & Sayed, 2024).

1.3.5 Profile of Saudi Entrepreneurs in the Fashion Sector

Saudi fashion entrepreneurs have become central actors in the Kingdom's economic diversification efforts, leveraging a fusion of cultural heritage, global fashion influences, and digital technologies. Despite their growing significance, academic literature on their comprehensive demographic and professional profiles remains limited, necessitating further empirical investigation. Demographically, fashion entrepreneurs in Saudi Arabia are predominantly young, with approximately 67% under the age of 35, reflecting the nation's youthful population structure (General Authority for Statistics, 2023). The sector is also markedly female-driven. Recent findings report that over 90% of surveyed MSMEs in fashion are led by women, indicating the positive impact of post-2018 reforms that expanded women's participation in the labour market (Alshibani et al., 2024). In fact, women now lead nearly 45% of all SMEs in the fashion industry, supported by targeted government initiatives such as Monsha'at's Women Entrepreneurship Centres (Miralam et al., 2025).

In terms of education, many fashion entrepreneurs possess university degrees in fashion design, business, or marketing often obtained from international institutions which informs their global brand positioning and cross-cultural market strategies (Alblowi & Henninger, 2024). However, national curricula often lack adequate training in entrepreneurship and management, resulting in skill gaps that hinder business growth and operational scalability (Alahmari & Alkulaib, 2024; Abdelwahed et al., 2022). Motivational drivers for entering the fashion sector include creative autonomy, economic independence, and a desire to contribute to cultural preservation and modernization (Alblowi & Henninger, 2024). Many entrepreneurs seek to reinvent traditional garments such as abayas and thobes by blending them with contemporary styles, a strategy aligned with national cultural branding goals (Fashion Commission, 2023).

Despite their creativity and market potential, entrepreneurs often face constraints in scaling their businesses beyond home-based models. These include limited access to capital, warehousing, and formal retail infrastructure, particularly among women-owned firms (Maghrabi & Abalkail, 2024; Alrefai & Hareesh, 2024).

Sustainability is becoming an increasingly important component of entrepreneurial identity in this sector. Around 30% of new Saudi fashion brands now adopt sustainable practices such as ethical sourcing, circular design, and reliance on local materials (Alshibani et al., 2024; Boston Consulting Group, 2023). Entrepreneurs frequently integrate sustainability messaging through symbolic product design and storytelling, reflecting broader global fashion trends (Alblowi & Henninger, 2024). However, sustainability is not yet a universal business priority. For many early-stage entrepreneurs, sustainable practices remain secondary to profitability unless driven by strong consumer demand or regulatory incentives (Abdelwahed et al., 2022).

1.3.6 The Saudi Fashion Entrepreneurship Ecosystem (FEE)

The Saudi FEE has undergone significant transformation since the establishment of the Ministry of Culture in 2018 and the subsequent launch of the Fashion Commission. These initiatives align with Saudi Arabia's Vision 2030 strategy, which prioritises economic diversification and positions creative industries, particularly fashion, as cornerstones of development. By integrating traditional crafts with modern design principles, these efforts aim to enhance the global competitiveness of Saudi MSMEs while preserving the nation's rich cultural heritage (Ministry of Culture, 2023, 2024).

To support creative industries, the Cultural Development Fund was established to generate an annual revenue of SAR4.6 billion and create 70,000 jobs by 2030. Initiatives such as the Saudi 100 Brands Program provide local designers with essential resources, training, and opportunities for international exposure to the global market. For example, Saudi designers showcased innovative collections that blended traditional and contemporary elements at the Paris Fashion Week in 2024 (Ministry of Culture, 2023, 2024). Additionally, collaboration with international organisations, such as the Memorandum of Understanding between the Fashion Commission and Adidas, has strengthened capacity-building efforts through joint projects and research initiatives, elevating the global presence of Saudi fashion (Ministry of Culture, 2023, 2024).

The sector is projected to grow annually by 11.62% between 2024 and 2029, reaching a market value of approximately US\$7.57 billion by 2029 (Statista, 2024). This growth is driven by a young population, fashion-conscious consumers, high disposable income, and a stable economy (MarketLine, 2020). By 2030, the anticipated influx of over 150 million tourists annually is expected to boost demand for fashion products, creating new opportunities for local

designers (Ministry of Tourism, 2024). Events such as Riyadh Fashion Week and Saudi Seasons showcase local talent and foster retail and fashion sector growth, while digital transformation facilitates the overcoming of traditional distribution barriers and the expansion of access to markets through e-commerce and digital platforms (Euromonitor International, 2023; Ministry of Culture, 2023). Digital transformation has also played a pivotal role in enabling fashion entrepreneurs to overcome traditional distribution barriers. The rise of e-commerce and digital platforms has expanded market access for local designers both domestically and internationally. This shift underscores the importance of evaluating infrastructural and regulatory reforms that have facilitated these advancements (Roomi et al., 2023).

Despite the positive trajectory of the FEE, critical structural challenges continue to undermine its ability to foster inclusive entrepreneurial growth. A major barrier is the sector's heavy reliance on imported raw materials and finished products, which limits local manufacturing infrastructure, particularly in textile and apparel production, further constrains sector competitiveness (Maghrabi & Abalkail, 2024). These dependencies create high entry barriers and discourage financially constrained start-ups from entering the market (Ministry of Culture, 2023; Elmonshid & Sayed, 2024).

Persistent difficulties in accessing finance continue to limit their ability to scale operations or adopt innovative technologies. While the market potential and economic benefits of the Saudi fashion industry are evident, inadequate fashion education in Saudi Arabia is a significant barrier to realising this potential. The education system has developed significantly in recent years, mainly due to government initiatives aligned with Vision 2030 that emphasise promoting cultural and creative industries, including fashion (Mitchell & Alfuraih, 2018).

However, there remains a significant disconnect between university education and entrepreneurial preparedness. Currently, 12 public universities offer Bachelor programs in fashion design, and five institutions provide postgraduate studies in fashion design and manufacturing (Ministry of Culture, 2023). The government has allocated SAR195 billion toward educational development, including scholarships to prestigious international universities, to enhance technical skills and improve labour market integration (Ministry of Finance, 2024).

The absence of globally aligned faculty and industry partnerships within Saudi fashion education limits students' ability to acquire international business knowledge. A key issue is the lack of internationally trained faculty, which limits the ability of educational programs to provide students with global perspectives and technical expertise needed in the fashion industry

(Othayman et al., 2021). Additionally, bureaucratic hurdles, such as delays in accreditation processes, impede the pace of educational reform (Othayman et al., 2022). Compounding these issues are heavy teaching workloads and insufficient professional development opportunities for academic staff, which strain universities' capacity to innovate (Eid et al., 2023). To address these issues, the Saudi government initiated strategies such as the King Abdullah Scholarship Program and curriculum improvements (Alfawaz et al., 2014).

Fashion education is increasingly incorporating entrepreneurship into its curriculum. For instance, nations such as Australia and the United Kingdom have made substantial investments in fashion entrepreneurship education by integrating industry partnerships, internships, and international collaborations into their education and training programs (Lazzeretti & Capone, 2020; Shi et al., 2012). These countries serve as exemplars of how Saudi institutions can enhance their programs by providing entrepreneurial training that fosters business creation and innovation. In contrast, Saudi Arabia remains predominantly focused on design and technical skills (Alahmari & Alkulaib, 2024; Alajaji & Alqudairi, 2018). This creates a significant gap in entrepreneurial training for graduates seeking to navigate the complexities of the fashion industry.

This educational misalignment reflects a broader systemic issue within the FEE. Saudi fashion graduates often find themselves ill-prepared for industry demands because of insufficient entrepreneurial education and practicality (Alzahrani & Kozar, 2017). The fashion sector has not yet fully benefited from the broader entrepreneurial resources available in the country. To address this gap, greater collaboration with the private sector, more industry partnerships, and more international academic exchanges are essential. These collaborations could integrate entrepreneurship into fashion education programs, enabling graduates to navigate the competitive global market and contribute meaningfully to Saudi Arabia's creative economy. Comprehensive vocational training that combines technical expertise with entrepreneurial skills is crucial for producing graduates capable of meeting fashion industry demands both locally and globally (McMahon, 2012). From a workforce perspective, the sector lacks skilled professionals in key areas such as design, production, and supply chain management. Although initiatives to establish specialised educational and training programs are underway, these efforts must be scaled up to meet the growing demand for expertise in the industry (Ministry of Culture, 2023).

Although numerous initiatives exist, they are often undermined by bureaucratic inefficiencies that limit their ability to grow and compete effectively (Maghrabi & Abalkail, 2024). These challenges include limited access to finance, narrow market opportunities,

inadequate business networks and insufficient entrepreneurial training (Alrefai & Hareesh, 2024). Additionally, the Saudi fashion industry struggles with limited international exposure and market expertise (Almahdi, 2020). Insufficient financial support remains a significant obstacle for fashion entrepreneurs. The Saudi government has implemented programs such as the Kafalah program and Centennial Fund to provide financial support to small and medium-sized enterprises (Altokhais, 2018). However, these initiatives are often inadequately organised, leaving significant gaps in critical areas such as mentorship and resource availability (Al Mulla, 2022; Alshibani et al., 2024). However, bureaucratic inefficiencies often limit the effectiveness of such programs. For example, Al Mulla (2022) reported that while funding programs may be well funded, their impact is often reduced by disorganisation and a lack of strategic oversight.

While sustainability is an emerging priority, many fashion entrepreneurs lack the financial and technical resources to adopt eco-conscious practices. Many are eager to incorporate sustainable practices into their fashion businesses, in alignment with global trends and Vision 2030 goals (Government of Saudi Arabia, 2024; Moshashai et al., 2020). The global emphasis on sustainable production practices in the fashion industry is driven by efforts to reduce environmental impact, enhance ethical supply chains, and promote circular economy models (Back, 2017; Gwilt et al., 2019). However, Saudi fashion enterprises face increasing pressure to adopt eco-friendly technologies and processes, often requiring substantial financial investment (Alshibani et al., 2024; Chalhoub Group, 2021; Euromonitor International, 2023). Moreover, competition from well-established international brands with significant financial resources and marketing expertise continues to overshadow local designers, limiting their ability to establish strong market positions both domestically and globally (Euromonitor International, 2023).

Despite their entrepreneurial ambitions and cultural value, Saudi fashion entrepreneurs often operate within an ecosystem that inadequately supports their growth. The following section evaluates the Saudi FEE and explores its limitations in stimulating broader entrepreneurial participation. In summary, while the FEE reflects substantial policy ambition, it continues to fall short in enabling widespread participation due to systemic barriers in finance, education, sustainability, and institutional design. Addressing these gaps is essential to unlock the full potential of fashion entrepreneurship in Saudi Arabia (Elmonshid & Sayed, 2024).

1.4 Research Problem

The Saudi fashion industry is undergoing rapid growth, largely driven by the economic diversification goals of Vision 2030. However, this expansion faces persistent structural challenges that constrain the industry's development and its competitiveness in international markets. These challenges include fragmented and inconsistent support frameworks, as well as limited access to key resources such as financial capital, skilled labour, and advanced technological infrastructure (Alshibani et al., 2024; Maghrabi & Abalkail, 2024).

These structural deficiencies raise fundamental questions about the current configuration and effectiveness of the Saudi FEE. Of particular interest is how this ecosystem supports or fails to support the development of EO, which is widely recognised as a key driver of business competitiveness and adaptability, especially in emerging and dynamic (Putria & Gunawan, 2019; Rufaidah, 2019). EO, typically expressed through innovativeness, proactiveness, and risk-taking, enhances a firm's capacity to identify market opportunities, innovate, and build sustainable competitive advantages (Akpoviro & Akanmu, 2021; Soares & Perin, 2019).

To better understand the relationship between EE support and entrepreneurial performance, this study draws on the Resource-Based View (RBV) of the firm. From an RBV perspective, EO is treated as an intangible strategic resource—valuable, rare, inimitable, and non-substitutable (Barney, 1991; Wernerfelt, 1984). This framing allows us to interpret the resource gaps faced by Saudi fashion MSMEs as limitations in their capacity to acquire, develop, and deploy EO effectively. EO acts as a mediating mechanism, transforming external ecosystem inputs into enhanced business performance (Mahmoud et al., 2023; Ferreira et al., 2011). By applying RBV, this study conceptualises EO not merely as an outcome of entrepreneurial behaviour, but as a core capability through which ecosystem resources are internalised and strategically leveraged.

In addition to RBV, this study employs a multilevel analysis framework to better understand the layered and systemic nature of the Saudi FEE. Multilevel analysis enables a disaggregated examination across three dimensions: the macro level (e.g., policies, regulations, societal culture), the meso level (e.g., business networks, support institutions), and the micro level (e.g., individual EO and firm behaviour) (Chandler & Lyon, 2001; Davidsson & Wiklund, 2001; Autio et al., 2014). This approach offers a more complete understanding of how structural factors interact with entrepreneurial agency to shape business outcomes.

Despite the sector's accelerated development, there remains a clear lack of empirical research on how the Saudi FEE influences EO and BP, particularly among MSMEs in the fashion industry. This gap hinders the development of tailored interventions, policies, and

institutional mechanisms that could support long-term growth in the sector.

While research on entrepreneurial ecosystems has grown globally, it continues to be primarily shaped by Western theoretical models and often fail to reflect the socio-cultural, institutional, and gendered specificities of non-Western contexts such as Saudi Arabia (Haddadi, 2023; Kemppainen, 2019; Miralam et al., 2025). They commonly overlook cultural dimensions such as gender norms, religious frameworks, and the local understanding of legitimacy factors that significantly shape EO in the Saudi context (Algumzi, 2022; Coduras et al., 2023; Aljarodi et al., 2024; Meeralam & Adeinat, 2021).

A particularly salient gap exists in relation to the cultural divide between traditional tailoring and modern fashion design. Tailoring, often performed by expatriates and viewed as low-status, domestic work, is frequently contrasted with fashion design, which is increasingly associated with innovation, prestige, and global aesthetics (Alajaji & Alqudairi, 2018; Alshibani et al., 2024). This dichotomy shapes not only how fashion ventures are socially perceived but also how entrepreneurs construct brand identity and seek legitimacy in a culturally conservative environment (Algohari et al., 2025; Algumzi, 2022). Although Vision 2030 has prioritised the development of creative industries, few empirical studies have explored how Saudi entrepreneurs navigate these cultural tensions or reconcile traditional values with aspirations for global competitiveness (Khoirunnisa & Nurhaliza, 2024; Haddadi, 2023; Anderson & Almuzel, 2020). Without this contextual lens, policies and education frameworks risk failing to address the lived realities of Saudi entrepreneurs (Algohari et al., 2025; Elnadi & Gheith, 2021).

Moreover, much of the literature treats culture as a passive backdrop, rather than as an active force shaping entrepreneurial identity, legitimacy, and strategic decision-making (Prado et al., 2024; Donaldson, 2020). This oversight is particularly critical in Saudi Arabia, where fashion entrepreneurship is highly gendered and women entrepreneurs must navigate unique social and institutional constraints (Alshibani et al., 2024; Maghrabi & Abalkail, 2024). This thesis addresses these theoretical and contextual gaps by applying a culturally grounded, multi-level framework. It explores how Saudi fashion entrepreneurs interpret, adapt to, and at times challenge dominant cultural narratives. This approach provides a more nuanced understanding of the Saudi fashion entrepreneurship ecosystem and contributes to the development of more inclusive and context-specific policy and support mechanisms.

Accordingly, this study aims to systematically examine the Saudi FEE, identifying its key elements and their impact on EO and BP among fashion enterprises. In doing so, it contributes to both theoretical advancement in entrepreneurship research and practical strategies for

strengthening the creative economy in Saudi Arabia.

1.5 Rationale for the Research

This study was undertaken for three interrelated reasons: (1) to explore the emerging field of fashion entrepreneurship in Saudi Arabia, (2) to understand the configuration of the Saudi FEE within its specific sociocultural and institutional context, and (3) to apply an integrated theoretical framework combining the RBV and multilevel analysis to explain how ecosystem elements shape EO and BP in fashion MSMEs. These aims respond directly to the national economic agenda articulated in Vision 2030, which calls for structural diversification, entrepreneurship development, and the expansion of the creative economy (Government of Saudi Arabia, 2024).

First, the fashion sector is increasingly recognised as a key pillar of Saudi Arabia's creative economy. However, despite its potential to contribute to economic diversification, the industry remains underdeveloped, facing challenges such as heavy reliance on imports, limited local production capacity, and the dominance of international brands (Abu Nab, 2019; Murad & Raju, 2019; Alshibani et al., 2024). As Alzahrani (2012) notes, the full economic potential of the Saudi fashion industry cannot be realised without the development of globally competitive local firms. In comparison, mature fashion economies such as the United Kingdom have demonstrated the significant economic impact of a well-supported fashion industry—contributing £28.9 billion and over 800,000 jobs in 2024 alone (British Fashion Council, 2024). This study seeks to identify factors within the FEE that could similarly enable Saudi fashion MSMEs to scale and compete globally.

Second, studying the Saudi FEE is critical because of the context-dependent nature of EEs. Research suggests that each EE develops within its own sociocultural, economic, and political framework, making the direct replication of strategies from other regions unlikely to be effective (Isenberg, 2011a, 2011b). Therefore, insights from established EEs in Western Europe and other global hubs cannot be directly applied to Saudi Arabia because of significant local differences (Ali et al., 2019; Almuzel & Anderson, 2020). Western Europe remains a dominant force in the global fashion sector because of its highly developed market infrastructure and competitive advantages (Gonda & Fekete, 2020; Zhang, 2024).

Comparative research on FEEs in Toronto, Canada (Brydges & Pugh, 2021) and Scandinavian (Colliander, 2015; Salonoja, 2013; Tervilä, 2015), shows that policies, gender norms, and institutional configurations deeply influence entrepreneurial outcomes. For

example, Denmark's ecosystem benefits from government-backed financing and support for women-led businesses (Dahl & Sorenson, 2012). while Sweden's FEE is driven by private venture capital (Salonoja, 2013). Saudi Arabia, in contrast, faces limitations in entrepreneurial education, mentorship, and market access compared to Western European EEs, necessitating policy-driven support and tailored strategies (Almuzel & Anderson, 2020; Content et al., 2020; Schrijvers et al., 2023). Research suggests that entrepreneurship education in Saudi Arabia remains largely theoretical, with limited hands-on training in business strategy, financial management, and global market positioning (Almuzel & Anderson, 2020; Dixit et al., 2024).

In contrast, studies of FEEs in global hubs such as London, Paris, and Stockholm indicate that integrating experiential learning, mentorship programs, and university-industry partnerships may contribute to more effective entrepreneurial training (Henry et al., 2016; Simmons et al., 2024). These differences highlight how sociocultural and economic factors shape EEs and MSME performance, reinforcing the need for tailored approaches to different markets (Purbasari et al., 2019). Consequently, a localised and culturally grounded model of the FEE is essential to address the specific needs of Saudi entrepreneurs and create an enabling environment for fashion businesses.

Third, this research adopts a multilevel analysis framework to address the limitations of previous studies that tend to focus solely on individual-level entrepreneurial traits (e.g., creativity, innovation, risk-taking) (Aakko & Niinimäki, 2018; Hodges et al., 2016; Kusumawardani & Triyono, 2017). Instead, it investigates how entrepreneurial outcomes are shaped by interactions across the micro (EO and firm-level strategy), meso (support institutions, education providers, incubators), and macro (finance, policy, culture) levels (Chandler & Lyon, 2001; Davidsson & Wiklund, 2001; Tyldsley, 2012 Autio et al., 2014). This systemic lens is particularly important in Saudi Arabia's top-down policy environment, where entrepreneurial activity is heavily influenced by government regulation, cultural norms, and institutional design.

For instance, research in China by Zeng (2020) found that ecosystem shortcomings such as weak mentorship and limited funding access not just individual entrepreneurial deficiencies led to high failure rates among fashion ventures. Similarly, Brydges and Pugh (2021) found that mapping Toronto's fashion ecosystem revealed previously overlooked constraints in support structures. Applying a multilevel lens to the Saudi context allows this thesis to uncover both enablers and inhibitors of entrepreneurial activity and provide policy recommendations tailored to each layer of the ecosystem.

In parallel, this study draws on the RBV to theorise how internal firm capabilities interact

with external ecosystem resources. According to RBV, firms achieve sustained competitive advantage through the strategic use of VRIN resources—those that are valuable, rare, inimitable, and non-substitutable (Barney, 1991; Wernerfelt, 1984). In this context, EO comprising innovativeness, proactiveness, and risk-taking is treated as a core intangible resource that enables firms to exploit opportunities and navigate uncertainty (Lumpkin & Dess, 1996; Sarsiti, 2024). EO acts as a mediator through which external ecosystem inputs—such as access to finance, policy support, and entrepreneurial education—are transformed into improved business performance (Ferreira et al., 2011; Hughes & Morgan, 2007).

By linking EO to RBV, this research conceptualises Saudi fashion MSMEs not as passive recipients of ecosystem support, but as strategic actors capable of converting external resources into competitive capabilities. This perspective is particularly useful in understanding why some MSMEs thrive in resource-constrained environments while others fail despite access to similar supports. In doing so, RBV complements the multilevel framework by focusing analytical attention on the strategic behaviours of firms within their structural context.

Together, the RBV and multilevel frameworks enable a holistic exploration of how entrepreneurial ecosystems function in the Saudi fashion sector. They also align directly with Vision 2030's dual mandate: to reform systemic structures (e.g., policy, regulation, education) and to build firm-level capabilities that support innovation and international competitiveness. This integrated approach provides both theoretical contributions to entrepreneurship literature and practical insights for policymakers and industry stakeholders seeking to foster sustainable, inclusive growth in the Saudi fashion industry.

1.6 Research Aim and Objectives (RO)

The overall aim of this study was to develop an effective FEE framework for Saudi Arabia. The proposed framework seeks to establish, promote, and reform ecosystems to enhance the EO and BP of fashion MSMEs. This study comprised two distinct studies.

Study 1 is a quantitative study utilising Partial Least Squares Structural Equation Modeling (PLS-SEM) to examine the direct and indirect effects of the FEE on the EO and BP of fashion MSMEs. Specifically, it addresses the following research objectives.

RO1: Examine the impact of FEE on the EO of fashion MSMEs.

RO2: Examine the direct impact of FEE on the BP of fashion MSMEs.

RO3: To investigate the mediating role of EO in the relationship between FEE and BP of fashion MSMEs.

Study 2 is a qualitative study based on interviews with key stakeholders in the Saudi fashion industry. It focuses on mapping and exploring the current landscape of the Saudi FEE and how each domain influences EO and BP. This study addresses the following research objectives:

RO4: To map the current landscape of FEE in Saudi Arabia.

RO5: To explore the impact of each domain of the Saudi FEE on EO and BP

1.7 Research Questions (RQ)

RQ1: To what extent does FEE affect the entrepreneurial orientation of fashion MSMEs?

RQ2: To what extent does FEE affect the business performance of fashion MSMEs?

RQ3: How strongly does EO mediate the relationship between FEE and the business performance of fashion MSMEs?

RQ4: What is the current landscape of Saudi FEE?

RQ5: How does each domain of the Saudi FEE impact EO and BP?

1.8 Thesis Outline

This thesis is organised into seven chapters. Chapter One (Introduction): This chapter presents an overview of the study, focusing on the Saudi Fashion Entrepreneurship ecosystem and its relevance to Saudi Arabia's economic and cultural landscape. It includes the country profile of Saudi Arabia, an economic overview, and a discussion of Vision 2030, which serves as a blueprint for economic and social reforms. The significance of this research is outlined, emphasising its academic and practical contributions. This chapter also presents the main research questions and objectives for exploring the relationship between the Saudi FEE, EO and BP. Finally, it provides an overview of the research approach and a summary of the subsequent thesis chapters.

Chapter Two (Literature Review and Hypotheses Development): This chapter reviews the key concepts relevant to entrepreneurship, entrepreneurship ecosystems, and the fashion industry, focusing on MSMEs. It begins with an overview of MSMEs as drivers of essential economic development and entrepreneurship. This chapter explores the concepts of EO and BP and examines the factors influencing the success of fashion MSMEs. Additionally, it discusses the unique challenges and opportunities in the Saudi entrepreneurship ecosystem, leading to the development of a conceptual framework that guides this study's analysis. This chapter outlines the study's theoretical framework. This explains the relationship between the fashion

entrepreneurship ecosystem, EO, and the BP of MSMEs. The framework draws on the existing literature to construct a model that will be tested using quantitative and qualitative research methods.

Chapter Three (Methodology): This chapter details the research philosophy, methodology, and study design. It discusses the exploratory sequential mixed methods approach, which integrates both quantitative and qualitative phases of research. This chapter explains the data collection methods, sampling techniques, data analysis procedures, and ethical considerations that guided the research process.

Chapter Four (Quantitative Results and Discussion): This chapter analyses the quantitative data collected in the study and presents the findings related to the research objectives, particularly RO1 (impact of Saudi FEE on EO) and RO2 (impact of Saudi FEE on BP). The data were analysed to explore the relationships between the components of the entrepreneurship ecosystem and their influence on entrepreneurial orientation and business performance among fashion MSMEs.

Chapter Five (Qualitative Findings and Discussion): This chapter explores the findings from the qualitative phase of the research, focusing on the perspectives of fashion entrepreneurs in Saudi Arabia. It presents a thematic analysis based on interviews with entrepreneurs, examining the enabling and constraining factors within the Saudi FEE. This chapter addresses RO3 (the mediating effect of EO) and RO4 (factors enabling or constraining fashion MSMEs), offering profound insights into the experiences of fashion entrepreneurs.

Chapter Six (Conclusion): This chapter summarises the key findings, addresses all research objectives, and offers concluding remarks on the study's academic and practical contributions. It discusses the implications of this research for policymakers, fashion entrepreneurs, and the broader entrepreneurship ecosystem in Saudi Arabia. This chapter concludes with reflections on the study's limitations and recommendations for future research on the Saudi fashion entrepreneurship ecosystem.

1.9 Chapter Summary

This chapter introduces the significance of the fashion industry and MSMEs in diversifying the Saudi economy. This chapter presents the research aim, objectives, and primary research questions, focusing on the impact of Saudi FEE on EO and BP of fashion MSMEs in Saudi Arabia. The rationale for this study was described as being based on the need for a better understanding of the fashion industry in Saudi Arabia, owing to the scarcity of

research on this topic. Hence, this study makes a significant contribution to the limited body of literature on the specific challenges and opportunities faced by fashion MSMEs in developing economies, particularly Saudi Arabia. Moreover, the importance of this research is underscored in the context of Saudi Arabia's Vision 2030, which aims to diversify the economy away from oil dependence while fostering the growth of the non-oil private sector, with a particular emphasis on creative industries such as fashion.

The following chapter presents an overview of the academic literature and government and industry publications on entrepreneurship, entrepreneurship ecosystems, and the Saudi fashion industry. It explores key concepts such as entrepreneurial orientation, business performance, and factors influencing the success of MSMEs in the fashion sector.

CHAPTER 2: LITERATURE REVIEW AND HYPOTHESES DEVELOPMENT

2.1 Chapter Overview

This chapter reviews the body of published literature in four interconnected research streams: entrepreneurship ecosystems (EEs), fashion entrepreneurship ecosystems (FEE), entrepreneurial orientation (EO), and Business Performance (BP). These themes are widely discussed in the literature, as they enhance the growth and competitiveness of MSMEs, making them critical to the development of the fashion industry, particularly in Saudi Arabia.

This review explores EEs, highlighting their systemic nature and key components, such as human capital, policy, culture, market dynamics, support, and finance. It then introduces FEE as an effective framework tailored to the creative and dynamic fashion industry.

In this literature review, EO was explored as a mediating factor between FEE and BP, with its dimensions of proactiveness, innovativeness, and risk-taking playing a critical role in driving business success. This chapter presents a conceptual model grounded in the RBV theory, positioning EO as a strategic capability that leverages external ecosystem factors to improve BP. This model provides the foundation for the empirical analysis in the subsequent chapters. The chapter concludes by summarising the significant findings and identifying the research gaps that this study addresses, particularly in the Saudi context.

2.2 The Entrepreneurship Ecosystem (EE)

The concept of EE has become increasingly significant in published academic research since the beginning of the 21st century (Isenberg, 2010; Maroufkhani et al., 2018; World Economic Forum, 2013). Scholars view entrepreneurship as an ecosystem-driven phenomenon shaped by complex interactions among individuals, institutions, cultures, and environmental factors. This systemic approach represents a shift from earlier research prior to the 1990s, which emphasised individual entrepreneurial traits. Van De Ven (1993) first proposed a systemic view of entrepreneurship, suggesting that entrepreneurial success depends on interactions within diverse networks of ecosystem components. Building on this foundation, recent studies, such as those by Acs et al. (2014), Autio and Levie (2017), and Wadichar et al. (2022) along with Wurth et al. (2023), have highlighted the dynamic and systemic nature of EEs. These studies illustrate the influence of broader socioeconomic, political, and cultural

contexts on entrepreneurial outcomes.

Spigel (2017) and Stam and Spigel (2016) define an entrepreneurship ecosystem as an interconnected network of stakeholders, resources, and conditions that collectively foster innovation, risk-taking, and business growth. Moore's (1993, 1998) foundational work in this field applied ecological principles to business, introducing the concept of an "economic community" in which companies co-evolve capabilities through shared innovation (Moore, 1993; Moore, 1998). This early framework was expanded by researchers such as Spigel (2017), who described EEs as comprising a set of interrelated cultural, social, and material attributes that support entrepreneurial activity and success. Spigel and Harrison (2018) proposed a process-based view, emphasising how EEs, clusters, and regional innovation systems interact to support new venture creation through the evolution and transformation of ecosystem structures. Moreover, Leendertse et al. (2020) developed metrics to quantify and qualify regional economies across the European Union, demonstrating the pivotal roles of formal institutions and physical infrastructure in EEs. Furthermore, Wurth et al. (2023) synthesised the findings from 181 empirical studies, identifying critical mechanisms such as the interdependence of ecosystem elements and processes of "upward" and "downward" causation that influence entrepreneurial outcomes. Their work highlights the dynamic and multifaceted nature of EEs, which is essential for understanding the mechanisms that shape entrepreneurial processes and outcomes in regional economic development strategies.

Key components of EEs profoundly influence entrepreneurial activities, particularly in specific regional or industrial contexts. Isenberg (2010, 2011) identified critical domains, such as leadership, culture, finance, human capital, and policies, that interact to promote entrepreneurship. Similarly, Feld (2020) describes EEs as geographically bound networks of stakeholders, businesses, governments, and civil society collaborating to support new venture creation. This regional perspective underscores the importance of local resources, policies, and cultural settings in shaping EEs (Alarifi et al., 2019; Porter, 2000; Stam & Spigel, 2016). Recent scholarship has highlighted the relevance of EEs in emerging economies, such as Saudi Arabia, where government policies and initiatives, such as Vision 2030, aim to foster entrepreneurship (Alarifi et al., 2019; Cao & Shi, 2021; D'Andrea et al., 2023). The EE framework continues to evolve with the distinction between external and internal factors. De Brito and Leitão (2020) described EEs as an intersection of external (e.g., cultural and institutional) and internal (e.g., entrepreneurial skills and attributes) conditions. Scholars such as Spigel and Harrison (2018) and Wurth et al. (2021) have emphasised the interplay between these elements, illustrating how EEs adapt and evolve over time to changing conditions. These

contemporary developments reflect the dynamic nature of EEs and their ability to adapt to global challenges and opportunities (Elia et al., 2020; Hameed et al., 2022; Kapturkiewicz, 2022; Mago & Van Der Merwe, 2023). This updated understanding of EEs provides a comprehensive and dynamic perspective, emphasising the importance of integrating contemporary insights and regional contexts into future research and practices. Table 2.1 synthesises definitions and components from key scholars, offering insights into how various configurations contribute to entrepreneurial success.

Definitions of EEs vary significantly in their conceptualisation of outcomes, reflecting evolving perspectives in the literature. Foundational works, such as those by Van De Ven (1993) and Isenberg (2010), remain integral to understanding the origins of EE definitions. Van De Ven (1993) early emphasis on innovation as a core outcome, and Isenberg's (2010) "domains of the EE" framework, laid the groundwork for contemporary research, providing baseline definitions that continue to inform and shape the field. By contrast, others focus on "productive entrepreneurship", which emphasises high-growth, high-impact ventures (Stam, 2015). While these older works predate more recent developments, their comprehensive and multidimensional nature ensures their lasting relevance. Subsequent studies, such as those by Audretsch and Belitski (2017), built on these frameworks to focus on systems that foster entrepreneurial opportunities, underscoring the enduring significance of earlier foundational ideas. Recent research complements these foundational theories and introduces new metrics and perspectives to refine our understanding of EEs. For example, Leendertse et al. (2020) argue for measuring EEs based on "productive entrepreneurship", which is a more meaningful indicator of ecosystem success than simply counting new ventures. Similarly, Wurth et al. (2023) provided an updated research program for conceptualising and measuring EEs, aligning with ongoing theoretical advancements. These recent contributions ensure that foundational insights are integrated with contemporary developments to provide a holistic view of entrepreneurship ecosystems.

This thesis adopts Spigel's (2017, p. 50) definition of EE as a combination of social, political, economic, and cultural elements within a region that support the development and growth of innovative start-ups and encourage nascent entrepreneurs and other stakeholders to take the risks of starting, funding, and otherwise assisting high-risk ventures". This definition aligns with this study's focus on the fashion entrepreneurship ecosystem, where innovation and entrepreneurial orientation are essential for growth (Porter, 2000).

Table 2.1: Entrepreneurship Ecosystems: Definitions and Key Components

Author (Year)	Definition	Components or Elements
Moore (1993)	“In a business ecosystem, companies co-evolve capabilities around an innovation: they work cooperatively and competitively to support new products, satisfy customer needs, and eventually incorporate the next round of innovations”.	Companies, innovation, customer needs, cooperative and competitive dynamics
Moore (1998)	“An economic community supported by a foundation of interacting organisations and individuals which produce goods and services of value to customers. Over time, they co-evolve their capabilities and roles and tend to align themselves with the directions set by one or more central companies. The ecosystem leader enables members to move toward shared visions to align their investments and to find mutually supportive roles”. (p. 26)	Organisations, individuals, goods and services, ecosystem leader, shared vision
Cohen (2006)	“An interconnected group of actors in a local geographic community committed to sustainable development through the support and facilitation of new sustainable ventures”. (p. 3)	Local community, sustainable ventures, support, facilitation
Isenberg (2010)	“The entrepreneurship ecosystem consists of a set of individual elements such as leadership, culture, capital markets, and open-minded customers that combine in complex ways”. (p. 44)	Leadership, culture, capital markets, customers
Isenberg (2011b)	“Entrepreneurship ecosystem consists of a dozen or so elements (which we consolidate into six domains including policy, finance, culture, supports, human capital, and markets) that, although they are idiosyncratic because they interact in very complex ways, are always present if entrepreneurship is self-sustaining”. (p. 6)	Policy, finance, culture, support, human capital, markets
Feld (2020)	Four principles for entrepreneurship ecosystems: “1) Entrepreneurs must lead the startup community. 2) The leaders must have a long-term commitment. 3) The startup community must be inclusive of anyone who wants to participate in it. 4) The startup community must have continual activities that engage the entire entrepreneurial stack”. (p. 23)	Entrepreneurs, leadership, inclusivity, continual activities
Qian et al. (2013)	“Those economic, social, institutional and all other important factors that interactively influence the creation, discovery and exploitation of entrepreneurial opportunities”. (p. 561)	Economic factors, social factors, institutional factors
Vogel (2013)	“An interactive community within a geographic region, composed of varied and interdependent actors (e.g., entrepreneurs, institutions and organisations) and factors (e.g., markets, regulatory framework, support setting, entrepreneurial culture), which evolves over time and whose actors and factors coexist and interact to promote new venture creation”. (p. 6)	Entrepreneurs, institutions, organisations, markets, regulatory framework, culture
Mason and Brown (2014)	“A set of interconnected entrepreneurial actors (both potential and existing), entrepreneurial organisations (e.g., firms, venture capitalists, business angels, banks), institutions (universities, public sector agencies, financial bodies) and entrepreneurial processes (e.g., the business birth rate, numbers of high-growth firms, levels of 'blockbuster entrepreneurship', number of serial entrepreneurs, degree of sell-out mentality within firms and levels of entrepreneurial ambition) which formally and informally coalesce to connect, mediate and govern the performance within the local entrepreneurial environment”. (p. 5)	Actors, organisations, institutions, entrepreneurial processes, performance mediation

Table continues...

Table 2.1 Continued

Author (Year)	Definition	Components or Elements
Auerswald (2015)	“An entrepreneurship ecosystem implies cooperative and productive relationships among different organisations. In many countries, these relationships are between startups, established companies, universities, and research institutions. In a vibrant ecosystem, people and ideas flow between these organisations, starting new ventures, joining existing ones, and linking innovations together”. (p. 10)	Startups, companies, universities, research institutions, ideas, people flow
Mack and Mayer (2016)	“Entrepreneurship ecosystems consist of interacting components, which foster new firm formation and associated regional entrepreneurial activities”. (p. 3)	Interacting components, firm formation, regional activities
Stam and Spigel (2016)	“A set of interdependent actors and factors coordinated in such a way that they enable productive entrepreneurship within a particular territory”. (p. 1)	Interdependent actors, factors, coordination
Acs et al. (2017)	“Entrepreneurship ecosystems are fundamentally interaction systems consisting of multiple, co-specialised, yet hierarchically independent stakeholders, many of which may not even know one another”. (p. 64)	Stakeholders, interactions, independence
Audretsch and Belitski (2017)	“Systems of entrepreneurship as institutional and organisational as well as other systemic factors that interact and influence identification and commercialisation of entrepreneurial opportunities”. (p. 2)	Institutional factors, organisational factors, systemic interactions
Autio and Levie (2017)	“Entrepreneurship ecosystems are regionally embedded interaction systems that drive the allocation of resources towards productive uses through the creation and scale-up of new ventures”. (p. 23)	Embedded systems, resource allocation, venture creation and scaling
Spigel (2017)	“Combinations of social, political, economic, and cultural elements within a region that support the development and growth of innovative start-ups and encourage nascent entrepreneurs and other actors to take the risks of starting, funding, and otherwise assisting high-risk ventures”. (p. 50)	Social elements, political elements, economic elements, cultural elements
Sussan and Acs (2017)	“Dynamic institutionally embedded interaction between entrepreneurial attitudes, abilities, and aspirations, by individuals, which drives the allocation of resources through the creation and operation of new ventures”. (p. 55)	Attitudes, abilities, aspirations, resources
Theodoraki and Messegheem (2017)	“The entrepreneurship ecosystem includes three dimensions: actors who form it and their interactions (formal and informal network), physical infrastructure, and culture”. (p. 50)	Actors, interactions, physical infrastructure, culture
Roundy et al. (2018)	“An entrepreneurship ecosystem is a self-organised, adaptive, and geographically bounded community of complex agents operating at multiple, aggregated levels, whose non-linear interactions result in the patterns of activities through which new ventures form and dissolve over time”. (p. 5)	Complex agents, non-linear interactions, patterns of activities
Lux et al. (2020)	“A set of inter-related actors and components, within a geographic region, that enable productive entrepreneurship”. (p. 1012)	Actors, components, geographic region
Cao and Shi (2021)	“A community of multiple co-evolving stakeholders that provides a supportive environment for new venture creations within a region”. (p. 75)	Co-evolving stakeholders, supportive environment, new venture creation

Viewing FEE as a network that supports innovation and risk-taking, this study examines how specific EE components, such as supportive policies, access to finance, and a strong cultural emphasis on creativity, contribute to entrepreneurial outcomes in the fashion sector (Gupta, 2024a; Porter, 2000; Spigel, 2017; Zeng et al., 2024).

The concept of EEs encompasses a variety of terms and frameworks discussed in scholarly literature that highlight different dimensions of environments conducive to fostering entrepreneurship. A review of the literature reveals multiple terminologies in use that have similar meanings, such as business ecosystems (Harraf et al., 2018; Moore, 1993), focusing on competitive and cooperative business networks and innovation ecosystems (Autio et al., 2014), emphasising tech-driven collaboration, but which may not address broader entrepreneurial needs. Startup Ecosystems (Stam, 2015) focus on early-stage venture support, whereas conducive (Isenberg, 2010) and Supportive Environments (Spigel, 2017) cover socioeconomic conditions and resource networks that enable entrepreneurship. Additionally, the External Environment (Spigel, 2017) captures the influence of market conditions, support, and talent access, although this focus risks undervaluing the internal ecosystem interactions. By defining these terms and explaining their limitations, as shown in Table 2.2, this study situates EEs within a broader academic context, establishing a foundation for rigorous analysis.

Table 2.2: Key Ecosystem Terminologies in Entrepreneurship Identified in the Literature

Terminology	References	Meaning	Limitations Compared to EEs
Business Ecosystems	Moore (1993)	Interconnected networks where businesses create mutually beneficial environments with competition and cooperation	Often limited to business-to-business relationships within industries, potentially overlooking broader entrepreneurial needs, such as diverse funding and policy support
Innovation Ecosystems	Autio et al. (2014)	Environments that drive technological advancement and product development through collaboration	Primarily focused on tech-driven innovation, which may exclude non-tech sectors and the specific needs of diverse entrepreneurs.
Startup Ecosystems	Stam (2015)	Structures designed to support early-stage ventures, including incubators, accelerators, and venture capital	Narrow focus on startups may miss long-term ecosystem dynamics and exclude support for established businesses
Conducive Environments	Isenberg (2010)	Socioeconomic conditions and policy support that favour entrepreneurship	Broad socioeconomic view lacks detailed interaction structures among ecosystem stakeholders, such as networking and mentorship.
Supportive Environments	Spigel (2017)	A comprehensive network of resources, guidance, and support available to entrepreneurs	Broad focus on support can miss critical factors such as policy, regulation, and cultural influence on entrepreneurship.
External Environment	Spigel (2017)	Emphasis on external factors such as market conditions, talent, and regulatory support	Over-reliance on external resources may undervalue internal ecosystem interactions, such as collaboration among local entrepreneurs.

2.3 Criticisms of Entrepreneurship Ecosystem Research

Recent research has highlighted several criticisms of EE regarding its theoretical and empirical foundations. One of the primary criticisms is the lack of a cohesive analytical framework. Many EE studies lack a structured model for analysing components and interactions within ecosystems, limiting clarity and comparability across different settings (Alvedalen & Boschma, 2017; Stam & Spigel, 2016). This inadequacy has led to conceptual ambiguity, as the rapid adoption of the EE concept has resulted in diverse definitions and inconsistent applications, often oversimplifying the field in policy discussions (Autio et al., 2014; Brown & Mason, 2017; Mack & Mayer, 2016). Another central issue is the methodological limitations of the study. Many studies rely heavily on descriptive case studies that focus on successful ecosystems, limiting insights into ecosystems facing unique challenges or those in their early-stages of development (Cavallo et al., 2019; Mason & Brown, 2014; Stam & Spigel, 2016). This emphasis on qualitative approaches can produce subjective insights, reducing generalisability (Acs et al., 2014; Brown & Mason, 2017). Additionally, research often highlights the positive elements of ecosystems, such as support networks and funding, while frequently overlooking negative aspects such as competition, inequality, and failure rates (Mason & Brown, 2014; Wright et al., 2017). This selective focus creates an overly optimistic perspective of EEs. Moreover, EE frameworks are further criticised for their static perspective, often providing only a snapshot rather than capturing the ongoing evolution of ecosystems. Current models rarely address the dynamic interactions and feedback loops that characterise EEs, thus lacking an understanding of how ecosystems develop and transform over time (Mack & Mayer, 2016; Spigel & Harrison, 2018; Stam, 2015). However, scholars have argued that process-oriented approaches capture these complex dynamics (Alvedalen & Boschma, 2017; Cavallo et al., 2019).

Another significant criticism of the literature is the lack of contextual and comparative analyses between developed and developing countries. Most EE research has been conducted in advanced economies, with a limited focus on emerging economies and regions that often face distinct resource constraints and institutional challenges (Audretsch & Belitski, 2017; Brown & Mason, 2017; Wright et al., 2017). Without broader contextual perspectives, the findings may not be applicable to diverse economic settings, thus impeding the development of universally applicable frameworks (Autio et al., 2014; Isenberg, 2011b). Finally, measurement challenges and limited practical relevance continue to pose problems in EE research. The absence of standardised metrics for evaluating EE performance complicates comparisons and limits the reliability of the findings (Leendertse et al., 2020; Malecki, 2018;

Spigel, 2017). This lack of measurement rigor makes it difficult for policymakers to generate actionable insights, especially in regions where resources and institutional support differ from those in developed economies (Spigel, 2017; Wright et al., 2017).

In response to these critical assessments, scholars have advocated for interdisciplinary, theory-driven approaches to address the diverse and evolving needs of entrepreneurs across ecosystems. This involves integrating network theory insights (Alvedalen & Boschma, 2017; Autio et al., 2014), exploring causal relationships (Spigel & Harrison, 2018; Stam & Van De Ven, 2021), and adopting comparative models that better capture the complex theoretical frameworks of entrepreneurship ecosystems (Cavallo et al., 2019; Malecki, 2018).

2.4 Entrepreneurship Ecosystem Frameworks

This section offers a comprehensive discussion of EE frameworks, focusing on their theoretical foundations, core components, and diverse factors influencing their evolution. EE frameworks aim to capture the complex, interrelated elements that support entrepreneurial activity and foster innovation within specific regions (Maroufkhani et al., 2018; Purbasari & Rahayu, 2019). Since the early 1990s, several influential frameworks have emerged, each highlighting different aspects of entrepreneurship ecosystems based on theoretical perspectives and empirical observations (Dai et al., 2014; Mason & Brown, 2014).

There are four prominent EE frameworks, proposed by Isenberg (2011a), World Economic Forum (2013), Vogel (2013), and Stam and Van De Ven (2021). These frameworks have shaped the debate on entrepreneurship ecosystems but differ in focus, ranging from cultural and regulatory factors to the local context. The following section compares these methods and highlights their strengths, weaknesses, and practical applications. Van de Ven's (1993) framework identifies four key elements supporting new firm creation: Institutional Arrangements (regulatory policies), Public Resources (funding and infrastructure), Market Demand (consumer-driven opportunities), and Business Ownership (legal structures). While it effectively highlights industrial infrastructure, it overlooks cultural and social dimensions, which are crucial in contemporary ecosystem models (Van De Ven, 1993).

2.4.1 *Domains of the Entrepreneurship Ecosystem by Isenberg (2011)*

Daniel Isenberg (2011a), who founded the Babson E-Ecosystem Project (BEEP), developed an EE Framework composed of elements grouped into six domains: Policy, Culture, Finance, Markets, Support, and Human Capital (see Figure 2.1). This model emphasises the interconnected nature of these domains, each playing a unique role in supporting

entrepreneurial activities. Policy involves establishing a conducive regulatory environment in which government leadership provides stability and support rather than restrictive oversight. Culture focuses on societal attitudes toward risk and entrepreneurship, and a positive cultural outlook can inspire and sustain entrepreneurial initiatives. Finance highlights the importance of accessible funding sources, such as venture capital and angel investors, which are crucial for startup growth. For example, angel investors invest capital to fund promising early-stage businesses that need a startup boost. In return, these investors receive equity stakes in the company (Isenberg, 2010). Markets stress the need for such investors, along with networks, early customer support, and market access to validate new ventures and create opportunities. Support includes infrastructure, mentorship, and professional services that assist entrepreneurs at different stages of their journeys. Finally, human capital involves building a skilled workforce and educational systems that nurture entrepreneurial talent and innovative thinking (Isenberg, 2011a; Isenberg, 2010).

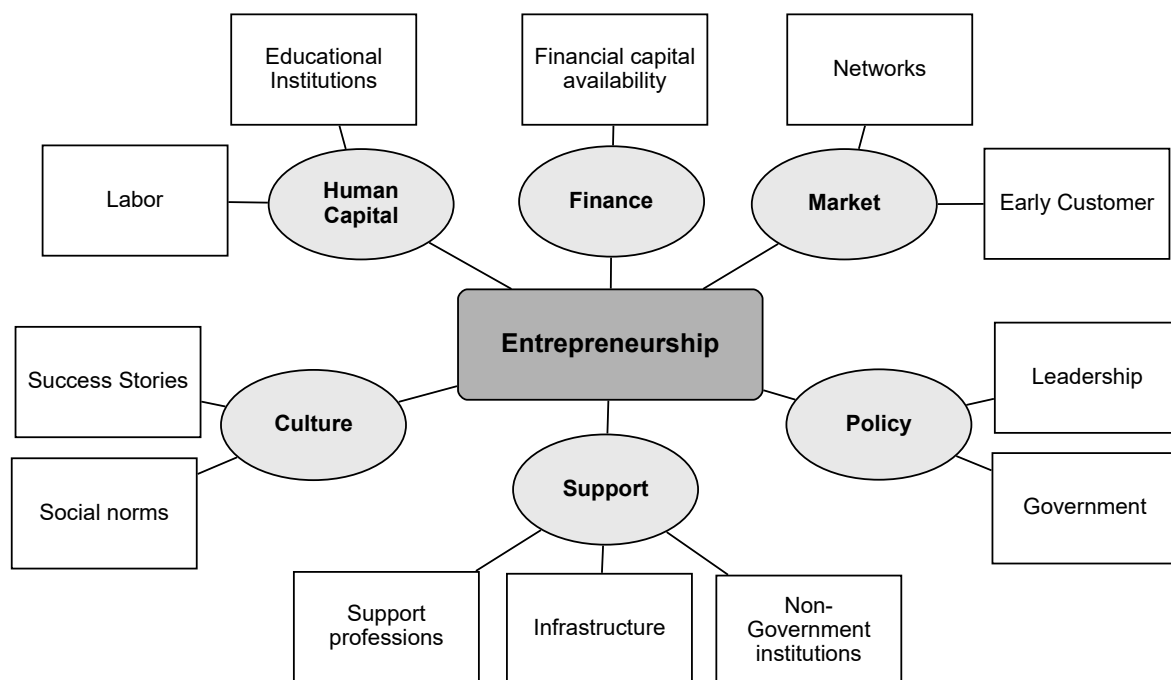


Figure 2.1: Isenberg's Entrepreneurship Ecosystem Framework (Isenberg 2011a, p. 7)

Several studies have employed this framework to map EE in Saudi Arabia by examining how various ecosystem factors influence entrepreneurial activities (Ali et al., 2019; Almuzel & Anderson, 2020; Balawi, 2021; Khan, 2013). Similarly, in the fashion industry, Brydges and Pugh (2021) and Colliander (2015) utilise Isenberg's framework to examine the structure and dynamics of the entrepreneurship ecosystem, underscoring the distinct challenges and opportunities faced by fashion entrepreneurs.

2.4.2 WEF Entrepreneurship Ecosystem Framework by the World Economic Forum (2013)

The World Economic Forum (2013) introduced an EE Framework that emphasises systemic integration, identifying eight essential pillars: Access to Markets, Human Capital, Finance, Support Systems, Culture, Policy, Infrastructure, and Innovation Capacity. The strength of this framework lies in its focus on the dynamic interplay between these components, offering a holistic approach to understanding the complexity of ecosystems. However, the global outlook of the EE framework may overlook specific regional differences and local contexts (see Figure 2.2).

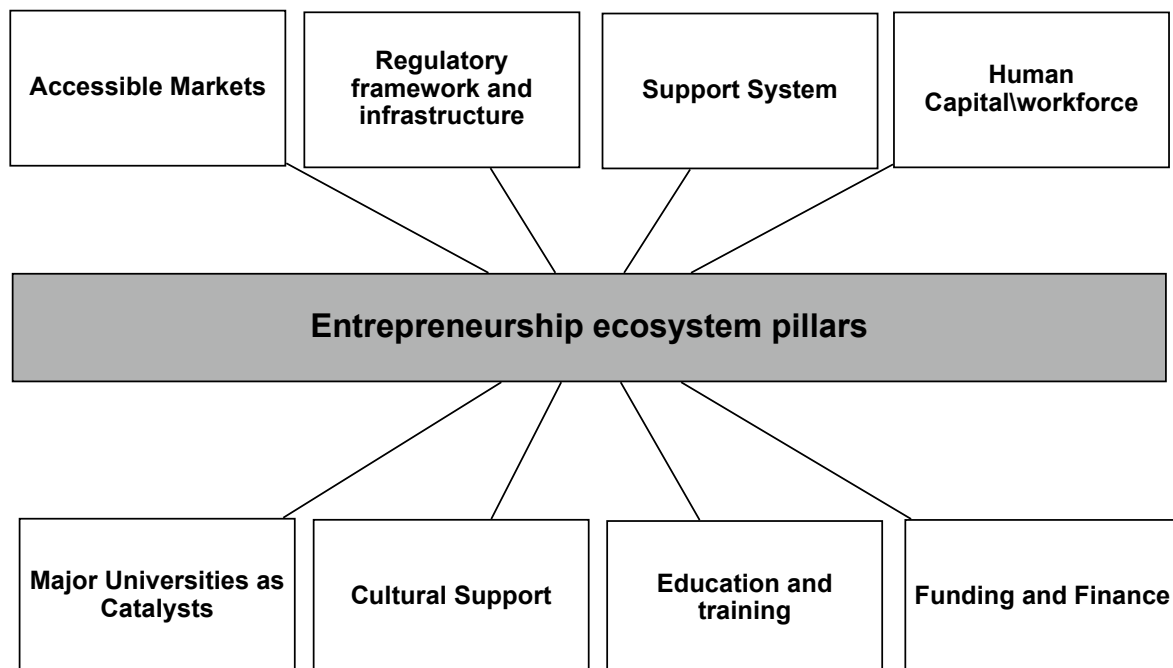


Figure 2.2: The WEF Entrepreneurship Ecosystem Framework (World Economic Forum, 2013)

2.4.3 Entrepreneurship Ecosystem Framework by Vogel (2013)

Vogel's (2013) framework, depicted in Figure 2.3, places entrepreneurs at the core of the entrepreneurship ecosystem. The framework distinguishes between two levels of elements that influence entrepreneurship: non-entrepreneur-specific and entrepreneurship-specific levels. The non-entrepreneur-specific level includes broader contextual factors such as Innovation, Markets, Infrastructure, Geographic Location, and Government & Regulations. In contrast, the entrepreneurship-specific level encompasses elements directly tailored to support entrepreneurial activities, including Support Systems, Culture, Visibility, Financing, Networks, and Education, as shown in Figure 2.3. Notably, Geographic Location and Visibility as distinct

elements set this framework apart from others in the literature, highlighting the unique contextual and promotional aspects influencing entrepreneurial success (Vogel, 2013).

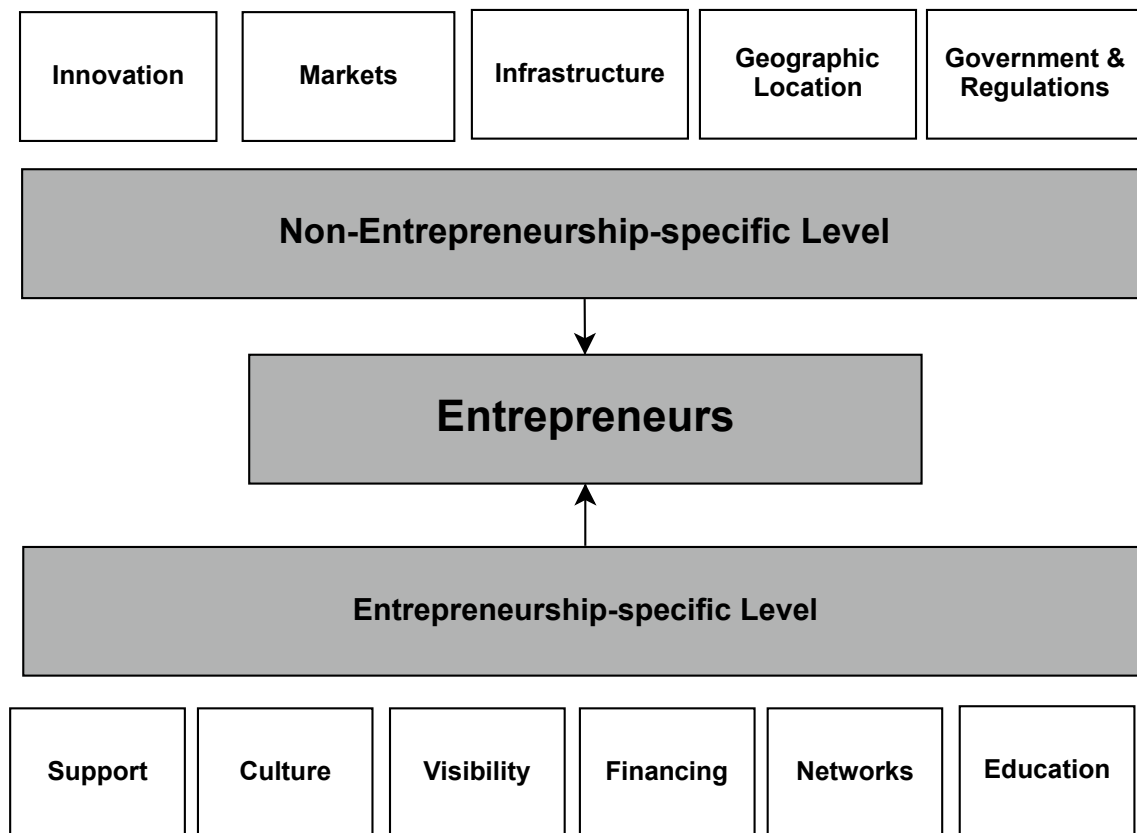


Figure 2.3: Vogel's Entrepreneurship Ecosystem Framework (Vogel, 2013)

2.4.4 Entrepreneurship Ecosystem Framework by Stam & Van de Ven (2021)

Stam (2015) proposed a revised framework that explicitly includes measurable outputs and establishes a clear causal link between the elements of an ecosystem and its outcomes. Building on this, Ven's (2019) EE framework (see Figure 2.4) offers a more structured approach by categorising ten key elements into two main components: Resource Endowments (comprising infrastructure, intermediaries, talent, knowledge, finance, demand, and leadership) and Institutional Arrangements (encompassing culture, formal institutions, and networks). These components function as inputs that collectively drive productive entrepreneurship, ultimately leading to the creation of new economic value. By explicitly linking these inputs to outputs, the framework provides a clearer understanding of how ecosystem elements interact and contribute to entrepreneurial success, addressing the key shortcomings of earlier models.

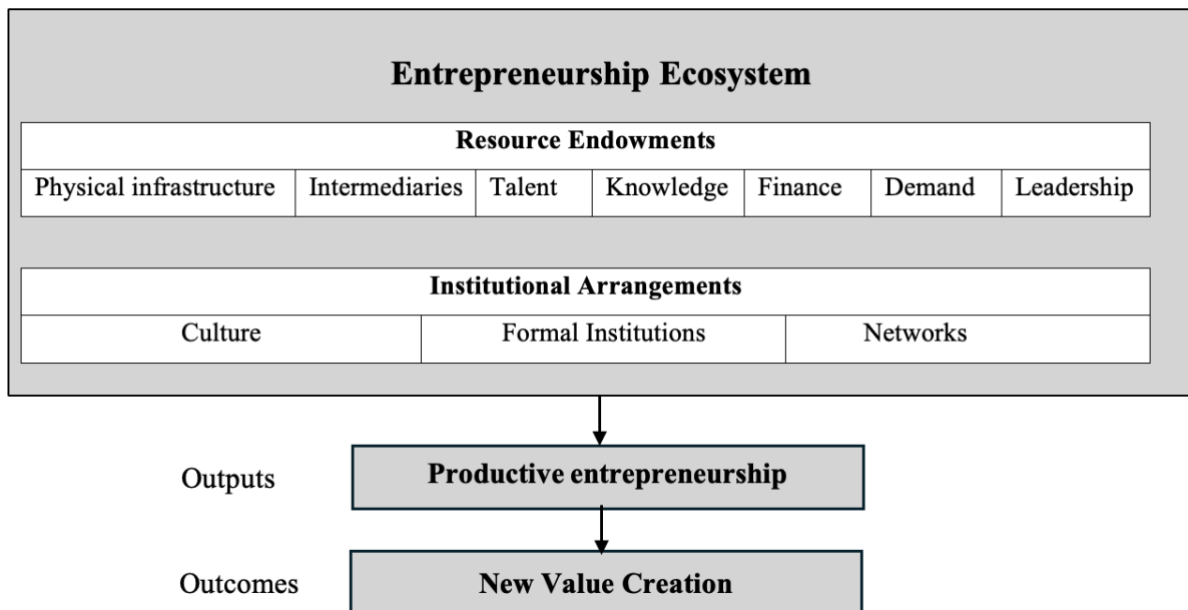


Figure 2.4: Entrepreneurship ecosystem (Stam & Van De Ven, 2021)

A review of the existing EE frameworks reveals similarities and limitations, suggesting areas for further research (see Table 2.3 for a synthesis of the main EE frameworks). Most frameworks include core elements such as policy, finance, culture, support systems, human capital, and market access (Isenberg, 2011a; Stam & Van De Ven, 2021; Vogel, 2013; World Economic Forum, 2013). This alignment reflects a shared understanding of the essential components required to support entrepreneurial activities. For instance, all four models acknowledge the influence of cultural factors, recognising that societal attitudes towards entrepreneurship can significantly impact the level of entrepreneurial engagement within a region.

Table 2.3: Synthesis of the Main EE Framework

Approach	D. Isenberg	The world economic forum (G. Foster)	B. Spigel	P. Vogel	and Van De Ven E. Stam (2019)
	Domains of entrepreneurship ecosystem	Entrepreneurship ecosystem pillars	Attributes of entrepreneurship ecosystems	Building entrepreneurship ecosystems	Entrepreneurship ecosystem elements
Human capital	Labour	Human capital workforce	Worker talent	Entrepreneurial actors	Talent
	Serial entrepreneurs Later generation family Skilled and unskilled	Management talent, technical talent, Entrepreneurial company experience, Outsourcing availability, and Access to immigrant workforce	Presence of skilled workers who are willing to work at startups.	Novice entrepreneurs Serial entrepreneurs	
	Educational institutions	Education and training	Universities	Entrepreneurial education	Educational institutions
	Specific entrepreneurship training General degrees (professional & academic)	Available workforce with university education Entrepreneur specific training Available workforce with pre-university education	Universities and other higher education institutions which both train new entrepreneurs and produce new knowledge spillovers	Entrepreneurship degree Skill training & certificates	Entrepreneurship degree, skill training & certificates
	Infrastructure	Infrastructure	Physical infrastructure	Infrastructure	Physical infrastructure
	Telecommunications Transportation and logistics Energy zones, incubation, centers, clusters	Access to basic infrastructure Access to telecommunications/ broadband Access to transport	Availability of sufficient office space, telecommunication facilities, and transportation infrastructure to enable venture creation and growth	Physical infrastructure, Educational institutions (e.g. Universities), Energy, telecom Transport & logistics, and Workspace	
	Support professions	Support systems	Support services	Support	Support services
	Legal, Accounting, Investment bankers, technical experts, and advisors	Mentors/advisors Professional services Incubators/accelerators Network of entrepreneurial peers	Firms and organisations that provide ancillary services to new ventures, for example, patent lawyers, incubators, or accountancies	Accounting & legal, Mentors & coaches, Experts & consultants, Export support, Labor & talents, Information hubs, Cluster/tech parks, and Foundation	Intermediaries
	Non-government institutions				
	Entrepreneurship promotion in nonprofits, Business plan contests, Conferences Entrepreneur friendly associations	N/A	N/A	N/A	N/A

Table continues...

Table 2.3 Continued

	D. Isenberg	The world economic forum (G. Foster)	B. Spiegel	P. Vogel	Van De Ven E. Stam (2021)
Market	Early customers	Accessible markets	Open markets	Markets	Demand
	Early adopters of proof-of-concept Expertise in productising Reference customers First reviews Distribution channels	Domestic market (SMS companies as customers; governments as customers) Foreign market (SMS companies as customers; governments as customers)	Presence of sufficient local opportunities to enable venture creation and unimpeded access to global markets.	Customers (including beta users and early adopters), Competitors, Distribution channels, Suppliers, Large corporations (as customers or strategic partners)	
	Networks		Networks	Networks	Networks
	Entrepreneur's network Diaspora network Multinational corporation	N/A	Presence of social networks that connect entrepreneurs, advisors, investors, and workers and allow the free flow of knowledge.	Group networks, Informal networks: friends, families, colleagues, Formal networks: organisations, institutions	N/A
Finance	Financial capital	Funding and finance	Finance	Financing	
	Private equity, public capital markets, Debt, Venture capital funds, Micro-loans, Angel investors, friends and family, and Zero-stage venture capital	Private equity, Access to debt Angel investors, Friends and family, and Venture capital	Availability of angel investors, investment capital from family and friends, and venture capitalists	Smart capital Loans & grants, Debt Crowdfunding, Microfinancing Private equity, Accelerators Business angels	N/A
Culture	Societal norms	Cultural support	Supportive culture	Culture	Culture
	Tolerance of risk, failure, mistakes Innovation, creativity, experimentation, social status of entrepreneur, Wealth creation, Ambition, drive, hunger	Tolerance of risk and failure Preference for self-employment Research culture Positive image of entrepreneurship Celebration of innovation	Cultural attitudes which support and normalise entrepreneurial activities, risk taking, innovation, Histories of entrepreneurship, Prominent local successful entrepreneurial ventures	Self-promotion skills Tolerance of failure & risk Tolerance towards success Social status of entrepreneur Mindset, ambition, drive, creativity	
	Success stories	Success stories/role models	Role models	Role models	
	Visible successes Wealth generation for founders International reputation		Businesspeople and local successful entrepreneurs who provide advice for younger entrepreneurs		N/A

Table continues...

Table 2.3 Continued

Policy	Government	Government and regulatory framework	Policy and governance	Government and regulations	Formal institutions
	Financial support (e.g. For R&D, jump start funds), Regulatory framework incentives (Tax benefits), Institutions (Investment, research and Support institutions), Venture-friendly legislation (bankruptcy, contract enforcement, property rights and labour)	Ease of starting a business Tax incentives Business-friendly Legislation/ policies	State-run programs or regulations that either support entrepreneurship through direct funding or remove barriers to new venture creation	Policy framework Immigration & labor law Property rights Freedom of people Regional economic development	
	Leadership				
	Unequivocal support, Social legitimacy, Open door for advocate, Entrepreneurship strategy, Urgency, crisis and challenge				
Others		Major universities as catalysts		Geographic location	New knowledge source
	N/A	Promoting a culture of respect for entrepreneurship Playing a key role in idea-formation for new companies Playing a key role in providing graduates for new companies	N/A	Liveability in the area Cost of living	
				Innovation	
				Knowledge & skill creation, Research & development, Ip Published scientific papers, Technology transfer, new processes and methods	
				Visibility	
				Events & meet-ups, Conferences Startup awards/labels, Startup-related internet portals, Media/newspapers	

Several limitations were identified in this review. First, distinct differences emerge in terms of their focus and applicability. Isenberg's (2011) model prioritises flexibility, making it effective for regional and context-specific analyses, but its qualitative approach limits its use in cross-regional comparisons. In contrast, the World Economic Forum's (2013) framework offers a global perspective with standardised metrics for international comparisons. Although useful for benchmarking, this macrolevel approach can overlook critical local dynamics, especially in developing regions where unique economic conditions may prevail.

The second critique of the current EE frameworks is the lack of clear success metrics in the literature. Many models emphasise the number of startups as an indicator of EE success; however, this approach is insufficient (Rezaei et al., 2018; Stam & Spigel, 2016). Simply counting new businesses overlooks factors such as quality, sustainability, and economic contributions. Stam (2015) argues that focusing on startup numbers ignores the broader impact of ventures on the economy, as many may fail without adding value, thereby weakening the ecosystem's overall effectiveness. To address this gap, productive entrepreneurship has emerged as a more meaningful measure of success. Productive entrepreneurship focuses on activities that boost economic growth, such as innovation, productivity, and job creation. High-growth firms, rather than the sheer number of startups, provide a better indication of an ecosystem's capacity to generate substantial regional economic benefits (Stam & Van De Ven, 2021). This shift from quantity to quality helps identify ventures that drive long-term economic impact.

Third, there is a lack of an explicit representation of the impact of EE on enterprise performance or outcomes. Many existing models assume that enterprise success is primarily driven by external ecosystem support, often neglecting the significant role of a company's internal resources and capabilities. However, research suggests that business success depends on both external factors and an enterprise's internal strengths. High-quality ecosystems may offer essential support; however, internal resources such as innovation, risk-taking, and proactive decision-making are crucial in determining how effectively a business leverages this support. RBV theory, as highlighted by Gueler and Schneider (2021), suggests that ventures with strong internal assets, including unique resources, strategic capabilities, and adaptability, are better positioned to effectively utilise ecosystem support. This perspective underscores the need for an EE framework that considers internal capabilities of the firm. Consequently, the framework proposed in this study combines EO and EE to enhance business performance comprehensively. The adaptability of EE frameworks across different regional and sectoral contexts also reflects the relevance of older publications. For example, Isenberg's (2010)

foundational model continues to be applied and critiqued in diverse environments, including emerging markets and creative sectors. Studies such as those by Brydges and Pugh (2021), which focus on sustainability and entrepreneurship in the fashion industry, and Alarifi et al. (2019), which evaluate entrepreneurship policies in Saudi Arabia, build on Isenberg's framework to address modern challenges in EEs. The general EE framework provides a comprehensive understanding of the external factors influencing entrepreneurial activity. However, it lacks specificity when applied to industry-specific contexts.

2.5 The theoretical frameworks

2.5.1 *Overview of Theories in Entrepreneurship Ecosystems*

Nine theories have been employed in the research literature to investigate and predict the dynamics of EE, each offering unique insights into ecosystem characteristics and their impact on entrepreneurship. These theories include Cluster Theory (Motoyama, 2008; Porter, 2000; Vorley, 2008), Systems Theory (Bertalanffy, 2008; Checkland, 1982; Luhmann, 1995), Social Network Theory (Burt, 2003; Coleman, 1988), Stakeholder Theory (Donaldson & Preston, 1995; Freeman, 2010; Mitchell et al., 1997), Institutional Theory (DiMaggio & Powell, 1983; North, 1990), Dynamic Capabilities Theory (Eisenhardt & Martin, 2000; Helfat & Peteraf, 2003; Teece et al., 1997), Resource Dependence Theory (Babely  -Labanausk  , 2022; Pfeffer & Salancik, 2015), and the Resource-Based View (RBV) (Kraaijenbrink et al., 2009; Peng, 2001; Peteraf, 1993). A detailed overview of these theories is presented in Table 2.4.

Despite the breadth of these theoretical frameworks, they often fall short of fully capturing the intricate and dynamic nature of EEs. Recent studies highlight a need for new approaches that focus on dynamic, self-reinforcing relationships capable of driving long-term, high-growth entrepreneurship (Cantner et al., 2020; Wurth et al., 2023). Recent developments have introduced an EE theoretical framework. However, this framework often lacks empirical validation and theoretical depth, highlighting the need for further development into a robust and comprehensive theoretical construct (Duan et al., 2021). Consequently, EEs are frequently regarded as conceptual frameworks rather than fully developed theories. To address these limitations, this study integrates insights from the RBV theory. RBV provides a valuable lens for analysing the impact of EEs on entrepreneurial outcomes, with a specific focus on how internal resources, such as entrepreneurial orientation, enable firms to leverage ecosystem benefits effectively and adapt to evolving challenges. Among these, the RBV offers a particularly useful firm-level perspective that aligns closely with the focus of this study, and is therefore adopted as the principal theoretical framework.

Table 2.4: Summary of Relevant Theories in Entrepreneurship Ecosystems Research

Theory	Theory Description	Theory's Relationship to Entrepreneurship Ecosystem	Key Concepts	Key References
Cluster Theory	Focuses on geographical aggregation of interconnected businesses and institutions to enhance innovation and economic vitality.	Highlights competitive and collaborative advantages resulting from geographical proximity and sectoral interconnectivity among ecosystem participants.	Proximity, collaboration, sectoral advantage	Porter (2000), Motoyama (2008) and Vorley (2008)
Systems Theory	Conceptualises ecosystems as complex systems where interconnected components work collectively, shaping overall outcomes.	Advocates for a holistic analysis of ecosystems, where feedback loops and interdependencies contribute to sustainable entrepreneurial outcomes.	Systemic interdependence, feedback, holistic view	Bertalanffy (2008), Luhmann (1995) and Checkland (1982)
Social Network Theory	Investigates structure and quality of relationships within networks and how these impact resource access and collaboration.	Highlights how entrepreneurs and stakeholders in ecosystems benefit from network ties, which facilitate resource sharing, knowledge flow, and innovation.	Network ties, relational embeddedness, knowledge flow	Burt (2003) and Coleman (1988)
Stakeholder Theory	Focuses on role of various stakeholders (e.g., investors, customers, regulators) in shaping organisational strategies and goals.	Emphasises collaborative and interdependent relationships among ecosystem stakeholders, essential for ecosystem cohesion and entrepreneurial success.	Stakeholder influence, multistakeholder collaboration	Freeman (2010), Donaldson and Preston (1995) and Mitchell et al. (1997)
Institutional Theory	Analyses influence of formal (e.g., regulations) and informal (e.g., cultural norms) institutions on organisational behaviours.	Examines how institutional contexts, including policies and cultural norms, shape and direct entrepreneurial actions within the ecosystem.	Institutions, regulatory frameworks, cultural norms	North (1990) and DiMaggio and Powell (1983)
Dynamic Capabilities Theory	Emphasises a firm's adaptive capacity to reconfigure resources in response to environmental shifts.	Explains how firms within an ecosystem can dynamically adapt and realign resources to maintain competitive advantage and foster ecosystem resilience.	Adaptability, resource integration, agility	Teece et al. (1997), Helfat and Peteraf (2003) and Eisenhardt and Martin (2000)
Resource Dependence Theory	Posits that firms are reliant on external resources, and that survival depends on managing these dependencies.	Examines how ecosystem actors mitigate dependencies by securing essential resources, fostering interdependence and collaborative resilience within the ecosystem.	Resource reliance, dependency management	(Babelytė-Labanauskė, 2022; Nguyen, 2023; Pfeffer & Salancik, 2015) (Hillman et al., 2009)
Resource-Based View (RBV)	Suggests firms achieve competitive advantage through unique, valuable, and inimitable internal resources.	Demonstrates how ecosystem enables firms to leverage unique resources strategically, contributing to long-term sustainability and competitiveness.	Unique resources, strategic advantage	Peteraf (1993), Peng (2001) and Kraaijenbrink et al. (2009)
Entrepreneurship Ecosystem Theory	Emphasises intricate dynamics among various ecosystem actors, such as policy, finance, and culture, collectively impacting entrepreneurial activity.	Provides a systemic framework that underscores how each component (e.g., policy, finance, culture, human capital, and markets) interacts synergistically to drive and sustain entrepreneurship.	Systemic support, network of actors, ecosystem synergies	(Acs et al., 2014; Isenberg, 2010; Stam & Spigel, 2016)

adapted from Fubah and Moos (2021)

2.5.2 *Resource Based View (RBV) Theory*

Recent studies have aimed to integrate EO with other constructs, such as innovation management and the business environment, using theories such as RBV and contingency theory. These frameworks seek to explain the direct and moderated effects of EO on organisational performance (Surin et al., 2023). The RBV has been a foundational theory in strategic management, offering a framework to explain how firms achieve sustainable competitive advantage through effective utilisation of internal resources. Originally developed by Barney (1991), RBV posits that resources must possess valuable, rare, inimitable, and non-substitutable (VRIN) attributes to provide a competitive edge. Over time, RBV has evolved to find applications in various contexts, particularly in conjunction with the EO. EO, characterised by innovativeness, proactiveness, and risk-taking, complements RBV by enabling firms to manage resources strategically in dynamic and competitive markets.

Recent studies have explored the interplay between EO and firm performance using RBV as a theoretical lens. For instance, Dionysus and Arifin (2020) demonstrated that EO significantly influences SME performance, whereas Market Orientation does not. This underscores the critical role of EOs in leveraging internal resources for superior outcomes, aligning with RBV's emphasis on VRIN attributes of resources. Similarly, Ferreira and Ferreira (2024) identified EO as a pivotal thematic cluster in RBV research, highlighting its importance in explaining firm dynamics and strategic behaviour.

To extend the RBV, Zahra (2021) investigated resource management in startups and proposed that integrating RBV with entrepreneurship frameworks could address resource allocation challenges while enhancing competitive advantage. Additionally, Surin et al. (2023) introduced a novel framework combining RBV with contingency theory to better capture the effects of EOs on organisational performance. Such integrative approaches demonstrate the adaptability and relevance of RBVs in explaining complex firm behaviours.

Despite its widespread application, RBV has faced criticism for its static assumptions, particularly in rapidly evolving business environments. For example, Kraaijenbrink et al. (2009) highlighted the need for a more adaptable framework to address these limitations and advocated dynamic models that reflect contemporary challenges. In response, Foss and Ishikawa (2007) incorporated Austrian capital theory into the RBV, aligning it with the EO's emphasis on innovation and adaptability. These refinements address criticisms while maintaining the foundational RBV principles. Agrawal et al. (2024) further explored barriers

to sustainable entrepreneurship and innovation, such as financial constraints and regulatory challenges, demonstrating the ongoing relevance of RBV in addressing modern business complexities. These studies collectively emphasise the need to continually refine the RBV to ensure its alignment with contemporary strategic imperatives.

The integration of the RBV with the EO extends beyond traditional strategic management. Abdullah et al. (2024) employed RBV alongside experiential learning theories to cultivate entrepreneurial mindsets in educational settings, fostering career readiness and adaptability. This perspective highlights the versatility of the RBV, as it contributes to entrepreneurship education and workforce development, particularly in preparing students for dynamic business environments. Empirical research further supports the applicability of the RBV and EO integration. For instance, Arifin and Dewi (2020) found that EO positively influences innovation through adaptive capabilities, aligning with the premise that effective resource utilisation of RBV drives superior performance. Similarly, Doria et al. (2021) emphasised the role of managerial actions in orchestrating VRIN resources to maximise strategic value, which is a critical extension of the RBV theoretical foundations.

Mahmoud et al. (2023) highlighted the strategic importance of combining internal resources with external networks, such as social media capital and business ties, to navigate competitive markets. These findings resonate with the RBV's emphasis on leveraging VRIN resources in conjunction with external opportunities to achieve sustainable growth. Abdullah et al. (2024) demonstrated that integrating RBV with experiential learning fosters entrepreneurial resilience, which is a critical consideration for Australian businesses operating in resource-constrained environments. These insights suggest that RBV and EO have substantial potential for addressing the unique challenges of Australia's business ecosystem.

Although RBV remains a cornerstone of strategic management, its static assumptions necessitate ongoing refinement to address dynamic business complexities (Bertram, 2016). Integration with EO offers a promising avenue for enhancing its practical and theoretical relevance (Lubis, 2022). Future research should prioritise empirical studies across diverse contexts, including Australian SMEs, while exploring how emerging technologies, such as artificial intelligence and blockchain, reshape resource value (D'Oria et al., 2021). Furthermore, hybrid frameworks combining RBV with digital transformation and sustainability theories could address critiques while retaining the foundational strengths of the RBV (Furr & Eisenhardt, 2021).

In support of its application in this study, the RBV is used exclusively due to its capacity to explain how internal resources, especially EO, enable firms to leverage ecosystem

advantages. While theories such as Dynamic Capabilities Theory (Teece et al., 1997), Institutional Theory (DiMaggio & Powell, 1983; North, 1990), and Resource Dependence Theory (Pfeffer & Salancik, 2015) offer valuable insights, they were excluded because they focus predominantly on external pressures, institutional environments, or require longitudinal data. These characteristics make them less suitable for the current study's firm-level, cross-sectional design. Moreover, Entrepreneurship Ecosystem Theory, although systemic in nature (Stam & Spiegel, 2016; Acs et al., 2014), lacks the granularity to analyse how individual firms mobilise internal capabilities in response to ecosystem conditions.

RBV addresses this gap by offering a focused lens on strategic resource utilisation. It has shown empirical value in explaining how EO drives innovation and agility in dynamic sectors like fashion (Arifin & Dewi, 2020; Dionysus & Arifin, 2020; Doria et al., 2021). Furthermore, the combined application of EO and RBV helps counteract the RBV's static orientation by introducing adaptive and proactive dimensions. As Mahmoud et al. (2023) argue, this integration is especially effective in resource-constrained environments, where entrepreneurial firms must optimise both internal capabilities and external opportunities.

RBV provides a robust framework for understanding how firms leverage resources to achieve competitive advantages. Its integration with EO enhances its applicability in dynamic and competitive markets, and ongoing refinements address the identified weaknesses of its static assumptions (Beamish & Chakravarty, 2021). By incorporating dynamic resource frameworks and exploring new applications, the RBV remains a valuable lens for examining strategic management in both traditional and contemporary contexts (Sabourin, 2020).

Understanding the RBV in strategic management clarifies how fashion entrepreneurs can leverage internal resources to navigate rapid industry transformation and cultural influences. The fashion sector's constant evolution, diverse products, and complex supply chains set it apart. This section explores these dynamics, emphasising the need for entrepreneurs to strategically manage resources for sustained success in competitive markets.

EO was selected as the sole internal resource construct due to its strong empirical linkage with MSME performance across diverse contexts (Kriseka et al., 2024; Hidayat et al., 2023; Kaushik & Dangwal, 2023), its strategic relevance in volatile industries such as fashion (Kharisma et al., 2020; Bražinskaitė & Durand, 2014), and its theoretical compatibility with the RBV as a behavioural mechanism for leveraging valuable internal capabilities (Khan et al., 2021; Lee & Chu, 2013; Stone, 2016; Kiyabo & Isaga, 2020). The exclusion of additional micro-level variables, such as innovation capabilities or entrepreneurial self-efficacy, was a

deliberate design choice to ensure model parsimony and has been acknowledged as a limitation, with directions for future research outlined accordingly.

2.6 Fashion Industry Characteristics

The fashion industry is distinguished by its unique characteristics that define its dynamic nature. It is characterised by rapid changes, diverse product offerings, and complex supply chains. The following section outlines the four key fashion industry characteristics identified in the literature. Understanding these characteristics is essential for examining how FEE influences EO and BP, particularly in the context of fashion entrepreneurship.

First, fashion is characterised by short-lived trends influenced by cultural changes, social media, and shifting consumer tastes (Čiarnienė & Vienažindienė, 2014). It reflects social values and serves as a means for individuals to express themselves, making it responsive to societal changes and personal preferences (Craik, 2019). This subjectivity and cultural influence shape consumer choices and brand perceptions, with emotional reactions to design and brand stories playing a key role (Bhagya et al., 2024). Fashion's connection to identity and status means that entrepreneurs must stay flexible, predict trends, and adapt their products to meet cultural expectations, especially in culturally dominated markets such as Saudi Arabia. For example, fashion entrepreneurs face challenges in Saudi Arabian markets, such as aligning fashion products with social norms and expectations of tradition and conservatism, which influence their business strategies. This context emphasises the need for adaptability in meeting cultural expectations in Saudi Arabia (Alshibani et al., 2024). The unusual and complex nature of the fashion industry has become more evident as it has adapted to changes driven by shifting consumer preferences, rapid technological advancements, and evolving cultural values (Jiju et al., 2024). Contemporary research increasingly highlights the significance of sustainability and ethical practices, reflecting heightened societal expectations and emerging business opportunities in the sector (Gazzola et al., 2020).

Second, the fashion industry is driven by creativity and a continuous push for innovation. The fashion industry is inherently creative and constantly seeks innovation to adapt to changing tastes and trends (Bernardoni & Ruppert, 2017). This drive for innovation is essential for maintaining the dynamic nature of the industry and ensuring that new styles are continuously introduced to the market (Rusli et al., 2023). The use of innovative materials and technologies is a key aspect of this creative process, allowing designers to explore new possibilities and enhance the artistic value of their designs (Antonina et al., 2021; Leshchyshyn et al., 2022).

Technological advancements such as smart textiles, 3D printing, and digital design tools play a key role in enhancing innovation capabilities (Aspers & Skov, 2006). The integration of technology has transformed both the design process and retail and marketing through e-commerce, social media, and augmented reality. This has enabled fashion brands to engage more effectively with global consumers and streamline their operations. This technological evolution supports sustainable practices and circular economy initiatives, helping businesses reduce waste and adopt ecofriendly methods (Brydges & Pugh, 2021).

Third, fashion is a global industry, with design, production, and distribution occurring across multiple countries. The industry's global reach has facilitated the rise of fast fashion, characterised by rapid production cycles, affordability, and quick responses to changing consumer demand (Şen, 2008). This model has intensified competition, creating a highly fragmented market with players ranging from luxury brands to small, independent designers (Aspers & Skov, 2006). The diverse nature of the industry, combined with varying local preferences, requires entrepreneurs to adapt their strategies to balance global trends and regional tastes. While globalisation has expanded market opportunities, it has also raised concerns about sustainability and ethical production practices, prompting fashion businesses to reconsider their sourcing and manufacturing strategies.

Finally, fashion operates on a seasonal basis or trends, with collections often tailored to specific weather conditions and events (e.g. resort and holiday collections). The demand for fashion items fluctuates with the seasons and during special events such as weddings and festivals (Şen, 2008). This cyclical nature affects production planning, inventory management, and marketing strategies, requiring fashion entrepreneurs to anticipate and respond to these variations effectively. Seasonality, coupled with the trend-driven nature of fashion, challenges businesses to manage resources wisely and align their product offerings with consumer expectations throughout the year.

Given the unique characteristics of the fashion industry, there is a clear need for further research on FEE to better understand how these dynamics influence entrepreneurial activities and outcomes. The following section provides an overview of the existing literature on fashion entrepreneurship, specifically examining its alignment with various domains of entrepreneurship ecosystems.

2.7 Fashion Entrepreneurship

Fashion entrepreneurship has emerged as a significant driver of growth and innovation within the industry, characterised by the integration of innovation, creativity, and strategic business acumen. These elements collectively contribute to competitive advantages and enhance organisational performance (Hosseini, 2019; Ünay & Zehir, 2012). Fashion entrepreneurship includes building successful brands, often shaped by the entrepreneur's values and innovative strategies in design, branding and retail (Brydges et al., 2014; Karra, 2021). Specifically, fashion entrepreneurship focuses on designers applying their creative vision and industry experience to launch new ventures (Fernandes et al., 2020). The ability of fashion entrepreneurs to introduce innovative ideas, designs, and marketing strategies is crucial for sustaining competitiveness and meeting the demands of a rapidly changing global market (Ünay & Zehir, 2012).

The body of literature on fashion entrepreneurship has expanded significantly, reflecting the industry's evolution and a deeper understanding of the factors underpinning success in this dynamic sector. Initial studies primarily focused on the individual characteristics of entrepreneurs, emphasising traits such as creativity, innovation, and a propensity for risk-taking (Moloi & Nkhahle-Rapita, 2014; Nwaozuru et al., 2024). These traits are essential for managing challenges and leveraging the advantages of being both a designer and an entrepreneur (Aakko & Niinimäki, 2018; Van Aardt & Van Wyk, 2011). However, focusing on individual characteristics often overlooks the broader structural challenges faced by fashion entrepreneurs, including shifts in market trends, limitations in accessing financial resources, and the pivotal role of industry networks (Zeng, 2020).

Recognition of this complexity has led scholars to adopt the concept of entrepreneurship ecosystems, framing fashion entrepreneurship as part of a broader set of factors that significantly influence the success and viability of fashion ventures (Stam & Van De Ven, 2021; Wurth et al., 2023). For instance, the lack of a supportive entrepreneurial environment has been identified as a key reason for high failure rates among fashion startups, especially those founded by emerging designers and recent graduates (Zeng, 2020; Zeng et al., 2024). The following section discusses existing research on fashion entrepreneurship ecosystems.

2.8 Fashion Entrepreneurship Ecosystem Studies (FEE)

The concept of the Fashion Entrepreneurship Ecosystem encompasses various facets of the structure and dynamics of the fashion industry, particularly focusing on interactions and

support systems that foster growth and innovation (Brydges & Pugh, 2021; Salonoja, 2013). In the Scandinavian context, studies such as Salonoja (2013) in Finland focus on the critical financing and entrepreneurship gaps within the FEE, particularly when compared to neighbouring countries such as Sweden and Denmark. This research employed a qualitative case study approach, using interviews with investors and industry experts to examine the structural barriers faced by Finnish fashion entrepreneurs. It highlighted a significant shortfall in business skills and institutional support, which hindered the international expansion of Finnish fashion firms. The findings underscore the need for more robust training programs and enhanced institutional support, targeting the development of entrepreneurial and managerial capabilities within the Finnish fashion sector. Similarly, Tervilä (2015) explored the lack of private capital in Finland's fashion ecosystem, identifying a fundamental disconnect between the strategic thinking of entrepreneurs and investors. This study utilised a qualitative methodology, including semi-structured interviews with entrepreneurs and investors, revealing that while entrepreneurs favoured effectual logic, investors preferred causal approaches. This strategic divergence was a major impediment to securing investment, indicating the need for greater alignment and understanding between these key stakeholders.

The approach by D'Itria and Colombi (2023) contrasts with the narrower research methodologies often seen in similar studies. By adopting the Quadruple Helix model, their study expands the traditional Triple Helix framework by including civil society as a key player alongside academia, government, and industry (D'Itria & Colombi, 2023). This broader perspective is essential for comprehensively addressing sustainable innovation in the fashion industry, as it emphasises the importance of collaborative efforts across all sectors of society.

However, while their methodological choice enriches the understanding of ecosystem dynamics, the omission of quantitative measures to assess the direct impact of these dynamics on business performance is a significant limitation. This gap may hinder policymakers and industry stakeholders' ability to implement specific strategies effectively, as there is less empirical evidence to support the causal relationships between ecosystem integration and firm-level outcomes. Such a quantitative assessment could provide clearer insights into how these collaborative environments contribute to tangible business improvements, particularly in the context of sustainability in fashion.

In Portugal, a study by Cunha et al. (2023) on a digital project focused on digital transformation within the textile and fashion industries, aiming to enhance production flexibility and responsiveness to the increasing demand for customisation. This study adopted a quantitative approach, drawing on industry reports to evaluate the impact of digitalisation

initiatives. It is recommended to adopt microfactories and digital printing technologies to enhance production efficiency. This study highlights the potential of digital transformation to increase responsiveness and customisation capabilities. However, it did not provide an in-depth sociocultural analysis of how these changes might be received or implemented in different market contexts, limiting the applicability of its findings beyond an industrial focus.

In contrast, Brydges and Pugh (2021) explored the application of the EE concept within cultural and creative industries, specifically examining the fashion sector in Toronto. The study employed a qualitative methodology, involving extensive mapping of institutions and organisations, and found that FEE in Toronto is fragmented, with limited coordination among key stakeholders. This lack of integration has been identified as a significant barrier to innovation and growth within the sector. Although the study provides detailed insights into the structural challenges faced by fashion entrepreneurs, it does not include a comprehensive assessment of the impact of these challenges on firm performance, instead focusing primarily on descriptive analysis and proposing improvements for institutional support.

In China, Zeng et al. (2024) offered a different perspective by proposing an “innovation ecosystem” framework tailored for the fashion industry. This study focused on the role of government policy in driving innovation, highlighting the importance of knowledge transfer between designers and the broader industry. An integrative literature review approach was adopted, allowing for a comprehensive synthesis of existing theories and models related to fashion innovation management. Despite the detailed theoretical framework presented, this study did not include any quantitative measures of the impact of these policies on business performance, thereby limiting its ability to provide concrete evidence of the framework’s effectiveness in improving firm outcomes.

Overall, these studies share a common goal of understanding the factors influencing the development of FEEs. However, a significant gap persists in assessing their impact on firm performance. These studies often emphasise challenges and proposed solutions rather than quantifying the ecosystem’s direct effects on EO and BP. Most studies rely on qualitative methods, such as case studies and interviews, which offer insights but lack robust quantitative analyses. Additionally, most existing research on fashion entrepreneurship focuses on developed markets such as the United Kingdom (Conlon, 2019; Shi et al., 2012), the United States (Relyea, 2016), Singapore (Fatt, 2001), and Finland (Colliander, 2015; Salonoja, 2013). A summary of these studies is presented in Table 2.5.

Table 2.5: Summary of Literature on Fashion Entrepreneurship Ecosystems

Author & Year	Research Topic	Research Type	Research Context	Framework/Theory Used	Key Findings
Salonoja (2013)	Bridging equity and entrepreneurial gaps in the Finnish fashion industry.	Comparative Case Study	Finland, Sweden and Denmark	Fashion Entrepreneurship Ecosystem Model	- Identified significant equity and entrepreneurial gaps in Finland when compared with Sweden and Denmark. - Emphasised importance of strategic government support to bridge these gaps. - Highlighted role of private-public collaboration in enhancing growth and performance of MSMEs.
Colliander (2015)	Mapping Finnish fashion entrepreneurship ecosystem.	Mixed-Methods Study	Finland	Adapted Isenberg's EE Framework	- Mapped key stakeholders and services within Finnish fashion ecosystem. - Revealed a mismatch between needs of designers and available support services. - Identified critical gaps in business development and financial support for fashion entrepreneurs.
Tervilä (2015)	Investigating scarcity of private capital in Finnish fashion ecosystem.	Qualitative Case Study	Finland	Effectuation Theory & Isenberg's EE Framework	- Found a disconnect between effectual logic of entrepreneurs and causal logic preferred by investors. - Highlighted strategic differences: entrepreneurs focus on adaptive decision-making, while investors demand structured planning. - Demonstrated how these conflicting approaches hinder investment deals and limit access to private capital.
Brydges and Pugh (2021)	Expanding Entrepreneurship ecosystem concept to creative industries.	Qualitative Case Study	Canada (Toronto)	Extended Entrepreneurship Ecosystem (Isenberg's model)	- Demonstrated fragmented nature of Toronto's fashion ecosystem, characterised by isolated designers. - Highlighted overlapping institutional support and need for tailored assistance. - Emphasised importance of ecosystem mapping to address specific needs of creative industries.
Cunha (2023)	Exploring impact of digitalisation on fashion ecosystem and MSMEs.	Conceptual and Empirical Analysis	Portuguese Textile Cluster	Next Generation Fashion Ecosystem	- Highlighted transformative role of Industry 4.0 technologies in the fashion sector. - Emphasised increased flexibility, customisation, and local production capabilities. - Demonstrated how digitalisation reshapes the supply chain, enabling faster responses to consumer demands and promoting sustainable practices.

Table continues...

Table 2.5 Continued

Author & Year	Research Topic	Research Type	Research Context	Framework/Theory Used	Key Findings
D'Itria & Colombi (2023)	Fostering sustainable innovation in fashion industry using a collaborative model.	Conceptual Study	European context (Italy)	Quadruple Helix Model	- Demonstrated that integration of academia, industry, government, and civil society drives sustainable innovation. - Emphasised collaborative efforts in achieving green and digital transitions. - Provided a model for multistakeholder engagement to address systemic sustainability challenges.
Hardabkhadze (2023)	Analysing fashion industry ecosystem dynamics, including regulatory impacts.	Qualitative Analysis	Ukraine	Fashion Industry Ecosystem Framework	- Examined market access and impact of regulatory policies on MSMEs. - Highlighted significant role of government policies in shaping industry dynamics. - Emphasised need for MSMEs to apply adaptive strategies to effectively navigate regulatory environments.
Zeng (2024)	Fashion innovation through an innovation ecosystem framework.	Conceptual Study	China (Hong Kong)	Innovation Ecosystem Framework	- Identified government policy as a critical driver of innovation within the fashion sector. - Emphasised strong collaboration between government and industry actors. - Suggested policy-driven approaches to enhance sustainability and integrate technological advancements.

This focus of previous research on developed markets leaves a gap in the existing research literature that would contribute to a greater understanding of how fashion entrepreneurship operates in emerging markets, including those in the Middle East. This study on the Saudi FEE aims to address this gap in the literature through a comprehensive multilevel analysis that combines quantitative and qualitative methods. The sociocultural and economic contexts of emerging markets in these less developed regions have remained underexplored, pointing to the need for this study and for future research to develop frameworks tailored to specific challenges and opportunities in developing markets.

The following section discusses FEE domains, which tailor the EE framework to the fashion sector's unique characteristics, accounting for its creative and trend-driven nature. This adaptation is crucial for exploring how tailored ecosystem components affect entrepreneurial outcomes in creative industries.

2.9 The Domains of a Fashion Entrepreneurship Ecosystem

This section reviews the fashion entrepreneurship literature based on the seven EE domains that are most common among the four frameworks previously discussed. These are Entrepreneurship Education, Human Capital, Policy, Market, Finance, Culture and Social Norms, and Support.

2.9.1 *Entrepreneurship Education*

Educational institutions are central to shaping entrepreneurship ecosystems by providing critical resources, fostering knowledge, and nurturing talent (Dif et al., 2019; Lehmann et al., 2022; Spigel, 2017). In response to rapid transformations in the fashion industry, global fashion schools have introduced specialisations focusing on sustainability, craftsmanship, technology and entrepreneurship (Murzyn-Kupisz & Hołuj, 2021; Yezhova et al., 2018). These institutions enhance entrepreneurial competencies through innovative curricula, industry collaborations, and initiatives such as incubators and accelerators (Arranz et al., 2017; Marniati & Witcjaksono, 2020).

Despite these advances, studies have revealed that many fashion programs prioritise creative and technical skills over essential business knowledge, such as financial management, marketing, and strategic planning. This focus leaves students underprepared for entrepreneurial roles (Matlay & Mills, 2012; Radclyffe-Thomas et al., 2018; Tilson-Scoble, 2018; Williams, 2018). For example, Finnish programs are recognised for their design training but lack comprehensive education in business management, limiting graduates' abilities to scale

operations or secure investment (Salonoja, 2013; Tervilä, 2015). Similarly, in South Africa Nana et al., (2021) identified the need for comprehensive training in business practices, including the development of business plans, basic bookkeeping, accurate pricing, preparing quotations and invoices, budgeting, market research, and effective advertising.

In fashion education, integrating entrepreneurship training is crucial, given the industry's diverse specialisations, each requiring distinct skills and approaches. Unlike other industries, fashion entrepreneurship spans several unique domains, such as haute couture, streetwear, and textiles, each demanding a tailored educational curriculum. For instance, haute couture emphasises bespoke craftsmanship and luxury branding, necessitating a high level of precision and unique design skills. According to research on fashion design entrepreneurs, traditional fashion education often lacks the necessary business training, requiring supplementary entrepreneurial education to bridge this gap (Fernandes, 2019). Conversely, streetwear focuses on trend analysis and brand identity, appealing to a broader and more varied market. Research suggests that entrepreneurship education within fashion merchandising curricula is critical for fostering innovation and helping students navigate the competitive retail landscape (Davis et al., 2019). Meanwhile, the textiles segment demands a thorough understanding of materials and a commitment to sustainable production techniques, reinforcing the importance of sustainability in fashion entrepreneurship training (Lang & Liu, 2019). Therefore, fashion entrepreneurship education must be designed to accommodate these variances, thereby equipping students with the specialised entrepreneurial skills required in each specific niche of the fashion industry.

Tailored pathways for haute couture, streetwear, and textile-focused entrepreneurs address specific needs, ranging from craftsmanship and luxury branding to mass production and digital marketing. Research advocates for curricula to incorporate soft skills, such as creativity, communication, and leadership, alongside sustainability principles, balancing entrepreneurial acumen with creativity to better prepare students for industry challenges (Agarwal, 2023; Fry et al., 2019; Rana & Ha-Brookshire, 2018). Emphasising digital and experiential learning equips graduates with relevant skills (Bremner, 2018; Lazzeretti & Capone, 2020; Zeng, 2020). Digital platforms and remote learning opportunities have further helped overcome geographical and resource barriers (Brydges & Hracs, 2019). Research underscores the significant role of entrepreneurship education in enhancing students' entrepreneurial intentions and outcomes, and a synthesis of practical training with mentorship has proven to be effective (Marniati & Witejaksono, 2020; Worrall & Linda, 2024). Programs

focusing on business management, digital marketing, and production techniques equip students to meet multifaceted industry demands (Jamaluddin et al., 2019).

Integrating practical training, internships, and robust industry connections fosters innovation and equips students with diverse skill sets. Real world experiences, such as internships, mentorship programs, and collaborative projects, enhance students' preparedness for industry challenges while deepening their understanding of business dynamics (Brydges & Pugh, 2021; Sousa et al., 2019). Addressing skill gaps requires collaboration between technical vocational education and training institutions and industry stakeholders, fostering exposure to practical business challenges and developing entrepreneurial capabilities (Colliander, 2015; Quang et al., 2020; Zeng et al., 2024). Collaborative models, such as the Quadruple Helix framework, which unites academia, industry, government, and civil society, have been shown to support innovation and entrepreneurship (D'Itria & Colombi, 2023). Industry-university interactions and knowledge transfer are significant in entrepreneurial activities and fostering innovation (Wurth et al., 2023). However, many fashion programs fail to deliver critical contemporary practices, such as e-commerce, digital marketing, and virtual design technologies (Cunha et al., 2023; Zeng et al., 2024). Sustainability education also demands greater focus, with competencies such as ethical sourcing, circular economy practices, and sustainable production methods becoming increasingly critical for aligning academic outcomes with industry demands (Cunha et al., 2023; Hardabkhadze et al., 2023).

In Saudi Arabia, education and training in the fashion industry can be pursued at universities or vocational and technical colleges. However, Saudi fashion education faces significant challenges, including outdated curricula, limited industry alignment, and a reliance on traditional teaching methods (Alshibani et al., 2024; Alzahrani & Kozar, 2017). The centralised education system enforces a uniform curriculum design and tertiary courses, limiting input from educators and industry experts. Abu Nab (2019) criticised continued use of an Egyptian syllabus from the 1970s, which excluded modern topics such as sustainability and ethical fashion. Furthermore, fashion design education remains restricted to females because of traditional gender role expectations, with males directed towards fields such as art, architecture, and design (Alzahrani & Kozar, 2015). Weak academic-industry partnerships further exacerbate skill shortages in the fashion industry, limiting graduates' workforce readiness (Ali et al., 2019; Maghrabi & Abalkail, 2024). These studies recommend strengthening collaborations with local and international stakeholders to create specialised training and certification programs. These educational system failings are further compounded by the insufficient integration of business practices and industry-relevant skills into curricula

(Alshibani et al., 2024; Alzahrani & Kozar, 2015, 2017). Nonetheless, opportunities exist, such as the Saudi Fashion Commission's "100 Saudi Fashion Brands" initiative, which provides practical engagement for fashion students and emerging designers. Additionally, exposure to international markets through study abroad programs enhance creative and business skills (Maghrabi & Abalkail, 2024).

Emerging markets, such as Saudi Arabia, continue to struggle with aligning education and industry needs (Mishrif et al., 2023; Yusuf & Albanawi, 2016). Higher education programs often fail to incorporate practical business skills, creating a mismatch between graduate competencies and employer requirements (Othayman et al., 2021; Othayman et al., 2022; Sharma & Paliwal, 2021). Researchers recommend embedding employability skills into curricula, enhancing academia-industry collaborations, and integrating sustainability education (Aljohani et al., 2022; Singh et al., 2022). Saudi Vision 2030 aims to transform education by redesigning curricula, improving skill development, and aligning graduate competencies with the labour market demands (Abdullateef et al., 2023; Ryan, 2023). While entrepreneurship courses have been shown to enhance managerial skills and entrepreneurial interests, these programs often lack interactive learning methods such as games, competitions, and role-playing, which are essential for developing psychosocial skills (Azim & Hariri, 2018). Despite these shortcomings, no study to date in the research literature has specifically examined the role of educational institutions in shaping entrepreneurship ecosystems or entrepreneurship education in the Saudi fashion industry. Therefore, this study aims to fill this gap by examining the alignment and functioning of the Saudi education system in relation to EE and the fashion industry.

2.9.2 Human Capital

A skilled workforce is widely acknowledged as a cornerstone of entrepreneurship ecosystems, facilitating innovation and driving entrepreneurial activity (Isenberg, 2011b; Spigel & Harrison, 2018; Stam & Van De Ven, 2021). In the fashion sector, skilled graduates, experienced designers, and serial entrepreneurs play crucial roles in enhancing production processes, developing high-quality products, and achieving strategic market positioning, thereby improving the performance of MSMEs (Brydges & Pugh, 2021; Castro et al., 2020; Rieple et al., 2018). This sector, shaped by globalisation, technological advancements, and shifting consumer preferences, demands a workforce that combines foundational skills such as drawing, designing, and sewing with artistic competence and technological expertise to meet

evolving industry requirements (Kampakaki & Papahristou, 2020; Takhirovna, 2024; Zhao & Leiby, 2020).

Emerging technologies, including advanced materials and digital tools, are increasingly integral to sustaining competitiveness in the fashion industry. Professionals are required to anticipate future trends while combining technical expertise with adaptability, collaboration, and problem solving skills to succeed in fast-paced environments (Leiby, 2021; Montagna et al., 2022; Torán et al., 2018; Varra, 2021). As digitalisation and sustainability reshape the industry, there is an increasing demand for roles requiring intellectual creativity and technical proficiency, particularly in digital skill sets (Gangoda et al., 2023).

The significance of workforce skills for entrepreneurial success has been well documented. Studies have consistently demonstrated that employees' expertise and skill levels strongly influence their entrepreneurial orientation and ability to deliver competitive products (Aakko & Niinimäki, 2018; Gurova & Morozova, 2018). Moreover, well-trained employees enhance productivity and quality, strengthening MSMEs' performance in both local and global markets (Cunha et al., 2023). In regions prioritising sustainability, such as Europe, key focus areas include resource-efficient manufacturing and the use of recyclable materials (D'Itria & Colombi, 2023).

Despite these advancements, skill shortages persist in areas such as financial management, branding, and strategic planning, which are critical for business success. Comparative studies have revealed disparities in fashion ecosystems. For example, Cravo et al. (2018) found that Sweden and Denmark benefit from private sector collaboration and government-led initiatives, respectively, whereas Finland faces challenges owing to limited business skills and industry collaboration. Similarly, the Turkish ready-to-wear industry and Egyptian technical schools experience shortages of qualified personnel and technical expertise, hindering industry growth (Abughattas, 2016; Özbek & Esmer, 2018). To address these human resource deficiencies, fashion companies often rely on freelancers or outsource labour to manage costs and regulatory challenges, with many shifting productions to lower-cost countries such as Bangladesh (Beerepoot & Lambregts, 2015; Kuruvilla, 2024; Seifried et al., 2023). However, these strategies can lead to job losses in local markets, reduced product quality, and limited consideration of ethical and environmental concerns (Chavan, 2018; Khan, 2018).

To counteract workforce challenges, research emphasises the importance of upskilling employees through structured and tailored training programs which enhance professional development and productivity (Aldhuhoori et al., 2022; Surikova, 2024). Such programs are

particularly crucial for addressing skill mismatches in MSMEs, which often lack the financial and organisational capacity to compete with larger companies offering more attractive training and employment opportunities (Herrera, 2020; Imran et al., 2024; Kapila & Dhillon, 2019). Accessible and cost-effective training initiatives, supported by targeted policies, can help align workforce capabilities with long-term organisational goals, thereby strengthening the competitiveness of MSMEs (Bilan et al., 2020; Rauch & Hatak, 2016). This underscores the need for targeted policies and support mechanisms to address skill shortages and improve HR efficiency in MSMEs (Castro-Silva & Lima, 2023). Furthermore, inefficient human resource practices, regulatory barriers, and local talent shortages exacerbate hiring challenges, particularly for technical roles in the industry. Although experienced foreign professionals bring expertise, their high costs present affordability issues (Sharma, 2024). Addressing these inequalities through strategic initiatives to promote entrepreneurial skills and bridge skill shortages can foster a more equitable and competent workforce (Manske, 2021).

In the Saudi Arabian context, significant discrepancies between educational outcomes and labour market demands have contributed to high unemployment rates and reduced productivity. To bridge these inconsistencies, studies have highlighted the importance of aligning higher education outputs with industry needs (Alanazi & Benlaria, 2023; Eid et al., 2023). Many graduates feel unprepared for fashion industry roles, which may limit their confidence, productivity, and career paths (Alzahrani & Kozar, 2017; Maghrabi & Abalkail, 2024). This misalignment between educational outcomes and market demands has been consistently identified in the literature as a critical issue affecting workforce availability and start-up growth (Alsolaim, 2019).

E-learning has emerged as a valuable tool for equipping individuals with the skills necessary for the development of MSMEs, particularly when supported by government policies and cultural norms (Alshibani et al., 2024; Singh et al., 2022). Such initiatives can enhance adaptability and skill levels of MSME workforces, particularly in addressing industry-specific challenges (Singh et al., 2022). Supportive policies can enhance the effectiveness of e-learning initiatives, thereby benefiting MSMEs by creating a more skilled and adaptable (Singh et al., 2022). However, government policies, including Saudization and visa restrictions, have created additional barriers to accessing skilled foreign labour, further exacerbating workforce shortages for startups and MSMEs (Alsolaim, 2019). For instance, the Nitaqat policy, a nationalisation employment scheme, has increased employment opportunities for Saudi nationals, including women, while also aligning with Vision 2030's economic diversification goals (Al Abdulkarim, 2018; Basahal et al., 2021). Nevertheless, these policies have

inadvertently limited access to highly skilled foreign workers, impacting productivity and contributing to higher SME exit rates (Cortes et al., 2020). Immigrant labour is widely recognised as essential to the Saudi economy; however, restrictive policies have reportedly led to significant departures of foreign workers, contributing to challenges in the labour market. Research highlights that Saudi nationals are generally less willing to work under the same conditions as foreign workers (Alsolaim, 2019). The preference for public sector jobs among Saudi workers, due to better wages and job security, further restricts the talent pool available to private sector MSMEs and raises concerns about retention and competition from former employees (Alqahtani et al., 2023).

The Saudi fashion industry, in particular, faces pronounced skills shortages in design, production, and management due to outdated curricula that fail to meet global market demands (Alshibani et al., 2024; Alzahrani & Kozar, 2015, 2017). Employers often rely on internal training programs or external providers to bridge these gaps, which can increase costs and constrain innovation (Maghrabi & Abalkail, 2024). Digital and remote learning platforms have been increasingly adopted to address barriers to training; however, challenges remain in the remote delivery of industry-relevant skills (Alshibani et al., 2024; Alzahrani & Kozar, 2017; Brydges & Hracs, 2019). Aligning educational curricula with industry requirements and fostering targeted training initiatives are crucial for improving workforce capabilities and enhancing the performance of MSMEs in the Saudi fashion sector (Maghrabi & Abalkail, 2024; Potnuru & Arora, 2022).

2.9.3 Policy

Government policies and regulations play a critical role in shaping EEs across various sectors, including the fashion industry. Effective initiatives often include simplifying business registration processes, streamlining municipal approvals, addressing labour and logistics challenges, and enhancing infrastructure to support entrepreneurial activities (Fkun et al., 2023; Pulka et al., 2021; Sipakoly, 2024). Financial incentives and educational programs are also widely recognised as essential for fostering an entrepreneurial culture and encouraging innovation. Dai and Si (2018) argue that well-designed government policies enhance entrepreneurial participation and strengthen firms' entrepreneurial orientation. Similarly, Campos et al. (2021) emphasised the importance of financial support, regulatory reforms, and education in promoting entrepreneurial ecosystems.

In the fashion sector, government intervention is critical for addressing issues such as equity and scarcity of entrepreneurial opportunities. For instance, Colliander (2015) and

Salonoja (2013) stress the need for education, business training, and investment incentives tailored to fashion MSMEs. Initiatives such as the “Next-Generation Fashion Ecosystem” exemplify how government efforts can foster digital transformation, micro factories, and localised production, enhancing both sustainability and customisation capabilities within the sector (Cunha et al., 2023). Such initiatives address the growing demand for eco-innovation and bolster industry competitiveness. Despite these positive developments, significant challenges remain. Delays in municipal approvals hinder the timely execution of fashion projects, affecting production and recycling processes (Sugeng et al., 2022). Inconsistent regulatory frameworks across regions exacerbate inefficiencies, creating confusion and increasing compliance burdens for businesses (Dobos & Éltető, 2023; Mizrachi & Tal, 2022). For example, Toronto’s fashion ecosystem has suffered from incoherent policies and a lack of cohesive government support, while Sweden and Denmark have successfully leveraged export-oriented strategies and strategic alliances to position their fashion industries as globally competitive (Brydges & Pugh, 2021; Colliander, 2015; Salonoja, 2013).

Inadequate incentives for innovation and research in the textile sector further exacerbate these challenges. Many firms face difficulties in implementing sustainable technologies because of limited government support (Khurana & Ataniyazova, 2020; Sugeng et al., 2022). Policy effectiveness is also influenced by how fashion entrepreneurship is categorised in the creative industries. For instance, Finland’s classification of fashion alongside video games and IT limits targeted policy support, whereas Sweden’s independent categorisation facilitates more tailored interventions (Salonoja, 2013). Labour policies that fail to support access to high-quality human resources undermine growth and competitiveness in the sector (Sugeng et al., 2022; Wilson, 2020).

Market and supply chain policies are equally significant in shaping EEs. Wilson (2020) observed that policies that improve the broader market environment yield more substantial benefits than narrowly focused support for individual companies. Intellectual property (IP) protection is another area that requires policy attention. Rusadi et al. (2023) argue that robust IP frameworks are crucial in the digital era, where designs are easily copied and distributed. However, León et al. (2016) found that the high costs associated with formal IP rights can act as barriers to entry for small enterprises, particularly in developing regions. Strengthening IP protection is essential for fostering creativity and innovation in the fashion sector (Rusadi et al., 2023).

In Saudi Arabia, recent policy amendments have created a more conducive environment for entrepreneurship. Studies have noted that these amendments include financial support,

regulatory reforms, and educational initiatives aimed at fostering an entrepreneurial culture (Akinwale et al., 2020; Almahdi, 2020). Measures such as enhancing R&D transfers and promoting post-school entrepreneurship training have been identified as critical for building an ecosystem that supports innovation-driven entrepreneurship (Akinwale et al., 2020; Saberi & Hamdan, 2019). However, persistent challenges remain, including limited access to resources, bureaucratic hurdles, and inadequate entrepreneurial skills (Almuzel & Anderson, 2020; Khan, 2013, 2016). The Saudi government's approach to charging fees prior to business start-up, rather than after income generation, has been identified as a significant constraint for entrepreneurs (Alsolaim, 2019). High compliance costs, lengthy licencing procedures, and unclear regulations create uncertainty and hinder start-up success (Alsolaim, 2019). Addressing these barriers through targeted reforms, particularly those aligned with Saudi Vision 2030, presents an opportunity to develop a robust FEE (Almuzel & Anderson, 2020).

The Saudi fashion industry also faces unique policy-related challenges. Outdated regulations, insufficient incentives for innovation, and high start-up costs limit the growth potential of the sector. Employers in the sector report difficulties in accessing skilled workers due to restrictive labour policies, while fragmented regulations hinder the development of sustainable practices (Alshibani et al., 2024; Sugeng et al., 2022; Wilson, 2020). Despite these obstacles, ongoing reforms aligned with Vision 2030 provide opportunities to advance innovation, sustainability, and competitiveness in the Saudi fashion industry (Almahdi, 2020; Almuzel & Anderson, 2021).

2.9.4 Markets

The market domain is a fundamental pillar of EEs, encompassing networks, collaborations, consumer behaviour, distribution channels, and market research. Collectively, these elements foster EO, drive BP, and catalyse innovation by enabling market opportunities (Spigel & Harrison, 2018; Stam & Van de Ven, 2018; Wurth et al., 2023).

Effective ecosystems rely on tailored, collaborative stakeholder support (Bischoff et al., 2018) and strong connections among entrepreneurs, support organisations, and other ecosystem components (Motoyama & Knowlton, 2017). Strong networks with potential partners and stakeholders are pivotal for MSMEs, facilitating access to essential resources, market opportunities and shared knowledge. For example, Salonoja (2013) attributes the success of the Swedish and Danish fashion ecosystems to robust collaborations among designers, industry stakeholders, and private investors. Similarly, online innovation networks, such as Openwear, have demonstrated their value in fostering creativity and sustainable practices in the fashion

sector (Eckert et al., 2022; Giusti & Alberti, 2018). Networks play a pivotal role in the configuration and evolution of EEs, supporting knowledge transfer, innovation and entrepreneurial performance (De Brito & Leitão, 2020; Secundo et al., 2020; Wurth et al., 2021). Advocating initiatives such as coworking spaces, tailored entrepreneurial events, and alternative networking environments beyond traditional chambers of commerce reflects an increasing emphasis on fostering collaboration and community building in EEs (Dollar, 2021). Additionally, large and accessible markets, supported by robust distribution channels and early adopters, are critical for entrepreneurs to achieve market traction and sustained growth (Almuzel, 2022; Herrera, 2020).

In the fashion industry, both formal and informal networks are vital for fostering entrepreneurial growth and sustainability in the fashion industry. These networks enhance supply chain relationships, professional connections, and sustainability practices, particularly for small and micro businesses (Eckert et al., 2022). Ethnic fashion designers and family businesses also rely on cultural networks to support internationalisation and knowledge transfer, leveraging traditional designs, colours, and fabrics (Singh, 2024). The literature further emphasises the importance of mentorship opportunities, stakeholder engagement, and knowledge-sharing platforms in strengthening entrepreneurial ecosystems (Bischoff & Volkmann, 2018). For example, studies illustrate the importance of networking among stakeholders for mentorship and knowledge sharing as critical support in the Finnish fashion ecosystem (Salonoja, 2013; Tervilä, 2015). Conversely, Brydges and Pugh (2021) indicate fragmented networks can undermine market engagement, as seen in Toronto's fashion ecosystem where inadequate collaboration has limited designers' access to distribution networks and market intelligence. In Saudi Arabia, Elnadi and Gheith (2021) emphasised the importance of social networks in creating business opportunities, whereas Ali et al. (2019) highlighted how networks provide critical resources such as market intelligence and distribution channels. However, Almuzel & Anderson (2020) observed that bureaucratic challenges and limited networking platforms impede effective collaboration, underscoring the need for structured support mechanisms in Saudi Arabia.

Market research is consistently recognised in the literature as a crucial tool for aligning products with consumer needs, guiding marketing strategies and informing strategic decision-making. Effective market research supports product positioning, quality control, and customer service, thereby enhancing brand performance (Gengler & Mulvey, 2017; Gul et al., 2021a; Gul et al., 2021b). Additionally, market research supports the development of new products and the formulation of content marketing strategies by providing valuable insights which

enhance decision-making processes and refine product positioning (Bode & Geiger, 2021; Wei et al., 2023). For example, in Russia, brands such as LIME leverage market research to understand target audiences and maintain real-time communication with customers, ensuring strategic decision-making and customer satisfaction (Mikryukov et al., 2023).

However, in Saudi Arabia, the lack of comprehensive market insights has hindered local fashion brands from competing effectively with their international counterparts (Maghrabi & Abalkail, 2024). An insufficient understanding of consumer behaviour and emerging trends undermines entrepreneurial success and market positioning (Abdulghaffar & Akkad, 2021). Although Vision 2030 initiatives aim to enhance market research capabilities, cultural and operational factors remain underexplored. Practical frameworks for integrating consumer insights into entrepreneurial strategies are needed to improve understanding (Ali et al., 2019).

Salonoja (2013) identified challenges such as limited access to business service providers, fragmented information sharing, inadequate distribution channels, and a scarcity of reliable market data. Understanding consumer behaviour is integral to seizing market opportunities and driving EO. Shifts in consumer preferences, such as the growing demand for sustainable fashion, have opened new markets and inspired the creation of sustainable fashion ventures (Busalim et al., 2022). Research has revealed a growing shift in Saudi consumer awareness towards sustainability, with many customers seeking products that align with environmental and ethical values. Thus, studies such as Alshibani et al. (2024) have highlighted that marketing strategies emphasising sustainability foster strong customer loyalty and enhance brand image, and leverage social media as a crucial platform for promoting ethical brands, as consumers increasingly rely on these channels to discover responsible products. Similarly, the rise of online shopping has expanded the reach of MSMEs, emphasising the importance of digital presence and strong brand engagement in boosting sales (Rahman & Mannan, 2018).

In Saudi Arabia, consumer attitudes towards local and foreign brands are complex. While ethnocentric sentiments promote local industries, international brands are often favoured for their perceived quality and prestige (Almatrudi et al., 2023; Alsulami, 2018; Šapiæ, 2018). Studies indicate that local brands can improve their competitiveness by enhancing quality, consumer satisfaction, and brand image (Klabi, 2020; Mkedder et al., 2021). Furthermore, leveraging national identity through targeted marketing strategies aligned with consumer ethnocentrism can significantly boost local brand success (Šapiæ, 2018). However, the dominance of foreign brands necessitates ongoing improvements in local product quality to maintain consumer trust and foster loyalty (Alsulami, 2018; Šapiæ, 2018). Effective distribution strategies are critical for expanding consumer reach, improving brand value, and

achieving sustainable business performance. Multichannel distribution allows fashion brands to cater to diverse consumer preferences through physical stores, online platforms, and mobile applications, thereby enhancing convenience and accessibility (Ko et al., 2017; Ryu et al., 2019).

Digital technologies have enabled brands to recreate offline distribution models online, aligning with evolving consumption patterns and increasing operational efficiency (Mosca & Chiaudano, 2020). In Toronto, Brydges and Pugh (2021) found that independent fashion designers rely on e-commerce platforms such as Shopify and social media such as Instagram to bypass traditional retail barriers and reach consumers directly. However, online sales often remain geographically tied to designers' physical locations or stockist areas, and systemic issues, such as high operational costs and limited local support, continue to hinder scalability (Al-Mushayt et al., 2022).

In Saudi Arabia, digital platforms have emerged as cost-effective tools that enable designers to navigate regulatory barriers and reduce overheads (Al-Mushayt et al., 2022). However, an overreliance on social media limits market reach and scalability. Physical retail faces additional challenges, including high operational costs and licencing hurdles, which deter many entrepreneurs from establishing traditional outlets (Alateeg & Alhammadi, 2023). Vision 2030 initiatives aim to address these barriers by improving logistical infrastructure and distribution networks; however, inconsistent implementation has left many entrepreneurs struggling to optimise supply chains (Abdulghaffar & Akkad, 2021).

2.9.5 Finance

Finance is a critical component of EEs, providing the foundational resources that enable startups and small businesses to innovate, grow, and sustain their operations (Ogujiuba et al., 2023; Sharma, 2021). According to Bramwell et al. (2019) and Lange and Schmidt (2020), the availability and accessibility of financial support are essential for fostering EO and enhancing BP. Different sources of finance, such as venture capital, private equity, debt financing, and alternative funding models, play distinct roles in supporting businesses across various stages of development (Katic & Vasic, 2020; Sharma, 2021; Tenca et al., 2020). For instance, Katic and Vasic (2020) highlight that venture capital and private equity provide financial resources and offer strategic guidance, mentorship, and networking opportunities that are crucial for high-growth startups. Likewise, venture capitalists absorb the risks associated with inexperienced entrepreneurs, fostering an environment conducive to innovation and risk-taking (Lehtonen & Lahti, 2009).

The literature also underscores the complementary role of public sector financial mechanisms, such as grants, equity funding, debt financing, and crowdfunding, in addressing early-stage financial gaps (Dhewanto et al., 2022; Frimanslund et al., 2023; Owen et al., 2018). These mechanisms are particularly significant for startups navigating the “Liabilities of Newness” as they alleviate the pressure of traditional loans and enable businesses to focus on growth and innovation (Li, 2023; Reddy, 2015). Crowdfunding, as discussed by Schlattau (2021) and Isenberg (2011b), has further democratised access to capital by allowing entrepreneurs to engage potential customers and validate market demand. However, Schlattau notes that while crowdfunding facilitates early-stage funding, its scalability for larger ventures is limited.

In the context of FEE, financial support is particularly important for enabling the recruitment of skilled employees, investment in marketing, and acquisition of infrastructure (Hodges et al., 2016; Mills, 2018). Scholars such as Yagoubi and Tremblay (2015) and Van Den Berg and Almanza (2016) argue that adequate financial resources are closely linked to the growth and sustainability of fashion businesses. However, financial constraints in FEEs often limit these benefits, particularly when compared to ecosystems with stronger financial backing, such as FinTech ecosystems (Colliander, 2015; Tervilä, 2015). For example, Salonoja (2013) contrasts robust private-sector financing in Sweden which has supported the global success of brands like H&M, with Finland’s limited private investment which has restricted the international competitiveness of its fashion industry. This disparity underscores the importance of private sector involvement in establishing effective financial ecosystems.

Studies have found that barriers to financial access within FEEs include a lack of dedicated financial programs, an underdeveloped culture of fashion investment, and limited entrepreneurial financial literacy. Entrepreneurs often struggle to find capital, especially in the early-stages, when investors prefer more established ventures with deals in the pipeline (Herrera, 2020). Raising seed funding in Australia can be challenging compared to other regions, such as the US, where investors are more willing to take risks on early-stage ideas (Herrera, 2020). Brydges and Pugh (2021) suggest that a lack of understanding of the fashion sector discourages potential investors. Furthermore, Sarkar and Karim (2019) and Min and Wilson (2019) highlight that fashion designers often encounter difficulties in presenting compelling business cases or establishing strong connections with investors. Heim (2018) critiques incubators for prioritising profitability over meaningful support, which further limits their financial accessibility. The literature also identifies financial literacy as integral to overcoming these challenges. Akinyede (2023) argues that entrepreneurs with advanced

financial knowledge make more informed decisions and manage resources more effectively, while Anshika and Singla (2022) observe that limited financial literacy exacerbates difficulties in securing investments and preparing business plans.

The inherent volatility of the fashion industry compounds its perception as a high-risk sector. Rapid product cycles, shifting consumer preferences, and market uncertainty discourage financial institutions from offering tailored support (Ningsih, 2023). Structural barriers, including insufficient financial planning and a lack of sector-specific training, further hinder the stability of fashion enterprises (Nwaozuru et al., 2024). Studies have noted that in emerging markets, fashion entrepreneurs frequently rely on informal funding sources, including personal savings and family contributions, to address these gaps (Mefful et al., 2023; Omri, 2020). When these resources are depleted, entrepreneurs often turn to social and political networks to secure formal funding (Nwaozuru et al., 2024; Khan & Jang, 2020).

Financial support also plays a critical role in enhancing EO dimensions, such as proactiveness, innovativeness, and risk-taking, which contribute to improved business outcomes (Davis et al., 2024; Lang & Liu, 2022). Aguilar et al. (2019) suggest that sector-specific financial solutions, such as short-term funding for new collections or marketing campaigns, are particularly effective in addressing the fashion industry's unique demands. Self-funding and venture capital offer distinct advantages, with Ermawati (2024) noting that self-funding promotes autonomy and resourcefulness, while venture capital provides both capital and expertise, albeit within a competitive environment.

In Saudi Arabia, the financial challenges faced by fashion MSMEs are shaped by institutional inefficiencies, gender-specific barriers, and sectoral dynamics (Abou-Moghli & Al-Abdallah, 2019; Albasri et al., 2016). Alsolaim (2019) identifies limited funding access, high regulatory costs, and inefficient institutional support as key obstacles for small businesses in Saudi Arabia. These challenges are compounded by the reliance on government initiatives, which are often hindered by bureaucratic inefficiencies (Abdulghaffar & Akkad, 2021; Albasri, 2020). While Vision 2030 initiatives aim to improve access to financing, unclear processes and limited awareness continue to impede progress (Almahdi, 2020). Additionally, Alshibani et al. (2024) identified high costs associated with sustainable materials and weak connections between designers and investors as key constraints on growth. To address these challenges, Alsaiani (2022) called for systemic reforms, including reducing bureaucratic hurdles, improving transparency in funding processes, and increasing private sector involvement. Enhanced financial literacy and targeted support for fashion entrepreneurs could mitigate gender-specific barriers and foster a more inclusive financial landscape.

2.9.6 *Culture*

The cultural domain within EEs encompasses societal attitudes towards entrepreneurship, collaboration, tolerance for risk and failure, innovation norms, and the celebration of entrepreneurial success. Collectively, these factors shape the ecosystem's support for entrepreneurial activity (Isenberg, 2016; E. Stam, 2018). Cultural values are critical in shaping entrepreneurial behaviour, with distinctions between individualistic and collectivist cultures. In individualistic cultures, failure is often perceived as a learning opportunity that promotes resilience and innovation. Conversely, collectivistic cultures prioritise stability and social harmony, which may moderate risk-taking and entrepreneurial activity (DeScioli, 2024).

In the context of fashion entrepreneurship, cultural attitudes significantly influence collaboration, mentorship, and sustainable practices. A prevalent culture of “going it alone” among fashion entrepreneurs often limits opportunities for collaboration and mentorship, both essential for entrepreneurial success (Salonoja, 2013, p. 87). Scholars have also underscored the significance of cultural shifts towards sustainability in driving innovation and collaboration in the fashion sector. For example, in Europe, the adoption of circular fashion practices has catalysed collaborative efforts and eco-conscious consumption behaviours, highlighting the importance of adapting to evolving cultural norms to ensure the success of fashion enterprises (D'Itria & Colombi, 2023).

Regional differences further illustrate the impact of cultural barriers on entrepreneurial outcomes. In Finland, limited integration between design and business hinders growth despite the country's strong design talent (Colliander, 2015; Tervilä, 2015). Similarly, Toronto's fragmented cultural identity among designers and institutions undermines cohesion and innovation (Brydges & Pugh, 2021). In Sri Lanka, geographical and cultural disconnections from Western markets pose challenges for designers, although targeted learning and exposure have helped to bridge these gaps (Gopura et al., 2019). Across these regions, entrepreneurs face significant challenges in forming peer connections due to insufficient social skills, which are critical for effective networking, mentorship and collaborative innovation (Dimitriadis & Koning, 2022).

Gupta (2024b) expands on these sociocultural dynamics in emerging markets, emphasising the need for culturally tailored interventions. Her research highlights cultural resilience and social networks as critical factors enabling entrepreneurial adaptability. Across these contexts, integrating cultural elements with entrepreneurial goals through frameworks such as the Quadruple Helix and Effectuation Theory is essential for fostering collaboration, sustainability, and ecosystem synergy. Addressing cultural fragmentation and promoting

global cultural adaptation are critical for enhancing the performance of fashion ecosystems worldwide (Brydges & Pugh, 2021).

The media also plays a crucial role in FEEs. Platforms such as fashion magazines and blogs increase industry visibility and strengthen investors' confidence. For instance, Rinkinen and Laakso (2023) highlighted that local media in Finnish regions disseminate entrepreneurial success stories, which foster ambition, risk-taking, and innovation, thereby strengthening the entrepreneurial culture. Similarly, the Swedish fashion industry has benefited from media coverage, which has increased investor confidence and visibility (Salonoja, 2013). Similarly, Colliander (2015) notes that the increased visibility of successful Finnish fashion companies has enhanced their international reputation.

Moreover, role models featured in the media significantly impact entrepreneurial intentions through constructs such as subjective norms and perceived behavioural control (Choukir et al., 2019). Dollar (2021) suggested emulating successful local business models to strengthen the business's culture. highlighted that role models significantly impact entrepreneurial intentions through constructs such as subjective norms and perceived behavioural control. Media attention helps normalise entrepreneurship, especially for women, by portraying successful female entrepreneurs. However, admiration alone does not guarantee increased entrepreneurial activity; tangible support and mentorship are essential (Alshebami & Seraj, 2022). Societal pressure for quick success discourages risk-taking and hinders innovation, as entrepreneurs fear societal backlash for failure (Almesahri et al., 2023; DeScioli, 2024). This aligns with the broader literature, which highlights the importance of leveraging local success stories to strengthen the entrepreneurial culture and foster collaboration (Donaldson, 2020; Schlattau, 2021).

In Saudi Arabia, cultural perceptions significantly shape the development of the fashion industry and the opportunities available to entrepreneurs. Alzahrani and Kozar (2015) noted that fashion is often perceived as a low-status academic and employment field, predominantly pursued by women. This perception stems from deep-rooted cultural and gender norms that position fashion as a “feminine” discipline, misaligned with societal expectations of men's roles (Kareem, 2014). Patriarchal frameworks reinforce this perception, creating barriers for male designers and limiting the field's prestige (Aldossari & Chaudhry, 2024).

Furthermore, social conventions create additional barriers for male designers, who may feel unwelcome or unwilling to design for a predominantly female market (Syed et al., 2018). However, Saudi Vision 2030 seeks to reshape these attitudes by promoting entrepreneurship as a prestigious career path, empowering women, and fostering the cultural acceptance of

fashion as a legitimate industry (Albasri, 2020; Almuzel, 2022; Khan, 2016). Saudi culture strongly influences product design, particularly by emphasising modesty and practicality, which aligns with conservative values. Clothing is often designed to respect religious and social norms, which is a priority for many Saudi brands (Zain et al., 2016). Simultaneously, Saudi brands are appealing to younger generations by incorporating innovative designs that merge traditional aesthetics with modern elements (Alshibani et al., 2024). This blending of tradition and modernity has gained wide consumer acceptance in China, highlighting the potential for similar success in Saudi Arabia to gain acceptance.

Cultural norms have a profound influence on fashion design in Saudi Arabia. Entrepreneurs must align their products with cultural values that emphasise modesty and tradition, which can limit creative expression but also create niche markets for modest fashion (Amalanathan & Reddy, 2024). Cultural heritage is a source of inspiration for Saudi designers, who incorporate traditional motifs into modern fashion, creating culturally resonant products appealing to local and international audiences. Fashion education in Saudi Arabia is evolving to preserve cultural authenticity while integrating global standards, although balancing these priorities remains challenging. As Alfawaz (2024) found, entrepreneurs are reviving traditional practices in the Saudi textile sector using modern techniques, such as computer-aided design and digital printing, to suit contemporary markets. However, in a globalised market, the local industry faces competition from imported, mass-produced textiles and fashions.

Empowerment initiatives under Vision 2030 are helping women overcome these barriers by enhancing their entrepreneurial self-efficacy and aspirations (Badghish et al., 2023). Changing family and community perceptions further encourage women to pursue entrepreneurship and recognise their contributions to society. Religious values also shape entrepreneurial activities, framing them as both economic and spiritual responsibilities. For Saudi entrepreneurs, aligning business practices with Islamic principles enhances cultural legitimacy and acceptance, particularly among women. The findings reveal that Saudi youth perceive entrepreneurship as a promising career option because of supportive institutional frameworks and targeted policies, which positively influence entrepreneurial intentions by increasing perceived desirability and feasibility (Aloulou, 2021).

Social norms have a dual impact, providing encouragement for entrepreneurship while simultaneously imposing constraints. For instance, social norms have a dual impact on entrepreneurship, encouraging innovation while simultaneously imposing constraints, as high societal expectations for perfection often deter risk taking (Blaese, 2019; Newth & Woods, 2014). fashion (Colliander, 2015). highlighted the necessity of tolerating risk, mistakes, and

failure in the fashion industry Fashion entrepreneurs lack entrepreneurial ambitions and are often seen as risk-averse (Salonoja, 2013). Nonetheless, these constraints have driven designers to innovate within conservative preferences, leading to unique adaptations (Fahrati, 2023).

In Saudi Arabia, women's entrepreneurship is extensively discussed as a significant cultural factor influencing entrepreneurship. Gender norms pose additional challenges to female entrepreneurs in the Saudi fashion industry (Basaffar, 2017). Female entrepreneurs in Saudi Arabia exhibit slightly higher growth-oriented attitudes than males, despite social norms influencing the entrepreneurial process (Sabri & Thomas, 2019). This underscores the importance of targeted policies to enable greater participation by women. Cultural, social, and financial constraints influence women's entrepreneurial participation; consequently, training and institutional support are critical for their success (Abou-Moghli & Al-Abdallah, 2019).

2.9.7 Support

The support domain within EE constitutes a fundamental pillar for enabling the establishment, growth, and sustainability of ventures. It encompasses resources, services, and infrastructure tailored to the needs of entrepreneurs, while informal networks and shared norms foster collaboration and innovation (Isenberg, 2011; Schmidt et al., 2024). Support mechanisms shape individual entrepreneurial behaviours and influence broader ecosystem dynamics by facilitating access to resources and fostering competitive capabilities (Biru et al., 2020). This literature review explores the unique challenges and opportunities of entrepreneurial support, with a particular focus on the fashion industry.

The fashion sector demonstrates a pressing need for specialised support structures, such as fashion incubators, design associations, and industry-specific clusters. These initiatives play a pivotal role in enabling innovation and international competitiveness. Research underscores the significance of fashion incubators in providing tailored technical and market assistance, as evidenced by studies spanning diverse regions (Harvey, 2013; Patnaik & Isaacs, 2024; Wahyuningsih, 2019). However, in Finland, the absence of comprehensive support frameworks, particularly mentorship and accelerator programs, has created significant barriers to entrepreneurial success. Fragmentation among existing organisations, such as Finatex and Ornamo, further compounds this issue, prompting discussions about establishing a Finnish fashion institute to unify industry efforts (Relyea, 2016; Salonoja, 2013).

In contrast, countries with robust fashion ecosystems, such as Italy and France, exemplify the positive impact of coordinated efforts. Annual events such as Italy's Pitti Uomo and France's Haute Couture Week set global trends and demonstrate the value of sustained industry

collaboration (Aiello et al., 2016; Majima, 2008). Furthermore, the integration of digital technologies enhances these ecosystems, as seen in the proliferation of online platforms that facilitate the expansion of fashion markets (Borgioli, 2021). Emerging markets also offer innovative approaches, such as creative hubs that integrate local knowledge and provide context-specific training for artisan entrepreneurs. For instance, the ZIPhouse in Moldova bridges the gap between manufacturing and design by fostering collaboration between universities and industry while emphasising sustainability and innovation (Nistor, 2024). Similarly, proposed initiatives, such as the techno hub in Cape Town, aim to equip emerging entrepreneurs with advanced technology and sustainable business practices (Patnaik & Isaacs, 2024).

Challenges to effective support mechanisms are not confined to Finland alone. For example, Australia exhibits significant regional disparities, with some areas providing better infrastructure and support than others. Herrera (2020) highlights the uneven resource allocation between industries, noting the concentration of support in established sectors such as technology and pharmaceuticals while overlooking creative industries such as fashion. Southeast Queensland's ecosystem illustrates the importance of tailoring support frameworks to regional needs and acknowledging the interdependencies between industries and regions (Scheepers et al., 2018). Similarly, in Ethiopia, support programs that do not prioritise innovation and competitiveness inadvertently stifle entrepreneurial risk-taking and growth (Biru et al., 2020).

The lack of adequate mentorship and industry expertise has been consistently identified as a critical deficiency in ecosystems across regions. For Saudi fashion entrepreneurs, limited access to expert guidance in areas such as brand development and design innovation hinders growth (Colliander, 2015). The need for cohesive leadership, credible network organisations, and knowledge-sharing platforms is evident in various ecosystems, with calls for creating dedicated spaces for entrepreneurial education and collaboration (Chahine, 2018; Dollar, 2021). Finland, in particular, exemplifies the detrimental effects of fragmented support systems, where a lack of collaboration and trust among existing organisations hinders long-term strategic progress (Salonoja, 2013).

2.9.8 Contextualising the Saudi FEE: A Comparative Perspective

While general EE frameworks such as those developed by Isenberg (2011), Stam and van de Ven (2021), and Spigel (2017) provide valuable conceptual foundations, they are largely shaped by Western institutional, cultural, and market assumptions. These models tend to presume regulatory maturity, liberal cultural norms, and sector-agnostic support systems. However, such assumptions are often misaligned with the institutional realities and industry-specific dynamics of emerging economies. In this context, the Saudi FEE presents a set of distinctive features that necessitate a tailored, context-specific analysis.

Table 2.6 outlines the principal differences between global EE assumptions and the Saudi FEE across key domains, informed by both international literature and regionally grounded studies. This comparison highlights substantial gaps between the assumptions underpinning global EE models and the lived realities of fashion-focused entrepreneurs in Saudi Arabia. These differences support the development of a distinct FEE framework that reflects local institutional limitations, socio-cultural dynamics, and sector-specific challenges.

2.10 The Research Gap

There is a growing body of literature on EEs and their relevance to the fashion industry. However, substantial gaps remain in our understanding of how these ecosystems function holistically in emerging markets such as Saudi Arabia. Existing studies underscore critical challenges, including the misalignment between education and industry requirements, workforce skill deficiencies, regulatory hurdles, and a lack of integration across the market, financial, and cultural dimensions. As outlined in Table 2.6, challenges are particularly acute in Saudi Arabia, where structural barriers, antiquated policies, and cultural norms constrain the growth and potential of MSMEs. This study seeks to address these gaps by adopting a holistic framework to investigate the interplay among key domains within the Saudi FEE: entrepreneurship education, talent development, regulatory policy, market dynamics, institutional support, financial resources, and cultural influences. By synthesising these elements, this study aims to elucidate their collective impact on entrepreneurial orientation and business performance, ultimately proposing a framework to foster the development and long-term sustainability of fashion MSMEs in Saudi Arabia.

However, the entrepreneurship ecosystem supporting EO in the fashion sector remains insufficiently examined in Saudi Arabia (Abdulrab et al., 2021; Aloulou, 2018). This knowledge gap limits understanding of whether the current ecosystem facilitates or obstructs

EO-driven growth in these businesses (Abdulrab et al., 2021; Aloulou, 2018). Given the nascent state of Saudi Arabia's entrepreneurship ecosystem, particularly in the fashion sector, it is imperative to investigate the challenges and opportunities MSMEs face in effectively implementing EO (Alblowi et al., 2021; Almuzel & Anderson, 2020; Balawi, 2021).

EO significantly influences BP by driving innovation, enhancing proactive market strategies, and enabling businesses to take calculated risks, all of which are critical for achieving competitive advantages and sustained growth (Lumpkin & Dess, 1996; Lumpkin & Dess, 2001; Sorama & Joensuu-Salo, 2022). Business performance includes both financial and non-financial outcomes (Lumpkin & Dess, 1996; Rauch et al., 2009). Financial measures, such as revenue and profit growth, and non-financial factors, such as brand reputation and customer satisfaction, play a role in assessing the success of fashion MSMEs (Kropp et al., 2008; Soininen et al., 2015). In fashion MSMEs, EO plays a crucial role in driving business success by enabling firms to innovate and remain competitive (Juliani & Nuvriasari, 2024; Mulyana & Sutapa, 2016). While high EO often leads to better business performance, a broader entrepreneurship ecosystem influences the relationship between EO and BP (Fayomi, 2022; Kharisma et al., 2020; Mulyana & Sutapa, 2016). Factors such as access to resources, financial support, and market infrastructure can either enhance or limit a firm's ability to achieve strong performance outcomes (Putria & Gunawan, 2019; Rufaidah, 2019; Wongmuek, 2018).

Considering the characteristics of the Saudi fashion industry and the influences of culture, policy, and social domains, fashion MSMEs must be flexible and responsive to changing market trends and consumer demands. EO is key to helping fashion entrepreneurs navigate this complex and evolving environment. By being innovative, proactive, and willing to take risks, entrepreneurs can respond quickly to changes, use new technologies, and adapt products to meet cultural and seasonal trends or demands. Understanding how the FEE impacts EO is essential for success in the industry. This study highlights EO as a key driver of business performance, helping fashion entrepreneurs make the most of their ecosystem and achieve success.

Table 2.6: Literature Review Key Findings (Comparison between general EE and the Saudi FEE)

Domain	General EE	Saudi FEE
Entrepreneurship Education	Global education integrates business, sustainability, and technology within creative industries (Murzyn-Kupisz & Hołuj, 2021; Yezhova et al., 2018).	Fashion-related education in Saudi Arabia is largely disconnected from entrepreneurship. Curricula remain outdated, gender restrictions persist, and business training is limited (Alshibani et al., 2024; Alzahrani & Kozar, 2017). Reforms via Vision 2030 and the Fashion Commission are underway (Maghrabi & Abalkail, 2024; Ryan, 2023).
Human Capital	Ecosystem success relies on talent with interdisciplinary expertise across creative, digital, and business domains (Isenberg, 2011a; Spigel & Harrison, 2018; Zhao & Leiby, 2020).	Saudi Arabia experiences skill shortages in branding, financial management, and strategic planning. Labour rigidity and limited training inhibit the development of fashion-specific human capital (Alanazi & Benlaria, 2023; Alsolaim, 2019; Singh et al., 2022).
Policy and Regulation	EEs generally benefit from coherent policies, IP protections, and efficient licensing structures (Campos et al., 2021; Dai & Si, 2018).	While Vision 2030 promotes entrepreneurship, fashion MSMEs encounter policy fragmentation, unclear IP mechanisms, and limited incentives for creative sectors (Almuzel & Anderson, 2020; Sugeng et al., 2022; Alshibani et al., 2024).
Market Access	Competitive EEs typically include access to digital platforms, consumer insights, and international markets (Secundo et al., 2020).	Saudi MSMEs lack market research infrastructure, face difficulties scaling digital businesses, and operate in a supply chain heavily reliant on imports (Al-Mushayt et al., 2022; Rahman & Mannan, 2018; Maghrabi & Abalkail, 2024).
Finance	Mature EEs feature a mix of angel investment, venture capital, and crowdfunding (Dhewanto et al., 2022; Katic & Vasic, 2020).	Saudi MSMEs often rely on informal funding due to restricted access to formal finance. Gender-based financial barriers and unclear funding pathways persist despite reform initiatives (Abdulghaffar & Akkad, 2021; Rahatullah, 2023; Almahdi, 2020; Alsaiani, 2022).
Culture	Cultures that support EEs typically celebrate innovation and tolerate entrepreneurial failure (Isenberg, 2016; DeScioli, 2024).	In Saudi Arabia, cultural perceptions of fashion entrepreneurship remain mixed. Tailoring is often viewed as low-status, and women face entrenched social barriers, although Vision 2030 is gradually shifting these narratives (Zain et al., 2016; Badghish et al., 2023; Sabri & Thomas, 2019).
Support Systems	Successful EEs offer sector-specific incubators, industry mentoring, and collaborative hubs (Harvey, 2013; Nistor, 2024).	Saudi support infrastructure is generally geared toward tech sectors. Fashion entrepreneurs report limited access to relevant mentoring and fragmented support services, especially in non-urban areas (Almuzel, 2022; Alshibani et al., 2024).

2.11 Entrepreneurial Orientation in the Fashion Industry

Entrepreneurial orientation is a critical framework for examining how firms strategically engage in entrepreneurial activities. According to Balasubramanian et al. (2020), EO is the organisation's overall strategic posture towards entrepreneurship, capturing a sustained pattern of entrepreneurial behaviour and new entry initiatives. Foundational conceptualisations by Miller (1983) and Covin and Slevin (1991) define EO as a unidimensional construct comprising innovativeness, proactiveness, and risk-taking. However, Lumpkin and Dess (1996) proposed a multidimensional view that includes autonomy and competitive aggressiveness, providing a more nuanced understanding of EO's influence on firm performance (Covin & Wales, 2012; Yuefen et al., 2024). EO is considered both an individual and firm level construct, reflecting the firm's ability to identify and exploit international opportunities. (Covin & Wales, 2019; Wach, 2015) emphasise that EO captures behaviours, structures, and processes that empower firms to innovate, take risks, and proactively pursue opportunities for competitive advantage.

In the fashion industry, EO is particularly relevant because of the sector's demand for rapid innovation, adaptability, and responsiveness. Studies have consistently demonstrated that the synergy of innovativeness, proactiveness, and risk-taking enhances the adaptability and competitiveness of fashion enterprises (Juliani & Nuvriasari, 2024; Kharisma et al., 2020). EO enables firms to sustain creativity and maintain their edge in dynamic and competitive markets.

2.11.1 Innovativeness

Innovativeness is a core aspect of entrepreneurship that facilitates the conversion of creative ideas into distinctive and valuable outcomes. Fu et al. (2024) and Wang and Dass (2017) defined it as an organisation's ability to develop and implement innovations by creating, integrating, and applying knowledge, often driven by research and development efforts. This capability enables organisations to address challenges and seize new opportunities, thereby enhancing entrepreneurial performance, as supported by Dej et al. (2013), Emami and Dimov (2016), Oliinyk (2021), and Stauffer (2016). Moreover, recent research, such as Anning-Dorson (2018) and Chatzoglou and Chatzoudes (2018), highlights the vital role of innovation in achieving competitive advantages.

The role of innovativeness in fostering creativity and experimentation is particularly evident in the fashion industry, where differentiation through unique designs and trendsetting is critical for sustaining a brand identity. Rufaidah (2019) demonstrated that innovativeness

among Indonesian creative fashion SMEs enables them to adapt swiftly to evolving market demands and produce consumer-aligned products. Similarly, Putra and Gunawan (2019) found that innovative practices in the Muslim fashion sector in Bandung, Indonesia, allowed businesses to refresh their market offerings and maintain relevance in a highly competitive landscape. Mulyana and Sutapa (2016) further linked EO with enhanced innovation capabilities which drive creation of distinctive designs and a sustained competitive advantage in creative industries such as fashion. Additionally, Juliani and Nuvriasari (2024) highlighted the significant positive impact of innovativeness on marketing performance among Muslim fashion SMEs in Yogyakarta, Indonesia, emphasising its importance in responding to shifting market trends. By prioritising research and development and fostering a culture of innovation, enterprises in the fashion industry can remain aligned with emerging consumer preferences, ultimately achieving stronger financial performance and an elevated brand reputation.

2.11.2 Proactiveness

Proactiveness in an organisation refers to the ability and readiness to seek opportunities to introduce new products and services to achieve customer satisfaction and gain a competitive edge. It involves a forward-looking perspective in which companies actively anticipate opportunities to develop and introduce new products, thereby obtaining first-mover advantages and shaping their operational environment (Bateman & Crant, 1999; Parker et al., 2010; Parker et al., 2019). First-mover advantages refer to benefits an organisation gains by being first to introduce a new product, service, or idea into the market. By doing so, the organisation can establish a strong market presence, build brand recognition, and secure customer loyalty before competitors enter the field. Anticipating and acting on future market needs is a crucial aspect of proactiveness, enabling firms to secure a first-mover advantage over competitors. This strategic foresight allows businesses to capitalise on emerging opportunities, thereby enhancing their competitive edge (Salim et al., 2021). Proactive firms are adept at leveraging their capabilities to innovate and adapt to dynamic market conditions, which is essential for maintaining sustainable competitive advantage (Alonso-Almeida et al., 2015; Mishra & Yadav, 2020). Furthermore, proactiveness can be applied across various domains, including product development, technological advancement, market exploration, and consumer demand anticipation, thereby fostering a culture of continuous innovation and strategic agility (Salim et al., 2021; Siedel & Haapio, 2010). This approach aligns with the concept of proactive maintenance, which is increasingly adopted in industries facing continuous change owing to factors such as global competition and technological advancement.

Proactiveness, which involves anticipating and acting on future needs, has been identified as the most impactful dimension of EO on MSME performance in the fashion industry. Proactiveness reflects a forward-looking approach that enables fashion enterprises to anticipate market trends and act ahead of their competitors. Proactiveness emphasises a forward-thinking orientation in which firms anticipate future market demands and capitalise on emerging opportunities ahead of their competitors. This dimension is especially important in fashion, where being the first to adopt new trends before they become mainstream often determines market leadership. Kharisma et al. (2020) revealed that proactiveness directly improves strategic flexibility, allowing fashion SMEs in Malang, Indonesia, to respond effectively to market uncertainties. Wongmuek (2018) emphasised that while proactiveness is crucial in the embryonic stages of entrepreneurship, its development is often underemphasised in educational programs. Kiani et al. (2022) noted that proactiveness among SMEs significantly strengthened their technological innovation capabilities, enabling them to maintain competitiveness in rapidly evolving markets. Similarly, Fayomi (2022) highlights that proactiveness was instrumental in the survival of Jordanian fashion SMEs by enhancing their readiness for e-commerce and digital transformation. By acting pre-emptively, fashion enterprises can lead market trends, ensuring that they remain relevant and agile in responding to consumer preferences and global dynamics.

2.11.3 Risk-taking

Risk-taking involves the willingness to allocate resources and pursue market opportunities, accepting potential losses as part of the process (McCarthy et al., 2018; Putniņš & Sauka, 2019). The link between risk orientation and innovation is evident, as innovation inherently involves risk; thus, more innovative organisations tend to face greater risks (Kreiser et al., 2013; Tang et al., 2009). Risk orientation is defined as the level of risk an individual is willing to accept to achieve their goals, reflecting the readiness of managers to commit to high-risk decisions and identify resources that can sustain significant losses when outcomes are uncertain (Kreiser et al., 2013; Linton & Kask, 2017). Risk-taking reflects the acceptance of uncertainty and risks inherent in innovation, typically characterised by a commitment to uncertain outcomes and activities. This is supported by research (De-Juan-Ripoll et al., 2021) indicating that risk-taking is a component of decision-making processes in uncertain situations, where cognitive and emotional processes significantly influence how risks are perceived and

addressed. Risk-taking reflects a firm's willingness to allocate resources to uncertain but potentially rewarding opportunities. This characteristic is especially important in the fashion industry, where market volatility and consumer preferences frequently change.

Risk-taking involves committing resources to uncertain ventures, a necessity in the volatile fashion market where trends and consumer preferences change rapidly. Mulyana and Sutapa (2016) highlighted that risk-taking fosters innovation capabilities, which are critical for maintaining competitive advantage and performance. However, Kharisma et al. (2020) noted that while risk-taking influences strategic flexibility, its direct impact on SME performance in the fashion sector may be less pronounced than that of other EO dimensions. Zehir et al. (2015) demonstrated that risk-taking enhances innovation performance by encouraging firms to invest in advanced technologies and strategies, ultimately leading to superior results. Additionally, Surriadi et al. (2019) observed that risk-taking behaviour in the Indonesian veil fashion industry helped firms navigate volatile market conditions and leverage digital transformation opportunities. Risk-taking allows enterprises to pioneer new markets and adopt bold strategies, which, despite uncertainties, can lead to breakthrough success if managed effectively, despite uncertainties. By embracing risks, fashion enterprises can enter new markets, explore unconventional designs, and adopt innovative technologies, securing their position at the forefront of industry trends.

These EO dimensions collectively create a powerful framework for enhancing the performance of fashion enterprises. Innovativeness drives creativity and differentiation; risk-taking supports bold investments and the exploration of untested opportunities; and proactiveness ensures strategic foresight and adaptability. Empirical evidence shows that these elements are interrelated, collectively enabling fashion enterprises to adapt to dynamic environments, achieve sustainable growth, and maintain a competitive edge. This synergy allows firms to meet immediate market demands and strategically position themselves for long-term success in an ever-changing industry landscape.

Research has further indicated that EO is instrumental in marketing and improving competitiveness. Rufaidah (2019) further noted that EO evolves with business maturity, with younger SMEs displaying different dimensions of EO compared to more established firms. Studies show that newer SMEs (operating for less than a year) and those with more experience (1-5 years) exhibit different EO dimensions, which influence their business strategies and outcomes. This differentiation underscores the importance of understanding EO in the context of business maturity. Brazinskaite and Durand (2014) have reported that firms with strong EO focus on market innovation, strategic alliances, and experimental strategies, which enhance

competitiveness. However, these studies are limited in their ability to address the broader challenges of global competition.

Nonetheless, much of this research generalises findings across creative industries and fails to adequately address fashion-specific challenges, such as seasonality and fluctuating consumer preferences. However, significant gaps remain in the literature. Studies such as those by Rufaidah (2019), Putra and Gunawan (2019), and Fayomi (2022) focused on specific regional contexts, such as Indonesia and Jordan, without considering broader market dynamics or globalisation pressures. In particular, there is little research on rapidly developing markets such as Saudi Arabia, where Vision 2030 emphasises entrepreneurship, sustainability and cultural identity.

Existing studies also appear to lack a detailed exploration of how EO dimensions, such as innovativeness, risk-taking, and proactiveness, are operationalised, for instance, through research and development (R&D), risk management, or consumer-driven strategies. Non-financial outcomes, such as brand equity and social impact, are rarely addressed, and reliance on quantitative methods has been noted as a limitation in understanding the underlying mechanisms of EO (Pal & Shalender, 2021).

2.12 Business Performance (BP)

Measuring BP is a critical aspect of organisational management, providing insights into how effectively an enterprise achieves its strategic goals. To fully understand BP, research highlights the importance of integrating both subjective and objective metrics, as each provides unique insights into organisational success (Bedi & Puri, 2019; Bedi et al., 2023). BP evaluation commonly includes financial and non-financial indicators, reflecting the multidimensional nature of performance (Rauch et al., 2009; Venkatraman & Ramanujam, 1986)

Traditionally, financial metrics such as sales growth, profitability, and return on investment (ROI) have been widely used because of their objectivity, ease of quantification, and relevance to stakeholders (Murphy et al., 1996; Tangen, 2004; Sarsiti, 2024). However, these measures are often criticised for focusing primarily on historical performance, limiting their utility in predicting future outcomes (Neely et al., 2005). For instance, a company may report strong profitability in one year but fail to sustain it due to declining customer satisfaction or lack of innovation.

Modern approaches address this limitation by incorporating non-financial metrics, such as customer satisfaction, employee engagement, and innovation. These measures capture both the internal and external dimensions of performance, offering a more holistic understanding of organisational success (Arifeen et al., 2014; Hult et al., 2004). Recent research emphasises their growing importance in fostering long-term strategic success, particularly in dynamic and competitive environments (Garengo et al., 2005; Santos & Brito, 2012). For example, in the Australian fashion industry, leading firms have successfully integrated non-financial metrics such as creativity and adaptability to enhance market competitiveness (Agustina & Dewi, 2021; Yang & Jang, 2020).

The debate over subjective and objective metrics centres on their reliability and practicality. Objective measures, such as financial data, are typically more reliable but can be difficult to access because of confidentiality issues (Aranda et al., 2014). Subjective measures, such as managerial satisfaction or customer feedback, are easier to collect but often face criticism for their potential bias and inconsistency (Wiklund et al., 2011; Wiklund & Shepherd, 2005). Despite these concerns, studies have demonstrated that subjective measures can strongly correlate with objective outcomes, making them valuable tools when used judiciously (Santos & Brito, 2012).

Performance measurement approaches also vary depending on the organisational size. Small businesses often emphasise subjective and non-financial metrics, as these directly impact owner-manager satisfaction and business continuity (Maduekwe & Kamala, 2016; Simpson et al., 2004). In contrast, large corporations prioritise financial metrics to meet shareholder expectations and regulatory requirements (Morin, 2024). For example, a small Australian startup may focus on customer feedback and employee morale to foster growth, whereas a large multinational firm may concentrate on financial ratios to satisfy global investors.

Entrepreneurial and incubated firms further complicate BP evaluation because these entities often combine financial goals with non-financial milestones, such as patents, technology transfer, or market disruption. Such hybrid approaches reflect their innovative and growth-oriented nature (Santos & Brito, 2012). Emerging research suggests that these blended metrics are crucial for capturing value creation processes inherent in these firms, particularly in competitive markets such as technology and biotech (Garengo et al., 2005).

The Balanced Scorecard (BSC) is a prominent framework that integrates financial and non-financial metrics, offering a structured approach to performance measurement. This model ensures strategic alignment across dimensions such as financial, customer, innovation, and internal processes (Kaplan, 1992; Neely et al., 2005). Recent adaptations of BSC have further

highlighted its relevance in addressing sustainability goals and digital transformation, making it a valuable tool for modern organisations (Pereira & Oliveira, 2020).

Ultimately, in fast-paced industries such as fashion, achieving superior BP requires a multifaceted approach that combines financial and non-financial metrics with adaptability and innovation. This integrated performance measurement enables organisations to navigate complex and competitive markets effectively, aligning operational practices with strategic objectives (Agustina & Dewi, 2021; Yang & Jang, 2020).

2.13 Hypotheses Development

2.13.1 Relationship between FEE and EO.

Numerous studies have extensively investigated the relationship between EEs and EO (Rizvi & Iqbal, 2024; Scheidgen, 2020; Wadichar et al., 2022). It has been shown that EEs provide essential resources, including financial support, government backing, market access, and policy frameworks, which foster development of EO in MSMEs (Cantner et al., 2020). This fosters a proactive, innovative mindset and improves decision-making capabilities (Wurth et al., 2021). Recent studies have revealed that ecosystems rich in entrepreneurial resources enhance EO, even in resource-scarce contexts, by driving startup activation and fostering resilience (Wei et al., 2023). Research has also demonstrated that factors within EEs, such as business protection, infrastructure, financial accessibility, and technological advancement, play major roles in shaping the EO. These factors create conducive environments that support risk-taking and proactiveness by leveraging local resources, institutions, and networks (Igwe et al., 2020; Malecki, 2018; Olutuase et al., 2018; Spigel & Harrison, 2018). Acs et al. (2018), using data from 46 countries, found that the combination of institutions and entrepreneurship within an ecosystem context significantly affects economic growth and EO, positioning ecosystems as a critical framework for cross-country differences.

Entrepreneurship education has been identified as a critical component of EEs that significantly impacts the EO. Studies have found that entrepreneurship education enhances individual EO by fostering innovation and risk-taking, which in turn increases entrepreneurial intention (Efrata et al., 2021). Furthermore, research has shown that the mindset developed through such education contributes to proactiveness and improved entrepreneurial performance (Sutanto et al., 2021). Integrating education with technology and social networks within EEs has also been demonstrated to strengthen these dynamics and amplify both individual and organisational EO (Güven & Yildirim, 2022; Lazzeretti & Capone, 2020).

Several studies have explored the role of government policies and institutional frameworks in influencing EO. It has been proven that effective entrepreneurial policies enhance EO by supporting high-growth entrepreneurial activities, with innovation-friendly policies having a direct and positive impact (Etemad, 2021; Okanović et al., 2023). Research has established that mission-oriented policies in conducive environments enable sustained entrepreneurial growth (Etemad, 2021). Additionally, the cumulative and path-dependent nature of EO within ecosystems has been highlighted, with practices such as “blockbuster entrepreneurship” and entrepreneurial recycling fostering proactivity and innovation (Mason & Brown, 2014). Connectivity within ecosystems has been shown to drive innovation by enabling entrepreneurs to leverage diverse resources and capabilities (Turkina et al., 2019).

Networking opportunities within EEs are consistently linked to the development of EO. Studies have found that collaboration within ecosystems promotes shared learning, co-innovation, and proactive entrepreneurial behaviour (Kusa et al., 2021). Support programs such as mentoring, financial assistance, and networking with experienced entrepreneurs significantly enhance entrepreneurs’ confidence and ability to successfully lead startups (Nate et al., 2022). Moreover, research has revealed that cultural and social components within ecosystems strengthen EO by fostering relationships between individuals and institutions, thereby enabling risk-taking and opportunity recognition (Spigel, 2017).

The impact of EEs on EO has been explored in cross-cultural and institutional contexts. It has been found that variations in these contexts shape EO differently across nations (Firman et al., 2024). For instance, ecosystems that foster knowledge exchange, cooperation, and innovation have been shown to empower entrepreneurs to identify opportunities and undertake risks (Wadichar et al., 2022).

While these findings provide a robust theoretical foundation, most of the cited studies address EO within general EE frameworks, not within the fashion sector. However, the applicability of these insights to the FEE is supported by existing literature demonstrating similar resource dependencies in creative industries. Studies in the Saudi context confirm that fashion entrepreneurs face unique institutional and cultural barriers that directly influence EO development (Alshibani et al., 2024; Alsaiani, 2022; Maghrabi & Abalkail, 2024). For instance, limited access to finance and gendered perceptions of tailoring constrain risk-taking and innovation. Fashion-specific challenges such as market saturation, design originality and global competitiveness demand high EO, and their development depends on the quality of the FEE. Similarly, policy interventions under Vision 2030 have targeted fashion start-ups with dedicated initiatives such as the Saudi 100 Brands Programme, which enhances EO by

facilitating international exposure and fashion-relevant mentorship (Ministry of Culture, 2024). Educational constraints within Saudi fashion programs, such as the lack of entrepreneurship integration, further illustrate how weak institutional FEE components hinder proactiveness and innovation (Alshebami, 2022; Alahmari & Alkulaib, 2024). Complementing these regional insights, Bražinskaitė and Durand (2014) found that small fashion businesses in Sweden developed a competitive advantage through EO-driven marketing strategies, including product innovation, customer engagement, and strategic collaboration with competitors. This affirms the critical role of EO in enhancing competitiveness in fashion markets. Furthermore, Bohnsack and DiVito (2019) identified distinct EO patterns among sustainable fashion entrepreneurs, demonstrating that EO varies based on founders' motivations and sustainability goals. These findings underscore the necessity for ecosystem structures tailored to fashion-specific entrepreneurial behaviour. Therefore, although the positive relationship between EE and EO is established across contexts, recent findings have emphasised the need for more research on the interaction between ecosystem components and entrepreneurial cognition, particularly focusing on mentorship and peer support, which have been shown to foster innovative and proactive behaviours (Wurth et al., 2021).

This research addresses this call by examining the FEE–EO relationship specifically within the Saudi fashion industry, using both quantitative and qualitative methods to contextualise how sector-specific ecosystem domains shape EO.

Therefore, it is hypothesised that:

H1: The Fashion Entrepreneurship Ecosystem positively affects the Entrepreneurial Orientation of fashion MSMEs in Saudi Arabia.

2.13.2 Relationship between FEE and BP.

Previous studies have consistently demonstrated the significant role of EEs in shaping EO, with various factors influencing this dynamic. Well-integrated ecosystems provide businesses with critical resources that enhance entrepreneurial activity and drive BP. For instance, Isenberg (2011a) and Acs et al. (2014) emphasised that robust ecosystems improve financial outcomes and support business growth by ensuring access to funding and markets.

In the Saudi context, the relationship between the FEE and BP is shaped by several socio-institutional and sector-specific dynamics. Ali et al. (2019) demonstrated that ecosystem support, including government initiatives, significantly improves the revenues and long-term viability of women-led ventures. This is pertinent to Saudi Arabia, where over 90 per cent of fashion MSMEs are operated by women (Alshibani et al., 2024). These ventures often face

additional constraints related to finance, market access, and institutional bias (Maghrabi & Abalkail, 2024). Thus, gender-inclusive policy design is central to ensuring the positive influence of FEE on BP.

Hechavarría and Ingram (2019) further emphasised that regulatory simplification improves both financial and non-financial business outcomes. In line with Saudi Vision 2030, which prioritises deregulation and creative sector development (Government of Saudi Arabia, 2024), various reforms have been implemented. However, recent findings suggest that bureaucratic inefficiencies and poorly tailored incubator services persist, limiting the relevance and accessibility of support services for fashion entrepreneurs (Al Mulla, 2022; Maghrabi & Abalkail, 2024).

Aliyari (2024) found that ecosystem elements such as innovation hubs and financial networks enhance startup survival and SME growth. Similarly, Tapsi et al. (2024) noted that cultural factors, financial inclusion, and peer networks are central to fostering a productive entrepreneurial climate. These elements not only encourage EO but also improve operational sustainability, which is directly linked to BP. This is especially critical in Saudi Arabia, where fashion entrepreneurs often rely on informal financing and navigate cultural attitudes that can constrain innovation and risk-taking (Alzahrani et al., 2015; Algohari et al., 2025).

Sahu and Panda (2024) also noted that supportive ecosystems contribute to the adaptability and resilience of SMEs in emerging economies, especially under uncertain conditions. This resilience enables businesses to maintain innovative trajectories and respond proactively to changing consumer and policy environments, which are essential for long-term BP. In the Saudi fashion sector, resilience is often tested by external dependencies on imported raw materials, limitations in retail infrastructure, and an evolving consumer base (Elmonshid & Sayed, 2024).

Salimi (2022) provided empirical evidence that comprehensive ecosystem support is associated with improved metrics such as profitability, sales growth, and employment generation. Similarly, Davari and Najmabadi (2018) highlighted the role of R&D, infrastructure, and government facilitation in enhancing SME performance in comparable contexts such as Iran. These findings reinforce the relevance of a sector-sensitive ecosystem approach in non-Western, emerging economies. Lux et al. (2020) further demonstrated that EEs moderate the relationship between internal entrepreneurial capabilities and business success. In the Saudi fashion ecosystem, this is illustrated by the presence of entrepreneurs with strong EO who are nevertheless hindered by misaligned support systems, such as technology-centric incubators that fail to address the specific needs of fashion enterprises

(Braun & Suoranta, 2024; Roomi et al., 2023). In this context, even high levels of EO may not be sufficient to improve BP in the absence of well-matched ecosystem elements.

Autio et al. (2014) proposed that EEs enhance access to market and technological resources, thereby improving business outcomes. This is particularly applicable to Saudi Arabia, where recent programs such as the Saudi 100 Brands initiative and international collaborations by the Fashion Commission have enabled global exposure and digital expansion for local designers (Ministry of Culture, 2023). However, institutional inefficiencies, education-to-employment misalignments, and a lack of international faculty in fashion education still constrain the full realisation of business performance potential (Alshebami, 2022; Othayman et al., 2021).

In summary, the relationship between FEE and BP in Saudi Arabia is highly contingent on the alignment between ecosystem domains (finance, education, policy, infrastructure, culture) and the unique operational needs of the fashion sector.

Hence, it is posited that:

H2: The Fashion Entrepreneurship Ecosystem positively affects the business performance of MSMEs in Saudi Arabia's fashion sector.

2.13.3 Relationship between EO and BP.

Previous research has extensively highlighted the relationship between EO and BP in various industries, including the fashion sector. Studies such as those by Lumpkin and Dess (1996) and Wiklund (1999) consistently show that EO dimensions, including innovativeness, proactiveness, and risk-taking, enable firms to maintain a competitive edge and achieve superior performance (Wahyuni & Sara, 2020). Meta-analyses conducted by Rosenbusch et al. (2013) further confirm these findings, indicating EO contributes significantly to enhanced BP across different regions and sectors. Similarly, a meta-analysis of 78 studies confirms this direct and positive impact, with stronger effects for multi-item and revenue-based performance measures (Soares & Perin, 2019). These findings have consistently been validated and expanded upon by more recent research, underscoring their enduring relevance and reliability in shaping the contemporary understanding of EO's impact (Lin & Chung, 2023; Soares & Perin, 2019).

In the fashion sector, EO plays a pivotal role in driving performance. Brazinskaite and Durand (2014) found that EO dimensions significantly contributed to success of small fashion businesses in Sweden. Similarly, Mulyana and Sutapa (2016) highlighted the critical role of social media in fostering innovation in Indonesia's creative fashion industry. These findings

were confirmed in the context of Bandung by Putria and Gunawan (2019), who emphasised the need for fashion enterprises to enhance their EO to improve BP. Similarly, DiVito and Bohnsack (2017) link EO configurations to sustainability decision-making in small sustainable fashion firms in the Netherlands, demonstrating how EO contributes to broader goals, such as aligning entrepreneurial reflexivity with sustainability objectives. This was further reinforced by Bohnsack and Divito (2019). These studies collectively underscore the multifaceted contributions of EO to performance in the fashion industry (see Table 2.7).

Despite these consistent findings, variations in the impact of EO on BP have been reported, influenced by contextual factors and differences in EO dimensions. For example, Hossain and Azmi (2021) found that innovativeness and proactiveness positively influenced export performance, while risk-taking showed no significant effect in their study of Bangladeshi apparel exporters. Similarly, Kharisma et al. (2020) identified strategic flexibility as a mediator between EO dimensions and BP, particularly emphasising the role of proactiveness in fashion SMEs.

However, other studies challenge the assumption of a universally positive relationship between EO and BP. Khalili et al. (2013) and Wiklund and Shepherd (2005) reported no significant direct relationship between EO and BP, suggesting contextual moderators, such as market conditions or industry dynamics, may influence the relationship. Idawati and Sumartini (2020) further complicated this narrative by observing a negative and insignificant effect of EO on the performance of fashion and handicraft SMEs in Denpasar, Indonesia, highlighting how local market dynamics can constrain EO benefits. In other sectors, such as farming, studies by Nybom et al. (2021) and Veidal and Flaten (2014) found negative relationships between EO dimensions and financial performance, particularly risk-taking, which was noted by Rezaei and Ortt (2018) noted sometimes adversely impacts BP. Additionally, several studies have shown no relationship between entrepreneurial orientation and business performance (Asad et al., 2021; Madison et al., 2014).

The need to view EO as a multidimensional construct rather than a single entity has been highlighted in the literature. For instance, Aloulou (2019) found that proactiveness significantly and positively affects BP in family firms, whereas innovativeness and risk-taking show no notable impact. Similarly, Tajeddini and Mueller (2019) and Cho and Lee (2020) demonstrated that the dimensions of EO can have varying effects depending on context, reinforcing the argument that EO's dimensions should be analysed independently to fully understand their contributions.

Table 2.7: Research on Entrepreneurial Orientation and Business Performance in Fashion Industry

Year/Authors	Title	Context	Main Findings
Brazinskaite & Durand (2014)	Entrepreneurial Orientation in Marketing: An empirical study of small businesses in the Swedish fashion industry	4 small companies in the Swedish fashion industry, located in the cities of Malmö, Stockholm, and Helsingborg.	The subdimensions of EO, including innovativeness, proactiveness, and risk-taking, significantly impact BP and marketing activities.
Mulyana & Sutapa (2016)	The Impact of Entrepreneurial Orientation and Collaborative Networks on Creative Industries Performance	114 participants from creative industries focused on the fashion sector from various cities on the north coast of Central Java, Indonesia.	EO significantly affects innovation capability and performance.
DiVito & Bohnsack (2017)	Entrepreneurial orientation and its effect on sustainability decision trade-offs: The case of sustainable fashion firms	24 small sustainable fashion firms, in the Netherlands	There are three sustainability decision making profiles based on EO, the correspondence between different EO configurations and these profiles, and interaction of EO reflexivity with sustainability orientation.
Bohnsack & Divito (2019)	Entrepreneurial orientation of SMEs in the creative fashion industry	328 SME owners in the creative fashion industry from 27 cities and regencies in West Java, Indonesia	EO has a significant effect on BP. No significant differences in EO were found between entrepreneurs operating for one year and those operating for five years.
Putria & Gunawan (2019)	The Impact of Entrepreneurial Orientation and Business Environment Towards Business Performance	60 Muslim fashion SMEs in Bandung, Indonesia.	EO has a significant positive effect on BP.
Kharisma et al. (2020)	Entrepreneurial Orientation and Its Effect Toward SMEs Performance Through Strategic Flexibility on Fashion Industry in Malang	31 owners of fashion industry SMEs in Malang, East Java, Indonesia	Only proactiveness had a significant direct effect on SMEs' performance. Innovativeness, risk-taking, and proactiveness significantly influenced strategic flexibility.
Mulyana & Hendar (2020)	Market and entrepreneurial orientation on business performance: role of networks innovation agility	302 owners or leaders of fashion SMEs in Central Java, Indonesia.	Market orientation, EO, and network innovation agility significantly impact BP. Network innovation agility acts as a mediator between market orientation and EO on BP.
Idawati & Sumartini (2020)	The role of development entrepreneurial orientation and market orientation in improving the performance of creative industry SMEs in Denpasar	40 creative industry SMEs (fashion and handicraft) in Denpasar City, Indonesia.	EO has a negative and insignificant effect on the BP of SMEs in Denpasar's fashion and handicraft subsector.
Hossain & Azmi (2021)	The effect of entrepreneurial orientation on the export performance of apparel industry	339 high-level managers and owners of apparel manufacturing MLEs, Bangladesh	Innovativeness and proactiveness positively and significantly impact export performance in the apparel industry. Risk-taking dimension does not affect export performance.
Fayomi (2022)	The Moderating Role of the Entrepreneurial Orientation on the Relationship Between E-Commerce Readiness and SMEs Survival: Evidence from Jordan Fashion Sector	Clothing merchants and companies in the fashion sector in Jordan.	Found an effect of e-commerce readiness on survival and continuity of SMEs, and also found EO has a modifying effect on this relationship.

The interaction between the EO and external factors further illustrates its complexity. Mulyana and Hendar (2020) demonstrated that network innovation agility mediates the relationship between EO and market orientation, showing how EO interacts with external networks to drive BP. Fayomi (2022) extended this perspective by examining the moderating role of EO in the relationship between e-commerce readiness and SME survival in Jordan's fashion sector, showing that EO enhances adaptability to contemporary business challenges, particularly in digital contexts. Building on these insights, the following hypothesis is proposed:

H3: Entrepreneurial Orientation positively affects the business performance of fashion MSMEs in Saudi Arabia.

2.13.4 The Mediating Effect of EO in the FEE and BP Relationship.

Studies have demonstrated that EO serves as a critical mediator in explaining the relationships between elements of EEs and BP across various contexts. Early studies have shown that proactiveness, innovativeness, and risk-taking enable firms to effectively leverage resources and opportunities within ecosystems, such as supportive policies, market conditions, finance, and infrastructure, thereby achieving a competitive advantage (Miller, 1983). Similarly, Burt (2003) demonstrated that business networks enhance BP by providing improved access to resources, markets, and knowledge, with EO playing an indirect role in this process.

This study identifies EO as crucial in translating the influence of EE factors, such as policy frameworks, market conditions, access to finance, and infrastructural support into tangible business performance improvements (Rauch et al., 2009; Spigel & Harrison, 2018). Consequently, this enables firms to undertake and manage innovative and risky projects more effectively, leading to improved business performance (Isenberg, 2010). Furthermore, infrastructural support provided by entities such as incubators and accelerators enriches the entrepreneurial climate, facilitating an EO conducive to business success (Cohen, 2006). This comprehensive support system creates favourable conditions and actively enhances the EO of firms, thereby directly impacting their operational and strategic outputs.

A recent study by Wahyuni and Sara (2020) finds that EO partially mediates the relationship between EEs and business performance, suggesting that other factors, such as environmental conditions or additional mediating and moderating variables, also shape the influence of EEs on firm outcomes. Wei et al. (2023) emphasised that entrepreneurial resources significantly influence the activation of startups, with EO acting as a vital mediator. This

underscores the importance of enhancing both entrepreneurial resources and orientation to foster an active entrepreneurial ecosystem.

A study by Syofya (2024) on coffee entrepreneurs in Indonesia suggests that EO plays a critical mediating role in enhancing business performance, which in turn positively impacts community and local economic empowerment. Oluwambe and Samuel (2020) highlight that EO and entrepreneurial intentions are significantly influenced by ecosystem variables, thereby improving entrepreneurial activity.

Auerswald (2015) emphasises the necessity for entrepreneurs to think broadly and act swiftly to fully maximise benefits of the entrepreneurial ecosystem approach for business performance growth. According to Etemad (2021), the success and sustained high growth rates of entrepreneurial start-ups are significantly influenced by a supportive entrepreneurial ecosystem, including factors such as attractive locations, robust support systems, financial incentives, conducive legal frameworks, and industrial specialisation. Studies by Miao et al. (2017) and Poon et al. (2006) provide evidence of EO's partial mediating role, demonstrating how human and social capital, as well as self-efficacy, influence firm outcomes.

Farsi et al. (2013) observed EO of staff influences organisational performance, confirming EO's mediating effect between social capital and organisational performance. Calabrò et al. (2017) noted that governance structures enhance the internationalisation of companies, mediated by firms' international entrepreneurial orientation. EO also mediates the influence of government intervention and tax policies on firm performance (Hendou, 2019). Similarly, Nurdin and Haryanti (2020) found that the role of government has no significant effect on increasing product value, but it has a significant effect when mediated by EO.

In the tourism sector, EO functions as a strong mediator, elucidating how the institutional environment influences firm performance (Bedi & Puri, 2019). Mehralian (2022) illustrated that the Internet of Things (IoT) enhances marketing and PB through EO. The effects of exploration and exploitation innovation on firm performance, with competition intensity moderating the relationship between EO and firm performance, were highlighted by Hou et al. (2019). Tsering (2015) and Wakjira (2023) found that the effect of market orientation on firm performance is mediated through high EO.

While some studies, such as Zainol & Wan Daud (2011), demonstrate that EO does not mediate the relationship between cultural background and company performance in Malay firms, this is not universally applicable to all contexts. In contrast, numerous other studies have highlighted the significant mediating role of EO in the relationships between different internal factors and business performance. For instance, EO mediates the relationship between

personality traits (Imran et al., 2019), firm-level factors (Maina et al., 2018), cultural intelligence (Kadam et al., 2019), organisational ambidexterity (Aftab et al., 2022), leadership style (Aziz et al., 2010; Paudel, 2020), and business performance. Furthermore, Khedhaouria et al. (2020) found that EO mediates the influence of adhocracy and market cultures on small firm performance, enhancing innovative and proactive behaviour. Similarly, Park (2023) demonstrated how EO facilitates the translation of organisational learning into innovation performance. These examples underscore EO's critical function of EO in linking internal organisational characteristics with enhanced performance outcomes, thereby illustrating its varied roles in different organisational contexts.

Therefore, the following hypothesis is proposed:

H4: Entrepreneurial Orientation mediates the relationship between the Fashion Entrepreneurship Ecosystem and Business Performance of fashion MSMEs in Saudi Arabia.

2.14 The Conceptual Model

Based on the literature review and discussion in the previous section, four hypotheses were developed. The conceptual model in this study represents the relationships among the Fashion Entrepreneurship Ecosystem (FEE), Entrepreneurial Orientation (EO), and Business Performance (BP), grounded in the Resource-Based View (RBV) theory. The model includes the Entrepreneurial Ecosystem (EE) as an independent variable, which is essential to EO and BP in fashion MSMEs in Saudi Arabia. Figure 2.5 shows the proposed conceptual framework.

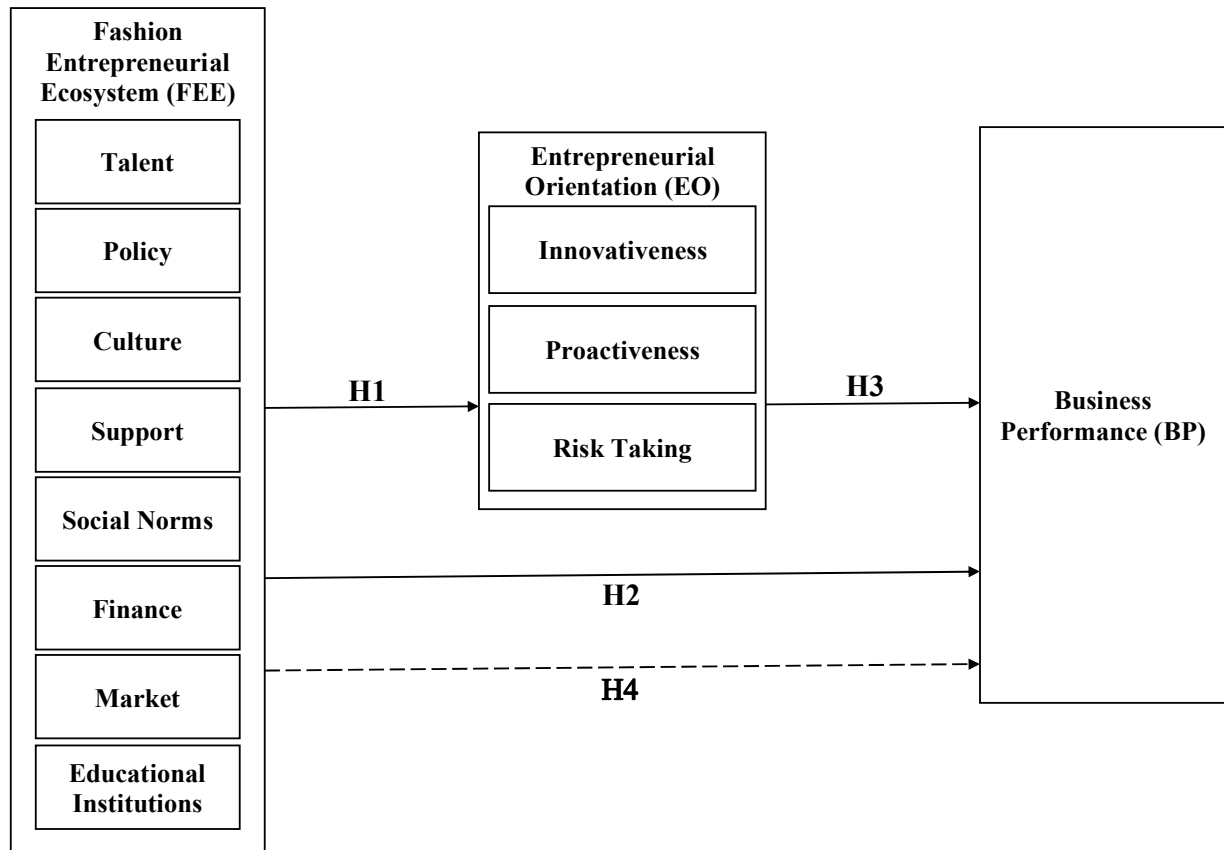


Figure 2.5: Proposed conceptual framework

2.15 Chapter Summary

This chapter provides a comprehensive review of the literature on EEs and their application within the fashion industry, focusing on EO's mediating role of EO in enhancing BP. It discusses the theoretical foundations underpinning this study, establishing a foundation for developing the research framework. The RBV was highlighted as a guiding theoretical framework, emphasising the dynamic interplay between internal firm resources and external ecosystem factors in driving entrepreneurship success.

This chapter identifies and analyses the key domains of EEs, including human capital, policy, markets, finance, culture, and support systems, with a particular focus on their relevance to the Saudi fashion industry. It also addresses the unique characteristics of the Saudi fashion sector. A review of the existing academic literature on the topic revealed several research gaps, which this study addressed by developing a tailored FEE framework that incorporates EO to enhance BP. The next chapter outlines the research design and methodology employed in this study.

CHAPTER 3: METHODOLOGY

3.1 Chapter Overview

This chapter outlines the philosophy and methodology guiding this research, examining EE and its impact on EO and BP of fashion MSMEs in Saudi Arabia. The chapter begins by presenting the research philosophy and emphasising the integration of quantitative and qualitative methods, providing a comprehensive approach to addressing the research objectives. It then details the research design and offers a rationale for adopting a sequential explanatory mixed methods approach. This design is particularly suited for exploring the complex nature of the entrepreneurship ecosystem being investigated. Next, the chapter discusses the development of survey instruments, including the piloting strategy, which informed the sampling techniques employed. Subsequently, the chapter describes the data collection and analysis methods utilised to achieve each research objective, with separate discussions on quantitative and qualitative approaches. This chapter also evaluates the reliability and validity of the methods and instruments to ensure that the study's findings are rigorous. Finally, it addresses the ethical considerations related to the research process, ensuring transparency and adherence to ethical standards throughout the study.

3.2 Research Onion Framework

The research onion framework was introduced by Saunders et al. (2015) to illustrate the stages of the research methodology. Figure 3.1 shows the research onion framework applied in this study, beginning with the research philosophy of pragmatism, which emphasises flexibility and practical problem solving and underpins the approach of this study.

The second layer involves a reasoning approach that integrates both deductive and inductive reasoning to support a mixed methods approach that effectively addresses research questions. The chosen methodological strategy follows the Sequential Explanatory Design, a mixed methods approach combining quantitative and qualitative data collection, to thoroughly examine the research problem. This design is particularly suitable as it enables initial quantitative data collection through questionnaires, followed by qualitative interviews to provide deeper insights into the findings.

To capture data from participants at a specific point in time, a cross-sectional time horizon was employed to provide a snapshot of the current state of FEE in Saudi Arabia. Data

analysis included Structural Equation Modelling (SEM) (Spigel, 2017) for the quantitative phase and thematic analysis for the qualitative phase, allowing a comprehensive exploration of relationships and patterns within the data. The following sections provide further details of these methods.

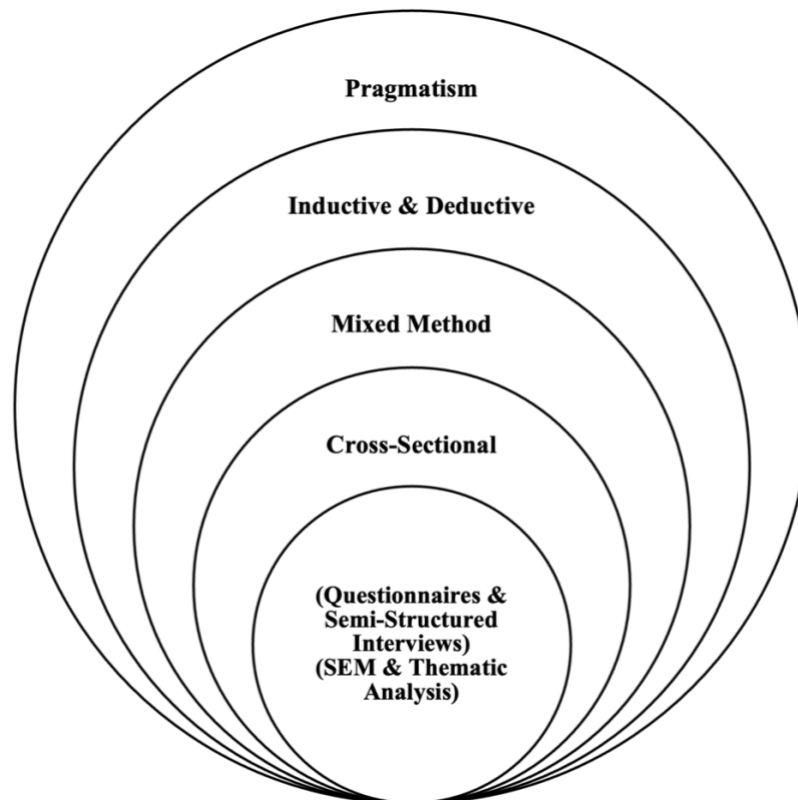


Figure 3.1: Research onion framework (adapted from Saunders and Lewis, 2012)

3.3 Research philosophy – Pragmatism

Research philosophy is defined by Bell et al. (2022) as a group of beliefs and principles that shape decisions about what topics warrant investigation, methodologies employed in conducting research, and approaches used to analyse and interpret resulting findings. These philosophical assumptions, ontological, epistemological, axiological, and methodological, guide key decisions throughout the research lifecycle, from formulating research questions to interpreting results (Saunders et al., 2015), particularly in entrepreneurial studies (Antwi & Hamza, 2015; Cope, 2005; Pittaway & Cope, 2007).

Ontology refers to the nature of existence or reality, posing fundamental questions regarding what constitutes reality and how it is perceived by humans. In business research, ontology typically involves a dichotomy between objectivism and constructionism. Objectivism posits that reality exists independently of human perceptions, whereas

constructionism argues reality is socially constructed through interactions (Bell et al., 2022). The relationship between ontology and epistemology is critical because it guides researchers in choosing methods that align with their worldview. Objectivist researchers prefer quantitative methods to measure objective reality, whereas constructionists are more likely to employ qualitative methods to explore subjective experiences. This alignment is particularly important in complex social research contexts, where multiple realities may coexist, requiring nuanced data collection and analysis approaches.

In social science, four dominant research philosophies are typically identified: positivism, interpretivism, realism, and pragmatism (Hughes & Sharrock, 2016). However, Creswell (2017) highlights three core paradigms: positivism, constructivism, and pragmatism, each of which offers a distinct approach to knowledge production and methodological design. Table 3.1 provides a summary of these paradigms and their associated approaches.

Table 3.1: Research Paradigm, adapted from (Creswell, 2017; Creswell & Clark 2011)

	Positivism	Constructivism	Pragmatism
Logic	Deductive	Inductive	Both inductive and deductive
Ontology The form of reality. What can be known about reality?	Reality does not exist in a single form; it exists in multiple forms.	Individuals create their own reality: there is no single reality.	Reality is determined by its usefulness; truth is that which works best to resolve a particular issue.
Epistemology The relationship between the investigator and what can be discovered.	Objective point of view. Reality can be measured.	Subjective point of view: reality constructed with participants.	Subjective and objective are both used.
Methodology How does the investigator do it?	Quantitative (QUAN)	Qualitative (QUAL)	Mixed method: both QUAL and QUAN
Data Collection Methods	Surveys, numerical methods, and quantitative observations	Interviews, audio, textual, visual data, qualitative observations.	Different sequences and priorities of a combination of qualitative and quantitative
Data Analysis Methods	Quantitative Descriptive and inferential (such as ANOVA, Correlation, Chi-square, regression), experimental	Coding, narratives, case studies, phenomenology, ethnography, grounded theory.	Combination of quantitative and qualitative at different stages

Positivism is grounded in the belief that reality is external and can be understood through empirical observation and measurement. Positivist researchers rely on scientific methods to gather objective data that can be statistically analysed and generalised to broader populations (Burrell & Morgan, 2019; Hathaway, 1995; Morgan, 2007). Positivism emphasises objectivity and neutrality, with researchers minimising bias to produce replicable and generalisable

findings (Creswell, 2017). Deductive reasoning, in which theoretical propositions are empirically tested to confirm or refute their validity, is central to this paradigm. This reliance on hypothesis testing and scientific rigor has made positivism the dominant framework in organisational and business research (Antwi & Hamza, 2015).

In contrast, constructivism posits that reality is subjective and shaped by social and cultural context. Constructivist researchers argue that individuals create their own understanding of the world based on their experiences and interactions (Guba & Lincoln, 1994; Howe, 1988). As a result, constructivism typically employs qualitative methods such as interviews, participant observation, and textual analysis to explore participants' subjective realities (Cope, 2005; Denzin & Lincoln, 2011). Theories in constructivist research often emerge inductively from data rather than being imposed from the outset, reflecting the socially constructed nature of knowledge (Guba & Lincoln, 1994; Scotland, 2012). This approach is invaluable for understanding individual or group perspectives, such as in entrepreneurship and interpretivism research (Leitch et al., 2010).

Pragmatism is often seen as a philosophical bridge between positivism and constructivism, emphasising practical problem-solving. Pragmatist researchers are flexible and often employ both quantitative and qualitative methods to gain a comprehensive understanding of complex phenomena. This paradigm allows methodological pluralism, encouraging researchers to choose the most appropriate tools based on their research objectives (Burrell & Morgan, 2019; Morgan, 2014). In fields such as entrepreneurship, pragmatism offers a valuable framework that blends objective data and subjective insights to address practical concerns (Anderson & Starnawska, 2008; Pittaway & Tunstall, 2016).

Pragmatism's ontology, which views reality as contingent and shaped by human experiences, is particularly relevant for studying complex dynamic systems, such as the entrepreneurship ecosystem. Within this framework, concepts such as EO and BP evolve and are influenced by various ecosystem factors (Moseholm & Feters, 2017). This pragmatic perspective highlights the fluid and interdependent nature of Saudi FEE, EO, and BP in a constantly shifting entrepreneurial landscape. Pragmatism's epistemological approach is well-suited to capturing these complexities by utilising both quantitative and qualitative data to explore the ecosystem's multifaceted realities.

A core principle of pragmatism is its advocacy for mixed methods, which is particularly valuable in fields such as business and social research, where multiple perspectives are essential to address complex issues. Pragmatism's openness to combining quantitative and qualitative methods allows researchers to leverage the strengths of each approach, leading to a

more nuanced understanding of phenomena (Clark & Creswell, 2011; Creswell, 2017). This methodological flexibility is particularly valuable in entrepreneurship and health research, where the practical implications of research outcomes are often paramount (Pathak & Thapaliya, 2022; Pittaway & Tunstall, 2016). By integrating diverse methodologies, pragmatists focus on the practical outcomes of research and their applicability to real-world challenges (Bowen et al., 2017; Howe, 1988).

Based on this review of research philosophies, this thesis adopts pragmatism as its guiding paradigm, integrating quantitative and qualitative methods to construct a robust framework for examining the interplay between Saudi FEE, EO and BP. Pragmatism's emphasis on real-world problem-solving and its adaptability make it a fitting foundation for investigating the complex and evolving dynamics of the Saudi FEE and its impact on entrepreneurial success in Saudi Arabia (Al-Mamary et al., 2020). The following section presents an overview of the research methods.

3.4 Study Approach (Mixed Methods)

Entrepreneurship research employs quantitative, qualitative, and mixed methods to understand complex entrepreneurial phenomena. Quantitative methods have dominated this field, relying on numerical data collection and statistical analyses to test hypotheses and identify patterns (Saunders et al., 2015). For instance, a recent systematic review of entrepreneurship research analysing 130 articles from 52 journals revealed that 40.77% of the studies employed quantitative methods, showing a continued emphasis on this approach despite earlier trends favouring qualitative methodologies (Nave & Ferreira, 2022). In EE studies, quantitative methods are used to assess ecosystem structures, such as funding and stakeholder interactions, providing evidence-based insights into entrepreneurial outcomes (Malecki, 2018; Roundy et al., 2018; Spigel & Harrison, 2018; Ziakis et al., 2022).

Alternatively, there has been a shift toward qualitative methodologies, as evidenced by research leveraging interviews and ethnographic case studies. For example, Spigel (2017) and Stam and Spigel (2016) utilised in-depth interviews to explore how regional cultural norms influence entrepreneurial behaviours, revealing aspects that purely quantitative methods might miss. Similarly, Roundy (2017) employed ethnographic techniques to delve into the relational dynamics within entrepreneurship ecosystems, providing a rich understanding of social interactions and their impact on innovation. These methods are essential for uncovering the nuanced social, cultural, and relational dimensions of entrepreneurship ecosystems. By

capturing the experiences and perspectives of ecosystem participants, qualitative research, such as that of Acs et al. (2014), offers invaluable insights into the unique factors and interactions driving entrepreneurship across diverse regional and cultural contexts.

However, the limitations of relying solely on either quantitative or qualitative methods have led to the increased recognition of mixed methods which integrate the strengths of both approaches. Mixed methods involve collecting, analysing, and integrating qualitative and quantitative data within a single study to achieve a more comprehensive understanding of the research problem (Bryman, 2006; Creswell, 2017; Teddlie & Tashakkori, 2011). Mixed methods research in entrepreneurship has gained prominence in offering a comprehensive view of complex ecosystems by combining quantitative precision with qualitative depth (McDonald et al., 2015). Early work by Gartner (1988) pioneered the mixed methods approach, integrating quantitative and qualitative techniques to explore opportunity identification. Recent studies have continued to highlight the advantages of mixed methods in capturing quantitative data for general trends and qualitative insights for contextual understanding. For example, (Bieńkowska & Sikorski, 2024), Mamabolo and Myres (2019), Adhikari and Timsina (2024), Pathak and Thapaliya (2022) ,and Molina-Azorín et al. (2012) demonstrate the effectiveness of mixed methods in exploring various aspects of entrepreneurship ecosystems. Recent studies illustrate how mixed methods can link ecosystem factors to business model selection in social entrepreneurship, offering a nuanced view of ecosystem interactions (Fkun et al., 2023; Scott et al., 2022).

Given the complexity of EEs, this study uses mixed methods approach to ensure both generalisability and contextual depth, ultimately enhancing relevance and applicability of findings. In this study, both quantitative (confirmatory) and qualitative (exploratory) insights were used to comprehensively address research questions related to Saudi FEE and its influence on EO and BP. By integrating both quantitative and qualitative methods, this study provides a comprehensive understanding of how Saudi FEE affects EO and BP. Quantitative research identifies broad trends and relationships, whereas qualitative research explores the underlying reasons and meanings of these trends (Bell et al., 2022; Creswell & Clark, 2017).

3.5 Research Design (Explanatory sequential mixed methods)

According to Teddlie and Tashakkori (2011), mixed methods research designs encompass approximately 40 different frameworks that integrate quantitative and qualitative approaches to provide comprehensive insights. Among these, six principal designs stand out: convergent parallel, explanatory sequential, and exploratory sequential designs. Each offers unique methods for integrating data to thoroughly investigate the research questions (Creswell, 2017).

A particularly common approach is the sequential explanatory design, in which research begins with quantitative data collection and analysis. This was followed by qualitative data collection, which aimed to provide deeper insights into the findings from the quantitative phase. Figure 3.2 illustrates this process, showing how qualitative research follows quantitative efforts, leading to a thorough interpretation of the combined analysis.

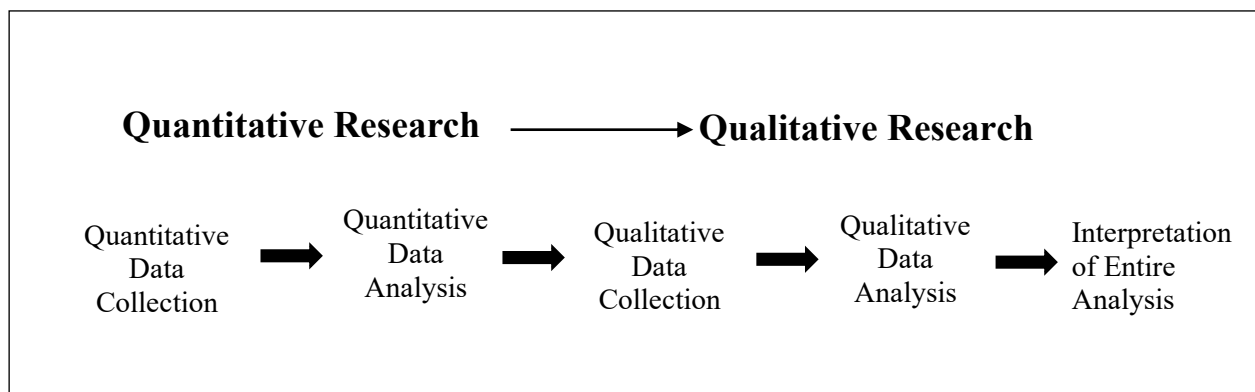


Figure 3.2: Sequential explanatory design (Cresswell et al., 2003)

The sequential explanatory research design is extensively used in social science research to explore complex phenomena (Bell et al., 2022). As illustrated in Figure 3.3, this study began with quantitative data collection (QUAN) to examine the impact of FEE on EO and BP among Saudi fashion entrepreneurs. Data were gathered through a survey and analysed using SPSS and PLS-SEM, focusing on the relationships between FEE, EO, and BP. Building on these quantitative findings, qualitative data (QUAL) were collected through semi-structured interviews with key stakeholders to further refine the framework. Thematic analysis was employed to provide deeper insights into the factors influencing ecosystem performance. The integration of both quantitative and qualitative phases allows for a comprehensive interpretation of the results, thus offering valuable implications for future research and practical applications within the Saudi fashion industry.

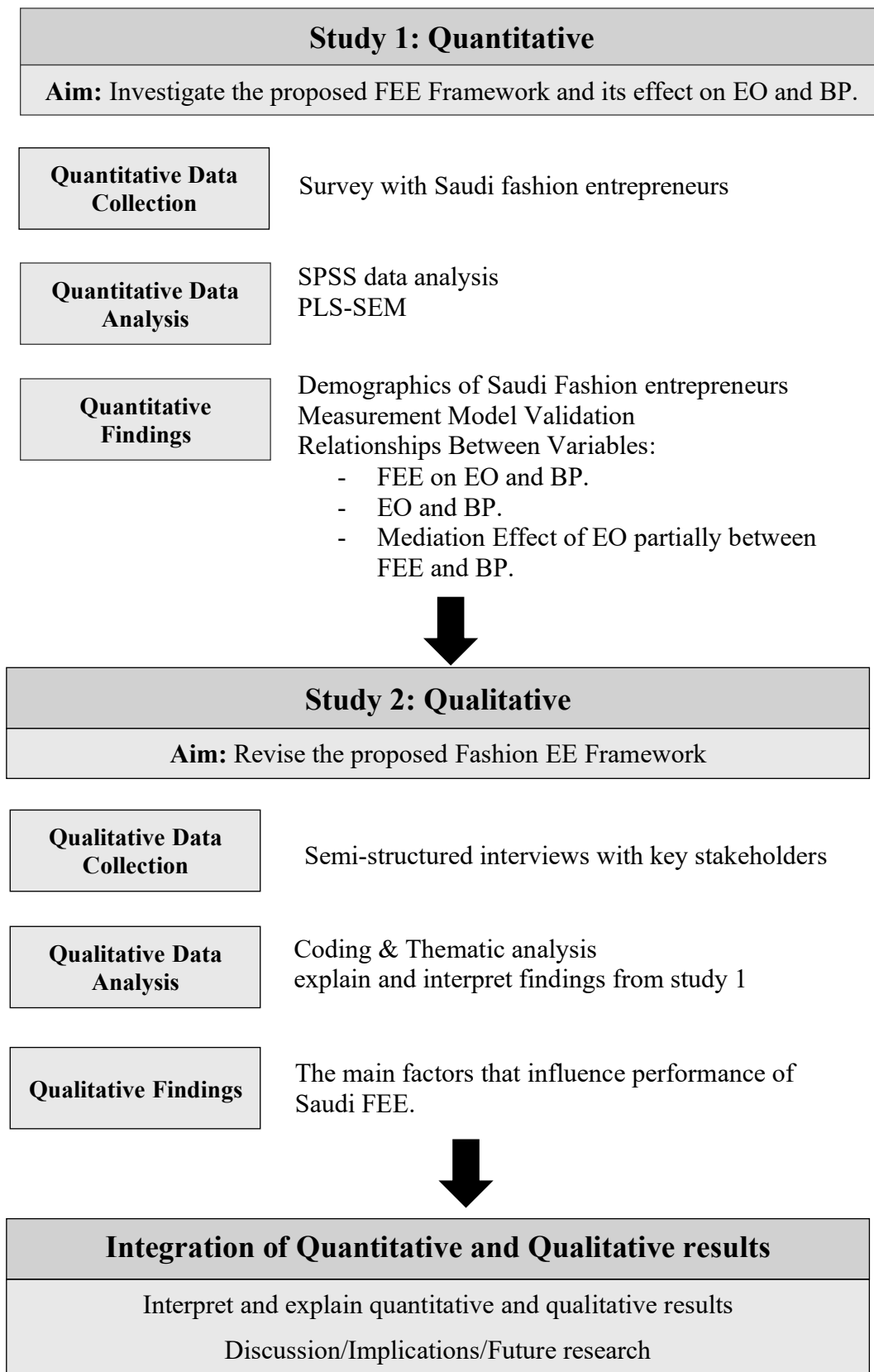


Figure 3.3: Explanatory sequential, design procedure model (Ivankova et al., 2006, p. 16)

To clarify the research design and notation used in this study, Table 3.2 provides an overview of the notation conventions for mixed methods research, as outlined by Morse (1991) and further elaborated by Creswell (2017). In this study, the notation (QUAN → qual) indicates a greater emphasis on the quantitative (QUAN) phase, which is prioritised to provide an extensive examination of the relationships within the FEE and their impact on EO and BP. The qualitative (qual) phase, with lesser emphasis, is conducted sequentially to offer contextual insights that enhance the interpretation of quantitative results.

Table 3.2 Notations used in mixed methods research (adapted from Cresswell et al., 2007, p. 280)

Method Indication	Example	Citation
Convergent methods	QUANT + QUAL	Morse (1991)
Sequential methods	QUANT → qual	Morse (1991)
Greater emphasis is given to a method	QUANT, QUAL	Morse (1991)
Lesser emphasis is given to a method	quant, qual	Morse (1991)
Embed within a design or framework	QUANT (qual)	Plano Clark (2005)
Recursive	QUANT →←QUAL	Nastasi et al. (2007)
Study within a series	QUANT → [QUANT + qual]	Morse & Niehaus (2009)

In line with the explanatory sequential design, the following joint display table illustrates how qualitative themes helped explain and enrich the interpretation of quantitative survey results, highlighting specific domains within the Saudi Fashion Entrepreneurial Ecosystem.

Table 3.3: Joint Display of Quantitative and Qualitative Findings

Quantitative Domain	Quantitative Result (Mean Score)	Qualitative Theme	Explanation/Integration
Finance	Mean = 2.88 (Lowest among domains)	Barriers in Accessing Finance	Participants described difficulty accessing targeted financial products for fashion businesses, noting limited awareness of programs and complicated procedures. These narratives help explain the low finance score in the quantitative data.
Market Access	Mean = 3.12	Market Saturation and Limited Visibility	Interviewees explained that it was challenging to penetrate the market due to high competition and lack of access to established platforms. This contextualises the relatively low market score.
Policy and Regulation	Mean = 3.08	Policy Gaps and Bureaucratic Complexity	Participants shared frustration with unclear or rapidly changing regulations. These insights align with the low quantitative score, suggesting systemic constraints.

3.6 Quantitative Research

3.6.1 Sampling

3.6.1.1 Target Population

In the present study, the target population is MSMEs in the fashion sector in Saudi Arabia. This research specifically focuses on MSMEs as defined by the categories they fall into based on two key criteria: the number of full-time employees and annual revenue. Micro-enterprises employ 1-5 people, small enterprises employ 6-49 employees, and medium enterprises employ 50-249 employees.

To align with the study's objectives, sample units were selected based on four key criteria: ownership or management of an MSME in the fashion sector; (2) the enterprise must be at least one year old; (3) the enterprise must be involved in womenswear, menswear, childrenswear, accessories, or jewellery; and (4) the enterprise must be located in Saudi Arabia. The rationale for selecting owners or managers as participants is grounded in the literature, as they are typically the key decision-makers and possess the most comprehensive understanding of their companies' operations, strategies, and business performance (Malem, 2008; Mills, 2011, 2018). Additionally, the criterion of having been in operation for at least one year was chosen to ensure that participants had sufficient experience to encounter obstacles and could provide valuable insights into the types of support required (Hodges et al., 2016; Zhao et al., 2019). To ensure that only eligible participants were included, a screening question was incorporated into the survey: "*Do you own or manage a fashion enterprise in Saudi Arabia?*".

3.6.1.2 Sampling Procedure

Sampling involves selecting organisations or individual units to gather information for a study, with the goal of ensuring result generalisability while minimising sampling errors (Lombardini, 2016; Saunders et al., 2003). In empirical research, sampling methods are typically categorised into probability and non-probability techniques. Probability sampling provides each individual in the population with an equal chance of selection, using methods such as stratified, random, cluster, and systematic sampling (Saunders et al., 2003; Vehovar et al., 2016). In contrast, non-probability sampling includes methods such as snowball, convenience, quota, and judgment sampling, where the selection probabilities are not predetermined (Bhardwaj, 2019). The selection of an appropriate sampling method depends on the study's objectives, sample accessibility, and practical constraints (Hair et al., 2014).

A critical first step in the sampling process was identifying a reliable sample frame, which is essential for the study's validity and generalisability (McEwan, 2020). A sample frame typically provides a comprehensive list of individuals or units in the target population from which participants are selected (Elangovan et al., 2016). In most studies, sample frames are sourced from official records or government agencies or directories. However, no official database or comprehensive list of fashion MSMEs in Saudi Arabia was available for this study, a challenge noted in previous research on Saudi entrepreneurship (Alkhaldi, 2019; Sabri, 2021). To address this gap, the researcher sought assistance from relevant bodies, including the Saudi Fashion Commission, Ministry of Commerce, and Chambers of Commerce in Riyadh, Makkah, Qassim, and the Eastern Province.

Given the limitations of the sample frame, various non-probability sampling methods were employed. The primary method for the quantitative phase was purposive sampling, selected based on the researcher's judgment due to the absence of a sufficiently reliable sampling frame (Bhardwaj, 2019). Potential participants were identified through professional and personal contacts in the Saudi fashion industry (Blanche et al., 2006; Creswell, 2017). Participants were also identified through web searches and social media using targeted keywords such as "Saudi fashion designers", "Saudi fashion entrepreneurs", and "Saudi fashion startups" Saudi fashion startups. Invitations to participate in the survey were sent via email and social media, with a direct link to the questionnaire provided upon informed consent.

To broaden participation, convenience sampling was used at the "100 Saudi Fashion Brand Exhibition" in Riyadh in August 2021. This allowed data collection from a wider array of fashion entrepreneurs and enhanced participant diversity.

3.6.1.3 Sample size and power

Determining an appropriate sample size is a critical step in ensuring reliable and valid results. Khalilzadeh and Tasci (2017) emphasised that factors such as data normality and missing data must be considered when determining sample size, particularly in complex models such as those used in SEM. While general guidelines for SEM provide a starting point, more precise methods are often needed to tailor the sample size to the study's specific requirements.

Many formulas have been used to calculate an appropriate sample size for various factors, such as confidence level, margin of error, and population size. However, as the current study employed SEM, specific guidelines were followed to ensure model robustness.

According to Beauducel and Herzberg (2006), a sample size of 200–400 is generally considered acceptable for SEM using Maximum Likelihood Estimation (MLE) to ensure sufficient statistical power and model stability, especially in more complex models with multiple parameters.

Furthermore, research suggests that sample sizes in this range provide better model fit and parameter estimation, particularly when dealing with multiple latent constructs and observed variables (Sayın, 2016). Additionally, confirmatory factor analysis (CFA) and covariance-based SEM techniques benefit from this sample size to achieve accurate estimation results (Rezaei & Valaei, 2018). This range aligns with SEM guidelines, confirming the adequacy of the sample size for this study.

While these SEM guidelines provide a baseline, researchers often turn to more advanced approaches, such as power analysis, to tailor the sample size more accurately to their model's needs. Power analysis, as recommended by Cohen (2013), determines the sample size based on the expected effect sizes, desired statistical power, and significance levels. It is particularly useful for ensuring that the sample size is sufficient to detect statistically significant effects in the regression model. Conducting a power analysis minimises the risk of sampling error and increases the probability of correctly rejecting the null hypothesis, which assumes no significant relationship between variables (Faul et al., 2007). As Snijders (2005) notes, achieving higher statistical power is essential, particularly with larger sample sizes.

For this research, G*Power 3.1 software was used to conduct a power analysis, ensuring an adequate sample size for detecting statistically significant effects (Faul et al., 2007). In social science research, benchmarks such as a medium effect size ($f^2 = 0.15$) and a statistical power of 0.80 (indicating an 80% probability of detecting a true effect) are commonly applied. The parameters set for this analysis included a power level of 0.80, an alpha level of 0.05, a medium effect size of $f^2 = 0.15$, and 11 predictor variables (representing dimensions of the FEE and EO). Figure 3.4 illustrates the results of the G*Power analysis which suggested that a minimum sample size of 44 participants would be required to achieve the desired statistical power of 0.80 (Faul et al., 2007).

Given that survey research rarely achieves a 100% response rate, adjustments were made to account for potential non-responses. Keyton (2023) highlights the importance of oversampling to ensure that the final sample size sufficiently represents the target population. Following Wong's (2013) recommendation to adjust for non-responses, an oversampling strategy was implemented by multiplying the required sample size by three ($44 \times 3 = 132$).

(Salkind, 2010). Consequently, a total sample size of 132 participants was chosen from the target population to account for potential dropouts or non-responses. Additionally, (Hair, 2010; Tabachnick et al., 2013) recommend a sample size between 120 and 150 for SEM studies to detect medium effect sizes with 0.80 power at an alpha level of 0.05. This sample size range further ensured model robustness and aligned with SEM guidelines, confirming the adequacy of the chosen sample size for this study.

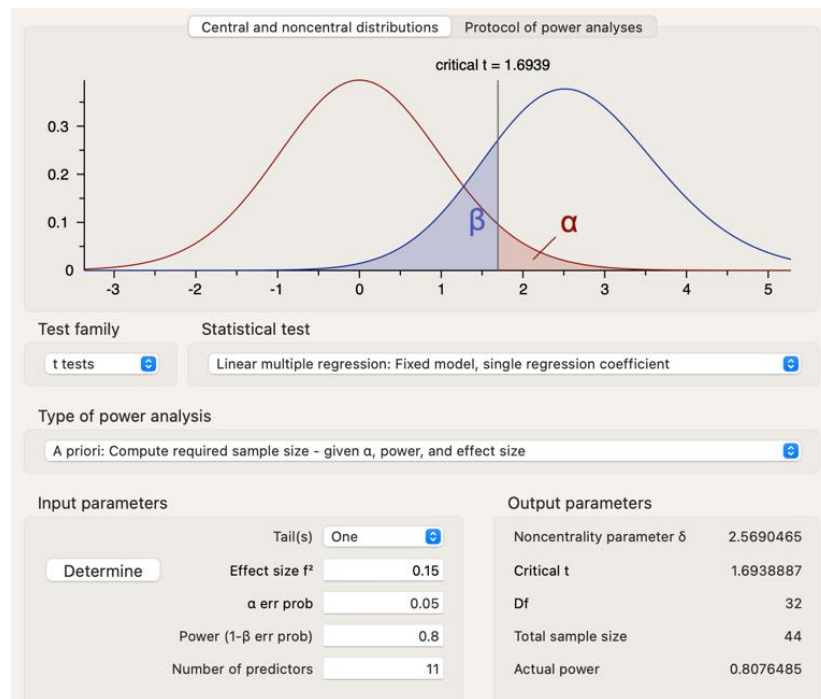


Figure 3.4: Power Analysis for Linear Regression (Faul et al., 2007)

3.6.2 The Instrument

A survey was chosen as an efficient tool for gathering diverse responses and generating insights into the characteristics and challenges of Saudi fashion MSMEs. Recognised for capturing wide-ranging experiences, surveys are ideal for identifying common patterns in entrepreneurship research (Dillman et al., 2014; Wiklund et al., 2011). An online questionnaire, administered via Qualtrics, allowed participants from across Saudi Arabia to respond at their convenience, facilitating broader coverage and higher participation than face-to-face methods (Evans & Mathur, 2018).

The questionnaire was structured into four main sections, each designed to align with the study's objectives and research questions (see Table 3.4 for a summary of the measurement levels and sources). These sections draw on both the theoretical frameworks

and empirical studies discussed in the literature review (see Appendix B, p.330 for the survey). The first section, comprising nine questions, focuses on demographic data, including age, gender, education, and experience in the fashion industry. The rationale for including these questions is grounded in research showing that demographic factors often influence the development of EO and BP (Carter et al., 2001; Moodley, 2003; Reynolds et al., 2005; Shah & Mehta, 2009; Shane, 2003). Example questions include: “What is your highest level of education?” and “Did you have business experience prior to starting this venture?”.

The second section aims to collect data on the characteristics of the participants’ fashion companies, such as their age, structure, and size. Firm characteristics are critical determinants of business performance and growth trajectories in the MSME sector (Begley & Boyd, 1986; Coad et al., 2017; World Bank Group, 2023). Questions included the following example: “How many equivalent full-time employees do you currently have?”. Participants were also asked to identify specific obstacles, particularly funding challenges, which are well-documented barriers for MSMEs in both developed and emerging markets (Kuez, 2010; Sarkar & Karim, 2019; Van Den Berg & Almanza, 2016). A sample question related to obstacles is, “What are the primary challenges your company faces in accessing financing?”. Understanding company characteristics and challenges is crucial for exploring how these factors impact the success of fashion MSMEs, providing essential context for examining the broader dynamics of the Saudi fashion industry.

The third section explores participants’ perceptions of the Saudi fashion entrepreneurship ecosystem, particularly in terms of market dynamics, policy, support, human capital, and cultural factors, based on World Bank Group (2023) and Global Entrepreneurship Monitor (2022). This section contains 39 items to measure perceptions essential for addressing RQ1 and RQ2, which investigate how the broader industry environment influences MSME EO and business outcomes in the Saudi context.

The final survey section aimed to assess the EO and BP of the participants’ enterprises. EO is measured using items derived from Covin and Slevin (1989) and Lumpkin and Pidduck (2021). BP was evaluated using statements adapted from Kaplan and Norton (2007), Murphy (2016), and (Neely et al., 1995, 2005). Both EO and BP were assessed using a 5-point Likert scale, a widely validated method for capturing variations in these constructs (Aliyu et al., 2015; Irikefe & Bagobiri, 2022; Shah & Ahmad, 2019). This section addresses Research Question 3 (RQ3), which examines the relationship between EO and business performance in the Saudi fashion sector.

While the online survey method offered the benefit of reaching a wide and diverse

sample of fashion MSMEs, it also introduced some limitations. Response bias, particularly self-reporting bias, is common in online questionnaires (Tian & Xiufang, 2017). To mitigate these issues, steps were taken to ensure anonymity and confidentiality, which encouraged more honest and accurate responses. Moreover, the survey design sought to minimise respondent fatigue by limiting the completion time to a manageable 15-20 minutes. Table 3.3 presents the descriptions of each section, including the corresponding levels of measurement.

Table 3.5: Measurement levels and survey sources

Sections	Number of questions	Measurement Level	Source
Entrepreneur profile	Q1 to Q9	Fixed choice: nominal and ordinal measurement	(Carter et al., 2001; Moodley, 2003; Reynolds et al., 2005; Shah & Mehta, 2009; Shane, 2003)
Business Profile	Q10 to Q22	Fixed choice: nominal and ordinal measurement	(Begley & Boyd, 1986; Coad et al., 2017; World Bank Group, 2023)
FEE in Saudi Arabia	39 items	5-point Likert scale: 1- Strongly disagree, 2- disagree, 3 - neutral, 4 - agree, 5 - strongly agree	(Global Entrepreneurship Monitor, 2022; World Bank Group, 2023)
EO	9 items	5-point Likert scale: 1- Strongly disagree, 2 - disagree, 3 - neutral, 4 - agree, 5 - strongly agree	(Covin & Slevin, 1989; Lumpkin & Dess, 1996)
BP	5 items	5-point Likert scale: answers range from 1 = much worse than competitors to 5 = much better than competitors.	(Kaplan, 1992; Kaplan & Norton, 2007; Murphy, 2016; Neely et al., 1995, 2005).

3.6.3 Variables measurement

In developing the measurement items, a comprehensive review of the literature was conducted, and items were adapted from validated measures. The scales used to measure the constructs are grouped and discussed in the following three categories.

3.6.3.1 Fashion Entrepreneurship Ecosystem Measurement

This study assesses the entrepreneurship ecosystem using the Global Entrepreneurship Monitor (GEM) National Experts Survey (NES), a widely used tool in global entrepreneurship research, with data from over 100 countries, providing key insights (Global Entrepreneurship Monitor, 2022, 2023). The credibility and multidimensional approach of the GEM framework make it ideal for evaluating ecosystem factors, including those affecting the Saudi fashion sector. Research by Hechavarria and Ingram (2019) and Ali et al. (2019) applied GEM to gendered entrepreneurship and ecosystem factors in emerging markets, while Ahmad and Xavier (2012) examined Malaysia, and Amorós and Bosma (2014) studied effects of education, policies, and finance in 70 economies. Other studies have highlighted the importance of

finance, government support, and cultural norms (Hechavarria & Reynolds, 2009; Leendertse et al., 2020). Given specific dynamics of the Saudi fashion sector, this study applies the GEM framework to assess factors influencing EO and business performance for fashion MSMEs in SFE, evaluating enablers and constraints shaping its entrepreneurial environment.

This study employs GEM's framework to assess seven core ecosystem factors, as outlined in Table 3.5 The support domain (S1–S6) evaluates access to subcontractors, mentors, and essential business services. The market domain (M1–M6) examines the ease of market entry and access to distribution channels. The policy domain (P1–P6) assesses government support and regulatory effectiveness. The education domain (HE1–HE2) measures institutional preparation for entrepreneurship. The talent domain (HT1–HT3) considers workforce availability and skill levels. The finance domain (F1–F6) evaluates access to diverse funding sources. The culture domain (C1–C5) reflects national support for risk-taking and creativity. Social norms (CN1–CN5) reflect societal perceptions of entrepreneurship's status and appeal. Each domain is measured on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree), allowing a structured analysis of the Saudi FEE and its impact on EO and BP of MSMEs. This comprehensive approach reveals how ecosystem factors influence entrepreneurial outcomes in the Saudi fashion sector.

Table 3.6: Fashion Entrepreneurship Ecosystem measurement

EE domains and indicators (1=Strongly Disagree, 5=Strongly Agree)
Support (availability and accessibility of subcontractors, mentors, and services for fashion businesses)
S1: In Saudi Arabia, there are enough subcontractors, mentors, qualified experts, suppliers, and consultants to support fashion businesses.
S2: In Saudi Arabia, fashion businesses can afford the cost of using subcontractors, suppliers, mentors, qualified experts, and consultants.
S3: In Saudi Arabia, it is easy for fashion companies to get good subcontractors, suppliers, and consultants.
S4: In Saudi Arabia, it is easy for fashion companies to obtain good, professional legal and accounting services.
S5: In Saudi Arabia, it is easy for fashion businesses to get good banking services (checking accounts, foreign exchange transactions, letters of credit, and the like)
S6: In Saudi Arabia, it is easy for fashion businesses to access the Business Incubator Accelerator.
Market (Ease of market entry and access to market data and distribution channels)
M1: In Saudi Arabia, fashion companies can easily enter new markets.
M2: In Saudi Arabia, fashion companies can afford the cost of market entry.
M3: In Saudi Arabia, fashion businesses can enter markets without being unfairly blocked by established fashion firms.
M4: In Saudi Arabia, antitrust legislation is effective and well-enforced.
M5: In Saudi Arabia, the availability of market information, such as fashion trends forecasting, is an obstacle for fashion businesses.
M6: In Saudi Arabia, access to distribution channels (retailers) is an obstacle for fashion companies.
Entrepreneurial Education (Quality of educational programs for launching fashion businesses)
HE1: In Saudi Arabia, colleges and universities provide good and adequate preparation to start and grow fashion businesses.
HE2: In Saudi Arabia, the vocational education system provides adequate preparation for starting and growing fashion businesses.

Table continues...

Table 3.5 Continued

Policy (Effectiveness of government programs and regulatory support)
P1: In Saudi Arabia, fashion businesses can obtain a wide range of government assistance through contact with a single government entity.
P2: In Saudi Arabia, there are adequate government programs to help fashion businesses.
P3: In Saudi Arabia, the people working for government institutions are competent and effective in supporting fashion businesses.
P4: In Saudi Arabia, almost anyone who needs help from a government program for a fashion business can find what they need.
P5: In Saudi Arabia, government programs aimed at supporting fashion businesses are effective.
P6: In Saudi Arabia, fashion hubs and business incubators provide effective support to fashion companies.
Talent (Availability, affordability, and skill level of labour for fashion)
HT1: In Saudi Arabia, there is no shortage of the types of employees you need for your fashion companies.
HT2: In Saudi Arabia, you can afford to hire the employees you need for your fashion business.
HT3: In Saudi Arabia, you are satisfied that the skill levels of people are sufficient for your fashion businesses' needs.
Finance (Access to various funding sources for fashion MSMEs)
F1: In Saudi Arabia, there is sufficient equity funding available for fashion MSMEs.
F2: In Saudi Arabia, there is sufficient debt funding available for fashion MSMEs.
F3: In Saudi Arabia, there are sufficient government subsidies available for fashion MSMEs.
F4: In Saudi Arabia, there is sufficient funding available from private individuals (other than founders) for fashion MSMEs.
F5: In Saudi Arabia, there is sufficient venture capitalist funding available for fashion MSMEs.
F6: In Saudi Arabia, sufficient funding is available through initial public offerings (IPOs) for fashion MSMEs.
Entrepreneurial Culture (National encouragement for success, creativity, and risk-taking)
C1: In Saudi Arabia, the national culture is highly supportive of individual success (fashion entrepreneurs) success achieved through their own personal efforts.
C2: In Saudi Arabia, the national culture emphasises self-sufficiency, autonomy, and personal initiative.
C3: In Saudi Arabia, the national culture encourages entrepreneurial risk-taking.
C4: In Saudi Arabia, the national culture encourages creativity and innovativeness.
C5: In Saudi Arabia, the national culture emphasises the responsibility that the individual (rather than the collective) has in managing his or her own life.
Social norms (Societal view on entrepreneurship's status and desirability)
CN1: In Saudi Arabia, the creation of fashion companies is considered an appropriate way to become rich.
CN2: In Saudi Arabia, most people consider becoming a (fashion) entrepreneur a desirable career choice.
CN3: In Saudi Arabia, successful entrepreneurs (fashion) have a high level of status and respect.
CN4: In Saudi Arabia, you will often see stories in the public media about successful entrepreneurs (fashion).
CN5: In Saudi Arabia, most people think of (fashion) entrepreneurs as competent and resourceful individuals.

3.6.3.2 Entrepreneurial Orientation Measurement

In this study, EO was measured using the framework established by Covin and Slevin (1989), which has been widely adopted and validated across various contexts. Covin and Slevin's model assesses EO through three key dimensions: innovativeness, proactiveness, and risk-taking (Covin & Slevin, 1989). These are critical for understanding the entrepreneurial behaviour of firms. The original scale has been modified by other scholars, such as Stam and Elfring (2008) and Dai et al. (2014), to enhance its applicability in different cultural and industrial contexts. Furthermore, additional dimensions such as competitive aggressiveness

and autonomy (Lumpkin & Dess, 1996) have been proposed, while others, such as future orientation (Tan, 2008), have been incorporated by subsequent studies.

For this study, the decision to use Covin and Slevin's (1989) scale was based on its proven reliability and specific relevance to entrepreneurial performance in dynamic environments. Despite the existence of other adapted scales, this framework remains one of the most widely used and tested in entrepreneurial research (Wales et al., 2013), providing a solid foundation for examining EO in the Saudi fashion industry. Additionally, to further align the EO construct with the specific context of fashion microenterprises in Saudi Arabia, the measurement approach draws on the work of Hughes and Morgan (2007), who provided refinements particularly suitable for smaller firms in emerging markets.

As Table 3.5 shows, the EO measurement in this study consists of nine indicators distributed across three core dimensions: innovativeness, risk-taking, and proactiveness. Each indicator was assessed by respondents using a 5-point Likert-type scale, where 1 represented "Strongly Disagree" and 5 represented "Strongly Agree". This scale allows for a nuanced understanding of the degree to which respondents engage in entrepreneurial behaviour across EO dimensions. By employing this structured and validated measurement approach, this study ensured that the EO construct was both theoretically grounded and practically applicable within the Saudi fashion industry.

Table 3.7: Entrepreneurial Orientation Measurement

Innovativeness (1=Strongly Disagree, 5=Strongly Agree)
EOI1: Our business favours a strong emphasis on R&D, technological leadership, and innovations. EOI2: Many new lines of products were marketed in the last 3 years. EOI3: Changes in product lines have usually been quite dramatic.
Proactiveness
EOP1: Typically, we initiate actions to which competitors then respond. EOP2: We are very often the first business to introduce new products, administrative techniques, processing technologies, etc. EOP3: Our company has a strong tendency to be ahead of others when it comes to introducing novel ideas or products.
Risk-taking
EOR1: The term 'risk taker' is considered a positive attribute for people in our business. EOR2: People in our business are encouraged to take calculated risks with new ideas. EOR3: Our business emphasises both exploration and experimentation for opportunities.

3.6.3.3 Business Performance Measurement

Evaluating BP is crucial for assessing entrepreneurial success, particularly in competitive sectors such as fashion, where both financial and non-financial outcomes are critical for long-term viability. As shown in Table 3.6, this study measures BP using five key indicators that reflect financial and non-financial outcomes: Return on Investment (ROI), sales increase, customer satisfaction, employee satisfaction, and overall firm results. These indicators align with the firm's objectives and offer a comprehensive performance assessment. ROI (BP1) assesses profitability relative to investment and serves as a key indicator of financial health (Marr, 2005; Neely et al., 2005). Sales growth (BP2) reflects market expansion and competitiveness, which are essential for MSME growth and sustainability (Hult et al., 2008; Murphy et al., 1996). Non-financial metrics, such as customer satisfaction (BP3) and employee satisfaction (BP4), strengthen brand loyalty, productivity, and retention, particularly in the competitive fashion sector (Harter et al., 2002; Jamal, 2021; Wong & Low, 2013). Overall firm performance (BP5) provides a comprehensive view of both financial and operational outcomes, encapsulating a company's health over time (Arifeen et al., 2014).

These indicators have been used in various academic studies. For example, Murphy et al. (1996) used similar metrics to evaluate business success, showing the importance of both financial and satisfaction-based metrics. Recent studies have confirmed the effectiveness of these metrics in capturing multidimensional performance, particularly in the fashion industry, where customer satisfaction plays a crucial role in sustaining competitive advantage. For instance, Harshita and Kavitha (2024) examined how customer relationship management (CRM) strategies enhance customer satisfaction and retention, ultimately strengthening long-term competitiveness in fashion retail. Similarly, Lubis et al. (2023) investigated the direct impact of service quality and employee performance on customer satisfaction, demonstrating their pivotal role in maintaining a competitive edge in the fashion sector. Thus, this measurement approach enables a nuanced assessment of Saudi fashion MSMEs, reflecting their success relative to the EO and broader market contexts.

Table 3.8: Business Performance Measurement

Please evaluate the performance of your business over the previous THREE years relative to your major competitors (1= <i>much worse than competitors</i> , 5= <i>much better than competitors</i>)
BP1: the level of ROI was
BP2: the level of sales increase was
BP3: the degree of customer satisfaction was
BP4: the satisfaction of your employees was
BP5: The overall results of the firm in the past year were

3.6.4 Pretesting and Piloting the Research Instrument

Numerous researchers have emphasised the importance of conducting a pilot study prior to full-scale data collection to develop and refine a survey instrument (Saunders et al., 2015; Van Teijlingen & Hundley, 2002; Wadood et al., 2021). In line with this recommendation, a small-scale pilot study was conducted to test the survey instrument designed to measure EE, EO, and BP. The pilot study followed a three-step process by Churchill Jr (1979), which involved developing the questionnaire, pretesting, and piloting (see Figure 3.5). Considering the essential elements introduced Mariel et al. (2020) and Rattray and Jones (2007) in developing and designing a questionnaire, this study developed scales for the fashion entrepreneurship ecosystem, EO, and BP.

The first step involved developing a questionnaire based on a comprehensive review of EE and EO literature to identify relevant measurement items. Seminal studies in these fields were reviewed to extract key measurement items that would ensure comprehensive coverage of the constructs being studied, and these items were adapted to the specific context of the Saudi fashion industry.

The second step focused on pretesting the questionnaire to evaluate the language clarity and comprehensibility of the questions. As Mariel et al. (2020) suggested, pretesting is critical to ensuring an instrument's suitability for large-scale implementation. To account for the diverse linguistic and cultural contexts of the participants, the questionnaire was administered in both English and Arabic. The pretesting process consisted of two stages. Initially, the English version of the questionnaire was reviewed by the supervisory team and a native English-speaking editor, resulting in minor modifications to improve its clarity and wording. In the second stage, the questionnaire was translated into Arabic with the help of language experts, ensuring that the terminology and phrasing were appropriate for the target audience. The most important elements considered during the questionnaire design stage were as follows: words

used in questions were simple and familiar; statements that included double-barrelled questions were avoided; the text made it possible to capture in-depth information; early questions were deemed easy to answer; questions on the same topic were grouped together and presented from a general to increasingly specific context; and questions were designed to avoid asking unrelated or irrelevant questions that could side-track respondents.

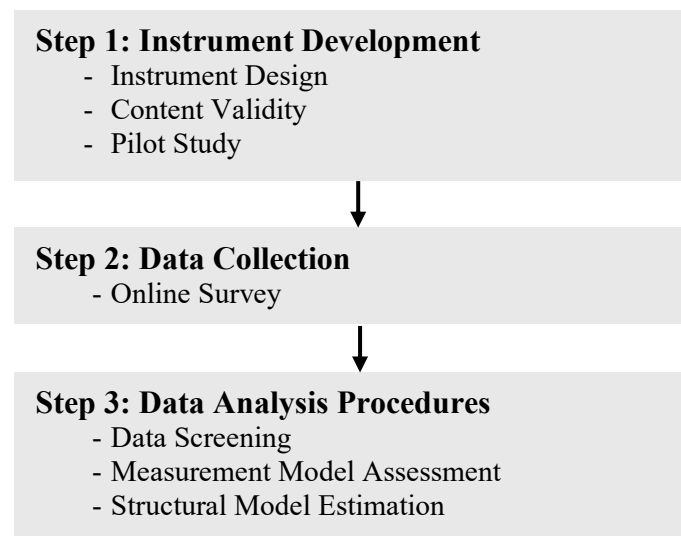


Figure 3.5: Data Collection and Analytic Procedures

Following these initial modifications, the questionnaire was reviewed by academics who were senior lecturers in the field of entrepreneurship and fluent in Arabic and English. These experts were selected based on their knowledge and experience in entrepreneurship and survey methodology. They provided feedback on several aspects of the questionnaire, including the clarity of questions, relevance of terminology, survey length, and appropriateness of technical terms. Based on these recommendations, several adjustments were made. Ambiguous or unclear items were revised to better align with the established literature in the EE field. Based on participant feedback, the 7-point Likert scale was replaced with a 5-point scale to simplify the response process. Research has shown that 5-point scales can reduce cognitive load, making it easier for participants to distinguish options, thereby enhancing clarity and response accuracy (Dawes, 2008; Revilla et al., 2014). Additionally, minor design issues were addressed, such as improving instructions, incorporating skip patterns for certain questions, and allowing multiple responses in “choose all that apply” questions. Further modifications included adding questions to capture objective measures of financial performance, as suggested by academic experts.

In the final step, a revised version of the questionnaire was administered to 56 fashion business owners in a pilot study. This sample size aligns with the recommendations of various

researchers that a pilot study should include approximately 10% of the total sample size (Hertzog, 2008) or at least 30 participants (Bujang et al., 2024). At the end of the survey, participants were asked to provide feedback on the overall clarity of the questions and to offer any additional suggestions for improvement. Most participants reported that the questions were clear and easy to understand, indicating that the instrument was well-suited for data collection. The feedback received from this pilot test was instrumental in finalising the research instrument for the main study.

3.6.5 Data Collection

Data collection was conducted using the Qualtrics Online Survey platform, which was selected for its secure data storage, automated response tracking, and ability to maintain confidentiality while enabling efficient electronic distribution. The survey was open for four months (August to December 2022). A total of 650 individuals accessed the survey, and 224 completed it, resulting in a 34% completion rate. Participants were recruited through multiple recruitment strategies to maximise reach and diversity of responses.

First, personalised email invitations were sent to 120 fashion brand entrepreneurs who had participated in the 2022 Saudi Fashion Brand Exhibition. These emails included the study's purpose, a secure survey link, and clear reassurances of confidentiality and voluntary participation (Creswell & Clark, 2017; Saunders et al., 2003). A follow-up reminder was issued two weeks later to encourage participation. Additionally, the researcher employed targeted social media advertising, including paid promotions on Instagram and Facebook, to reach a broader audience of Saudi fashion professionals (e.g., brand owners and designers) who may not have received the email invitation (Bennetts et al., 2019). This combined recruitment approach helped reach a wider audience. Past studies have shown that using social media for research recruitment is both effective and more affordable than traditional methods (Ellington et al., 2021; Gilligan et al., 2014).

Several challenges arose during data collection, such as participant hesitation, time constraints, and sensitivity about business-related information. To address these, the survey was designed to be concise (approximately 15 minutes) and included clear assurances of privacy throughout both email and social media messaging (Evans & Mathur, 2018). These strategies were intended to increase trust and improve engagement.

3.6.6 Data Analysis Procedures

The data analysis process comprised three key stages: data screening, measurement model validation and structural model assessment. Figure 3.6 presents an overview of the data analysis procedures. Two software packages were used for the quantitative data analysis phase: SPSS (used for frequency and descriptive statistics) and SmartPLS used for PLS-SEM technique.



Figure 3.6: Data Analytic Methods

3.6.6.1 Data screening

IBM SPSS Statistics (Version 28.0) was employed in the initial stage to prepare the dataset for inferential analysis. This phase included conducting response rate analysis, data coding, construct specification, and detecting presence of outliers. SPSS was instrumental in identifying and correcting errors, handling missing data, and testing statistical assumptions such as normality. In empirical social science research, missing data is a common issue that can introduce bias, reduce sample representativeness, and complicate analysis (Horton & Kleinman, 2007). Tabachnick et al. (2013) noted there are no strict guidelines regarding

acceptable amounts of missing data for a sample size. However, (Hair, 2010) and (Tabachnick et al., 2013) recommend the expectation-maximisation (EM) method in SPSS for managing missing values, provided at least 50% of the data is available for each case.

Outliers were identified using the Mahalanobis distance (D^2) method, which evaluates the distance of each observation from the centroid of all cases (Tabachnick et al., 2013). Observations with D^2 values exceeding the critical chi-square threshold, adjusted for the number of independent variables, were flagged as potential outliers. To ensure model accuracy and parsimony, any flagged observations were removed (Hair et al., 2012).

To evaluate normality, graphical analysis through residual plots was conducted, such as expected normal probability and detrended normal probability plots (Chantarangsi et al., 2018). Consistent with Pallant (2005), if these plots showed a normal distribution of residuals, further individual variable analysis for normality was deemed unnecessary (Suleiman et al., 2015). Descriptive statistics, including frequencies, means, and standard deviations, were applied to summarise the dataset (Martin et al., 2017). For categorical data obtained from nominal scales, frequencies and percentages were used, while measures of central tendency described the key study variables.

After completing the data screening procedure, the next step involved validating measurement models, followed by assessing structural models. Partial Least Squares (PLS) was used as the statistical technique for this analysis. IBM SPSS version 23 was utilised for data screening, while SmartPLS version 4.0 was employed to run PLS analysis (Hair et al., 2014). The following section provides detailed information on this statistical technique and steps taken to evaluate measurement and structural models.

1.1.1.1 Partial Least Squares Structural Equation Modelling (PLS-SEM)

PLS-SEM was chosen as the primary analytical method for this study due to its efficacy in handling complex models with multiple and higher-order constructs (Hair et al., 2023; Henseler et al., 2016). This aligns with the study's objective of examining the Saudi FEE and its influence on EO and BP among fashion MSMEs (Reinartz et al., 2009; Sarstedt et al., 2021). Rationales for using PLS-SEM include:

1. PLS-SEM is particularly suited for studies involving small to medium sample sizes, such as 223 Saudi fashion entrepreneurs analysed in this research. In emerging markets such as the Saudi fashion sector, accessing large datasets is often challenging, making

PLS-SEM's capacity to yield reliable results with smaller samples a significant advantage (Hair et al., 2023; Henseler et al., 2016).

2. PLS-SEM is designed to handle non-normal data, a frequent occurrence in social sciences, especially in entrepreneurial research where data distribution may not conform to normality (Gefen et al., 2000). This characteristic of PLS-SEM ensures validity and robustness of results, even when underlying data deviates from the normal distribution.
3. Saudi FEE and EO constructs in this thesis are inherently multidimensional, encompassing elements such as policy, market conditions, financial access, cultural dynamics, and entrepreneurial support mechanisms. PLS-SEM's strength lies in its capacity to effectively model such higher-order constructs, facilitating a detailed and comprehensive analysis of relationships between these dimensions and their impact on BP (Becker et al., 2012).
4. This study aims to explore and extend theoretical frameworks within the niche context of the Saudi fashion industry. PLS-SEM is particularly suitable for such exploratory and theory-building research as it does not impose stringent requirements for model fit, unlike covariance-based SEM (CB-SEM). This flexibility supports development of a new, context-specific theoretical framework tailored to the Saudi FEE (Chin, 1998; Hair et al., 2023).
5. A core objective of this thesis is to explain variance in BP among Saudi fashion MSMEs as influenced by Saudi FEE and EO dimensions. PLS-SEM is optimised for variance-based analysis, maximising explained variance (R^2) and ensuring predictive relevance (Q^2). This focus on prediction allows the study to generate actionable insights, which are particularly valuable for industry stakeholders and policymakers in the Saudi fashion sector (Hair et al., 2012; Sarstedt et al., 2021).

By employing PLS-SEM, this research ensures a rigorous, flexible, and contextually appropriate approach for analysing dynamic relationships within the Saudi fashion entrepreneurship ecosystem. This methodology enhances the robustness of findings and supports development of a theoretical model that aligns with the unique characteristics and needs of the Saudi fashion industry. The PLS-SEM model consists of two main components: the measurement model (outer model), and the structural model (inner model). The measurement model defines relationships between latent constructs and their corresponding indicators (Sarstedt et al., 2021). The structural model depicts connections between latent

variables or constructs. In developing the measurement model, it is crucial for researchers to base their decisions on theoretical justifications, ensuring each indicator effectively represents and predicts the underlying construct (Hair et al., 2020; Jarvis et al., 2003). Therefore, it is essential to understand characteristics of the two main types of unobservable variables, reflective and reflective measurement models (Hair et al., 2014), as elaborated in the following section.

3.6.6.2 Higher-Order Constructs (HOC)

Higher-order construct models (HCMs) are an advanced SEM approach, allowing researchers to explore relationships across multiple levels of abstraction, especially when a higher-order construct (HOC) comprises multiple lower-order components (LOCs) (Sarstedt et al., 2021). There are four main types of HCMs (Becker et al., 2012; Hair et al., 2019; Sarstedt et al., 2021).:

1. Reflective-Reflective HCMs: Both HOC and LOCs are measured using reflective indicators, indicating the HOC is a latent variable reflected through the LOCs
2. Reflective-Formative HCMs: The HOC is assessed with reflective indicators, while LOCs use formative indicators, implying LOCs are causal factors shaping the HOC.
3. Formative-Reflective HCMs: The HOC is formed by the LOCs, which are measured with reflective indicators.
4. Formative-Formative HCMs: Both HOC and LOCs are measured formatively, suggesting a causal and interactive relationship between them

The choice of HCM type depends on the theoretical framework and research objectives, as the nature of the constructs and their interrelationships will determine whether a reflective or formative approach is more suitable. This study adopts a reflective-reflective model, where both the higher-order construct (HOC) and its lower-order components (LOCs) are measured using reflective indicators. This approach captures the multidimensional nature of constructs such as Saudi FEE, where elements such as finance, culture, and support are integrated into a cohesive structure. By consolidating LOCs into a single HOC, reflective-reflective HCMs simplify model complexity, minimising relationships in the SEM model and enhancing parsimony and interpretability (Hair et al., 2019).

Reflective-reflective HCMs also resolve collinearity issues among correlated LOCs by combining them, which improves discriminant validity (Diamantopoulos et al., 2008). In this study, this approach is applied to operationalise two reflective-reflective HOCs: the

Entrepreneurship Ecosystem with eight dimensions, and EO with three domains (see Table 3.7 for a summary of HOCs and their respective LOCs) (Hair et al., 2019).

Table 3.9: Higher-order construct conceptualisation

Higher-Order Construct (HOC)	Respective lower-order components
Entrepreneurship Ecosystem	Finance, Government, Market, Culture, Social Norms, Entrepreneurial Education, Talent, and Support
Entrepreneurial Orientation	Innovativeness, Proactiveness, and Risk Taking

1.1.1.2 Reflective-Reflective Construct Specification

Reflective indicators are utilised to capture observable manifestations of latent constructs, representing all possible items that theoretically measure the underlying concept (Hair et al., 2017; Sarstedt et al., 2021). In the context of SEM, particularly when employing PLS-SEM, reflective models are critical as they provide a means to quantify latent variables. Reflective indicators assume changes in the latent construct will be mirrored by changes in all its indicators, signifying they reflect the underlying construct rather than cause it (Diamantopoulos et al., 2008; Henseler et al., 2016).

To ensure that reflective indicators accurately represent the construct, two primary criteria must be met. The first criterion is high intercorrelation: items must exhibit high correlations among themselves, confirming they consistently measure the same latent construct. High intercorrelations imply indicators are interchangeable and reinforce the construct's unidimensionality (Hair et al., 2019). The second criterion is interchangeability of indicators: indicators should be sufficiently interchangeable so that removing any single item does not substantially alter the construct's validity or integrity (Diamantopoulos & Siguaw, 2006; Fornell & Larcker, 1981). For example, in this study, the EO construct includes indicators such as risk-taking, proactiveness, and innovation. Removing the proactiveness indicator should not significantly alter the construct's overall meaning or validity, demonstrating the interchangeable nature of reflective indicators. This characteristic ensures each indicator is merely a manifestation of the underlying construct, thus preserving the model's reliability and accuracy.

In contrast, formative measurement models assume indicators combine to form the construct, suggesting that removing an item could alter the construct's meaning. This distinction is critical when specifying the model, as the theoretical definition of the construct

guides whether a reflective or formative model is appropriate (Diamantopoulos et al., 2008; Hair et al., 2014). Given the nature of constructs such as EO and BP, which manifest through observable characteristics rather than forming the construct itself, a reflective approach is justified in this study.

Reflective measurement models are particularly suited for this study's framework, where constructs such as EO and BP in the Saudi FEE are operationalised using multiple observable indicators (Becker et al., 2012; Ringle et al., 2020). Such an approach ensures indicators collectively capture the full scope of constructs without reliance on any single item, enhancing robustness of the model. The single-headed arrows in the model depicted in Figure 15 indicate that any changes in the construct will automatically reflect in all its indicators, underscoring their reflective nature (Sarstedt et al., 2021).

Estimating reflective-reflective HCMs can be complex due to the hierarchical nature of constructs. Two primary approaches are commonly used in PLS-SEM: the extended repeated indicators method and the two-stage approach. The repeated indicators approach estimates the HOC directly by using indicators for both LOCs and HOC simultaneously (see Figure 3.7 for conceptualisation of the Higher-Order Construct for EO). In contrast, the two-stage method first estimates LOCs before aggregating them into the HOC, which is useful for complex reflective-formative models. Since this study applies a reflective-reflective model, the repeated indicators method is particularly well-suited for ensuring consistency and accuracy in estimating the HOC (Hair et al., 2020).

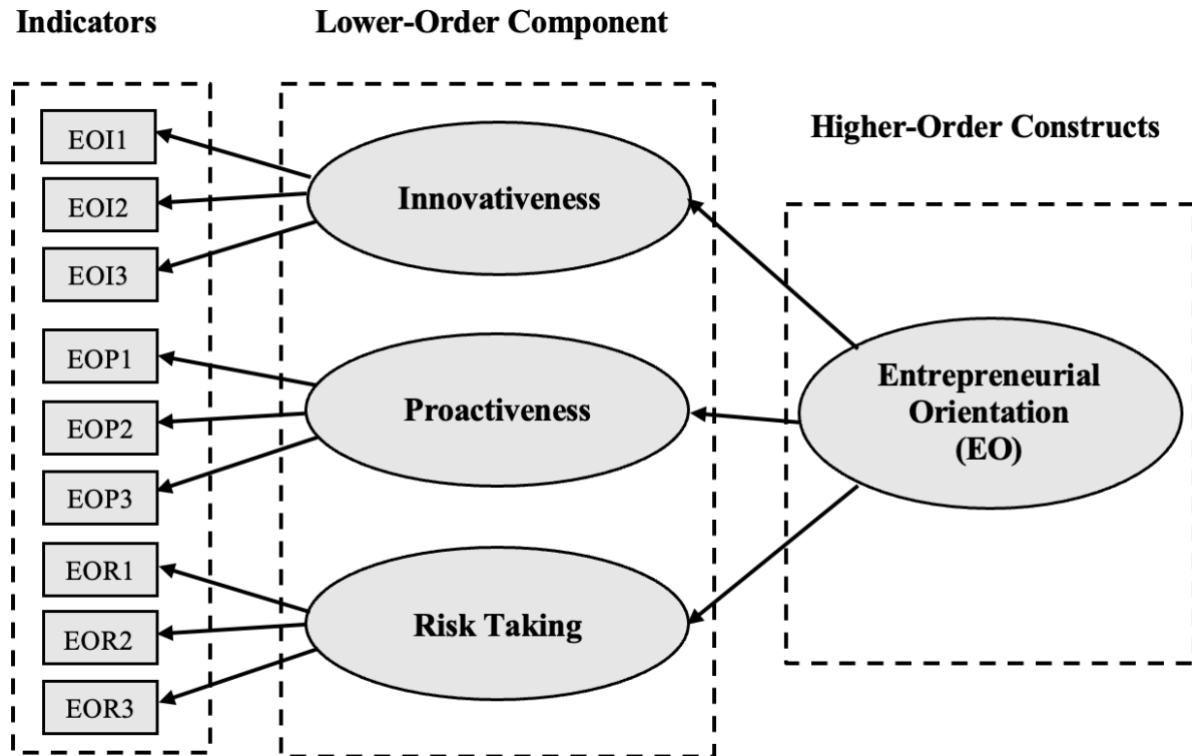


Figure 3.7: Conceptualisation of Higher-Order Construct EO, adapted from Hair et al. (2020)

3.6.6.3 Assessment of the Measurement Model

In this study, evaluation of the measurement model was conducted using SmartPLS (Version 4) and involved bootstrapping 5,000 resamples (Ringle et al., 2020). A two-step approach was followed to comprehensively assess the data (Hair et al., 2017). The first stage focused on evaluating the measurement model (outer model), specifically examining indicator loadings, internal consistency reliability, convergent validity, and discriminant validity, as recommended by (Hair et al., 2014). The evaluation process commenced by verifying reliability and validity of indicator items to ensure they accurately reflected constructs being measured (Ringle et al., 2020). For reflective measures, internal consistency was validated using Cronbach's alpha and composite reliability (CR), with both metrics exceeding the threshold of 0.70 (Nunnally & Bernstein, 1994). Convergent validity was assessed through Average Variance Extracted (AVE) values, while discriminant validity was verified using the Fornell-Larcker criterion and the HTMT ratio (Fornell & Larcker, 1981; Henseler et al., 2016).

Reflective indicators were applied to capture observable manifestations of constructs such as the FEE and EO. These indicators adhered to criteria of high inter-item correlation and

interchangeability, ensuring consistent measurement of latent variables (Diamantopoulos et al., 2008; Diamantopoulos & Siguaw, 2006).

Following evaluation of reflective measures, the second stage focused on validating Higher-Order Constructs (HOCs). To validate HOCs, the study applied factor and path weighting schemes rather than centroid weighting to reduce bias (Hair et al., 2020). When the number of indicators varied across lower-order constructs (LOCs), alignment was crucial to ensure unbiased results. A two-stage approach was used to compute parameters of HOCs accurately (Becker et al., 2012). This method, which captures the hierarchical nature of constructs such as the Saudi FEE and EO, enhances model precision by effectively balancing different dimensions within these complex constructs.

1.1.1.3 Assessment of the Structural Model

After validating the measurement model, the final stage involved assessing the structural model (inner model) to evaluate relationships among latent constructs and test hypotheses (see Figure 3.8 for the conceptual model illustrating these relationships). The structural model assessment aimed to estimate variance in endogenous constructs (e.g., Business Performance) explained by exogenous constructs (e.g., FEE and EO). This process involved examining path coefficients, R^2 values, and Q^2 values to confirm the model's predictive relevance (Chin, 2010; Geisser, 1974). Variance Inflation Factor (VIF) values were used to diagnose multicollinearity, which was confirmed as non-problematic (Sarstedt et al., 2021).

A two-stage approach was utilised, starting with computing the latent constructs' scores using the PLS algorithm. These scores were then saved and treated as first-order constructs representing the second-order constructs for further analysis. Bootstrapping was applied to estimate the inner model's outer weights and path coefficients, ensuring statistical inference accuracy by resampling with replacement (Hair et al., 2014; Hair et al., 2023).

Mediation analysis was also conducted to test indirect effects of EO on the relationship between FEE and BP, revealing that EO partially mediates this relationship, thereby enhancing the model's explanatory power (Hair et al., 2014; Hair et al., 2023). Results aligned with the study's objective of understanding and quantifying the impact of various components within the Saudi fashion entrepreneurship ecosystem.

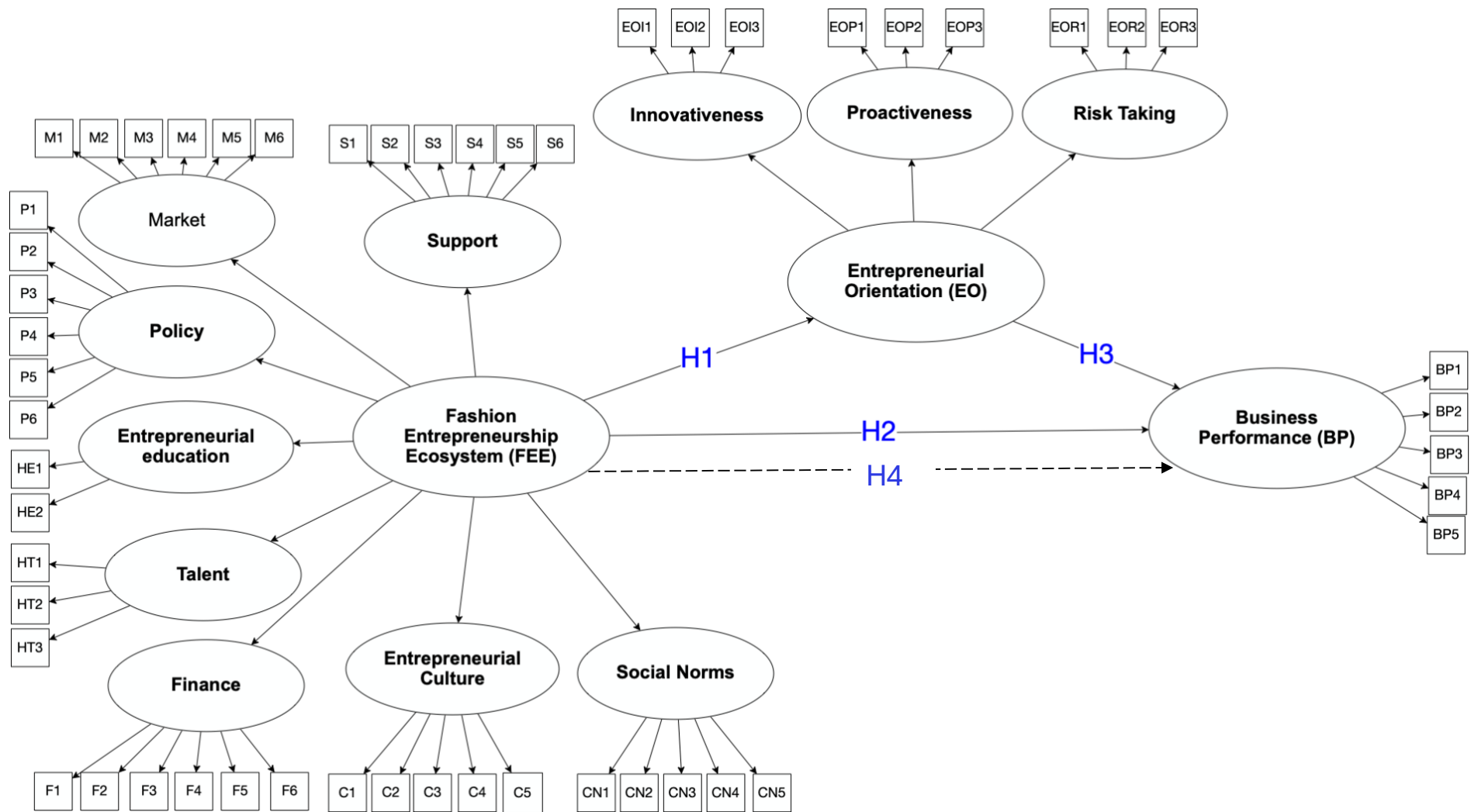


Figure 3.8: Conceptual Model of Saudi Fashion Entrepreneurship Ecosystem impacts on EO and BP in fashion MSMEs

3.7 Qualitative Research

This study used semi-structured interviews as a method for data collection, a common qualitative approach that allows for detailed exploration of topics while providing flexibility based on participants' responses (Bell et al., 2022; DiCicco-Bloom & Crabtree, 2006). Semi-structured interviews are a flexible and adaptable method for collecting detailed and meaningful qualitative data, making them suitable for gaining in-depth insights into the research subject (Creswell & Poth, 2016; Kallio et al., 2016). In Entrepreneurship Ecosystem research, the use of semi-structured interviews is prevalent as it allows researchers to gather in-depth insights from various stakeholders, such as entrepreneurs, policymakers, and support organisations within the ecosystem. For example, Spigel (2017) explored relational dynamics through interviews with ecosystem actors, Stam and Spigel (2016) mapped critical components influencing entrepreneurship development, Motoyama and Knowlton (2017) examined networks and resources in St. Louis, Roundy and Fayard (2020) investigated small-town ecosystems, and Mack and Mayer (2016) traced evolution of ecosystems over time through detailed stakeholder accounts.

3.7.1 *Interview Protocol*

The interview protocol served as a structured guide for the data collection process in this study (Clark & Creswell, 2011). As the current study uses a sequential exploratory mixed methods approach, the interview protocol was designed after analysing quantitative data, as these findings informed the structure and focus of the qualitative phase (Creswell & Poth, 2016). This section outlines the details of the interview protocol that supports the data collection process.

The interview protocol was divided into three main sections. The first section focused on understanding the participants' roles in supporting Saudi fashion MSMEs. It included background questions such as the participants' positions, years of experience, and responsibilities within the Saudi FEE. This section ensures that the participants selected are suitable and relevant for the study (Tremblay & Yagoubi, 2014). Relevant questions from previous studies were evaluated for inclusion; for instance, a question from Alkhaldi (2019) study, "What type of support do you offer Saudi fashion entrepreneurs?" was incorporated into the current study for its relevance and the possibility of comparing findings.

The second section of the protocol captured participants' perceptions and evaluations of the current state of Saudi FEE. It explores six key areas: politics, finance, human capital, culture, the market, and support. A question addressing the support aspect of the Saudi FEE

was adapted from previous studies (Chahine, 2018; Colliander, 2015; Gesellschaft für Internationale Zusammenarbeit, 2018; Herrera, 2020), The adapted question was: “How do you view the adequacy of support for fashion entrepreneurs at the present time?”

The final section examines the impact of FEE on EO and on BP. For instance, to gather insights into the direct effects of the FEE, participants were asked, “How does the Saudi Fashion Entrepreneurship Ecosystem influence your overall business performance?”. The complete interview protocol is available in Appendix C, p.339.

3.7.2 Sampling

3.7.2.1 Sampling Criteria

Participants were selected from diverse domains within the Saudi FEE, including fashion education experts, industry professionals, and entrepreneurs. This diversity facilitated a comprehensive exploration of key areas, such as policy impacts, market dynamics, and support mechanisms within the Saudi FEE, allowing for a nuanced understanding of how these elements are interconnected (Colliander, 2015; Lange & Schmidt, 2020; Salonoja, 2013). The sampling criteria reflect established practices in ecosystem research, where diverse stakeholder inclusion is essential for understanding ecosystem complexity and interactions (Isenberg, 2010; Mason & Brown, 2014; Spiegel, 2017). The inclusion criteria for fashion entrepreneurs were as follows:

1. Ownership or management of fashion MSMEs.
2. A minimum of five years of operational experience in the Saudi fashion industry.

The 5-year threshold ensured that participants had deep familiarity with ecosystem dynamics, consistent with research highlighting experienced entrepreneurs’ abilities to identify systemic challenges and opportunities (Brown & Mason, 2017; Hodges et al., 2016; Zhao et al., 2019). Selecting MSMEs participants to provide insights relevant to the typical scale and challenges of Saudi fashion enterprises, consistent with literature on the role of MSMEs in driving innovation and local development (Acs et al., 2014; Raghuvanshi et al., 2017).

3.7.2.2 Sample Size

The sample size for the semi-structured interviews in this study was determined using data saturation, a key criterion in qualitative research. Data saturation occurs when further interviews yield no new insights, ensuring that the sample adequately addresses the research questions (Guest et al., 2006). This approach aligns with purposive sampling principles, where

sample sufficiency is based on informational richness rather than statistical representativeness (Morse, 1995, 2015). In exploratory research, Denscombe (2017) highlights the flexibility in sample size, noting that findings may not generalise broadly. In entrepreneurship ecosystem studies, prior research demonstrates that 10-15 interviews can reveal core themes. For example, Pittaway et al. (2004) identified the importance of social capital through 14 interviews with entrepreneurs and network facilitators. Bliemel et al. (2019) identified mentorship and funding themes through 15 interviews on accelerator programs. Similarly, Wurth et al. (2021) used 12 interviews to uncover key ecosystem dynamics, supporting the use of smaller sample sizes in qualitative studies. For this study, 15 participants were interviewed; no new insights emerged beyond this point, indicating that data saturation had been reached. This choice reflects both the practical considerations of participant accessibility and alignment with qualitative ecosystem research practices (Patton, 2014). The sample size reflected both the practical considerations of participant accessibility and alignment with qualitative ecosystem research practices.

3.7.2.3 Sampling Procedures

The qualitative phase employed a combination of purposive (judgement), snowball, and maximum variation sampling techniques to recruit participants with relevant expertise in the Saudi fashion industry (Saunders et al., 2015; Saunders et al., 2015). As a former fashion educator in Saudi Arabia, the researcher used existing professional and academic networks to identify eligible participants and gain access to otherwise hard-to-reach individuals. Additional candidates were identified through online platforms such as LinkedIn and industry directories. Snowball sampling further supported the recruitment process, with initial participants referring others with specialised domain knowledge (Alkhaldi, 2019; Creswell & Clark, 2017; Sabri, 2021).

Additionally, during the quantitative phase, 112 survey respondents voluntarily indicated their willingness to participate in follow-up interviews and provided contact details. A pre-screening process was conducted based on the predefined inclusion criteria listed in Section 3.7.2.1. Of the 112 interested participants, only 42 met these criteria. From this eligible pool, 15 respondents were selected using maximum variation sampling to ensure a diverse representation across several key dimensions:

- Gender

- Geographic location (Northern, Western, Southern, Eastern regions, and Riyadh).

- Educational background (domestic vs. international fashion degrees).

Type of business activity (e.g., B2B, B2C, consultancy, training)

Domain expertise (entrepreneurship, education, advisory roles)

By integrating these approaches judgment, snowball, and leveraging professional networks, the study ensured a comprehensive and methodologically sound data collection process that aimed for thematic saturation, enhancing credibility and depth of qualitative insights obtained (Tashakkori, 2010; Teddlie & Tashakkori, 2011).

3.7.2.4 Participants Profile

Among the 15 participants, 83% were female, with educational levels ranging from high school to doctoral degree. Seven participants held a bachelor's degree, two had master's degrees, one held a doctorate, three had completed high school, and two held diplomas. Half of the participants had academic backgrounds in fashion-related fields, while the remainder specialised in fields such as mechanical engineering, finance and commercial law.

Experience levels varied, with the largest segment (33%) reporting 3–5 years of experience in their respective fields. Geographically, most participants were concentrated in the Western Region and the Capital City, representing approximately 75% of the total sample size. This geographical distribution reflects the centralisation of industrial professionals in key urban areas.

Interviews with fashion entrepreneurs, such as D4, D6, and D7, revealed engagement in both Business-to-Business (B2B) and Business-to-Customer (B2C) markets. Participants reported offering product design, consultancy services, and training programs as strategic approaches to diversify their income streams and strengthen their market positioning. This diversity in business models reflects the evolving nature of the Saudi fashion sector and the strategic adaptation of industry professionals. The participant profile demonstrates a balanced sample representing the key domains of the Saudi FEE, including business operations, advisory roles and educational backgrounds. This diversity provides a comprehensive understanding of the entrepreneurial ecosystem, contributing to the depth and methodological robustness of the study.

Table 3.10: Demographic profile of interviewees

Participant' s Code	Gender	Educational level	Education Major	Experience (years in the field)	Role	Location
D1	Female	High school	Self- taught sewing and tailoring	3-5	Fashion Entrepreneur	Western Region
D2	Female	High school	Self- taught sewing and tailoring	6-10	Fashion Entrepreneur	Riyadh
D3	Female	Bachelor	Fashion Design	Less than 3	Fashion Entrepreneur	Riyadh
D4	Female	Master	Fashion marketing	3-5	Fashion Entrepreneur	Western Region
D5	Female	Bachelor	Finance	6-10	Fashion Entrepreneur	Western Region
D6	Female	Master	Fashion marketing	3-5	Fashion Entrepreneur	Riyadh
D7	Male	Diploma	Mechanical Engineering	Less than 3	Fashion Entrepreneur (Retailer)	Riyadh
D8	Female	Diploma	Home economics and art education	More than 10	Fashion Entrepreneur (Manufacturer)	Western Region
D9	Female	Bachelor	Fashion Design	3-5	Fashion Entrepreneur	Northern Region
S1	Male	Bachelor	Finance	6-10	Fashion Business Advisor	Northern Region
S2	Female	High school	Self-learning sewing and tailoring	More than 10	Fashion Business Advisor	Eastern Region
S3	Female	Bachelor	Home economics and art education	More than 10	Fashion Business Advisor	Western Region
S4	Female	Bachelor	Commercial Law	3-5	Educator	Riyadh
S5	Male	Bachelor	English (Literary Studies)	3-5	Fashion Editor	Riyadh
S6	Female	Doctorate	Entrepreneurship	More than 10	Researcher	Riyadh

3.7.3 Data collection

Data collection was conducted over two months, from early October to late November 2023. Before data collection, a letter of invitation, written in English and Arabic, was emailed to potential participants recruited from the 100 Saudi Fashion Brands Expo. The letter outlined the purpose of the study and provided clarity on the nature and scope of the participation. It included the Participant Information Sheet and Consent Form as attachments (see Appendix D, p.341). Participants were required to read these documents and sign an informed consent form before participating in the interview, ensuring that they fully understood their rights and the scope of the study.

To reinforce ethical standards, oral consent was obtained from each participant at the beginning of the interview to confirm their voluntary participation and understanding of the study objectives. This process aligns with the established ethical guidelines for conducting

research involving human subjects (Creswell & Poth, 2016; Hammersley & Traianou, 2012). Interviews were conducted in Arabic using the participants' preferred video conferencing platform, Microsoft Teams. Conducting interviews in the participants' native language facilitated clearer communication and encouraged richer responses, consistent with best practices in cross-cultural qualitative research (Brislin, 1970; Van Nes et al., 2010). Each interview lasted between 45 and 75 minutes, providing sufficient time for an in-depth exploration of topics while maintaining participant engagement.

Following each interview, the researcher transcribed the recordings using Microsoft Word's transcription feature to capture both verbal and nonverbal elements, ensuring accurate and immediate documentation of responses in Arabic. Capturing nonverbal cues enhances the depth of qualitative data and strengthens interpretative accuracy (Walsham, 2006). The transcripts were then professionally translated into English by a certified translation service to ensure linguistic accuracy and to maintain the integrity of participants' responses. Translation accuracy is critical for preserving the original meaning and context, aligning with methodological recommendations for qualitative cross-language research (Squires, 2009; Temple & Young, 2004). A total of 15 interviews were conducted, consistent with qualitative research standards for achieving data saturation. Studies have demonstrated that 10–15 in-depth interviews are typically sufficient to reach data saturation in qualitative research, where further data collection yields no new insights (Guest et al., 2006; Hennink et al., 2017).

3.7.4 Data Analysis Methods (Thematic Analysis)

This study employed thematic analysis to systematically examine the qualitative data collected from semi-structured interviews with 15 fashion entrepreneurs and key stakeholders within the Saudi FEE. Thematic analysis is particularly appropriate for uncovering contextually rich insights and identifying patterns within complex social systems, such as entrepreneurial ecosystems (Braun & Clarke; Braun et al., 2016; Lochmiller, 2021; Majumdar, 2019). Drawing on established practices in ecosystem research (e.g., Mubarak et al., 2019; Yassin et al., 2024; Gonzalez & Dentchev, 2021; Roundy et al., 2018; Autio & Levie, 2017), the analysis combined inductive and deductive strategies. Preliminary themes were informed by existing literature on FEE domains, including finance, culture, education, policy, markets, and support systems. However, theme development remained grounded in the data to accommodate the emergence of contextually nuanced insights.

The analytical framework adopted in this study was guided by the six-phase process developed by Braun and Clarke (2006), a widely recognised and applied approach in qualitative

research. Each phase was systematically implemented to ensure transparency, reliability, and interpretive coherence. Table 3.10 outlines how these phases were operationalised in this study.

Table 3.11: Phases of Thematic Analysis (Braun & Clarke, 2006, p. 87)

Phase	Description	Application in the Study
1. Familiarisation	Immersing oneself in the data through repeated readings	Transcripts were reviewed and annotated to identify recurring concepts and expressions
2. Generating Codes	Systematic identification of meaningful textual elements	Open codes were manually identified in Excel, focusing on data related to the FEE domains
3. Searching for Themes	Grouping related codes into overarching patterns	Codes were clustered into six themes aligned with key FEE elements (e.g., finance, culture)
4. Reviewing Themes	Assessing coherence and consistency of themes	Themes were refined for internal consistency and relevance to the research objectives
5. Defining Themes	Clearly articulating and naming each theme	Themes such as “Support System Misfit” and “Cultural Resistance” were defined and contextualised
6. Producing the Report	Integrating themes into a coherent analytical narrative	Themes were illustrated in Chapter 5 using direct participant quotes to support interpretation

All interviews were audio-recorded, transcribed verbatim, and translated into English where necessary. Manual coding was conducted without using Microsoft Excel to manage, organise, and refine the codes. Initial open coding involved a line-by-line review of the transcripts to identify significant concepts. These were subsequently grouped into axial codes, and later consolidated into major themes through iterative comparison and reflection.

While the initial coding framework was informed by established literature on FEE domains specifically finance, culture, education, policy, markets, and support systems (e.g., Mubarak et al., 2019; Roundy et al., 2018; Autio & Levie, 2017; Gonzalez & Dentchev, 2021; Yassin et al., 2024) theme development remained grounded in the empirical data, allowing for contextually emergent insights.

A codebook was developed and refined iteratively throughout the process to ensure internal consistency and clarity in code application. A summary of the final code-to-theme mapping, consistent with the thematic structure reported in Chapter 5, is presented in Table 3.11.

Table 3.11: Code-to-Theme Mapping

Theme	Sub-theme	Sample Initial Codes
Finance	Accessibility of Financial Support	“Family funding”, “Social Development Bank support”, “Application delays”, “Risk aversion”
Culture and Norms	Social Perceptions of Fashion	“Fashion seen as low status”, “Tailoring is for foreigners”, “Fear of business failure”
Support Systems	Institutional Fit and Trust	“Tech-focused incubators”, “Irrelevant services”, “No fashion-specific mentors”
Education	Gaps in Practical Entrepreneurial Training	“Too theoretical”, “Lack of fashion business skills”, “Better overseas training”
Policy	Regulatory Environment	“Licensing barriers”, “Bureaucratic delays”, “Vision 2030 support”
Market	Customer Access and Trends	“Changing preferences”, “Limited distribution”, “Social media reach”

To enhance the credibility and dependability of the thematic findings, two validation strategies were employed:

Peer debriefing: Two academic colleagues with qualitative expertise independently reviewed a sample of the coded transcripts and corresponding themes. Their feedback informed minor revisions to theme definitions and boundaries, thereby strengthening analytical clarity.

Member checking: Six participants were invited to review the preliminary thematic framework and its interpretations. Their feedback confirmed that the themes accurately captured their lived experiences and viewpoints, contributing to the interpretive validity of the analysis.

This systematic and transparent approach to thematic analysis enhanced the trustworthiness, transferability, and confirmability of the findings (Lincoln & Guba, 1985). By clearly detailing how themes namely, Finance, Culture and Norms, Support Systems, Education, Policy, and Market were developed, validated, and integrated into the final report, this section addresses concerns regarding methodological clarity and reinforces the overall rigour of the qualitative inquiry.

3.8 Integration of Quantitative and Qualitative Data

To comprehensively analyse the data collected in this study, both quantitative and qualitative data were utilised to provide breadth and depth in understanding the Saudi FEE and its impact (Creswell & Clark, 2017). The integration of quantitative and qualitative data was achieved through a joint display table that aligned the numerical findings with the corresponding qualitative insights (James et al., 2024). This method facilitated the identification of consistencies and discrepancies, allowing for a nuanced interpretation of the

data. For example, while quantitative results indicate difficulties in accessing funding, qualitative data reveal specific bureaucratic hurdles and investor confidence issues faced by entrepreneurs, adding valuable context to the statistical patterns observed.

Additionally, the integrated findings were interpreted using Isenberg's and Stam's EE frameworks, selected for their comprehensive approach to integrating both the systemic and structural components of entrepreneurship ecosystems (Isenberg, 2010; Stam, 2015). By mapping the elements and domains of the Saudi FEE onto these frameworks, the analysis provides a deeper understanding of the ecosystem's current state and highlights potential areas for policy and structural improvement.

By integrating quantitative and qualitative data, this study effectively triangulated the findings, ensuring that the insights were both rigorous and contextually grounded. This approach allowed for a comprehensive understanding of the Saudi FEE, directly addressing the research objectives and providing evidence-based recommendations for enhancing the functionality and performance of the ecosystem.

3.9 Ethical Considerations

This research received ethical approval from the College Human Ethics Advisory Network (CHEAN) at the College of Design and Social Context (NHMRC Code: EC00237) for the ethical approval (see Appendix E, p 346). Recognising the importance of understanding ethical research fundamentals, particularly in direct engagement with participants, this study was guided by the insights of (Polonsky & Waller, 2018). Additionally, the research followed the ethical guidelines outlined by Pan (2020), which emphasise four key principles to guide every stage of research: straightforwardness, diligence, objectivity, and applicability. These principles ensured that all procedures, from data collection to result reporting, were conducted with the highest ethical standards.

Respect for participants' autonomy was paramount throughout the research process, aligning with the best practices outlined by (Diener & Crandall, 1978), which stress the importance of voluntary participation and informed consent in ethical research. Informed consent was obtained from all participants before data collection commenced, ensuring that they were fully aware of the study's purpose, their role, and their rights, including the option to withdraw without facing any negative consequences.

Confidentiality was strictly maintained throughout the study period. All identifiable information was anonymised and securely stored to protect participants' privacy, as

recommended by Corti et al. (2000). No personal or sensitive information was disclosed, and all data were processed using coded identifiers to ensure confidentiality and data integrity.

Although the research did not offer direct benefits to the participants, it is expected to contribute significantly to the further development of the Saudi FEE. The study was carefully designed to avoid sensitive questions that might cause discomfort, and the participants were informed of their right to refrain from answering any such questions. Ethical guidelines were followed rigorously, ensuring that the research prioritised participant welfare at every step. This approach reinforced the study's ethical foundation and supported its broader goals of contributing meaningful insights into environmental education practices by closely integrating these ethical practices into the research design. This study adhered to established ethical standards and proactively addressed potential ethical challenges, demonstrating a commitment to responsible and respectful research.

3.10 Chapter Summary

This chapter outlines the methodology used to investigate the Saudi FEE and its impact on the EO and BP of MSMEs. This study adopted a pragmatist approach, employing a sequential explanatory mixed methods design to provide a comprehensive analysis. The quantitative phase consisted of a survey conducted with 223 fashion entrepreneurs, followed by a qualitative phase involving semi-structured interviews with 15 stakeholders to further explore the survey findings. Sampling techniques included non-probability sampling for the quantitative phase, given challenges in accessing a complete sample frame, and judgment and snowball sampling to select key informants for the qualitative phase of the study.

This chapter details the development and refinement of research instruments, including piloting processes to enhance reliability and validity. PLS-SEM was used for quantitative data analysis, while thematic analysis was applied to interpret qualitative data. A joint display approach was used to integrate the findings, offering both statistical and contextual insights.

Ethical considerations were comprehensively addressed, ensuring participant confidentiality, informed consent, and adherence to RMIT University's ethics guidelines. This chapter also acknowledges the study's limitations and challenges, providing transparency and setting the groundwork for future research. In summary, this chapter demonstrates how selected methods and pragmatist philosophy align with the study's objectives, establishing a rigorous framework for exploring the Saudi FEE's influence on EO and BP. The following chapter presents quantitative results, detailing the findings from a survey conducted with 223 fashion entrepreneurs to examine the impact of the Saudi FEE on EO and BP.

CHAPTER 4: QUANTITATIVE RESULTS AND DISCUSSION

4.1 Chapter Overview

This chapter presents the data analysis of the quantitative study (Study 1), which aims to provide empirical insights into the proposed framework of the FEE and its impact on enhancing MSMEs' EO and BP. To achieve this aim, this study focused on the following objectives:

RO1: Examine the impact of FEE on the EO of fashion MSMEs.

RO2: Examine the direct impact of FEE on the BP of fashion MSMEs.

RO3: To investigate the mediating role of EO in the relationship between FEE and BP of fashion MSMEs.

The chapter is organised into three main sections. The first section is the pre-analysis process, which includes the response rate, data coding, construct specification, and data screening. The second section presents the descriptive statistics. The third section is devoted to structural equation modelling, which involves hypothesis testing.

4.2 Pre-Analysis Data Processing

4.2.1 Response Rate

Initially, 650 respondents accessed the survey. Nine participants refused to participate, and 192 respondents were excluded from the analysis because they did not have a fashion business in Saudi Arabia. In addition, 191 incomplete surveys were excluded from the analysis. A total of 224 participants completed the survey in full and were included in the analysis, representing a completion rate of 34% (see Table 4.1).

Table 4.1: Survey Completion Rate

Survey Completion Rate	<i>n</i>
Logged onto the survey	650
Logged into the survey and did not participate	35
Refused	9
Not eligible (do not have a business)	192
Questionnaires not completed	191
Questionnaires completed	224

4.2.2 Data Coding

This study used a survey as a data collection method, given the study's aims and objectives. The survey was designed and administered online using the Qualtrics platform. All categorical variables were coded as numeric values (e.g. male=1, female=2), multiple-choice questions were separated as binary variables (0 and 1), and all FEE and EO items had a 5-point Likert scale coded as 1 for strongly disagree and up to 5 for strongly agree. Similarly, BP variables were coded from 1 for "much worse than competitors" to 5 for "much better than competitors" (Table 3.6).

4.2.3 Data Screening

Data screening and preliminary analysis were conducted to explore and examine the interrelationships between the research variables (Tabachnick et al., 2013). Prior to conducting the multivariate data analysis, several checks were performed to address missing data, outliers, normality, and the presence of common method bias (Hair et al., 2010; Tabachnick et al., 2013). These screening procedures were performed using SPSS.

4.2.3.1 Missing Values

In this study, there were no missing data points. All survey questions were adjusted to ensure compulsory responses. Consequently, no completed responses (224) had missing data.

4.2.3.2 Assessment of Outliers

Methods for detecting outliers are frequently implemented by statisticians when analysing univariate data. A common method for detecting outliers in multivariate data analysis is the Mahalanobis distance (D^2), as recommended by Hair et al. (2010). The Mahalanobis measure calculates each observation's distance in multidimensional space from the mean centre of all observations, providing a single value for each observation regardless of the number of variables considered (Hair et al., 2010, p. 66). An observation was considered an outlier if the p-value was less than 0.001. Additionally, if the Chi-square value is less than the Mahalanobis value, it is flagged as an outlier. In this analysis, only one case with extreme responses to all questions was identified and subsequently removed from the dataset, resulting in a final sample size of 223 cases for further analysis.

4.2.3.3 The Common Method Bias (CMB)

An assessment of CMB was conducted, considering that this study used self-reported surveys for data collection (Podsakoff et al., 2003). Harman's single-factor test was applied by performing an unrotated principal component analysis (PCA) in IBM SPSS 28, including all

scale items that measure the variables in this study. The results indicated that multiple factors emerged, with the first factor accounting for less than 50% of the total variance. This outcome suggests that CMB is not a significant issue in the present study.

4.2.3.4 Normality Test

The normality test evaluates whether a dataset is properly modelled by a normal distribution and analyses how a random variable can be normally distributed within the dataset. This study employed the Kolmogorov-Smirnov and Shapiro-Wilk tests. A significance value greater than 0.05 in the Shapiro-Wilk test indicates that the data are normally distributed; conversely, a value below 0.05 suggests a significant deviation from normality (Shapiro & Wilk, 1965). Table 4.2 shows that all study variables were not normally distributed ($p < 0.05$). Based on these findings, non-parametric statistical techniques such as SEM-PLS were used.

Table 4.2: Normality test results

Constructs	Dimensions	Kolmogorov-Smirnov			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Fashion Entrepreneurship Ecosystem	Finance	0.105	223	0.000	0.963	223	0.000
	Policy	0.099	223	0.000	0.977	223	0.001
	Market	0.105	223	0.000	0.985	223	0.017
	Culture	0.118	223	0.000	0.945	223	0.000
	Social norms	0.119	223	0.000	0.940	223	0.000
	Entrepreneurship Education	0.149	223	0.000	0.953	223	0.000
	Talent	0.079	223	0.002	0.968	223	0.000
	Support	0.075	223	0.004	0.982	223	0.006
Entrepreneurial Orientation	Innovativeness	0.137	223	0.000	0.951	223	0.000
	Proactiveness	0.134	223	0.000	0.959	223	0.000
	Risk-Taking	0.150	223	0.000	0.941	223	0.000
Business Performance		0.077	223	0.003	0.974	223	0.000

4.2.4 Construct Specification

In this study, reflective constructs were used, as discussed in the previous chapter (see Table 18). The EE and EO constructs were multidimensional. Table 4.3 summarises the operationalisation details of the constructs used in this study.

Table 4.3: Operationalisation of Constructs

Variable Construct	Type of constructs	Code for Constructs	Code for indicators
The Independent Variable (Entrepreneurship Ecosystem)	Reflective- Reflective	EE	
Support	Reflective	S	S1, S2, S3, S4, S5, S6
Market	Reflective	M	M1, M2, M3, M4, M5, M6
Policy	Reflective	P	P1, P2, P3, P4, P5, P6
Entrepreneurial Education	Reflective	HE	HE1, HE2
Talent	Reflective	HT	HT1, HT2, HT3
Finance	Reflective	F	F1, F2, F3, F4, F5, F6
Culture	Reflective	C	C1, C2, C3, C4, C5
Social norms	Reflective	CN	CN1, CN2, CN3, CN4, CN5
The Mediating Variable	Reflective-Reflective	EO	EO
Innovativeness	Reflective	EOI	EOI1, EOI2, and EOI3
Proactiveness	Reflective	EOP	EOP1, EOP2, EOP3
Risk-taking	Reflective	EOR	EOR1, EOR2, EOR3
The Dependent Variable (Business Performance)		BP	BP1, BP2, BP3, BP4, BP5

4.3 Results

4.3.1 Demographic Profile

4.3.1.1 Saudi Fashion Entrepreneurs

Table 4.4 presents the demographic profiles of the participants. As shown in Table 4.4, the majority of the sample 91 % were female participants, and only 9% were male. The largest proportion of participants fell within the 31-40 age group, accounting for 40.8% of the sample, followed by 25-30, comprising 23.8% of participants, and the 41-50 age group at 19.3%. The lowest number of participants was found in the age groups of less than 25 and 51-60, comprising 11.7% and 4.5%, respectively.

Of the participants, 67.3% held a bachelor's degree, 16.6% had diplomas, 9.9% had completed a master's degree, 4.5% graduated from high school, and 1.8% possessed a Ph.D.

Of the respondents, 50.7% studied fashion, while only 4.5% majored in design, 17.5% in entrepreneurship, and 27.4% in other fields.

36.3 Of the participants, 36.3% had attended entrepreneurial or business courses, while 63.7% had not. Regarding previous experience, 22% of participants had less than 3 years' experience, and 14.3% had more than 10 years of experience. The majority of participants (48%) were owners/founders of their businesses, followed by 33.6% who were designers of their own brands. of the participants, 64.1% work for themselves, 17.5% have full-time employees, 11.7% are unemployed, 2.7% have part-time employees, and 4% are students.

Table 4.4: Demographic profile of participants

Variable	Indicator	Frequency (n=223)	Percent (%)
Gender	Male	20	9.0
	Female	203	91.0
Age group	< 25 less than	26	11.7
	25-30	53	23.8
	31-40	91	40.8
	41-50	43	19.3
	51-60	10	4.5
Education Level	High school and less	10	4.5
	Diploma	37	16.6
	Bachelor's Degree	150	67.3
	Master's degree	22	9.9
	PhD	4	1.8
Field of study	Fashion	113	50.7
	Design	10	4.5
	Entrepreneurship or Business Administration	39	17.5
	Other	61	27.4
Attended entrepreneurial or business course	Yes	81	36.3
	No	142	63.7
Previous years of experience	None	80	35.9
	Less than three years	49	22.0
	3-5 years	38	17.0
	6-10	24	10.8
	More than 10 years	32	14.3
Position in business	Designer of own brand	75	33.6
	Owner/Founder	107	48.0
	CEO (professional manager)	13	5.8
	All the above	25	11.2
	Other	3	1.3
Job situation	Full-time employee	39	17.5
	Part-time employee	6	2.7
	I work for myself	143	64.1
	Unemployed	26	11.7
	Student	9	4.0

4.3.1.2 Saudi Fashion SMMEs Profile

Approximately 43% of fashion enterprises are situated in the Central Province, which serves as the capital city. This location provides an extensive array of fashion and business support systems, including access to human capital and infrastructure. The next largest number of fashion businesses, 34.1%, is found in the Western Province, followed by 14.3% in the Eastern Province, 5.4% in the Southern Province, and the lowest number, 3.1%, in the Northern Province.

The results indicate that 44.4% of enterprises were established between 2017 and 2020, 30.5% between 2021 and 2022, and 25.1% between 1999 and 2016. In terms of legal structure, 56.8% of businesses operate as sole traders. For number of employees, 94.2% of businesses fall under the category of microcompanies with 1 to 5 employees. For distribution channels, 71.3% of participants use social media platforms (e.g., Facebook, Instagram). Most participants, 85.7%, are involved in the women's clothing category. The most common product types offered by participants are bridal or evening wear, representing 38.1% of the sample, followed by 35% for Abayas, which are floor-length, modest outer garments for women. Finally, for manufacturers, the highest percentage was 39% for "*made-to-measure*," followed by 36.3% who indicated they deal with tailors outside their establishment, and the lowest percentage, 15.2%, reported manufacturing at home (see Table 4.5).

Table 4.5: Fashion MSME Profiles

Variable	Indicator	Frequency (n=223)	Percentage
Business location	Northern Province	7	3.1
	Southern Province	12	5.4
	Central Province	96	43.0
	Eastern Province	32	14.3
	Western Province	76	34.1
Year established	1999–2016	56	25.1
	2017–2020	99	44.4
	2021–2022	68	30.5
Legal structure	Private Company	57	28.2
	Sole Trader	121	56.8
	Partnership	6	2.7
	Public company	4	1.8
	Family Trust	4	1.8
	Unregistered company	18	8.6
Number of employees	1–5 (micro)	210	94.2
	6–49 (small)	10	4.5
	50–249 (medium)	2	0.9
	250 and above	1	0.4

Table continues...

Table 4.5 Continued

Variable	Indicator	Frequency (n=223)	Percentage
Distribution channel	Social media (e.g., Facebook, Instagram)	159	71.3
	Company owned website	81	36.3
	Company-owned physical store or boutique	43	19.3
	Multibrand stores	39	17.5
	Online multibrand website (e.g., Amazon, Etsy)	8	3.6
Business category	Womenswear	191	85.7
	Menswear	28	12.6
	Childrenswear	37	16.6
	Accessories (footwear, bags, leather goods)	12	5.4
	Jewellery	15	6.7
	Fabric	18	8.1
Type of product	Activewear	10	4.5
	Ready-to-wear	66	29.6
	Casual wear	46	20.6
	Formal wear or Uniform	36	16.1
	Bridalwear or evening wear	85	38.1
	Abayas	78	35
	Traditional distribution	9	4.0
	Jewellery	6	2.7
	Other (please specify)	5	2.2
Manufacturers	Overseas clothing manufacturers	40	17.9
	Domestic clothing manufacturers	45	20.2
	Import and wholesale	11	4.9
	Made-to-measure	87	39
	Dealing with tailors outside your establishment	81	36.3
	Tailors work at your establishment	60	26.9
	At home	34	15.2

Note: Participants were allowed to select more than one option for this variable. Total responses may exceed the number of participants, and percentages may exceed 100%.

4.3.1.3 Motivations and funding resource of Saudi Fashion Entrepreneurs

Table 4.6 presents main motivations and obstacles reported by participants in starting their new businesses. Most participants, 43.9%, stated their main motivation to start a new business was an innovative new idea or product. Some 65.9% of participants indicated their main source of funding was personal savings. Participants reported difficult obstacles encountered in obtaining financial support. The highest percentage, 22%, nominated preparing

a feasibility study. This was followed by 21.9% who mentioned a lack of knowledge about financing options and financial institutions in Saudi Arabia. A small percentage of participants, 4.5%, reported encountering no obstacles in obtaining financial support.

Table 4.6: Start-up Characteristics

Variable	Indicator	Frequency (n=223)	Percentage
Main motivation to start a new business	Innovative new business idea or products	98	43.9
	Take advantage of a market opportunity	23	10.3
	Independence	31	13.9
	Increase income	37	16.6
	Escape unemployment	8	3.6
	To obtain prestige or to be socially recognised	14	6.3
	Passion for fashion	12	5.4
Main source of funding	Bank loan	23	10.3
	Savings	147	65.9
	Angel investor	6	2.7
	Family and/or friends	81	36.3
	Venture capital	1	.4
	Incubator	15	6.7
What obstacles did you face when accessing financial support?	The application procedures were complex	43	19.3
	Interest rates are high	32	14.3
	Collateral requirements were too high	39	17.5
	The size of the loan and the maturity were insufficient	22	9.9
	Difficulty preparing a feasibility study	51	22.9
	I cannot create relationships with prospective investors	43	19.3
	I am unaware of financing options and finance institutions in Saudi Arabia	47	21.1
	I do not have the capacity or skill to persuade an investor to invest in their project	25	11.2
	Did not apply	17	7.6
	None	10	4.5

4.3.1.4 Business Profitability

Table 4.7 shows the participants' business profitability. It outlines the annual sales in the last financial year, as reported by the participants. Most participants (74.9 %) reported annual sales of less than SAR 50,000. Furthermore, 19.3% reported sales between SAR 50,000 and 550,000, 3.6% reported sales between SAR 550,000 and 1 million, 0.4% reported sales between SAR 2 million and 3 million, and 1.8% reported sales of SAR 3 million or more. Regarding changes in annual sales compared to the previous year, 38.1% of the participants reported an increase in sales, and 25.1% reported a decrease. The remaining 36.8% reported that their sales remained steady. Among those whose sales increased, the highest percentage (34.1%) reported

an increase of between 0% and 5%. However, among those whose sales decreased, the highest percentage (28.6%) reported a decrease between -0% and -5%.

Table 4.7: Business Profitability

Variable	Indicator	Frequency (n=223)	Percent (%)
Last year annual sales	Less than SAR 50,000	167	74.9
	SAR 50,000–550,000	43	19.3
	SAR 550,00–1 million	8	3.6
	SAR 2 million–3 million	1	0.4
	SAR 3 million or more	4	1.8
Compared to last year, the annual sales in this year...	Increased	85	38.1
	Decreased	56	25.1
	Remained stable	82	36.8
Annual sales for this year have increased by... (n=85)	0 %–5%	29	34.1
	6%–10%	22	25.9
	11%–15%	14	16.5
	16%–20%	9	10.6
	More than 20%	11	12.9
Annual sales for this year have decreased by... (n=56)	(-0 %)–(-5%)	16	28.6
	(-6%)–(-10%)	14	25.0
	(-11%)–(-15%)	7	12.5
	(-16%)–(-20%)	4	7.1
	More than -20%	15	26.8

4.3.2 Descriptive Analysis

The descriptive analysis presented in Table 4.8 offers a comprehensive overview of the key variables: FEE, EO, and BP. This analysis included essential metrics such as the minimum, maximum, and mean values and standard deviations. Each variable was measured using a 5-point Likert scale, with values ranging from 1 (strongly disagree) to 5 (strongly agree), providing standardised comparisons across dimensions. The overall mean score for participants' responses about FEE registered at the medium level with a mean score of 2.96 (± 0.610). Within FEE, culture scored highest at 3.58 (± 0.698), while other components such as support (2.71, ± 0.866) and policy (2.98, ± 0.927) remained moderate. For EO, the mean score was 3.72 (± 0.673), reflecting a high level of entrepreneurial orientation. The dimensions of innovativeness, proactivity, and risk-taking exhibit similar values, indicating consistency across EO measures. BP had a mean score of 3.24 (± 1.006), suggesting moderate performance levels among SMEs.

Table 4.8: Descriptive Analysis for FEE, EO, BP

Constructs	Mean	Std. Deviation	Degree
Fashion Entrepreneurship ecosystem	2.960	.6103	Neutral
Finance	2.845	.7675	Neutral
Policy	2.984	.9274	Neutral
Market	2.926	.5962	Neutral
Supportive Culture	3.529	.8066	Agree
Social norms	3.583	.6980	Agree
EE	2.876	1.0296	Neutral
Talent	2.600	.9410	Neutral
Support	2.709	.8662	Neutral
Entrepreneurial Orientation	3.72	0.673	Agree
Innovativeness	3.74	0.788	Agree
Proactivity	3.67	0.786	Agree
Risk Taking	3.76	0.731	Agree
Business Performance	3.24	1.006	Medium

4.4 Assessment of the Measurement Model

The initial stage evaluates reliability and validity of the Measurement Model, which includes both lower-order constructs (LOC) and higher-order constructs (HOC). This study employs the repeated indicators approach to first assess reliability and validity of reflective LOCs before advancing to evaluation of reflective HOCs (Sarstedt et al., 2021). This approach involves examining factor loadings, composite reliability (CR), average variance extracted (AVE), and HTMT ratios to confirm discriminant validity, as outlined in Table 4.9.

Table 4.9: Measurement model analysis criteria

Objective	Measurement parameter	Criteria	Sources
Indicator Reliability	Factorial loading (PLS-SEM)	Factor Loading > 0.7	(Chin, 1998; Hair et al., 2020; Hair et al., 2023)
Internal Consistency Reliability	Composite Reliability (CR)	CR > 0.7	(Bagozzi & Yi, 1988; Sarstedt et al., 2021)
Convergent Validity	Average Variance Extracted (AVE)	AVE > 0.5	(Bagozzi & Yi, 1988; Sarstedt et al., 2021)
Discriminant Validity	Fornell-Larcker criterion, HTMT ratio	HTMT < 0.90, Cross-loadings	(Fornell & Larcker, 1981; Henseler et al., 2015)

4.4.1 Assessment of Lower Order Constructs (LOC)

4.4.1.1 Indicator Loadings

In line with standards outlined by Hair et al. (2019), indicator loadings in SEM should exceed 0.700 to ensure adequate construct representation. In this study, five indicators were removed due to insufficient factor loadings (less than 0.700). Specifically, one indicator from the “social norms” dimension (CN2), one from the “finance” dimension (F2), and three from the “market” dimension (M4, M5, M6) were excluded. Consequently, the refined model comprises 48 items, all with loadings above the 0.700 threshold. The highest loading observed was 0.937, and the lowest was 0.736 (see Figure 4.1).

4.4.1.2 Internal consistency reliability

The internal consistency reliability of each construct was assessed according to established cut-off points, using Cronbach’s alpha and composite reliability values as indicators (Hair et al., 2020; Hair et al., 2023). Both measures met the minimum threshold of 0.70 for all constructs, with Cronbach’s alpha values ranging from 0.752 to 0.912, confirming the adequacy of internal consistency for this reflective measurement model (See Table 4.10). Additionally, both composite reliability indicators (rho_a and rho_c) consistently exceeded the 0.70 threshold without approaching the upper boundary of 0.95, thereby ensuring that there was no redundancy among items within each construct (Hair et al., 2019; Sarstedt et al., 2021).

Table 4.10: Internal consistency reliability (Cronbach’s Alpha and Composite Reliability)

	Cronbach’s alpha	Composite reliability (rho_a)	Composite reliability (rho_C)
Support	0.882	0.885	0.911
Market	0.813	0.815	0.889
Policy	0.912	0.913	0.932
Entrepreneurial Education	0.845	0.849	0.928
Talent	0.767	0.780	0.865
Finances	0.852	0.854	0.894
Culture	0.851	0.856	0.894
Social Norms	0.807	0.818	0.873
Innovativeness	0.752	0.752	0.859
Proactiveness	0.786	0.787	0.875
Risk-taking	0.797	0.802	0.880
Business Performance	0.862	0.863	0.900

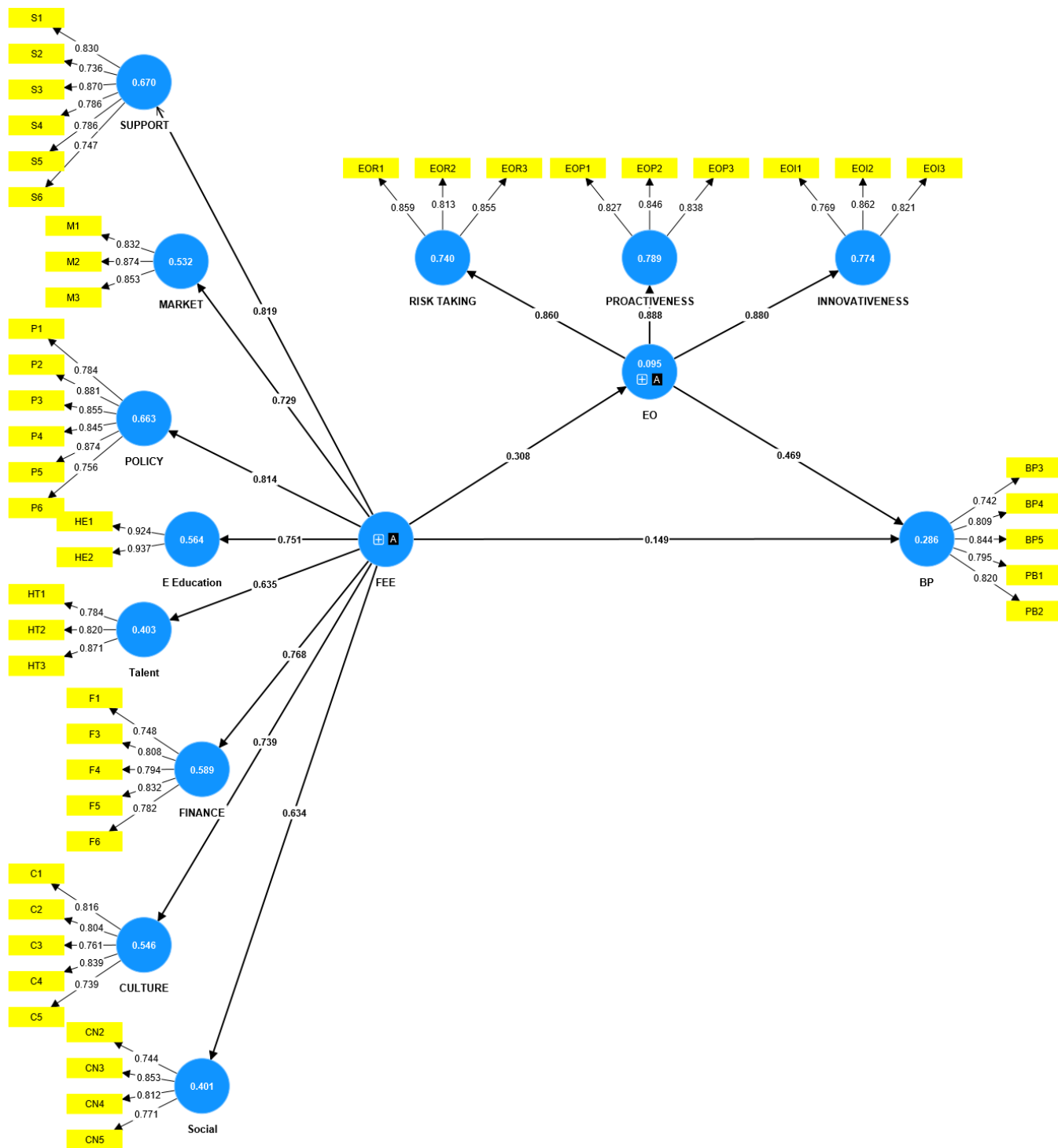


Figure 4.1: Conceptual model PLS-SEM, with 48 items

4.4.1.3 Indicator Reliability

Factor loadings indicate how well a particular item represents the underlying construct. The factor loadings of all constructs are all above the required value of 0.70 (Hair et al., 2017). Hence, all indicators are maintained for all constructs (see Table 4.11).

Table 4.11: Factor loading (The Saudi FEE, EO and BP)

Fashion Entrepreneurship Ecosystem	Loadings
Support	
S1 <- S	0.830
S2 <- S	0.736
S3 <- S	0.870
S4 <- S	0.786
S5 <- S	0.786
S6 <- S	0.747
Market	Loadings
M1 <- M	0.832
M2 <- M	0.874
M3 <- M	0.853
Policy (P)	Loadings
P1 <- P	0.784
P2 <- P	0.881
P3 <- P	0.855
P4 <- P	0.845
P5 <- P	0.874
P6 <- P	0.756
Entrepreneurial Education	Loadings
HE1 <- HE	0.924
HE2 <- HE	0.937
Talent (HT)	
HT1 <- HT	0.784
HT2 <- HT	0.820
HT3 <- HT	0.871
Finances	Loadings
F1 <- F	0.748
F3 <- F	0.808
F4 <- F	0.794
F5 <- F	0.832
F6 <- F	0.782

Table continues...

Table 4.11 Continued

Culture	Loadings
C1 <- C	0.816
C2 <- C	0.804
C3 <- C	0.761
C4 <- C	0.839
C5 <- C	0.739
Social Norms	Loadings
CN2 <- CN	0.744
CN3 <- CN	0.853
CN4 <- CN	0.812
CN5 <- CN	0.771
Entrepreneurial Orientation	Loadings
Innovativeness (EOI)	
EOI1 <- EOI	0.769
EOI2 <- EOI	0.862
EOI3 <- EOI	0.821
Proactiveness (EOP)	
EOP1 <- EOP	0.827
EOP2 <- EOP	0.846
EOP3 <- EOP	0.838
Risk-taking (EOR)	Loadings
EOR1 <- EOR	0.859
EOR2 <- EOR	0.813
EOR3 <- EOR	0.855
Business Performance	Loadings
PB1 <- PB	0.795
PB2 <- PB	0.820
BP3 <- PB	0.742
BP4 <- PB	0.809
BP5 <- PB	0.844

4.4.1.4 Convergent Validity

To establish convergent validity for a factor, Hair et al. (2014) recommended using the Average Variance Extracted (AVE) as a secondary cutoff, with a threshold of 0.50 or above. This threshold indicates that a factor's item loadings account for at least 50% of the variance, surpassing the variance attributed to measurement error for the construct. The results show that AVE values ranged from 0.628 to 0.865 (see Table 4.12).

Table 4.12: AVE (Measure of Construct Convergent Validity)

	Average variance extracted (AVE)
Support	0.630
Market	0.728
Policy	0.695
Entrepreneurial Education	0.865
Talent	0.682
Finance	0.629
Culture	0.628
Social Norms	0.634
Innovativeness	0.670
Proactiveness	0.701
Risk-taking	0.710
BP	0.644

4.4.1.5 Discriminant Validity

Assessing discriminant validity confirmed that each construct was statistically distinct from the others, ensuring that they were not overly correlated. This evaluation employed both the Fornell-Larcker criterion and the heterotrait-monotrait (HTMT) ratio. The Fornell-Larcker criterion establishes discriminant validity when the square root of each construct's AVE is greater than its correlations with other constructs (Fornell & Larcker, 1981). As shown in Table 4.13, the square root of the AVE for each construct (on the diagonal) exceeds any off-diagonal correlation values, supporting construct uniqueness. For example, the AVE for the support construct was 0.794, with correlation values between 0.135 and 0.629, all below the AVE, meeting the Fornell-Larcker threshold for distinctiveness.

Additional evidence of discriminant validity was provided through the HTMT ratio, where values below 0.90 indicate adequate construct separation (Henseler et al., 2015). Table 4.14 shows HTMT values ranging from 0.143 to 0.895, reinforcing the distinctiveness of each construct. Furthermore, the findings across both the Fornell-Larcker and HTMT criteria were validated using the SmartPLS bootstrapping algorithm with 10,000 subsamples. None of the confidence intervals included the value of 1, indicating strong statistical reliability and reinforcing the discriminant validity of the constructs within the measurement model.

Table 4.13: Discriminant Validity using the Fornell-Larcker Criterion

	Support	Market	Policy	Entrepreneurial Education	Talent	Finances	Culture	Social Norms	Innovativeness	Proactiveness	Risk-taking	BP
Support	0.794											
Market	0.574	0.853										
Policy	0.576	0.479	0.834									
Entrepreneurial Education	0.556	0.525	0.577	0.930								
Talent	0.629	0.492	0.400	0.463	0.826							
Finances	0.530	0.436	0.688	0.516	0.337	0.793						
Culture	0.485	0.542	0.454	0.505	0.378	0.461	0.793					
Social Norms	0.376	0.425	0.381	0.488	0.256	0.407	0.607	0.796				
Innovativeness	0.135	0.172	0.180	0.144	0.145	0.160	0.22	0.223	0.818			
Proactiveness	0.253	0.193	0.213	0.133	0.252	0.165	0.231	0.167	0.695	0.837		
Risk-taking	0.228	0.249	0.187	0.242	0.236	0.191	0.264	0.280	0.626	0.634	0.843	
BP	0.284	0.279	0.196	0.216	0.313	0.256	0.116	0.115	0.486	0.442	0.427	0.803

Table 4.14: Heterotrait ratio of correlations

	Support	Market	Policy	Entrepreneurial Education	Talent	Finances	Culture	Social Norms	Innovativeness	Proactiveness	Risk-taking	BP
Support												
Market	0.679											
Policy	0.639	0.557										
Entrepreneurial Education	0.641	0.630	0.656									
Talent	0.757	0.616	0.474	0.572								
Finance	0.612	0.523	0.774	0.606	0.419							
Culture	0.560	0.651	0.512	0.592	0.461	0.536						
Social Norms	0.437	0.518	0.439	0.581	0.313	0.482	0.730					
Innovativeness	0.170	0.218	0.218	0.180	0.203	0.2	0.274	0.290				
Proactiveness	0.304	0.244	0.252	0.183	0.320	0.202	0.281	0.212	0.895			
Risk-taking	0.272	0.308	0.215	0.295	0.301	0.228	0.317	0.349	0.803	0.796		
BP	0.328	0.334	0.222	0.250	0.385	0.299	0.143	0.143	0.597	0.534	0.506	

4.4.1.6 Cross-loadings

Cross-loadings measure how well an item is associated with its intended construct compared to other constructs. High cross-loadings indicate that an item is strongly associated with its intended construct rather than with other constructs, supporting discriminant validity. In this study, the cross-loadings of all items were examined to confirm that they loaded significantly on their respective constructs without high loadings on any other constructs, as recommended by Chin (1998) and Hair et al. (2014). The results, shown in Table 4.15, demonstrate that each item loaded highly on its respective construct, with no significant cross-loadings on other constructs. This finding supports the adequacy of discriminant validity, indicating that each construct measures a unique concept, ensuring that the study's results are both accurate and interpretable (see Table 4.15).

Table 4.15: Discriminant validity – Cross-loadings

	Support	Market	Policy	Entrepreneurial Education	Talent	Finances	Culture	Social norms	Innovativeness	Proactiveness	Risk-taking	BP
S1	0.830	0.468	0.498	0.520	0.629	0.426	0.373	0.349	0.130	0.240	0.190	0.247
S2	0.736	0.490	0.356	0.428	0.539	0.422	0.395	0.267	0.083	0.187	0.170	0.297
S3	0.870	0.460	0.484	0.510	0.552	0.414	0.402	0.319	0.139	0.235	0.166	0.214
S4	0.786	0.455	0.399	0.348	0.415	0.383	0.371	0.242	0.102	0.262	0.232	0.232
S5	0.786	0.447	0.522	0.389	0.432	0.471	0.353	0.299	0.058	0.137	0.151	0.189
S6	0.747	0.415	0.471	0.439	0.410	0.409	0.417	0.307	0.129	0.142	0.182	0.175
M1	0.465	0.832	0.365	0.381	0.336	0.337	0.515	0.399	0.099	0.108	0.169	0.197
M2	0.513	0.874	0.426	0.512	0.472	0.365	0.442	0.356	0.143	0.158	0.196	0.233
M3	0.489	0.853	0.434	0.447	0.445	0.414	0.434	0.335	0.196	0.226	0.271	0.283
P1	0.472	0.398	0.784	0.472	0.351	0.660	0.337	0.280	0.113	0.158	0.134	0.178
P2	0.540	0.399	0.881	0.462	0.313	0.652	0.393	0.342	0.215	0.256	0.222	0.176
P3	0.482	0.389	0.855	0.492	0.353	0.603	0.391	0.364	0.124	0.152	0.131	0.115
P4	0.411	0.344	0.845	0.450	0.288	0.539	0.382	0.263	0.173	0.177	0.147	0.140
P5	0.485	0.403	0.874	0.520	0.343	0.542	0.398	0.340	0.131	0.178	0.180	0.203
P6	0.482	0.466	0.756	0.488	0.353	0.432	0.366	0.310	0.145	0.141	0.113	0.170
HE1	0.500	0.462	0.502	0.924	0.401	0.469	0.443	0.420	0.119	0.086	0.194	0.195
HE2	0.534	0.514	0.569	0.937	0.459	0.490	0.494	0.485	0.147	0.159	0.255	0.206
HT1	0.430	0.313	0.304	0.405	0.784	0.286	0.275	0.179	0.010	0.155	0.159	0.183
HT2	0.554	0.449	0.279	0.307	0.820	0.239	0.273	0.186	0.163	0.208	0.211	0.295
HT3	0.566	0.448	0.397	0.433	0.871	0.307	0.377	0.261	0.173	0.253	0.212	0.289
F1	0.397	0.392	0.436	0.360	0.299	0.748	0.326	0.331	0.073	0.097	0.130	0.157
F3	0.416	0.335	0.686	0.434	0.213	0.808	0.413	0.374	0.133	0.104	0.120	0.165
F4	0.483	0.446	0.502	0.412	0.232	0.794	0.414	0.285	0.202	0.204	0.241	0.295

Table continues...

Table 4.15 Continued

	Support	Market	Policy	Entrepreneurial Education	Talent	Finances	Culture	Social norms	Innovativeness	Proactiveness	Risk-taking	BP
F5	0.392	0.283	0.542	0.394	0.257	0.832	0.333	0.351	0.151	0.163	0.152	0.211
F6	0.413	0.273	0.544	0.440	0.344	0.782	0.337	0.271	0.069	0.085	0.111	0.187
C1	0.398	0.399	0.365	0.400	0.265	0.361	0.816	0.451	0.151	0.178	0.167	0.066
C2	0.368	0.460	0.353	0.395	0.318	0.343	0.804	0.510	0.181	0.212	0.238	0.034
C3	0.378	0.443	0.293	0.372	0.324	0.360	0.761	0.419	0.148	0.124	0.199	0.161
C4	0.428	0.483	0.440	0.449	0.332	0.448	0.839	0.509	0.210	0.216	0.266	0.127
C5	0.343	0.355	0.335	0.379	0.252	0.304	0.739	0.519	0.177	0.178	0.168	0.071
CN2	0.191	0.250	0.261	0.321	0.116	0.243	0.448	0.744	0.179	0.131	0.205	-0.002
CN3	0.307	0.401	0.303	0.384	0.197	0.359	0.534	0.853	0.152	0.169	0.224	0.087
CN4	0.339	0.363	0.363	0.486	0.266	0.390	0.510	0.812	0.158	0.094	0.220	0.088
CN5	0.343	0.320	0.275	0.342	0.218	0.282	0.433	0.771	0.231	0.142	0.243	0.182
EOI1	0.198	0.130	0.214	0.154	0.135	0.181	0.304	0.281	0.769	0.545	0.577	0.349
EOI2	0.072	0.145	0.119	0.133	0.133	0.122	0.129	0.167	0.862	0.556	0.489	0.457
EOI3	0.062	0.146	0.110	0.066	0.088	0.090	0.107	0.099	0.821	0.603	0.469	0.387
EOP1	0.213	0.219	0.202	0.133	0.192	0.182	0.229	0.200	0.602	0.827	0.463	0.357
EOP2	0.272	0.218	0.198	0.181	0.264	0.145	0.182	0.107	0.556	0.846	0.561	0.392
EOP3	0.151	0.053	0.137	0.022	0.176	0.090	0.170	0.114	0.587	0.838	0.565	0.360
EOR1	0.228	0.254	0.153	0.247	0.223	0.144	0.215	0.182	0.535	0.571	0.859	0.363
EOR2	0.151	0.194	0.108	0.211	0.201	0.112	0.206	0.241	0.435	0.467	0.813	0.352
EOR3	0.193	0.182	0.204	0.158	0.174	0.220	0.246	0.285	0.601	0.559	0.855	0.364
PB1	0.179	0.189	0.126	0.095	0.234	0.207	0.044	0.015	0.343	0.344	0.275	0.795
PB2	0.310	0.296	0.202	0.206	0.334	0.242	0.114	0.038	0.337	0.339	0.271	0.820
BP3	0.148	0.163	0.117	0.143	0.145	0.197	0.068	0.130	0.461	0.384	0.429	0.742
BP4	0.247	0.250	0.145	0.202	0.213	0.170	0.158	0.169	0.400	0.320	0.397	0.809

4.4.2 *Assessment of Higher Order Construct (HOC)*

The second stage of the analysis assessed the measurement model for the reliability and validity of higher-order components, specifically FEE and EO (see Figure 18). Following established procedures for HOC evaluation, latent variable scores (LVS) for lower-order components were obtained using SmartPLS (Becker et al., 2012). These LVS served as indicators of reflective higher-order constructs. Support, market, policy, entrepreneurial education, talent, finance, culture, and social norms constitute the HOC FEE. Innovativeness, proactiveness, and risk-taking form the HOC of EO. Figure 4.2 illustrates the SEM path model, depicting these relationships within a higher-order framework.

The reliability and validity of the HOCs were verified using Cronbach's alpha and composite reliability, as shown in Table 4.16. Both constructs (FEE and EO) achieved Cronbach's alpha and composite reliability values exceeding the threshold of 0.70, with AVE values above 0.50, thus meeting the guidelines for internal consistency and convergent validity (Hair et al., 2023).

The Fornell-Larcker criterion was also employed to confirm discriminant validity, as shown in Table 4.17. The square root of each construct's AVE (displayed on the diagonal) was greater than the inter-construct correlations (off-diagonal values), confirming that FEE and EO are distinct and measure separate conceptual domains (Fornell & Larcker, 1981). These findings indicate that the higher-order constructs possess adequate reliability, convergent validity, and discriminant validity, supporting the validity of the measurement model. Table 4.18 shows that the HTMT values extracted for the HOC constructs in the model in the two-stage approach are less than the threshold 0.90. Therefore, discriminant validity has been achieved using the HTMT procedure (see Table 4.18).

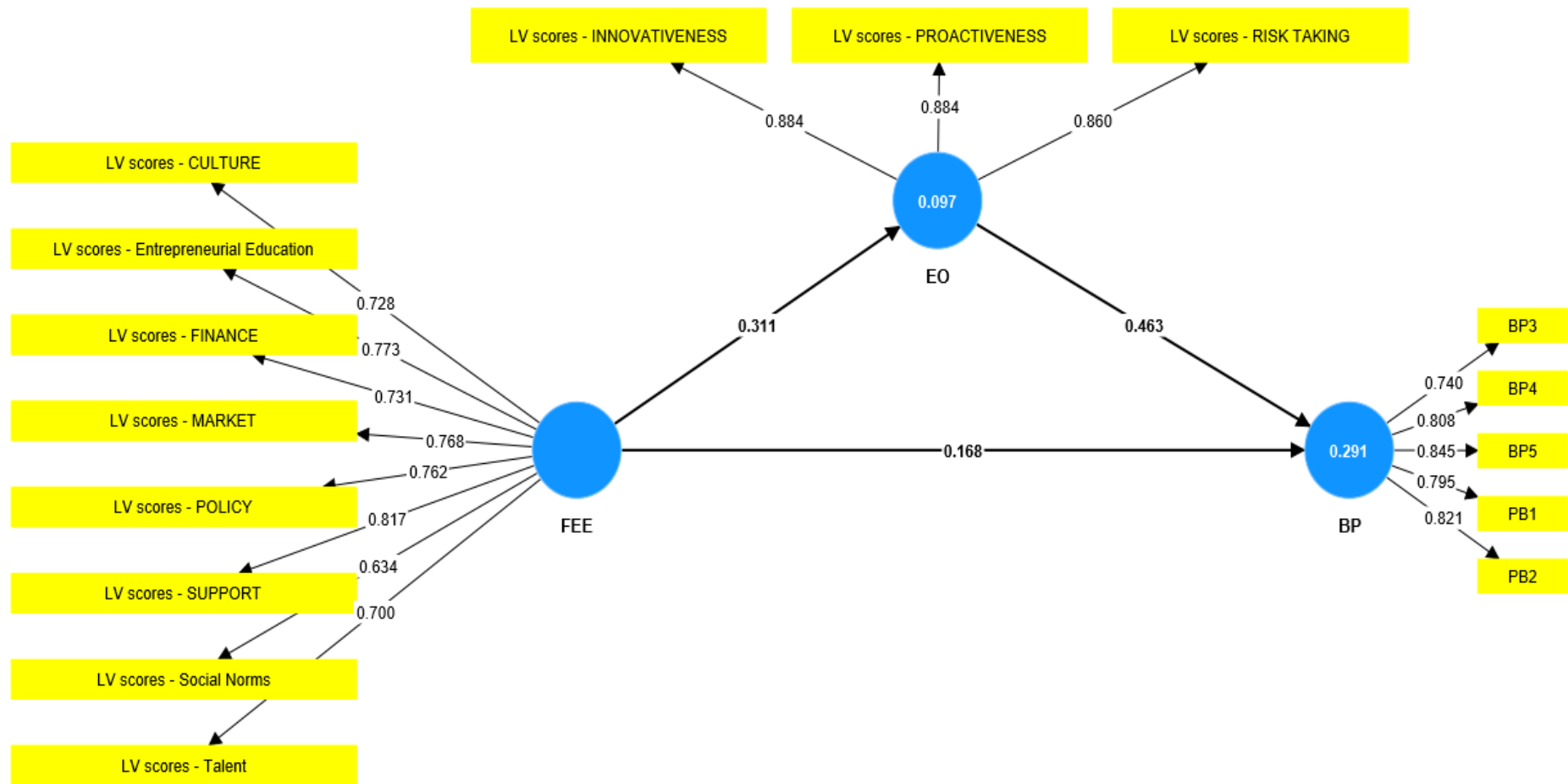


Figure 4.2: The two-stage approach for the higher-order reflective-reflective constructs

Table 4.16: Higher-order construct reliability and convergent validity

	Cronbach's alpha	Composite reliability (rho a)	Composite reliability (rho c)	Average variance extracted (AVE)
EO	0.849	0.849	0.908	0.768
FE	0.882	0.888	0.906	0.549
E				

Table 4.17: Higher-order construct discriminant validity (Fornell and Larker criterion)

	BP	EO	FEE
BP	0.803		
EO	0.515	0.876	
FEE	0.312	0.311	0.741

Table 4.18: Higher-order construct discriminant validity (HTMT)

	BP	EO	FEE
BP			
EO	0.596		
FEE	0.343	0.361	

Cross loadings for HCM using LVs are presented in Table 4.19. All items of the two constructs (FEE and EO) loaded higher than others (>0.70) on their respective constructs. Therefore, discriminant validity of HOC constructs is achieved using cross-loadings.

Table 4.19: Higher-order construct discriminant validity (cross-loads)

	FEE	EO
Support	0.817	0.234
Market	0.768	0.234
Policy	0.762	0.221
Entrepreneurial Education	0.773	0.198
Talent	0.700	0.241
Finances	0.731	0.196
Culture	0.728	0.272
Social norms	0.634	0.255
Innovativeness	0.228	0.884
Proactiveness	0.275	0.884
Risk-taking	0.315	0.860

4.5 Assessment of the Structural Model

After confirming reliability and validity of the measurement model, the next step involved constructing a structural model to test research hypotheses, and to evaluate the model's predictive capabilities. Following methodological guidelines of Hair et al. (2023), this evaluation included several critical stages: assessing collinearity, determining significance of structural relationships, evaluating coefficient of determination (R^2), and examining predictive relevance (Q^2). This systematic approach facilitated in-depth analysis of both direct and indirect effects within the hypothesised model, as summarised in Table 4.20. The resulting insights contribute to a robust understanding of the model's overall predictive strength and its capacity to capture complex interrelationships among variables.

Table 4.20: Structural model analysis criteria

Objective	Measurement parameter	Criteria	Sources
Collinearity Assessment	Variance Inflation Factor (VIF)	$VIF < 5$	(Hair et al., 2023; Hair et al., 2012; Ringle et al., 2020)
Path Coefficients	Path Coefficient Significance	$p\text{-value} < 0.05$	(Hair et al., 2023; Hair et al., 2012)
Coefficient of Determination	R-square (R^2)	Weak: 0.04-0.16, Moderate: 0.25-0.49	(Falk, 1992; Cohen, 2013; Hair et al., 2023)
Effect Size (f^2)	Cohen's f-square	0.02 weak effect 0.15 moderate effect 0.35 strong effect	(Cohen, 2013; Hair et al., 2023)
Predictive Relevance (Q^2)	Blindfolding Q^2	$Q^2 > 0$	(Cohen, 2013; Hair et al., 2023)

In this study, the hypothesised structural model positions the entrepreneurship ecosystem as the independent variable, fashion business performance as the dependent variable, and entrepreneurial orientation as the mediating variable. The assessment includes both first order and second-order constructs, as detailed in Table 4.21, and adheres to the guidelines of Becker, (Becker et al., 2012) regarding mode of measurement.

Table 4.21:Details on latent variables and mode of measurement for PLS structural model

First-order construct	Latent variable	Second-order construct	Mode of measurement
Support	S	FEE (Fashion Entrepreneurship Ecosystem)	Mode A
Market	M		
Policy	P		
Entrepreneurial Education	HE		
Talent	HT		
Finance	F		
Culture	C		
Social Norms	CN	EO	Mode A
Innovativeness	EOI		
Proactiveness	EOP		
Risk-taking	EOR		
Business Performance	BP		Mode A

4.5.1 Collinearity Assessment

Multicollinearity is a condition in which two or more independent variables in a regression model are highly correlated. This can lead to wider confidence intervals and imprecise coefficient estimates, complicating the reliability of significance testing and the interpretation of variable effects. Therefore, the detection and management of multicollinearity are essential for accurate regression analysis.

This study used the Variance Inflation Factor (VIF) to detect multicollinearity. According to Henseler et al. (2015), VIF values of 5 or below indicate minimal collinearity concerns, while Vittinghoff et al. (2012) suggest values up to 10 are acceptable, indicating no severe multicollinearity issues. Following these guidelines, (Sarstedt et al., 2021) and (Hair et al., 2023) also confirm that values below these thresholds support confidence in interpreting predictor effects without necessitating adjustments, such as higher-order constructs.

Applying these thresholds, all VIF values for the predictive constructs of EE and EO were below the threshold of 5, specifically ranging from 1.344 to 3.971 (see Table 4.22). This result confirms there are no multicollinearity issues in the model, reinforcing its stability and reliability in estimating structural relationships.

Table 4.22: VIF Variance Inflation Factor

	Items	VIF
Support	S1	3.572
	S2	2.541
	S3	3.485
	S4	2.219
	S5	2.604
	S6	1.812
Entrepreneurial education	HE1	2.150
	HE2	3.408
Talent	HET1	1.759
	HET2	1.594
	HET3	1.719
Finances	F1	1.885
	F3	3.069
	F4	2.395
	F5	2.212
	F6	1.840
Culture	C1	2.356
	C2	1.941
	C3	1.876
	C4	2.657
	C5	1.955
Social norms	CN2	1.842
	CN3	1.999
	CN4	1.619
	CN5	1.816
Market	M1	1.695
	M2	1.939
	M3	2.129
Policy	P1	2.969
	P2	3.971
	P3	2.994
	P4	3.029
	P5	3.577
	P6	2.492
Innovativeness	EOI1	1.344
	EOI2	1.932
	EOI3	1.652

Table continues...

Table 4.22 Continued

	Items	VIF
Proactiveness	EOP1	1.919
	EOP2	1.988
	EOP3	1.626
Risk-taking	EOR1	1.773
	EOR2	1.617
	EOR3	2.132
Business Performance	BP1	2.534
	BP2	2.980
	BP3	1.762
	BP4	2.130
	BP5	2.492

4.5.2 Path coefficients (hypothesis testing)

The path coefficient results, as shown in Table 4.23 and Figure 4.3, confirm the significance of each hypothesised relationship. Hypothesis H1 evaluates impact of FEE on EO. The results indicate a significant positive effect of FEE on EO ($\beta = 0.308$, $t = 4.145$, $p < 0.001$), thereby supporting H1. Hypothesis H2 assesses influence of FEE on BP, revealing a significant positive effect ($\beta = 0.149$, $t = 2.113$, $p < 0.05$), thus supporting H2. Hypothesis H3 examines the impact of EO on BP, with analysis confirming a strong positive effect ($\beta = 0.469$, $t = 8.078$, $p < 0.001$), supporting H3. Collectively, these results indicate EO's intermediary role between FEE and BP, highlighting its substantial influence in the model. The detailed results, including path coefficients, t -statistics, and p -values, are outlined in Table 4.23.

Table 4.23: Path Coefficients, t -statistics, and p -values for direct effects of relationships

	Coefficient	t -statistics	p -values	Decision
H1: FEE -> EO	0.308	4.145	0.000	Significant
H2: FEE -> BP	0.149	2.113	0.035	Significant
H3: EO -> BP	0.469	8.078	0.000	Significant

4.5.3 Mediation analysis

Hypothesis H4 examines the mediating role of EO in the relationship between FEE and BP. To test this hypothesis, (Hayes, 2009, 2017) recommended the bootstrapping approach to assess mediating effects by repeatedly resampling the data and estimating the indirect effect. This method further validated by (Dastgeer et al., 2020) and (Nitzl, Roldan, et al., 2016). Bootstrapping enables calculation of confidence intervals for the mediating effect, a capability not offered by the traditional (Baron & Kenny, 1986) approach, which lacks confidence intervals and is limited to single mediators. Studies highlight that bootstrapping provides more reliable results for mediation analysis by producing confidence intervals and reducing bias compared to traditional methods (Fossum & Montoya, 2021). The bootstrapping results for this analysis indicate the total effect of FEE on BP is significant ($\beta = -0.294$, $t = 4.677$, $p < 0.001$). However, when EO is included as a mediator, the direct effect of FEE on BP becomes weakly significant ($\beta = 0.149$, $t = 2.113$, $p < 0.05$), and the indirect effect remains significant ($\beta = 0.145$, $t = 3.576$, $p < 0.001$). This suggests EO partially mediates the FEE-BP relationship. A summary of total and indirect effects is presented in Table 4.24.

Table 4.24: Total and indirect effects

	Coefficient	<i>t</i>-statistics	<i>p</i>-values	Decision
Direct effect (FEE -> BP)	0.149	2.113	0.035	Supported
Indirect effect (FEE -> EO -> BP)	0.145	3.576	0.000	Supported
Total effect (FEE -> BP)	0.294	4.677	0.000	Supported

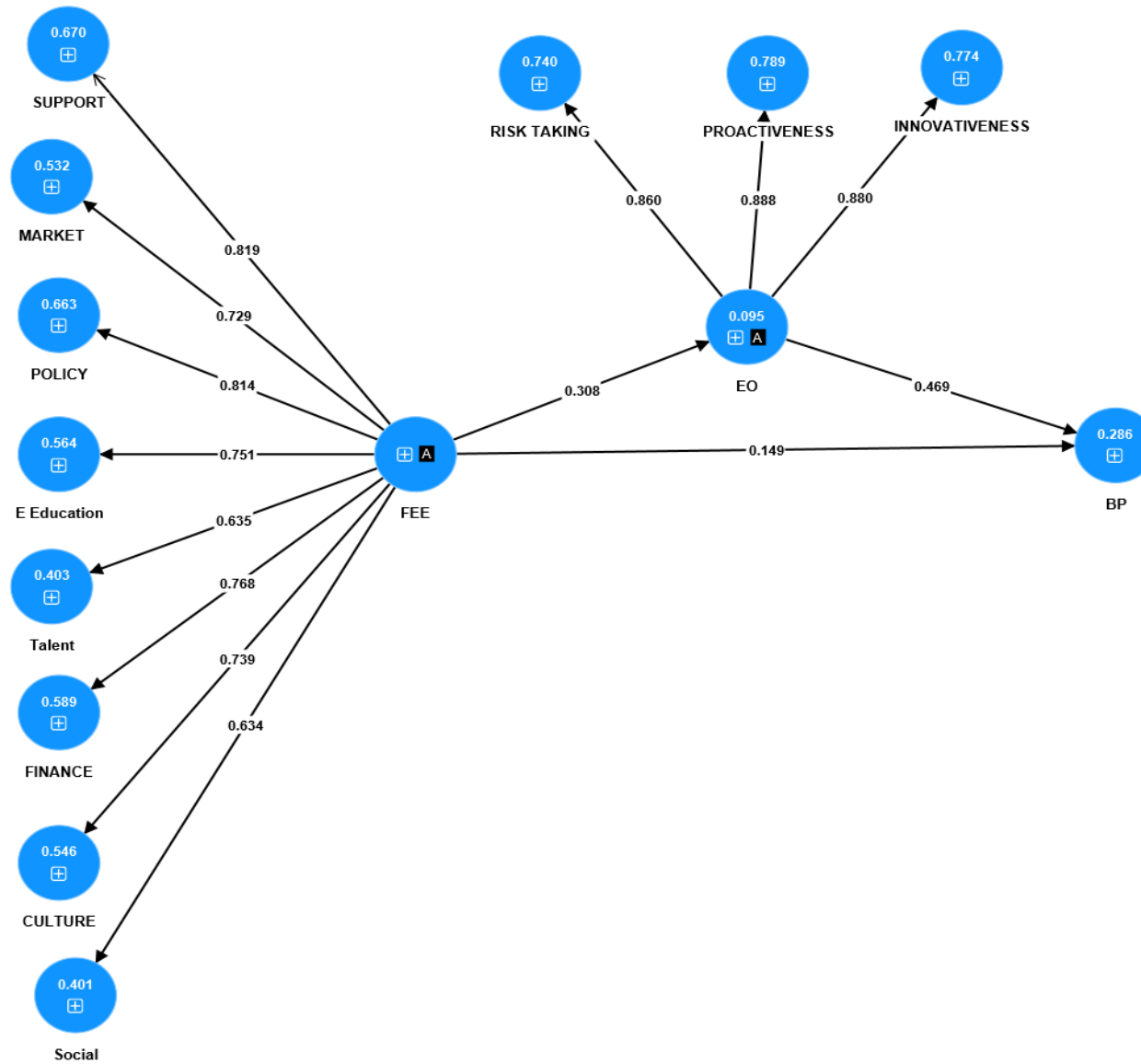


Figure 4.3: Hypothesised path coefficients of FEE, EO and BP

4.5.4 R-square of endogenous latent variables

The R-square values for endogenous latent variables are considered moderately weak when ranging between 0.04 and 0.16, and moderately strong when between 0.25 and 0.49 (Ozili, 2023; Ritchey, 2002). Additionally, these values comply with the threshold proposed by Falk (1992), which states R-square should be above 0.10. Based on these benchmarks, results of PLS-SEM analysis in this study reveal a weak R-square value for the mediator variable EO and a moderate R-square value for the dependent variable BP (refer to Table 4.25).

Table 4.25: Coefficient of determination R square (Variance Explained)

	R-square	R-square adjusted
BP	0.286	0.279
EO	0.095	0.091

4.5.5 Effect Size (f^2)

In SEM, endogenous variables may be influenced by one or more exogenous variables. Removing an exogenous variable can alter the dependent variable's outcome, depending on the model's specific structure. Effect size (f^2) measures this impact by observing change in the R-square value when an exogenous variable is removed. This metric provides insight into the extent of influence each exogenous variable exerts on the dependent variable (Hair et al., 2014; Hair et al., 2023). Cohen (2013) established the following guidelines for interpreting f-square values: values of ≥ 0.02 indicate a weak effect, ≥ 0.15 a moderate effect, and ≥ 0.35 a strong effect. In this study, the calculated effect size (f^2) is 0.279, representing a moderate effect size (see Table 4.26).

Table 4.26: Effect size of constructs (f^2)

	BP
BP	
EO	0.279

4.5.6 Predictive Relevance Q^2

Predictive relevance (Q^2) is an essential metric in SEM-PLS for evaluating a model's predictive validity, particularly in complex models. Calculated as the squared correlation between predicted and actual values, Q^2 helps assess predictive accuracy, complementing R-

square values (Geisser, 1974; Stone, 1974). The blindfolding technique, which divides data into training and test sets, is employed to calculate Q^2 (Akter et al., 2011; Becker et al., 2013). This approach omits blocks of data and then uses estimated parameters to predict omitted values. According to Cha (1994), a Q^2 value above zero indicates the model has predictive relevance. In this study, Q^2 for BP is 0.176, suggesting acceptable predictive relevance (see Table 4.27).

Table 4.27: Predictive Relevance

	SSO	SSE	$Q^2 (=1-SSE/SSO)$
BP	1115	918.565	0.176
EO	2007	1915.741	0.045

4.6 Discussion

This study investigated the influence of the FEE on fashion MSMEs' EO and BP in Saudi Arabia. Focusing on direct and indirect effects, this study contributes to the limited empirical literature on this subject, particularly in Saudi Arabia. As mentioned, this study examines the collective impact of various domains within FEE, including culture, social norms, market policies, finance, support, entrepreneurial education, and talent, on entrepreneurial orientation. EO encompasses innovativeness, proactiveness, and risk taking.

4.6.1 H1: The Fashion Entrepreneurship Ecosystem positively affects the Entrepreneurial Orientation of fashion MSMEs in Saudi Arabia.

The first hypothesis, that FEE positively affects EO in Saudi fashion MSMEs, was supported. The analysis revealed that FEE had a significant positive impact on EO ($\beta = 0.308$, $p < 0.001$), suggesting that a supportive entrepreneurial environment promotes key EO dimensions such as innovativeness, proactiveness, and risk-taking. This finding aligns with existing literature highlighting how entrepreneurial ecosystems foster EO development across varying national contexts (Acs et al., 2018; Olutuase et al., 2018; Wurth et al., 2023). For instance, Spiegel (2017) conducted case studies in Canada showing that regional ecosystems' cultural, social, and material attributes significantly influence entrepreneurial orientation and resource mobilisation. Similarly, Mai and Nguyen (2023) found that ecosystem factors such as market access, education, and support systems had statistically significant impacts on EO

among Vietnamese entrepreneurs, with effect sizes ranging from 0.2 to 0.4 depending on the domain.

Despite the statistical significance observed in this study, the explanatory power was relatively modest, with an R^2 value of just 0.095. This contrasts with findings from more mature ecosystems. For example, Stam and Van De Ven (2021) reported that the entrepreneurship ecosystem in the Netherlands accounted for 52.6% of the variance in entrepreneurial outcomes, with components like finance and business services showing strong correlations with high-growth entrepreneurship (often $r > 0.6$).

Several contextual factors may explain this difference. First, Saudi Arabia's entrepreneurial ecosystem particularly within the fashion sector is still developing and lacks the institutional maturity observed in advanced economies (Almuzel & Anderson, 2021; Dixit et al., 2024). Second, creative sectors such as fashion rely on unique inputs like artistic talent, design networks, and niche incubators, which may not yet be well established. These findings echo those of Nicotra et al. (2018), who underscored the importance of ecosystem maturity and structural integration for enhancing entrepreneurial outcomes.

Moreover, Stam and Van De Ven (2021) emphasised that the compounded effect of ecosystem elements such as leadership, networks, and finance is essential for achieving systemic impact. In the Saudi context, such synergies may still be underdeveloped, potentially limiting the strength of FEE's influence on EO. This study's limited R^2 is consistent with meta-analytic research showing that ecosystem factors typically account for no more than 20% of entrepreneurial variance (Queissner et al., 2024). This suggests that EO is influenced by a wider range of factors beyond ecosystem support alone. Previous studies have identified individual entrepreneurial intent (Wiklund & Shepherd, 2005), organisational learning capabilities (Autio et al., 2014), and sector-specific disruptions (Brazinskaite & Durand, 2014) as key contributors to EO. Additionally, Pugh et al. (2021) argue that in dynamic and trend-driven sectors like fashion, the manifestation and success of EO are highly dependent on contextual and industry-specific variables. Therefore, while the FEE offers essential foundational support, it represents only one of several interdependent layers influencing EO. Future models should incorporate complementary variables such as psychological traits, learning capabilities, and environmental dynamism to improve explanatory power and contextual accuracy.

These conclusions are reinforced by qualitative findings from Chapter 5, where participants highlighted persistent structural gaps within the ecosystem. These included fragmented policies, inadequate fashion-specific training, and limited access to relevant support institutions. Such issues underscore the need to refine the FEE construct to better

capture latent institutional inefficiencies and the unique requirements of the fashion sector, such as specialised incubators and mentorship tailored to creative entrepreneurship.

4.6.2 H2: The Fashion Entrepreneurship Ecosystem positively affects the Business Performance of fashion MSMEs in Saudi Arabia.

The second hypothesis was supported, indicating that the FEE has a statistically significant positive influence on the BP of Saudi fashion MSMEs ($\beta = 0.149$, $p < 0.05$). This aligns with prior research demonstrating how EE contribute to business outcomes by improving access to resources, fostering innovation, and enabling firm growth (Isenberg, 2011; Acs et al., 2014; Ali et al., 2019). Similarly, Mai and Nguyen (2022) found that five out of six ecosystem components positively influenced startup success in Vietnam, with ecosystem support playing a critical role in entrepreneurial performance.

Although the β value of 0.149 reflects a relatively modest effect size, it is still meaningful, especially in the context of an emerging sector. Creative industries such as fashion often operate under different dynamics than more conventional sectors, relying on intangible assets, aesthetic value, and trend responsiveness. These characteristics may partly explain the smaller effect sizes observed. Compared to studies in developed economies, where ecosystems are more mature and integrated, the FEE's influence in the Saudi context may be less pronounced (Lux et al., 2020; Mai & Nguyen, 2022).

The moderate impact of the FEE on BP warrants a closer examination of the individual ecosystem components. Descriptive analysis revealed that several domains scored below the midpoint on a 5-point scale namely Talent ($M = 2.60$), Finance ($M = 2.84$), and Support ($M = 2.71$) indicating underdevelopment in these critical areas. This is consistent with findings by Salimi (2021), who identified human capital and financial access as major bottlenecks for entrepreneurs in developing economies.

Qualitative data from Chapter 5 reinforce these concerns. Participants reported various structural and practical barriers to securing finance, including high collateral demands, limited awareness of available funding mechanisms, and difficulty in preparing feasibility studies. These challenges reflect broader systemic issues commonly observed in developing contexts such as Ethiopia and Pakistan (Shkabatur et al., 2021; Ullah, 2019).

These ecosystem limitations are often rooted in fragmented policy execution, poor institutional coordination, and weak links between education providers and entrepreneurial needs. In the context of creative sectors, generic or untargeted government support may further

dilute the effectiveness of the ecosystem. Research has shown that policy frameworks not tailored to sector-specific needs particularly in cultural and creative industries are less likely to translate into tangible business outcomes (Sipakoly, 2024; Tekic & Kurnosova, 2024).

Therefore, future research should move beyond aggregate ecosystem measures and adopt domain-level structural equation modelling to test which FEE elements exert the strongest or weakest influence on business performance. This would allow for more precise policy and strategic recommendations. From a practical standpoint, ecosystem reforms should focus on strengthening vocational pathways for creative talent and expanding financial literacy and accessibility initiatives for first-time and early-stage entrepreneurs.

4.6.3 H3: Entrepreneurial Orientation has a positive effect on the Business Performance of fashion MSMEs in Saudi Arabia.

Hypothesis 3 (H3) posits that EO has a positive effect on BP in fashion MSMEs in Saudi Arabia. The results support this hypothesis, revealing a substantial positive impact of EO on BP, quantified by a beta coefficient (β) of 0.469. This finding aligns with the existing literature that underscores the positive influence of entrepreneurial orientation on business performance across various sectors. For instance, studies have consistently shown that dimensions of EO, including innovativeness, proactiveness, and risk-taking, contribute significantly to enhancing a firm's competitive advantages and overall performance (Covin & Slevin, 1989; Lumpkin & Dess, 1996). Particularly in the fashion industry, as seen in studies such as those by Brazinskaite and Durand (2014) and Mulyana and Sutapa (2016), EO significantly contributes to success and innovative capacity of fashion businesses. This is echoed by the findings of Putra and Gunawan (2019) and DiVito and Bohnsack (2017), who link EO to business performance and sustainable decision-making within the fashion sector. The consistency of these findings across different countries from Sweden and Indonesia to the Netherlands highlights the universal applicability of EO's benefits in the fashion industry.

However, the variations in the strength and nature of the EO-BP relationship observed in this study compared to others, such as Hossain and Azmi (2021) and Kharisma et al. (2020), suggest that the impact of EO on BP can be influenced by contextual factors. These variations include different responses across EO dimensions and the mediation effects of strategic flexibility, highlighting the nuanced and context-dependent nature of EO's influence on business performance. Furthermore, contrasting findings in other studies, such as those of Khalili et al. (2013) and Idawati and Sumartini (2020), which reported non-significant or even negative effects of EO on BP, prompt a re-evaluation of the universally positive view of EO.

These studies emphasise the potential moderating roles of external conditions, such as market dynamics or industry-specific challenges, which may either inhibit or enhance EO's effectiveness of EO.

Moreover, the magnitude of the beta coefficient in this study suggests a robust relationship between EO and BP, which is slightly higher than that typically reported in the broader entrepreneurship literature. This may indicate unique aspects of the Saudi fashion sector, such as market dynamics, cultural influences, or specific challenges and opportunities, that might amplify the impact of EO on business outcomes. The literature's emphasis on EO as a multidimensional construct, as discussed by Aloulou (2019) and Tajeddini and Mueller (2019), further complicates this narrative. This suggests that the individual dimensions of EO may have varying impacts on BP depending on specific internal and external factors. This calls for a more granular analysis of how the different dimensions of EO interact with contextual variables to influence BP.

4.6.4 H4: Entrepreneurial Orientation mediates the relationship between the Fashion Entrepreneurship Ecosystem and Business Performance of fashion MSMEs in Saudi Arabia.

The results revealed that the EO mediated the relationship between FEE and BP. While FEE provides essential resources, EO is critical for transforming these resources into tangible improvements. Researchers have emphasised the importance of distinguishing between partial and full mediation to accurately interpret the role of a mediator between independent and dependent variables (Aguinis et al., 2017; Carrion et al., 2017). This study indicates that EO partially mediates the FEE–BP relationship, suggesting that while FEE independently contributes to BP, the strength and effectiveness of this contribution are significantly enhanced by the EO. These results suggest that the EO enhances the impact of certain aspects of the FEE on BP but does not influence the effects of all ecosystem factors. Previous studies have demonstrated that EO mediates the relationship between external factors and BP in various contexts. Burt (2003) identified EO as a mediator between business networks and BP, while Bedi et al. (2019) found a similar relationship with the institutional environment. Cohen (2006) reported that EO strengthens the effect of infrastructural support on BP, and Mulyadi et al. (2021) highlighted the mediating role of EO between educational support and BP. Similarly, Hendou (2019) identified EO's mediating influence of EO between government policies and BP, while Wei et al. (2023) demonstrated EO's role in enhancing the relationship between ecosystem support and resources and BP.

Furthermore, evidence supports EO's partial mediating role between human capital and BP. Miao et al. (2017) and Poon et al. (2006) identified EO as a partial mediator in this relationship, indicating that while EO facilitates the conversion of ecosystem resources into performance outcomes, other variables also influence this dynamic. Wahyuni and Sara (2020) similarly reported partial mediation by EO, reinforcing the idea that EO enhances but does not entirely account for the effect of ecosystem resources on BP. Conversely, Zainol & Wan Daud (2011) found no mediating effect of EO between cultural backgrounds and BP, highlighting the variability in the impact of EO in different contexts. Recent research by Salehe et al. (2024) demonstrated that while EO enhances the impact of entrepreneurial practices on performance, other contextual factors, such as market competition and policy support, also contribute to business success.

The context-dependent nature of EO influence has been well established. Studies have confirmed that EO's impact of EO is strengthened in dynamic environments (Bedi et al., 2023; Indriyani et al., 2023). This reinforces the notion that the effectiveness of the EO as a mediator is influenced by environmental conditions and the broader business landscape.

The mediating role of EO is consistent with the RBV, which posits that a firm's competitive advantage stems from its ability to strategically acquire, develop, and deploy valuable, rare, inimitable, and non-substitutable resources (Barney, 1991). EO is conceptualised as a strategic intangible resource that enables firms to leverage external ecosystem support to achieve superior business performance. Ferreira and Azevedo (2017) and Huang et al. (2010). confirmed that EO is a key resource that drives the growth of small firms, reinforcing the positive integration of EO and RBV. Jiang et al. (2018) found that EO enhances firm performance by facilitating the acquisition and integration of network resources. Kiyabo and Isaga (2020) further supported this by demonstrating that EO enhances competitive advantage through innovation and capability development. Siddiqui et al. (2024). Similarly, the EO's role in transforming external resources into actionable strategies is underscored, confirming that while FEE provides critical resources such as infrastructure and financial incentives, EO is necessary for converting these into effective performance outcomes. This is consistent with the RBV's assertion that the strategic value of resources depends largely on their contextual adaptability (Eisenhardt & Martin, 2000). Additionally, Chew and Idris (2022) and Crick and Crick (2023) demonstrated that the impact of EO is moderated by contextual factors like environmental dynamism and cultural influences, highlighting its contextual variability.

4.7 Limitations of this study

The findings of this study offer valuable insights into the effects of FEE on EO and BP in Saudi Arabia. However, this study had two limitations. First, this study primarily used subjective self-report measures to assess EO and BP, including metrics such as customer satisfaction. Although subjective measures are commonly employed in entrepreneurship research, they may introduce biases and fail to fully capture the multidimensional nature of these constructs (Jennings & Young, 1990). Research suggests that the dichotomy between subjective and objective measures is outdated because even seemingly objective measures are influenced by context and personal bias (Schachter, 2010). Reliance on subjective data risks reduced precision in capturing the true performance and orientation of businesses (Vij & Bedi, 2016). While subjective measures are sometimes necessary when objective metrics are difficult to obtain, they can impact the reliability and validity of findings (Schachter, 2010). Observations from Wiklund and Shepherd (2005) suggest that the type of performance measures employed can influence the outcomes of studies. Therefore, the limitations associated with subjective measures in assessing FEE, EO, and BP underscore the need to consider the potential biases introduced by such measures and strive for a more comprehensive understanding of these constructs (Hansemark, 1997).

In addition, the quantitative tool employed in this study, the survey, is validated and reliable; however, it has inherent limitations. Surveys are crucial for quantitatively assessing trends and correlations in domains such as FEE, EO, and BP; however, they are restricted in their ability to fully assess the complex and qualitative dimensions of entrepreneurship (Jabeen & Mahmood, 2014). Therefore, it is important to undertake qualitative studies employing mixed methods in subsequent research which could provide a richer, more nuanced understanding of the FEE, leveraging qualitative data to complement and deepen insights gained from quantitative surveys (Zoyza et al., 2011). Therefore, Chapter 5 addresses these limitations by incorporating a qualitative study to understand how the Saudi FEE influences EO and BP.

4.8 Chapter Summary

This chapter delves into the quantitative analysis of the Saudi FEE and its impact on the EO and BP of fashion MSMEs in Saudi Arabia. Data were gathered from 223 fashion business owners through an online survey and analysed using structural equation modelling. The study results indicate that FEE has a significant positive impact on EO and BP. It was also found that EO partially mediated the relationship between FEE and BP. However, these findings highlight the complexity of these relationships, suggesting the involvement of other variables and dynamics not fully captured by quantitative measures. Therefore, a qualitative study is necessary to gain a deeper understanding of these issues. The following chapter presents a qualitative study, which involves interviews with 15 actors and aims to provide valuable insights into the current landscape of the Saudi FEE and how each domain of the Saudi FEE impacts EO and BP.

CHAPTER 5: QUALITATIVE FINDINGS AND DISCUSSION

5.1 Chapter overview

As identified in Chapter 1, Study 2 aims to create a framework that explains the impact of Saudi FEE on EO and BP of fashion MSMEs in Saudi Arabia. This chapter presents a thematic analysis of qualitative data gathered from semi structured interviews, supported by survey findings from the quantitative study to enrich and contextualise the results. To achieve this aim, the following research questions were posed:

RQ4: What is the current landscape of Saudi FEE?

RQ5: How does each domain of the Saudi FEE impact EO and BP?

5.2 Thematic Findings

The thematic analysis highlighted key themes for elements in the entrepreneurship ecosystem, including Policy, Entrepreneurial Education, Finance, Market, Support, Human Capital, Culture, and Social Norms. In addition, the analysis investigated EO and BP. Analysis of the interviews revealed several themes regarding stakeholders' and fashion entrepreneurs' perceptions of the impact of the fashion entrepreneurial ecosystem on their businesses' entrepreneurial orientation and performance. These themes were organised and classified according to Six elements of the Saudi FEE: finance, policies, market, culture and social norms, support, and entrepreneurial education. Each theme provided relevant challenges and needs within these domains and enabled an examination of how they affected EO and BP of fashion MSMEs in Saudi Arabia.

5.3 Finance

This theme explores the financial landscape of Saudi FEE. The interviews identified three key themes: (1) Availability of Financial Support, (2) Accessibility of Financial Support, and (3) Impact of Financial Support on EO.

5.3.1 Availability of Financial Support

The interviews revealed that Saudi fashion entrepreneurs rely on informal funding sources despite the availability of formal funding options. Six of the nine entrepreneur participants utilised personal savings as their primary funding source. For example, D1 stated,

“I began this journey with around eight thousand riyals from my savings”. D9 said, *“I had to borrow from my family to cover the start-up expenses”*. This finding supports the survey results, as 66% of the participants depended on personal savings, 36% on family contributions, and only 10% on bank loans for financing their businesses. Moreover, these findings also align with studies indicating that entrepreneurs in emerging markets often depend on personal and familial resources due to systemic gaps in formal financial mechanisms (Mefful et al., 2023; Omri, 2020). In Saudi Arabia, small startups primarily rely on personal savings, family loans, credit cards, and home equity because of strict bank lending conditions and limited access to venture capital (Binzomah, 2016). Although Vision 2030 aims to improve financial literacy and increase private sector involvement (Almahdi, 2020; Alsaiani, 2022), most start-ups are dependent on personal and family contributions (Aljuhani & Balubaid, 2020). Reliance on informal funding sources may raise questions regarding the effectiveness and accessibility of formal financial support. For instance, Herrera (2020) highlights that stringent requirements for formal funding frequently push fashion entrepreneurs towards informal channels. Similarly, studies on Indian MSMEs (Athaide & Pradhan, 2020) and German firms (Regasa & Abraham, 2022) confirm that firms facing credit constraints frequently rely on family and friends for funding.

In contrast, formal funding channels, including venture capital and incubator support, were underutilised. Notably, none of the participants pursued funding through crowdfunding, competitive grants, angel investments, or private equity avenues. For example, D4 combined support from institutions such as the Majid Society and the Development Bank but noted that the funds were insufficient for comprehensive project development. D5 primarily relied on self-financing but received additional support from Dim Al-Manahil, a business incubator. Similarly, D8 used a mix of Social Development Bank funding and personal savings. S2 stated, *“Financial support is available, as I have been contacted by Majid Society designer Reem Al Faisal and the Development Bank, which provides various funding programs. However, there are ongoing discussions about how accessible and effective incubators and commissions truly are in supporting entrepreneurs”*. The survey supported these findings, showing that only 4% of participants accessed venture capital and 6% utilised incubator support.

The Social Development Bank was mentioned as an accessible loan program. S4 stated, *“The Social Development Bank is the primary source of financial support for designers. It offers accessible loans with favourable terms, such as a grace period before repayment begins”*. S1 added, *“This bank is an excellent option for MSMEs. It provides interest-free financing with minimal requirements, but the need for a government-employed guarantor can*

pose a barrier for some". Similarly, S5 said, *"Designers favour the Social Development Bank because of its hassle-free process. It offers loans of up to 120,000 Saudi Riyals with no administrative fees and a five-year repayment period"*.

These results show that despite the availability of financial support, fashion entrepreneurs relied on financing from family and friends or the Social Development Bank for easy access to financial support. These findings align with those of existing studies, confirming that fashion designers often rely on personal savings or family support because of difficulties in accessing formal financing (Brydges & Pugh, 2021; Tervilä, 2015). Research highlights several obstacles designers face when seeking funding, including limited access to bank loans, stringent credit conditions, and a lack of investor interest in the creative industries (Heim, 2018; Pinto-Gutierrez et al., 2023; Sági & Juhász, 2019). Similarly, Sarkar and Karim (2019) point to the lack of alignment between the offerings of financial institutions and the specific needs of the fashion sector. Alsaiani (2022) called for streamlined processes and enhanced transparency in government funding programs in Saudi Arabia. These challenges reinforce the need for alternative financing models and tailored support programs for fashion entrepreneurs in the region.

5.3.2 Accessibility of Financial Support

The interview findings revealed that fashion entrepreneurs encounter four key challenges in accessing financial support: procedural complexities, personal, cultural, and institutional biases, and industry-specific barriers. First, the participants identified procedural complexities as significant barriers to accessing formal funding. This highlights the rigorous and sometimes prohibitive standards that financial institutions impose on borrowers, which can include detailed business plans and the necessity of a government-employed guarantor. For example, D6 observed, *"Securing a bank loan requires a lot of papers, such as the submission of a comprehensive feasibility study"*. Similarly, D2 said, *I tried to get financial help from an incubator, but it was hard because of the strict rules, such as needing collateral and conducting feasibility studies, which also took a lot of time"*. This aligns with the findings of Akinyede (2023), who noted that strict application processes hinder access to financing for MSMEs. Additionally, D1 pointed out the necessity of a government-employed guarantor: *"Obtaining loans necessitates a government-employed guarantor, which poses a substantial obstacle for many"*. This aligns with the findings of studies that examined the impact of the Kafalah program on SMEs' access to finance and business development, demonstrating its role in enhancing financial accessibility (Altokhais, 2018). Similarly, Abdulghaffar and Akkad (2021)

and (Alsolaim, 2019) found that Saudi MSMEs encounter barriers such as limited funding access and unclear application procedures. These procedural challenges are not unique to the fashion industry but are prevalent across various sectors, as highlighted by the SME Finance Forum (2025), which advocates for streamlined processes and digital solutions to improve access to finance. Research consistently underscores the financial challenges faced by fashion designers in securing funding for their businesses. Studies conducted in Nigeria (Nwaozuru et al., 2024), Ghana (Mefful et al., 2023), and Finland Aakko and Niinimäki (2018) have revealed that fashion designers struggle to access finance and meet collateral requirements and complex loan procedures. Aakko and Niinimäki (2018) also highlighted the lack of investors in the fashion industry, further limiting growth opportunities. To address these challenges, scholars recommend tailored financial management training, the development of networking platforms, and simplification of loan processes for fashion entrepreneurs (Mefful et al., 2023; Nwaozuru et al., 2024). These measures aim to improve financial accessibility and sustainability in the fashion industry, fostering growth and innovation.

Second, personal challenges, such as limited financial literacy and a lack of awareness of available financial support, significantly affect fashion entrepreneurs. These personal barriers often stem from a background focused more on design and arts rather than financial management, leading to difficulties in managing finances, attracting investors, and understanding financial support mechanisms. Participants highlighted that Saudi fashion designers often lack financial literacy, a critical competency for effectively managing finances and attracting investors (Akinyede, 2023; Polisetty et al., 2021). For instance, participant D7 remarked, *“As fashion designers, we often come from a design and arts background, struggling with numerical concepts”*. Similarly, participant D8 stated, *“When I started my brand, I had no idea how to manage cash flow or budget for production costs”*. The interviews revealed that fashion designers struggle to establish relationships with potential investors or convince them to invest in their enterprises. As revealed by D3, *“Pitching my business idea to investors was tough. I have tried twice, and they were more interested in long-term goals and financial plans than my creative designs”*.

These findings are consistent with studies conducted in various countries, which indicate that fashion designers worldwide face similar difficulties in financial management and literacy. For example, research in Indonesia has demonstrated that financial literacy and personality traits significantly influence financial management behaviour among fashion entrepreneurs (Azib, 2021; Parastika & Hidayati, 2023). Similar challenges have been identified in Nigeria (Nwaozuru et al., 2024) and in Malaysia (Kamarulzaman & Shaari, 2024). Previous studies

suggest that financial knowledge is essential for conducting feasibility studies and effectively presenting projects. A lack of financial literacy can impede access to external funding and reduce the likelihood of securing investments (Akinyede, 2023; Polisetty et al., 2021). Similarly, Alshebami and Al Marri (2022) advocated for initiatives to enhance financial literacy and encourage effective financial behaviour.

Furthermore, the interviews indicated a general lack of awareness among fashion designers of the available financial resources. Participant S4 noted, *“Many financial support options exist, but entrepreneurs are often not aware of them, leading them to rely on personal savings or even discouraging them from starting a business”*. This finding aligns with Alsolaim (2019), who emphasised the pressing need for improved dissemination of information regarding financial support within entrepreneurial ecosystems in Saudi Arabia. Fashion entrepreneurs are often hesitant to seek partnerships for financial support because of concerns regarding design theft. For example, participant D9 stated, *“Investors want quick returns, but fashion is about building a brand with longevity. I have turned down funding opportunities because they came with too many conditions that did not align with my creative vision”*. Many fashion designers prefer to retain full brand ownership to safeguard their names and reputations. Aardt and Wyk (2011) found that numerous fashion entrepreneurs prioritise creativity, self-reliance, and risk tolerance over external financial partnerships, emphasising the need for control over their brands’ direction. Similarly, Zhao et al. (2018) highlighted that while financial support is a critical factor for success in the fashion industry, concerns over ownership and creative control often drive designers to seek alternative financing methods. However, other participants, such as D2, S3, and D5, perceived partnerships as preferable alternatives to loans. As S3 expressed, *“I favour partnerships to share financial risks and secure funding, which is vital for entrepreneurial success”*. This underscores the importance of collaboration in achieving financial stability and business growth (Kariv & Coleman, 2015).

Third, female entrepreneurs face additional challenges because of cultural and institutional biases. As noted by participant D9, *“Securing loans or investments that necessitate a guarantor or collateral proved difficult as a female entrepreneur”*. This finding aligns with existing research, which highlights the financial barriers that disproportionately affect women in business. Research by Sabri (2021) and Ali et al. (2019) supports this claim, indicating that women are at a disadvantage because of stringent guarantor requirements and limited networking opportunities. A study conducted by Iqbal (2020) in Jeddah, Riyadh, and Khobar found that financial challenges are a primary obstacle for Saudi women entrepreneurs, significantly hindering their business growth. Similarly, Elnadi and Gheith (2021) suggest that

while financial support programs exist, many female entrepreneurs in Saudi Arabia remain unaware of these resources, and bureaucratic obstacles continue to impede access. The authors recommend implementing more gender-inclusive policies to ensure equitable opportunities for female entrepreneurs. Furthermore, although the Saudi Chamber of Commerce has made efforts to develop female-focused business programs, participation rates remain low. This is largely due to persistent cultural barriers and a lack of financial literacy among women (Sabri & Thomas, 2019). These findings underscore the necessity of targeted interventions aimed at improving financial accessibility and equipping female entrepreneurs with the necessary resources to overcome systemic challenges.

Finally, industry-specific barriers, such as the unpredictable nature of the fashion industry, significantly discourage investment. This includes the perception of fashion as a high-risk sector, economic volatility, and shifting consumer behaviour, all of which contribute to a cautious approach from financial institutions and investors. For example, S5 observed, *“The immediate commercial feasibility of fashion projects is frequently questioned”*. Similarly, S6 stated, *“Investors do not trust investing their money in fashion projects because they perceive the industry as highly volatile and trend-dependent, which makes it difficult to predict long-term returns”*. This concern is corroborated by Ningsih (2023), who notes that financial institutions often favour sectors with predictable returns. This is further supported by McKinsey's report on the fashion industry, which highlights economic volatility and shifting consumer behaviour as significant challenges for fashion businesses (McKinsey & Company, 2025). This underscores the need for tailored risk assessment frameworks to enhance support for fashion entrepreneurs.

Additionally, the World Economic Forum (2020) emphasises the importance of creating conditions for manageable risk and attractive returns to encourage investment in the fashion industry. Moreover, participants consistently advocated for financial institutions that cater to the unique demands of the fashion industry. For example, S2 stated, *“The current financial support system does not adequately address the specific needs of the fashion industry, often overlooking the unique challenges and cycles of fashion trends, which results in insufficient tailored financial solutions for fashion entrepreneurs”*. Other studies, for example Aguilar et al. (2019) and Ermawati (2024), echo this sentiment, suggesting that targeted funding solutions, such as short-term loans for new collections or marketing campaigns, could effectively meet industry-specific needs. Boston Consulting Group (2020) underscores the importance of innovative financial models, such as strategic partnerships between investors and fashion brands, to provide sustainable funding solutions. Furthermore, studies have found that fashion

entrepreneurs often rely on social capital to secure funding (Braun et al., 2016; Purba & Irawan, 2024). Expanding professional networks and leveraging business relationships can help mitigate financial constraints by attracting investors and lenders willing to support industry-specific ventures (Jonsson & Lindbergh, 2013).

5.3.3 *The impact of financial support on EO*

Financial support plays a pivotal role in enabling fashion entrepreneurs to invest in the latest technologies and sustainable materials, which are essential for remaining competitive. For instance, Participant D3 highlighted the transformative impact of obtaining a loan: *“After obtaining a loan, I was able to import sustainable fabric to produce clothing suitable for kidney disease patients, which marked a transformative phase for our project”*. Similarly, participant D6 attributed a significant sales increase to the financial support received: *“The financial support from Dim Al-Manahil allowed me to shoot our collection in Georgia with professional models; our sales doubled within a year”*. Previous research has confirmed that financial support empowers Saudi fashion MSMEs to adopt sustainable production methods and improve their market competitiveness (Alshibani et al., 2024). Moreover, a lack of early-stage funding has been identified as a significant barrier to innovation and business growth (Siddiqui et al., 2021). Similarly, Mathkur (2023) shows that improved access to funding increases the likelihood of adopting sustainable business practices, developing innovative product lines, and creating opportunities for international expansion. Access to seed capital and investment opportunities is crucial for fashion businesses to achieve innovation and economies of scale (Agarwal, 2023).

Access to financial resources is critical for initiating innovative projects and sustaining proactive market strategies. In the fast-paced fashion industry, staying on trend is imperative, and financial support facilitates investment in expensive trend-forecasting services. As D6 explained, *“For each collection, I allocate funds to study fashion trends to ensure my designs are market-leading and align with future directions”*. Similarly, D7 stated, *“I utilised funding to analyse customer preference and predict seasonal trends such as Ramadan and Eid”*. This highlights that financial support is key to staying competitive and launching new designs ahead of trends. Studies from Saudi Arabia confirm that good financial systems are important for businesses to adapt quickly to market changes (Almawishir, 2018). In addition, research shows that financial flexibility helps businesses quickly react to shifts in the market and customer needs, improving their competitiveness and profits (Alsolamy, 2023). Moreover, access to various funding sources allows companies to proactively forecast markets, improve supply

chains, and tweak products, thereby enhancing their resilience and competitive edge (Siddiqui et al., 2021). Additionally, research in the Saudi apparel sector indicates that female entrepreneurs with solid financial backing are more likely to adapt their business models to changing market conditions (Alshibani et al., 2024). The relevance of these findings extends beyond Saudi Arabia, as illustrated by international research. For example, in Kashmir, Shah and Lala (2021) found that entrepreneurs with reliable financial support were better equipped to navigate changing market conditions. In Sweden, enterprises backed by sufficient funding are more likely to introduce new ideas, experiment with various marketing strategies, expand into new markets, and adapt supply channels, all of which enhance their competitive edge (Brazinskaite & Durand, 2014). Similarly, in Indonesia, effective financial strategies were pivotal in enabling the El-Zatta Muslim fashion industry to adjust its business models swiftly in response to market changes (Edwina & Wahdiniwati, 2020).

Several interviewees expressed that access to external funds allowed them to take on more ambitious and high-risk projects. For example, D1 stated, *“Designers from wealthy families tend to take more creative risks and grow their businesses more confidently, knowing they have financial backing to fall back on”*. Similarly, D7 remarked, *“I took fewer risks before obtaining a loan. Now I can afford to take creative risks and explore new markets”*. These findings are corroborated by academic research, which suggests that access to external funds can significantly shift an entrepreneur's capacity to undertake risks. For instance, a study which surveyed 711 SME owners and managers in Saudi Arabia found that securing external funding significantly influences the willingness of these businesses to engage in higher-stakes strategies (Adam, 2024). This observation aligns with the findings of Warhuus et al. (2021), who observed that entrepreneurs generally increase their risk-taking behaviours when they obtain external financing. Furthermore, Krzepkowski (2014) highlights that businesses relying on external debt tend to engage in riskier projects to fulfil their obligations. Therefore, research suggests that entrepreneurs with reliable financial backing are more likely to take risks, particularly in the fashion industry (Tang, 2024).

Table 5.1: Summary of Thematic Findings of Finance Domain within Saudi FEE

Subtheme	Findings
Availability of Financial Support	- Most Saudi fashion entrepreneurs rely on personal savings or financial support from family and friends. - Only a small number of participants have accessed venture capital or incubator funding. - The Social Development Bank is regarded as a viable source of accessible loans with favourable terms.
Accessibility of Financial Support	- Complex procedures and strict requirements from financial institutions pose significant barriers to funding. - Entrepreneurs are often required to present detailed business plans and secure government-employed guarantors. - A lack of financial literacy and limited awareness of available financial support options further restrict access to funding. - Entrepreneurs with backgrounds in design and the arts frequently struggle with financial management and attracting investors. - The fashion industry is perceived as high-risk due to economic instability and changing consumer behaviour, which discourages potential investors. - Female entrepreneurs face additional challenges due to cultural and institutional biases, including stricter guarantor requirements and limited networking opportunities.
Impact on EO	- Loans and financial support enable innovation projects, such as sustainable clothing lines and professional marketing campaigns, which enhance sales and market visibility. - Access to external funding increases entrepreneurs' confidence to undertake higher-risk, innovative projects.

5.4 Culture and social norms

This theme examines the role of culture and social norms in Saudi FEE. The interviews identified five key themes: (1) Social and Religious Perceptions of Fashion Entrepreneurship, (2) Local Success Stories, (3) Culture of Collaboration, (4) Influence of Religious and Cultural Values on Fashion Design, and (5) Impact of Culture on EO.

5.4.1 Social and religious perceptions of fashion entrepreneurship

The interviews revealed that entrepreneurship is widely encouraged in Saudi society. Participant S6 stated that “*Entrepreneurship and the establishment of new businesses are highly encouraged in Saudi society. Saudis tend to regard individuals who initiate a project alongside their regular employment as accomplished and independent*”. This view is supported by survey results showing that 60% of participants agree that Saudi culture supports entrepreneurial success, emphasising independence and personal initiative, while 71% agree that Saudi culture fosters creativity and innovation. These findings are consistent with the

Global Entrepreneurship Monitor (2024b), which considers Saudi Arabia's entrepreneurial culture to be above average compared to peer nations. Hakami (2021) emphasised that parents often perceive entrepreneurship as a mark of independence and success, aligning with S6's insights about familial views.

Moreover, the interviews revealed that regional and cultural traditions have a significant influence on entrepreneurial activities, which vary according to the region. As participant S2 noted, *"In regions like Qassim, trade is not just a business, but a heritage passed down through generations. Each region, whether the Eastern or Southern regions, has its own unique commercial flavour that influences how businesses are managed there"*. Another participant, S3, stated, *"The Eastern Province is well known for its commercial role. The community actively supports trade, and there are numerous family run businesses. I believe that this entrepreneurial spirit is deeply rooted in its culture"*. This observation aligns with academic research, as the Eastern Province is a major contributor to Saudi Arabia's economy, largely because of its strategic location and industrial development (Mahmud, 2020). Furthermore, a study found that entrepreneurial orientation, including innovation, proactivity, and risk-taking, is deeply ingrained in local business practices (Alshagawi et al., 2021). Alrubaishi et al. (2021) emphasised the role of Islamic capital and cultural heritage in shaping entrepreneurial behaviours, particularly in family businesses. They found that entrepreneurial legacies in Saudi family firms are sustained through strong cultural ties and religious values, which foster business continuity across generations. (Alarifi & Alrubaishi, 2018) found that Islamic teachings influence social entrepreneurship in Saudi Arabia. This suggests that cultural factors impact not only traditional businesses but also newer forms of entrepreneurship, including socially driven enterprises.

Participants highlighted the importance of religious principles in Saudi culture for initiating business ventures in Saudi Arabia. Participant D9 commented, *"In the Kingdom, we have a culture of encouraging entrepreneurship and business because it is also embedded in our religion. The Prophet Muhammad (peace be upon him), who was a trader, encouraged commerce"*. This participant further explained, *"Prophet Yunus was a tailor, and Prophet Muhammad (peace be upon him) used to sew his own clothes. However, society still struggles to accept the idea of a Saudi man working as a tailor or a fashion designer"*. Research findings support this view, highlighting that Islamic principles such as integrity, reliability, and social welfare are crucial to entrepreneurial achievement in Saudi Arabia (Ukil et al., 2024). For example, Saudi entrepreneurs often find inspiration in the Prophet Muhammad's (peace be upon him) involvement in commerce (Olaofe, 2023).

Despite positive views on societal perceptions of entrepreneurship, the interviews revealed social challenges related to fashion as a professional and academic field. The participants mentioned that Saudi society perceives tailoring as a low-status profession. For example, participants identified themselves as fashion designers or fashion entrepreneurs rather than tailors. As D8 stated, *“I prefer the term fashion designer because it carries a more professional and respected connotation than tailor, which is often viewed as merely technical or laborious work”*. D5 said, *“It is common to find that most tailors are foreign workers because in Saudi Arabia, we consider it a demanding job and of lower status compared to fashion designing or running a fashion business”*. Moreover, several participants noted that the field of fashion is still regarded as secondary compared to entrepreneurship or other fields, such as medicine. Participant D8 stated, *“In 2015, when I told my family I wanted to study fashion design, they did not welcome the idea because they believed this field does not reflect high status”*. Participant S6 added, *“The stereotype of tailoring exists, even among fashion design students. Some prefer designing and drawing over tailoring and mastering manual skills”*. This aligns with Alzahrani et al. (2015), who emphasised the prevalent perception that fashion is a “low status” field in the Saudi context and is “primarily a feminine choice”. Subsequent studies have corroborated that this societal view restricts the talent pool in the sector and impedes innovation (Alshibani et al., 2024). These findings do not align with the survey results of the current study, which indicated that 70% of participants believed that successful fashion entrepreneurs were highly respected in society, while 56% considered establishing ventures in the fashion sector a suitable way to achieve wealth. This perspective is comparable to the positive outlook on fashion design as a promising career in countries such as Finland (Colliander, 2015; Donaldson, 2020). The discrepancy between these views may be attributed to significant cultural changes in Saudi attitudes towards fashion design and craftsmanship following Vision 2030, or to regional differences in cultural support for entrepreneurship (Alzahrani & Al-Dabbagh, 2020).

5.4.2 Local success stories

Participants expressed concerns regarding the inadequate representation of Saudi designers' success stories in local media. For example, participant D1 remarked, *“I have not observed substantial media coverage of local fashion entrepreneurs; if it exists, it is minimal and lacks continuity”*, while D8 noted that *“Newspaper and television coverage often concentrates on a select group of established designers”*. D3 further commented, *“I wish that the media would begin to cover emerging fashion designers more extensively”*.

The Saudi Fashion Commission actively showcases the success stories of established designers by coordinating domestic and international fashion events. D4 stated, *“The fashion industry in Saudi Arabia has experienced significant growth and achieved international recognition. Our experiences have taken us to Paris, Riyadh, and Rome to attend various courses”*. These findings align with the present survey results, wherein 71% of participants believed that success stories of fashion entrepreneurs frequently appear in the media.

Participants emphasised that highlighting the accomplishments of local designers in the media would serve as a catalyst for motivating more young individuals to enter the industry. S1 stated, *“The global recognition of a Saudi designer would inspire numerous new designers and encourage investors to support the fashion sector”*. Media attention on successful entrepreneurs positively impacts the initiation of new enterprises, although admiration for these figures alone does not guarantee an increase in business creation (Alshebami & Seraj, 2022). This finding is consistent with research from Finland (Rinkinen & Laakso, 2023) and Sweden (Salonoja, 2013), highlighting that media plays a critical role in promoting entrepreneurial culture in the fashion industry. In Saudi Arabia, role models significantly influence entrepreneurial behaviour, particularly among female students, as noted by Satar (2024) and Choukir et al. (2019). However, the media’s impact is limited unless supported by robust institutional frameworks, including educational programs, incubators, and funding initiatives that can turn these inspirational stories into tangible role models for aspiring entrepreneurs (Alshebami & Seraj, 2022).

5.4.3 Culture of collaboration

According to the interview findings, Saudi fashion designers may prefer to work independently and have full ownership of fashion enterprises rather than collaborating with others. This preference arises from the desire to maintain their creative designs and brand identity. For example, D4 stated, *“I prefer working alone rather than sharing my creative designs with others”*. D7 said, *“As a fashion designer, I want my brand to represent my name, so I prefer to have full ownership of the brand”*.

Designers highlighted the lack of successful examples of partnerships and collaboration within the Saudi fashion industry, which reduces their interest in collaboration. As D6 explained, *“We often hear that designers achieve success through personal efforts; on the other hand, many partnerships fail, especially when designers feel constrained by others’ involvement”*. A strong competitive spirit prevails among designers, leading them to protect their trade secrets and maintain a level of distrust toward others. This atmosphere makes it

challenging to build strong partnerships. For example, S4 remarked, *“Competition is ingrained among Saudi fashion designers. Many choose to protect their trade secrets by working alone”*.

These findings align with Salonoja’s (2013) concept of the “working alone” culture among fashion entrepreneurs in various contexts, which inhibits the development of strong collaborative networks. Similarly, Patnaik and Isaacs (2024) and Dimitriadis and Koning (2022) noted that weak social and communication skills can hinder partnerships and collaboration, denying entrepreneurs the benefits of feedback and advice that enhance entrepreneurial performance. Literature highlights the importance of teamwork and effective communication as critical enablers of Saudi SMEs success (Alkhoraif & McLaughlin, 2018).

5.4.4 Influence of religious and cultural value on fashion design

The interviews revealed that the transformation in the cultural and social landscape resulting from Vision 2030 had a substantial impact on Saudi FEE. Participant S1 stated, *“Vision 2030 has indeed altered our cultural landscape. It extends beyond policies to foster a new paradigm where modernisation and cultural heritage coexist, enhancing both fashion and entrepreneurship approaches”*. The interviews indicated that the demand for modern and modest designs was the impetus for creating certain fashion projects that addressed this need. For example, Participant D9 stated, *“My journey in fashion design was driven by my desire to find attire that matched my taste, particularly modest clothing, which was scarce in the market”*. Researchers suggest that this combination serves as a source of strength and distinction, creating new markets for “modest” or “heritage-blended” fashion (Fahrati, 2023).

The interviews revealed that the integration of religious and cultural values in fashion design is crucial in Saudi Arabia and cannot be disregarded. Participant D3 noted, *“Fashion here is not just about aesthetics; it is about identity and values. We have to consider religious guidelines meticulously, which sometimes limits the scope for creative expression”*. Participant S2 stated, *“It is a delicate balance that maintains the rich traditions that define our cultural identity while integrating modern business practices. This balance is crucial for the evolution of the fashion industry”*. This is consistent with the findings of Abalkhail (2018b), who reported that religiosity and traditional cultural values significantly impact attitudes toward Islamic fashion, shaping consumer preferences and purchase intentions. Similarly, DeCoursey (2017) found that professional Saudi women perceive the abaya as a culturally authentic garment rather than merely a religious requirement, highlighting the intersection of fashion and cultural identity.

Cultural heritage is considered a significant market force, with participants affirming the evolution of Saudi fashion to meet international standards and using cultural heritage as a unique feature that enhances its global appeal. Participant S4 noted, *“The increased awareness of heritage fashion and the call for its revival have led to a radical change in the fashion design landscape, resulting in increased interest in designs inspired by heritage”*. Participant S6 stated, *“Our cultural heritage is our competitive advantage. This makes Saudi fashion unique. With support from cultural initiatives, we have been able to preserve this heritage while evolving to meet international tastes”*. This is supported by Alotaibi and Starkey (2020), who examined the integration of traditional Saudi embroidery into modern apparel to bridge the gap between historical and contemporary styles. Similarly, Hasanin (2021) found that incorporating traditional motifs into modern dress designs helps preserve cultural heritage while adapting to evolving consumer preferences.

There is significant appreciation for fashions that reflect and respect Saudi cultural identity, enhancing the market acceptance of local designs. Participant S5 stated, *“Regarding local culture, there is growing appreciation for fashions and designs that reflect Saudi cultural identity, allowing designers to explore this aspect”*. Participant D8 noted, *“As designers, our work thrives when it narrates the richness of our culture. The market's strong response to our locally inspired collections demonstrates a profound respect and growing desire for authentic Saudi fashion”*. This integration also represents an effective way to attract consumers, both domestically and internationally, by offering unique products that reflect authentic cultural identities (Alfawaz, 2024).

Several interviewees highlighted the significant regional differences in fashion taste in Saudi Arabia. For instance, D5 and D4 explained that while Qassim favours traditional black abayas, Jeddah embraces more diverse and colourful designs. Participant D5 elaborated, *“In Qassim, the cultural preference for black abayas means we integrate printing only subtly, as coloured abayas are less common. Conversely, in Jeddah, our work in abayas is highly appreciated, and in Riyadh, we target the market with formal and luxurious abayas, showcasing cultural differences”*. Similarly, D4 remarked, *“Different regions in Saudi Arabia have varied tastes and expectations regarding fashion. What works in Jeddah might not be well received in more conservative areas such as Najd. This regional variation requires us to be very strategic about our design and marketing approaches”*. This aligns with Algumzi (2022), who found that Saudi women's luxury fashion preferences vary depending on cultural and lifestyle influences, particularly in the context of the NEOM project. These findings suggest

that Saudi fashion designers must adopt region-specific strategies to cater to their clientele's diverse tastes and expectations in the Saudi Arabia.

While the integration of religious and cultural values remains central to Saudi fashion, the participants acknowledged the limitations imposed by conservative norms on design and marketing. Participant D1 noted the challenge of balancing tradition with innovation, while D9 highlighted marketing restrictions, such as the inability to use live models for abaya advertisements. D1 noted, *"Integrating religious and cultural values into design work is both a responsibility and a challenge. It is challenging to consider traditional elements while also gently pushing boundaries to maintain fresh and relevant designs"*. D9 commented, *"In terms of marketing, showcasing abayas on models was not feasible due to familial and societal disapproval"*. This observation underscores the need for fashion designers to be innovative in both their designs and promotional strategies to overcome the constraints of cultural acceptability. These challenges were reported by Zaynah and Steinberg (2017), who examined the impact of conservative religious values on marketing strategies in Saudi Arabia and noted the need for brands to adopt modest and culturally appropriate advertising techniques.

Despite these constraints, some participants, such as S3, argued that cultural norms did not necessarily hinder creativity or marketability. Instead, they suggested that the Saudi market is sufficiently diverse to accommodate both conservative and progressive design. S3 elaborated, *"At present, there is no justification for not pursuing or implementing a project because of concerns about conservative designs. Our large and diverse population encompasses all groups, making it straightforward to find a market for both conservative and more progressive designs"*. This perspective aligns with Alajaji (2022), who argued that sustainable traditional fashion designs can cater to both local and international audiences while maintaining cultural integrity. Amalanathan and Reddy (2024) highlight the necessity of reconciling modesty, a cultural and religious value, with innovation and renewal in the Saudi fashion world.

5.4.5 *The impact of Culture on EO*

The interviews revealed that Saudi culture does not influence innovation or proactivity. Nevertheless, the findings indicate that risk aversion is generally prevalent in Saudi culture, which can restrict entrepreneurial endeavours and impede business innovation. Participants noted that Saudi society tends to have a negative view of failure, which creates pressure to avoid taking risks. As D1 stated, *"If I fail in my project, I see it as a learning opportunity, but I am worried about what my family will think, as they might see it as a weakness in my*

character”. Similarly, D9 stated, *“My biggest fear was criticism from my community. I felt like I was constantly being watched in case I made a mistake or did not achieve success in my project”*. This tendency to avoid risk seems to originate from societal perceptions of failure, which are often seen as personal shortcomings rather than opportunities for development. This negative perception of failure is further reflected in the survey data, where only 43% of respondents believe that Saudi culture encourages entrepreneurial risk-taking. The Global Entrepreneurship Monitor (2024a) also reports that 58% of Saudi entrepreneurs hesitate to start new businesses because of fear of failure, a figure higher than the global average.

This cultural aversion to risk places additional pressure on entrepreneurs in the fashion industry, where innovation and creative risk-taking are critical to success. This aligns with broader patterns observed in collectivist societies, which tend to prioritise social stability over individual risk-taking, according to DeScioli (2024). Similarly, Almesahri et al. (2023) identified societal pressures in Saudi Arabia as a significant barrier to entrepreneurial ventures, highlighting that the stigma surrounding failure discourages innovation and limits the willingness of individuals to pursue business opportunities. Promoting the acceptance of failure as a learning opportunity and sharing success stories of entrepreneurs who have overcome setbacks could help foster a more supportive entrepreneurial culture (Isenberg, 2016; Rinkinen & Laakso, 2023). One factor contributing to this aversion to risk is the cultural preference for government employment over entrepreneurship. D3 noted, *“My parents didn’t encourage me because Saudi culture often favours government jobs or jobs in large companies, as they provide financial stability and job security”*. This observation reflects the findings of Al-Yazidi and Hassan (2022), who indicated that societal norms tend to undervalue entrepreneurship compared to stable jobs in the public sector or large corporations, further discouraging entrepreneurial risk-taking and innovation.

Additional research confirms that fear of failure remains a dominant factor shaping entrepreneurial behaviour in Saudi Arabia. Fridhi (2020) identified fear of failure as the most influential factor affecting entrepreneurial intentions among Saudi youth. Elshaer and Sobaih (2022) also found that while risk-taking positively affects entrepreneurial intentions among Saudi graduates, cultural barriers, such as fear of failure and social pressure, limit individuals' ability to act on business opportunities. Similarly, Al-Ajlouni and Saad (2024) found that fear of failure and social stigma are key cultural barriers to entrepreneurship in Saudi Arabia, reinforcing the idea that Saudi culture’s aversion to risk constrains economic and social development.

Table 5.2: Summary of Thematic Findings of Cultural and Social Domain within Saudi FEE

Subtheme	Findings
Social and Religious Perceptions	- Entrepreneurship valued as a symbol of independence. Religious ethics influence business practices, with the Prophet Muhammad as a model. - Fashion viewed variably in social status.
Local Success Stories	- Media coverage of local fashion entrepreneurs seen as inadequate, with a desire for more exposure to inspire and attract investment.
Culture of Collaboration	- Preference for individual ownership prevalent among designers to protect creative integrity; collaborative efforts limited due to competitive nature and distrust.
Influence of Religious and Cultural Values	- Vision 2030 enhances cultural heritage integration with modernization, affecting design creativity and market appeal with strong regional variations.
Impact of Culture on EO	- Minimal cultural influence on innovation; failure perceived negatively as a personal shortcoming, with a preference for social stability over individual risk-taking.

5.5 Support Systems

The analysis of interview data reveals critical themes about the support systems in the Saudi FEE. These themes are categorised into four main themes: (1) Specialised Support Structures, (2) Challenges in Accessing Support, (3) Formal Support Institutions, and (4) Impact of the Support System on EO.

5.5.1 Specialised Support Structures

The interviews revealed a significant lack of dedicated support institutions for emerging fashion entrepreneurs in Saudi Arabia, such as fashion incubators and design hubs. Participants expressed concerns regarding the relevance of available incubator resources. D8 noted, “*I joined an incubator expecting guidance on scaling my brand, but the programs were all about pitching to investors, which wasn’t my immediate priority*”. This observation aligns with research indicating that incubator effectiveness depends on the alignment between an entrepreneur's needs and the services provided (Dlamini et al., 2022). A mismatch between incubator offerings and business needs limits benefits, as startups often receive generic advice that fails to address industry-specific challenges.

Participants further emphasised the challenge of finding incubators suited to the fashion industry. D2 stated, *“Most incubators are focused on tech startups, and finding one that understands the fashion business is incredibly difficult”*. D7 echoed this sentiment, explaining, *“I needed help with supply chain management and sustainable sourcing, but the incubators I found primarily offered mentorship in software development and venture capital fundraising, which weren’t relevant to my business”*.

Survey results reinforced these concerns, with 50% of respondents disagreeing that sufficient subcontractors, mentors, qualified experts, suppliers, and consultants exist to support fashion businesses. These findings align with literature indicating that incubators often prioritise high-growth industries such as technology and biotechnology, limiting support for creative and manufacturing sectors (Kazhenov, 2023; Nicholls et al., 2020). Braun and Suoranta (2024) highlight that incubators frequently neglect business model innovation tailored to creative industries, exacerbating the mismatch in available support. This limitation is particularly problematic in fashion, where access to design mentorship, material sourcing, and retail partnerships is crucial (Ahmed et al., 2020).

International case studies illustrate the potential benefits of industry-specific incubators. Studies on South Korea’s fashion incubation sector demonstrate that effective programs combine financial support, workspace, and brand development strategies, which significantly benefit emerging designers (Lim et al., 2016). Similarly, Moldova’s ZIPhouse initiative integrates local craftsmanship with sustainable practices, fostering an ecosystem conducive to new designer success (Nistor, 2024). Research on Canada’s fashion sector indicates that systemic institutional and policy weaknesses hinder industry growth, rather than a lack of talent (Brydges & Pugh, 2017).

To address these gaps, scholars recommend developing industry-specific incubators or hybrid models that integrate sectoral expertise while maintaining a diverse portfolio of startups (Huang & Chen, 2024). Such approaches ensure that entrepreneurs across different industries receive relevant support.

Beyond institutional incubators, emerging designers often turn to private consultants for guidance. However, participants highlighted the limitations of current consultancy services, noting that these are frequently offered by fashion business owners who provide consulting as a supplementary income stream. D5 remarked, *“Many consultants have their own brands, so you always wonder if they’re truly helping or just looking out for themselves”*. Similarly, D3 stated *“I found a consultant, but she also has an abaya brand. How can I share my designs with someone who’s basically my competitor”*.

This dual role raises concerns regarding impartiality, as consultants may prioritise their own business interests over those of their clients. Studies indicate that professionals with competing financial interests may struggle to provide objective guidance, leading to diminished trust among clients (Cech & Leonard, 2001; Dipaola et al., 2014). Transparency and clear conflict-of-interest policies can mitigate these risks. Disclosure of financial or personal interests has been found to enhance professional integrity, particularly in industries where secondary interests may compromise judgment (Cech & Leonard, 2001).

Conversely, the absence of disclosure policies fosters scepticism, with clients perceiving advice as biased (Dipaola et al., 2014). Business ethics literature further underscores the risks of conflicts of interest in consulting services, with Sennewald and Baillie (2021) arguing that ethical conflicts undermine trust and decision-making. Similarly, Zaborovsky (2022) highlights conflicts of interest as a key determinant of trust in professional-client relationships, necessitating proactive management.

The participants mentioned scarcity of specialised consultants capable of providing evidence-based, strategic guidance tailored to their needs. Participants expressed frustration with consultants who relied solely on personal experience rather than structured methodologies. S4 remarked, *“Most consultants I met shared only their own stories. That’s not enough I need real, strategic advice”*. D4 noted, *“I paid a consultant a hefty fee only to realise they had no formal background in fashion business strategy. They just relied on their own past successes”*. This issue contrasts with best practices in business consulting, which emphasise the integration of industry experience with structured analytical tools and market data (Chia, 2018; Kubr, 2002). Professional consultants who use data and structured frameworks are more likely to provide actionable insights and improve client satisfaction than those relying solely on anecdotal experiences (Raja et al., 2024; Schein, 2016). Furthermore, studies indicate that formal training and certification improve consultancy quality. Certified consultants tend to adhere to structured methodologies, which enhances the strategic value of their advice (Haeusler, 2009). For example, structured problem-solving frameworks like the McKinsey six-stage methodology have been shown to improve consulting outcomes by guiding consultants to develop solutions in a logical and evidence-based way (Chia, 2018).

Research on business consulting effectiveness suggests that professional advisory services significantly impact business success when structured methodologies are employed (Bondarchuk et al., 2020). However, studies indicate that many consulting firms fail to provide data-driven strategic guidance, thereby offering limited value to clients (Lisowska & Stawasz, 2020). By contrast, well-structured consulting services in industries such as finance and

technology adhere to best practices that prioritise research-based, customised advice (Levishchenko, 2020). The growing demand for expert consultants further underscores the necessity for improved industry standards. Kopytko and Vereskliia (2021) found that businesses seeking advisory services prefer consultants with specialised training in strategy and market research over those who rely solely on personal experience.

5.5.2 Challenges Accessing Support

The interviews revealed that fashion entrepreneurs face three primary obstacles in accessing support: geographic limitations, high service costs, and a lack of awareness. These challenges significantly hinder business growth and innovation in the industry.

First, the physical location of available support services presents a major barrier to fashion entrepreneurs in rural areas. Most industry bodies, incubators, and networking events are centralised in Riyadh and Jeddah, leaving entrepreneurs in other regions with limited access to critical resources. This geographical disparity forces business owners to relocate or seek alternative solutions to their problems. For example, D9 highlighted this issue by stating, *“I am from the Northern Region, where support for fashion entrepreneurs is quite limited. This is why some designers relocate their businesses to Riyadh, as it offers abundant support services, business incubators, fashion shows, and networking opportunities”*. This aligns with the findings of recent studies, indicating that entrepreneurs in peripheral areas face challenges due to inadequate infrastructure and business ecosystems (Nana et al., 2021).

Second, financial barriers significantly affect entrepreneurs' ability to access necessary support services. The survey results revealed that 50% of respondents disagreed with the statement that fashion businesses can afford to hire subcontractors, suppliers, and industry experts. Interviews further reinforced this concern, with participants highlighting the high costs as a key challenge. For instance, S3 shared, *“Some incubators demand equity or substantial payments, which discourages many entrepreneurs”*. Similarly, D5 said, *“Currently, there are agencies with professional fashion models and photographers, but their costs are very high. Therefore, I travel to Georgia, where the costs of travel, photography, and models are less than a quarter of those in Saudi Arabia. I think the high service costs are due to scarcity”*.

These findings are consistent with the survey results which found that 53% of the respondents disagreed that it is easy to obtain reliable subcontractors, suppliers, and consultants, reinforcing the financial constraints faced by fashion businesses. These experiences reflect broader industry trends, where high service costs limit the opportunities for emerging designers. Research suggests that expensive service fees and equity demands by

incubators act as deterrents, particularly for startups that aim to maintain ownership while seeking external assistance (Qasim & Cirera, 2014).

A study on incubators in Turkey found that while incubators provide valuable advisory and administrative support, financial costs and equity demands often act as barriers to entry for startups seeking external assistance (Birden & Bastug, 2020). Equity demands can reduce the willingness of startups to engage with incubators (Wei et al., 2020).

Additional challenges include a lack of awareness of the available support. S4 stated, “*There are many support programs, but entrepreneurs struggle with awareness of available support programs and find the application processes overly complex*”. Effective communication strategies, tailored outreach, and simplified application procedures are essential for bridging this gap (Nana et al., 2021; Qasim & Cirera, 2014).

Language barriers also pose obstacles, particularly when accessing international grants and programs offered by the Saudi Fashion Commission. D 8 stated “*Many of these opportunities require proficiency in English as they are often led by foreign experts*”. Entrepreneurs with limited English skills are excluded from these programs, reducing their ability to gain global exposure and industry insights. Proficiency in English is crucial for accessing global business opportunities and fostering successful intercultural communication (Kumar, 2023).

5.5.3 Formal Support Institutions

The interviews revealed several formal institutions that currently support fashion MSMEs, including the Ministry of Commerce, Monsha’at, the Ministry of Culture, the Saudi Fashion Commission, the Social Development Bank, and the Ministry of Labor. These entities play an essential role in shaping the industry by providing fashion entrepreneurs with financial, regulatory and strategic support.

The Ministry of Commerce has been recognised as a pivotal player in developing fashion businesses in Saudi Arabia. Interviewees highlighted their significant role in simplifying administrative processes, making it easier for designers and entrepreneurs to establish businesses. S3 stated, “*The Ministry of Commerce has significantly supported general and fashion-related projects. They have simplified the registration process and made it easier to obtain permits online*”. Furthermore, the Ministry has implemented initiatives to ease regulatory requirements and reduce entry barriers for emerging fashion businesses in the fashion industry. S2 explained, “*The Ministry of Commerce’s efforts to reduce bureaucracy have streamlined the process of starting businesses and entering the fashion industry*”.

These findings align with recent studies highlighting the pivotal role of the Ministry of Commerce in business development in Saudi Arabia, including its support for e-commerce infrastructure and innovation (Ismail & Azab, 2023), facilitation of social commerce growth (Albelaihi et al., 2020), support for SME e-commerce adoption (Al-Tayyar et al., 2021), and its influence on financial and organisational performance in SMEs (Almuwallad & Alhumoudi, 2024). Government-supported chambers of commerce play a crucial role in providing a conducive environment for entrepreneurs. S4 stated, *“One of the initiatives that the government is undertaking is supporting chambers of commerce in providing a suitable environment for entrepreneurs”*. Policy amendments have further enhanced the role of chambers of commerce, enabling them to address the specific needs of fashion entrepreneurs. S3 elaborated: *“Being a board member at the Eastern Province Chamber of Commerce has opened up numerous opportunities and significantly expanded my industry knowledge and professional network. It has also enabled me to support other designers by understanding their requirements and advocating policy changes to assist them”*.

Research on the entrepreneurial ecosystem in Saudi Arabia underscores the importance of chambers of commerce in supporting business development, reducing bureaucracy, and fostering business education (Almuzel & Anderson, 2020). This finding echoes research highlighting the role of networking and policy amendments in supporting fashion entrepreneurs, indicating that government-backed initiatives help small business owners access resources and industry connections (Seraj et al., 2022).

Despite these advancements, some participants emphasised the necessity of establishing fashion-focused chambers of commerce in various regions. S5 expressed, *“The presence of fashion-specific chambers of commerce is imperative for facilitating operational processes and promoting collaboration between designers and influential figures in the industry. This will enhance the overall business environment for fashion designers and more effectively address commercial concerns”*. Studies on sector-specific business associations corroborate this viewpoint, showing that dedicated industry bodies lead to more tailored policy interventions and better advocacy for industry needs (Gamba, 2019; Irwin, 2015).

5.5.4 The impact of Support Systems on EO

The interview findings indicate a lack of support systems for innovation in the Saudi fashion sector, in contrast to the considerable innovation support for sectors such as technology and petrochemical. This disparity reflects an imbalance in access to resources and infrastructure within the broader innovation ecosystem of the Kingdom. Several participants highlighted that

Saudi Arabia's leadership in the petrochemical industry has not translated into significant advancements in the fashion sector. For instance, S6 noted, *"Saudi Arabia is one of the leading countries in the petrochemical industry, yet the fashion sector has not capitalised on these innovations to develop innovative fabrics and designs"*. The fashion industry in Saudi Arabia is often perceived as luxurious or culturally specific; however, it makes significant contributions to global innovation in the field. Participant S4 noted, *"A notable example within Saudi Arabia is the support extended by the Crown Prince to fashion designer Tariqah Al-Mutairi, who used innovative techniques in textiles and design for military applications"*. This example highlights the fashion sector's potential to develop more successful models with increased support. This observation aligns with Todeschini et al. (2020), who underscored the importance of cross-sector collaboration in driving fashion industry innovation. They noted that sustainable innovation in fashion depends heavily on effective cooperation among suppliers, designers, and consumers. Therefore, strengthening partnerships between the fashion industry and others, such as the military or petrochemical industries, could unlock new opportunities for developing innovations within these industries.

Participants consistently mentioned the absence of advanced resources and specialised centres for fashion design as a major barrier to innovation. For example, S4 stated, *"There is an abundance of business incubators for various sectors in Saudi Arabia, such as technology, software, and artificial intelligence, with incubators like Badir for Technology and KAUST. However, in the fashion sector, we do not see the same level of support"*. Participant S5 explained, *"We hear about collaboration in other industries where R&D centres help develop leading products. We need this kind of support in the fashion sector to truly innovate"*.

Participants pointed to successful models of specialised R&D centres and design facilities that provide designers with access to the latest manufacturing technologies, enhancing both product quality and market competitiveness. S1 stated, *"We need funding for research into smart and sustainable materials. The Massachusetts Institute of Technology in the United States and the Denkendorf Institute in Germany are successful examples of how R&D can support innovation in the fashion sector"*. Additionally, S2 remarked, *"Having R&D centres equipped with 3D printing technologies helps designers to experiment with innovative designs and improve them in a short time"*. Adopting similar funding and support structures in Saudi Arabia could accelerate innovation in the fashion industry. Nayak et al. (2022) emphasised that investing in sustainable materials and advanced manufacturing technologies strengthens company competitiveness and promotes environmentally responsible practices within the fashion industry.

This finding is supported by Zeng et al. (2024), who emphasised the critical need to build an innovation ecosystem within the fashion sector, backed by infrastructure and the development of specialised centres for innovation. Similarly, Rosário et al. (2022) found that innovation support in MSMEs enhances business performance and competitiveness, highlighting the importance of extending such strategic support to the fashion industry. According to the participants, fashion designers require access to R&D centres that facilitate collaboration and experimentation with advanced technologies to enable meaningful innovation. This view is reinforced by Luan et al. (2024), who highlighted that integrating modern technologies, such as digital design, artificial intelligence, and the Internet of Things into the fashion industry, can enhance the efficiency and creativity of product development. (Casciani et al., 2022) reinforced the importance of infrastructure development and adopting advanced technologies to enhance the competitiveness of the fashion sector.

Table 5.3: Summary of Thematic Findings of Support System Domain within Saudi FEE

Theme	Findings
Specialised Support Structures	- Mainly focus on tech and high-growth industries, services don't match their needs. - Lack of dedicated fashion incubators and design hubs.
	- Consultancy services often provided by competitors, raising concerns about impartiality. - Consultants is often rely on their personal experience rather than knowledge
Challenges in Accessing Support	- Entrepreneurs outside major cities (e.g., Riyadh) struggle to access support.
	- High costs of support services deter entrepreneurs. - Expenses for essential needs like fashion models and photographers add extra financial strain.
	- General lack of awareness among entrepreneurs about available support programs. - Complex application processes and language barriers hinder access to international grants and programs, limiting global exposure and industry insights.
Formal Support Institutions	- The Ministry of Commerce and Monsha'at simplify processes. - Chambers of commerce help with networking and policy changes for fashion. - Calls for fashion-specific chambers to better support the industry.
Impact of support system on EO	- The Saudi fashion sector lacks dedicated innovation support compared to industries like technology and petrochemicals. - Collaboration between fashion and industries like petrochemicals and military can drive innovation and new applications. - Unlike other sectors, the fashion industry in Saudi Arabia lacks specialised incubators, research centres, and funding. - Integrating AI, digital design, IoT, and sustainable materials can enhance the innovations and competitiveness of the Saudi fashion sector.

5.6 Educational Institutions

The interviews highlighted the distinct opportunities and challenges associated with the roles of educational institutions in supporting the Saudi FEE. These findings were distilled into three main themes: (1) the evolution of fashion education, (2) the contribution of educational institutions in supporting new company creation and talent development, and (3) the influence of educational programs on EO by promoting knowledge, research, innovation, and an entrepreneurial culture.

5.6.1 Evolution of Fashion Education

This theme highlights the development of fashion education in Saudi Arabia and the need for further improvements. Participants noted the development of the education sector, specifically fashion education, and the Saudi government's significant support for and contribution to this field. For example, D2 highlighted the benefits of government-funded education, stating, *“In comparison to other countries, the Saudi government provides tuition-free education at state universities, offering an excellent chance for individuals keen on studying fashion design to pursue their education without cost”*.

Participants reported that Since 2018, numerous Saudi universities have established fashion design programs. According to S3, *“In the past five years, several universities, including Tabuk University and Taibah University, have restructured their home economics bachelor, which includes courses such as arts, nutrition, childhood education, and design, to be more dedicated to fashion designs”*.

The interviewees mentioned efforts to establish international fashion academies and institutes in Saudi Arabia. For instance, D2 noted, *“We see lots of developments, including the recent opening of an Italian fashion school in Riyadh that offers valuable training opportunities for fashion designers”*.

Participants also highlighted the growth of overseas scholarship programs for creative and cultural disciplines, including fashion. D7 remarked, *“There are more scholarship opportunities now than in 2008, including scholarships from the King Abdullah Scholarship Program, the Ministry of Culture, and NEOM University”*.

However, despite recent developments in fashion education and government support, the participants emphasised that fashion education in Saudi Arabia has yet to achieve international standards and is falling behind in addressing the requirements of the global fashion industry. In this regard, S4 remarked, *“In Saudi Arabia, we do not have internationally recognised*

fashion schools, unlike countries such as Canada, the United States, and France, which are home to globally renowned fashion institutions”. Participants also highlighted the challenges in improving fashion education in public universities due to constraints such as regulations, required approvals for program accreditation, and administrative restrictions. S6 stated, *“Improving fashion education in public universities is difficult because of many rules and procedures. For example, creating a new program takes time because accreditation is slow, making it hard to keep up with the fast changes in the fashion industry”*. Participants, including S6, explained that achieving international standards requires the expertise of professionals in the field:

“Government funding is not enough; we need access to industry experts to guarantee that our fashion programs meet global standards. Currently, many faculty members have only traditional training in fashion or lack international experience in developing learning and teaching methods”.

The findings highlight that the Saudi government’s support has contributed to fashion education development in Saudi universities. These findings align with Ministry of Finance (2024) data showing that 20% of the national budget has been allocated to educational development, including scholarships to leading international universities, in contrast to countries like Lebanon, where financial constraints impede educational advancement (Chahine, 2018; Ministry of Finance, 2024).

However, in Saudi Arabia, a gap remains between the government’s funding support and the resources required to elevate fashion education in Saudi Arabia to internationally recognised standards (Gonaim, 2021). While crucial, government funding is insufficient to address several key challenges that hinder educational reform, including a lack of faculty with international expertise, delays in accreditation processes, and bureaucratic constraints (Aljendan, 2024). Studies suggest that organisational and financial restructuring within Saudi higher education institutions is necessary to resolve resource limitations (Othayman et al., 2021; Othayman et al., 2022).

Furthermore, heavy teaching workloads and a shortage of locally trained faculty continue to place considerable pressure on Saudi universities, underscoring the need for ongoing professional development and capacity building among academic staff (Abdullah et al., 2024; Aldosemani et al., 2018).

Beyond government support, collaboration with industry, private sector investment, international partnerships, and academic exchanges with leading international fashion schools are critical for improving the quality of Saudi fashion education. For instance, countries with recognised fashion programs benefit from strong industry ties, extensive internships, and international collaborations, which are areas where Saudi Arabia is still progressing (Chun & Niehm, 2010; Kohli, 2019; Zanu et al., 2024).

5.6.2 Role of Entrepreneurial Education

5.6.2.1 Supporting the Establishment of New Companies

Participants shared varying perspectives on the role of university support in launching their startups. Notably, D4, D5, and D6, who all studied abroad, confirmed that their education significantly contributed to establishing their businesses in the UAE. In contrast, participants who studied in Saudi Arabia expressed dissatisfaction with the level of support provided by their universities in starting fashion companies. Reflecting on their experiences, D4 stated, *“When I studied abroad, there were numerous resources and supportive systems to aid in starting a business. However, it feels like you are on your own”*. Similarly, D5 stated, *“In Saudi Arabia, universities did not really support my business ideas. I had to figure everything out on my own”*. D6 echoed these sentiments, saying, *“My experience abroad was different; the university really pushed me to develop our business ideas. I did not see the same level of encouragement when I returned to Saudi Arabia”*.

Furthermore, participants highlighted the absence of fashion incubators in Saudi universities and the difficulties in accessing incubators where they are available. D9 remarked, *“My university has some support, but it is hard to access. We need more help and information about how to use those resources to start our fashion businesses”*. Another key issue highlighted by the participants was the lack of guidance provided by their universities on navigating the market entry and regulatory frameworks. D7 said, *“I did not receive much guidance on how to enter the market or deal with regulations. It is something I had to figure out after graduating”*.

The interview findings highlight the limited role of Saudi fashion schools in supporting students and graduates in starting their businesses compared to international institutions. Globally, educational systems offer comprehensive entrepreneurship support, including financial aid, mentorship, and networking, which are essential for success in the fashion industry. For instance, entrepreneurship education programs in international fashion schools

equip students with critical business development skills, offering continued mentorship and industry connections that boost their entrepreneurial confidence and capabilities (Pierrakis et al., 2023).

However, in Saudi Arabia, the absence of such structures, particularly incubators and financial support systems, limits the potential of aspiring fashion entrepreneurs. Research on entrepreneurship education indicates that students benefit significantly from tailored courses and strategic market entry guidance; however, Saudi institutions fall short in offering these resources (Raju & Singh, 2023). This gap becomes more apparent when comparing international models that integrate entrepreneurship within fashion programs, enabling students to navigate regulatory frameworks, IP management, and investor relations (Worrall & Linda, 2024).

Moreover, studies emphasise the critical need for Saudi fashion schools to adopt a more holistic approach to entrepreneurship education, which combines theory and practical experience and offers crowdfunding as an educational tool, similar to schools abroad (Alshebami, 2022; Lee & Kwon, 2020). These programs promote long-term business sustainability and innovation, fostering an ecosystem that supports emerging fashion designers and established entrepreneurs (Bazyl et al., 2021; Strydom & Kempen, 2021). Addressing these gaps can empower Saudi fashion graduates to compete globally by providing the necessary infrastructure for entrepreneurship development. Research conducted by Almuzel and Anderson (2020) revealed that 80% of Saudi respondents felt that the nation's educational frameworks failed to promote entrepreneurship and that they perceived a current shortage of entrepreneurs in the country.

5.6.2.2 Nurturing Entrepreneurs and Skilled Talent

Educational institutions play a crucial role in shaping the fashion entrepreneurship ecosystem, especially by equipping students with the necessary entrepreneurial skills, resource knowledge, and mindsets for success in the industry. These graduates, upon entering the workforce with fresh perspectives, up-to-date knowledge, and eagerness to innovate, contribute as employees, entrepreneurs, and freelancers, thus becoming valuable assets for emerging fashion businesses. However, educational programs need to be tailored to industry-specific contexts, especially given the challenges in curriculum design, instructional approaches, and selection of role models (Fischer et al., 2019; Wadee & Padayachee, 2017).

Fashion design programs increasingly incorporate entrepreneurship education to foster entrepreneurial interest, intentions, and competencies among students, preparing them to

effectively tackle industry challenges (Jamaluddin et al., 2019; Sousa et al., 2019). However, the participants consistently mentioned the inadequacy of generic business courses. Despite efforts to introduce business knowledge, the content is often irrelevant to the fashion industry, creating a disconnect between entrepreneurship theory and practice in the fashion industry. D8 highlighted this problem: *“The university offers courses on small business administration, but they aren’t tailored to the fashion industry, and the background of lecturers is in general business, so much of the information isn’t relevant to our fashion students’ needs”*.

This gap is even more pronounced in the absence of fashion-specific business management education. Although courses in general entrepreneurship are offered, they do not cater to the fashion industry’s unique needs. S4 stated, *“There are fashion design courses available, but not fashion-specific ones”*. Consequently, such entrepreneurs are often poorly prepared to manage the business aspects of their ventures. Moreover, despite courses that increase students’ business skills, knowledge, and professional networks, they often fail to bridge the gap between theoretical understanding and entrepreneurial practice (Adelowo et al., 2018). Practical facilities and hands-on training opportunities are notably lacking, further emphasising the need for a more balanced approach to fashion-focused entrepreneurship education.

Universities must provide a comprehensive curriculum that includes both theoretical and practical aspects of entrepreneurship for individuals aspiring to become fashion industry business owners. The curriculum should encompass essential areas, such as business planning, financial management, marketing strategies, and operational efficiency. As S3 explained, *“The College of Design and Arts primarily focuses on imparting technical skills, but there are also small business and marketing courses”*.

D6 observed that although technical institutes also play a critical role in training professionals across industries, there remains *“A lack of oversight in the fashion sector to ensure that those diverse courses are accredited”*. The interviews revealed a lack of financial literacy and business acumen among fashion designers. Many participants expressed a need for a greater understanding of finance and explained that without such skills, they would struggle to secure investments or manage the financial aspects of their businesses. As D9 stated, *“We, fashion students, come from a background of design and arts, and many of us need help with understanding the language of numbers”*.

Such financial illiteracy creates significant barriers for designers seeking external financial support, underscoring the importance of incorporating comprehensive financial education into university-level curricula. Although essential, entrepreneurial education faces

numerous challenges owing to inconsistencies in teaching methods and content. Participants raised concerns about the lack of standardisation in entrepreneurship training, which can lead to confusion and ineffective learning outcomes. S5 emphasised this point by saying, “*The inconsistency in fashion design training and education poses a problem*”.

The absence of structured and standardised content delivery across institutions adds to the difficulty of fostering entrepreneurial skills in students, further exacerbating the gap between theory and practice.

Research strongly supports the need for a balanced fashion entrepreneurship curriculum. Sousa et al. (2019) emphasise that an immersive, hands-on approach is essential for developing entrepreneurial skills among fashion design students. Similarly, Davis et al. (2019) highlighted that while entrepreneurship programs in fashion education have been increasing, they often lack a cohesive understanding of how to teach entrepreneurship effectively. Lang and Liu (2019) suggest that providing a clear direction for curriculum development based on student needs could help address gaps in business skills and entrepreneurial motivation. Nana et al. (2021) confirm that many fashion entrepreneurs lack essential business skills, particularly in financial planning, market research, and pricing strategies. Adebisi and Abiodun (2022) further highlight that limited access to capital and inadequate business training are major barriers for fashion entrepreneurs.

The need for consistency in entrepreneurship education is also evident in the literature. Gustafsson et al. (2021) found that fashion graduates often have low confidence in starting businesses because of a lack of structured, consistent training. Chen (2024) supports this by advocating for a structured, project-based teaching model to enhance consistency and relevance. Practical learning plays a critical role. Jamaluddin et al. (2019) found that mentorship and experiential learning significantly improve entrepreneurial competencies among fashion students. Sousa et al. (2019) reinforce this, noting that “Learning by doing” through real-world applications enhances students’ ability to apply entrepreneurial knowledge effectively.

5.6.3 The impact of educational institutions on EO

5.6.3.1 Generating Knowledge, Research, and Innovation

The interview findings revealed several barriers that limited Saudi fashion schools’ ability to generate new knowledge and innovation. A key issue identified by the participants was the disconnect between academic research conducted by fashion students and its practical

application in the industry. As S6 noted, “*Fashion students produce good research, but we do not see enough of it being applied in the real world*”. Similarly, S5 stated, “*While students generate promising research on fashion design and sewing, much of it remains academic, focused on meeting graduation requirements rather than commercialising innovations*”. This gap between research and practical application results in missed opportunities for the fashion industry.

These findings align with the literature that underscores the role of higher education in generating knowledge and fostering innovation, which, in turn, drives entrepreneurial opportunities (Audretsch et al., 2021; Guerrero & Urbano, 2017; Lehmann et al., 2022). The Knowledge Spillover Theory of Entrepreneurship (KSTE) suggests that new knowledge created in universities leads to entrepreneurial opportunities (Acs et al., 2013). Furthermore, global fashion institutions actively engage in research and development, providing innovations in materials, design techniques, and sustainable practices that influence industry trends (Agarwal, 2020, 2021; Rao et al., 2020). These initiatives shape the fashion industry by contributing to new business ideas, technologies, and models that drive trends and entrepreneurial opportunities (Casciani et al., 2022; Sayem, 2022; Wood et al., 2022).

The interview data also highlight a significant barrier to innovation in the Saudi fashion industry: the lack of effective communication and collaboration between academia and industry. According to D4, “*No platform connects us, fashion students, with the industry during their research. During my master’s program, I witnessed a lot of new research from universities that often does not reach the industry because the students do not know who might adopt their research*”. Participants expressed frustration regarding the absence of structured mechanisms to facilitate the transfer of research outputs into industry applications. D7 remarked, “*There are great student fashion design projects, but it is slow and difficult to turn them into sellable products. We need better ways to bring our innovations to market*”.

This is consistent with studies indicating that weak connections between academic institutions and industry hinder knowledge transfer and innovation. The literature suggests that insufficient collaboration mechanisms between Saudi universities and the corporate sector limit students’ exposure to industry trends and market demands (Aloqayli & Alqahtani, 2019; Iqbal & Zenchenkov, 2014; Iqbal, 2020). Research also emphasises that strengthening the culture of entrepreneurship within academic institutions and aligning research with market needs is crucial for driving technological progress (Alhuthali & Sayed, 2022; Ghulam & Mousa, 2019). Numerous studies have highlighted that fostering industry-academia collaborations enhances innovation, particularly in fast-evolving sectors such as fashion (Bridgstock, 2013).

Additionally, the interview findings underscore the inadequate support and recognition for fashion researchers, which hinders their ability to conduct impactful research, such as inadequate supervisory expertise, the absence of dedicated research hubs, and the absence of IP. These concerns restrict students' ability to commercialise their innovations and develop impactful research in the fashion industry. D3 explained, *"As a fashion researcher, I designed clothing suitable for dialysis patients but faced challenges gathering accurate data. I wish the university would support me with data collection"*.

Participants highlighted the need for assistance in selecting research topics relevant to the fashion industry in Saudi Arabia. Participant S6 explained, *"Right now, universities are the primary institutions supporting research in fashion, but there needs to be continuous support, including with choosing research topics relevant to the Saudi industry, such as innovations in textiles and design or fashion entrepreneurship"*. D4 confirmed the need for more diverse research topics: *"We need to expand research areas and research centres and encourage the development of more diverse master's and doctoral theses to advance knowledge in various fields"*. Participants also highlighted how a lack of supervisors' knowledge of innovation and entrepreneurship poses a significant obstacle to providing innovative research that can benefit the industry. S3 stated, *"Students should be guided by experts in those fields instead of relying solely on supervisors whose expertise may be limited to traditional fashion design and sewing"*.

These findings are supported by the literature on entrepreneurial education, which emphasises the crucial role of industry partnerships in commercialising research (Lehmann et al., 2022; Spigel, 2017; Wilcox, 2015). Partnerships between universities and industries serve as key conduits for knowledge transfer, enabling the transformation of academic research into innovative commercial activities (Audretsch et al., 2012).

The "triple helix" model proposed by Etzkowitz and Leydesdorff (2000) advocates collaboration among universities, industries, and governments as a crucial factor in fostering innovation. This finding underscores the importance of collaboration, as seen in global examples where fashion education institutions have established strong connections with the industry and local communities. These connections offer students opportunities to collaborate on real-world projects, fostering innovation and addressing sociocultural issues (Agarwal, 2018; Atalay Onur, 2020; Khurana & Ryabchykov, 2017).

Significantly, global fashion education increasingly incorporates real-world case studies and industry partnerships to bridge gaps between academia and industry, focusing on sustainability, technology, ethics, and leadership (Alexander & Trigoni, 2019). Research confirms that fashion design education in Korea is becoming increasingly industry-focused,

with internships, industry-sponsored projects, and professional design critiques enhancing both student preparedness and faculty expertise (Choi, 2012; Choi, 2018). This approach aligns Korean fashion education more closely with the global industry standards (Yeonhee et al., 2011).

Similarly, fashion schools in the European Union, USA, and Japan emphasise international mobility programs, double diplomas, and international research projects, which broaden faculty and supervisors' expertise by exposing them to the best global practices (Yezhova et al., 2018).

Moreover, participants highlighted the absence of a dedicated research hub for arts and fashion as another primary concern. According to S4, *"There aren't any research hubs dedicated to arts and fashion where we can get consultations or network with experts from other schools or the industry"*. Participants also highlighted the need for better processes and practical training for commercialising IP. For example, D5 said, *"It is frustrating that the university does not have a clear process for turning our research into a business. We need more practical training on how to commercialise our ideas"*. In this regard, IP management within universities hinders their ability to fully capitalise on innovations. D9 said, *"I developed a design tailored for children suffering from rickets, but because I do not understand intellectual property rights, I could not proceed with production. I did not know how to protect my idea or navigate the complex process of bringing it to market"*.

These findings resonate with the literature that highlights the significance of research hubs in fostering collaboration, innovation, and commercialisation of new ideas (Dong et al., 2024; Sotirofski & Kraja, 2024). The literature also supports the notion that research centres serve as platforms for commercialising innovations in fashion-related research, such as sustainable fabrics and digital tools (Nyiringango, 2024). For instance, in the UK, the Centre for Fashion Enterprise conducted a feasibility study exploring the need for a high-end production hub, highlighting the importance of dedicated research centres in addressing educational and skills gaps in the fashion sector (Ingram et al., 2020; Malem, 2008).

These findings align with the literature, indicating that weak IP frameworks in Saudi higher education institutions hinder innovation and commercialisation. Similar challenges exist in India, where weak industry linkages and inadequate incentives slow the commercialisation of academic research (Patil & Sagar, 2024). Research hubs enhance collaboration, innovation, and commercialisation of ideas, with stronger IP policies and institutional support in countries like the United States and Kenya significantly improving the transition of research into market-ready products (Nyiringango, 2024; Sotirofski & Kraja, 2024).

Another key theme emerging from the interviews was the lack of diverse specialisations in fashion education in Saudi Arabia. S5 observed that *“Global fashion schools offer a wide range of specialisations in fashion education, including sustainable fashion, fashion technology, fashion entrepreneurship, textile science, luxury management, and digital fashion, to better equip students for diverse opportunities in the industry. However, in Saudi Arabia, we have fashion design”*. D9 added, *“There are limited areas of specialisation in fashion education that do not meet the evolving needs of the fashion industry. For example, in my business, I need a fashion designer with expertise in sustainability”*.

Participants also emphasised the importance of integrating entrepreneurial thinking and skills into the Saudi education system to foster a culture of innovation and creativity from an early age. S5 observed, *“Art projects and fashion designs focus on the execution and exhibition of pieces instead of innovating new designs or creating products that meet market demands”*.

Finally, the interview findings highlight the limited scope of research topics within Saudi fashion education. Current programs primarily focus on traditional areas such as fashion, jewellery, and textile design, with little emphasis on emerging fields such as digital fashion and sustainability (Ministry Of Education, 2020). Expanding these offerings would allow Saudi institutions to align more closely with the evolving industry trends and global best practices.

5.6.3.2 Promoting an Entrepreneurial Culture

Educational institutions are pivotal in fostering an entrepreneurial culture within the broader entrepreneurship ecosystem by providing various programs and initiatives that nurture entrepreneurial skills, mindsets, and innovations. However, the interviews revealed that many participants felt that these efforts were often lacking or insufficiently implemented in their educational institutions. D3 mentioned the absence of mentorship opportunities: *“I wish there were more opportunities to connect with experienced entrepreneurs who could guide us, fashion students; we had no real mentors”*. D6 stated, *“Though the university offered a course on entrepreneurship, it was very theoretical and did not provide any real-world experience or practical skills, which are crucial for starting a business”*. Likewise, networking opportunities were described as limited. As D8 put it, *“Networking events were rare, and when they happened, they did not connect us with the industry professionals we needed to meet”*.

Moreover, participants believe that more institutional support is needed to normalise entrepreneurship as a viable career path. D8 observed, *“In fashion, entrepreneurship is seen as a real career option, but there is no real push from the university to explore that path”*.

seriously". Finally, fostering a campus culture that celebrates entrepreneurial efforts, even in failure, by highlighting the learning experiences and growth is crucial. Participants believed that their universities did not encourage a risk-taking mindset. However, D9 pointed out, "*At university, failure is still seen as something bad. There is no focus on learning from mistakes, so most of us are afraid to take risks because we might lose grades*".

Higher education plays a critical role in shaping an environment in which students are equipped with essential competencies such as risk-taking, decision-making, creativity, and innovation, which are crucial for cultivating entrepreneurial ambition and creating business opportunities (Ncanywa & Dyantyi, 2022; Porkodi et al., 2023). These institutions adopt various strategies to encourage entrepreneurial thinking, including entrepreneurship-focused curricula that integrate theoretical knowledge with practical business skills to prepare students to confront real-world challenges (Gopinath & Sreekanth, 2024; Isenberg, 2010). Mentorship programs are also critical, providing students with access to experienced entrepreneurs and industry professionals who offer valuable guidance and support for business idea development (Al Ayyash, 2024 ; Yani et al., 2024). Networking events and partnerships with the business community allow students to connect with potential investors and collaborators, thereby expanding their professional networks (Eggleston & Rabb, 2020; Jorre, 2021). Moreover, although educational institutions have the potential to encourage students to pursue entrepreneurial careers, particularly in the fashion industry, participants agreed that more institutional support is needed to normalise entrepreneurship as a viable career path (Almahry et al., 2021; Boldureanu et al., 2020). Encouraging a mindset that views failure as a stepping stone to success helps students build resilience and remain motivated in their entrepreneurial pursuits (Breznitz & Zhang, 2021; Hahn et al., 2019)

Table 5.4: Summary of Thematic Findings of Entrepreneurial Education within Saudi FEE

Theme	Key Findings
1. Evolution of Fashion Education	- Since 2018, the establishment of fashion design programs at several universities has contributed to the growth of government-funded fashion education. - There are challenges such as the absence of fashion schools with international recognition, lengthy accreditation timelines, a shortage of faculty with global expertise, and outdated educational methods.
2. Role of Entrepreneurial Education	- Limited business incubators support for fashion startups in Saudi universities. - Absence of structured guidance on market entry and regulations. - General business courses are not tailored to the fashion industry. - Limited financial literacy among fashion students restricts business success. - Need for fashion-specific business courses, better financial training, and more industry engagement enhances entrepreneurial competence. - Lack of hands-on business training, limited industry exposure, and insufficient internship and mentoring programs.
3. Impact on EO	- Fashion students produce strong research but lack industry application. - Need for improved knowledge transfer between academia and industry. - No structured platform for industry-academia collaboration, lack of research hubs for fashion, and inadequate support for IP protection. - Participants report need for structured mentorship programs, limited availability of industry experts, and insufficient networking opportunities. - Need to normalised entrepreneurship in fashion education as a career path. - Need to foster a risk-taking culture and resilience to failure. - Lack of programs encouraging entrepreneurial mindsets. - There is fear of failure among fashion students due to academic penalties

5.7 Policy

Government policies play a crucial role in shaping the entrepreneurial landscape by facilitating or hindering the establishment of new businesses (Acs et al., 2018; Acs et al., 2014). The participants provided valuable insights into the complex policy and regulatory environments influencing MSMEs in fashion. Overall, they expressed a positive reception of the government's role in supporting entrepreneurial activities, including in the fashion industry. Their perspectives highlighted both opportunities and challenges, which were organised into three key themes and subthemes: (1) government support and Saudi Vision 2030; (2) policies and regulations, including subthemes of business registration, municipal approvals, labour,

logistics and infrastructure, and legal environment; and (3) the role of key government institutions.

5.7.1 Government Support and Vision 2030

Participants reported a marked increase in government support aimed at enhancing entrepreneurial activities across various industries, including the cultural sector, such as fashion. Participants mentioned specific initiatives, such as establishing business incubators, facilitating online business registration, and providing financial aid. As S6 noted, *“Since the launch of Vision 2030, entrepreneurship and the cultural and entertainment industries have grown significantly, with more events, business incubators, and project support services”*. S5 also observed, *“The government’s efforts to support entrepreneurship in general and in the fashion sector, specifically, are evident. There’s always a new government program to support small businesses, offering various financial incentives”*. D7 observed, *“Local cultural industries such as fashion have grown significantly since Vision 2030”*. The establishment of the Ministry of Culture and the Saudi Fashion Commission were viewed as pivotal steps in developing the fashion industry. Likewise, D8 reinforced this, stating, *“There has been noticeable improvement with the establishment of the Ministry of Culture and the Saudi Fashion Commission, showing a growing focus on fashion”*. This aligns with broader research indicating that Saudi government support plays a crucial role in fostering entrepreneurship and sustainable development, particularly within the framework of Vision 2030 (Alshuwaikhat & Mohammed, 2017; Moshashai et al., 2020). Vision 2030 was designed to reduce Saudi Arabia's reliance on oil by promoting non-oil industries and entrepreneurship. Government support, including policy changes and financial incentives, is crucial for this transition as it encourages investment and innovation in diverse sectors (Alam et al., 2023; Alarjani et al., 2021; Ayad et al., 2022).

Despite recent improvements, participants emphasised the need for greater government support, particularly in defining the fashion industry as an independent sector. Establishing fashion as a distinct industry would improve the accuracy of market data and enable more tailored policy development for the industry. S3 highlighted this issue, stating, *“A Ministry of Commerce report included 27,000 businesses unrelated to fashion, such as laundry and carpet businesses, making it hard to assess the market accurately”*. This misclassification reflects the broader challenges faced by emerging fashion industries worldwide. For example, similar issues with market data accuracy and policy alignment have been reported in Canada and

Finland, where the lack of industry-specific classification has hindered growth and strategic planning (Brydges & Hracs, 2019; Dal et al., 2016).

Participants also called for more comprehensive support for fashion entrepreneurs. D6 summarised this need, explaining, *“The support offered to fashion projects is vital but insufficient to meet all needs and achieve success”*. Research highlights that targeted government support, such as financial aid, business mentorship, and market entry assistance, significantly improves the survival rate and competitiveness of creative enterprises (Becksky-Nagy & Fazekas, 2024). A recent study from Malaysia showed that funding-related government support had a significantly positive impact on startup success, while policy-related support exhibited mixed effects, suggesting that tailored, industry-specific policy development is essential for sustainable growth (Azhar et al., 2024).

Moreover, the participants stressed the importance of clear and transparent regulations to protect business interests and ensure fair competition. S1 explained, *“Strict, transparent regulations for property owners and business operations would prevent financial losses and make home-based businesses more viable and professional”*. Research highlights that clear business regulations improve market confidence and reduce financial risks for small businesses, fostering a healthier business environment (Ibyzhanova et al., 2023; Usmany, 2024). Another challenge identified was the lack of accessible and structured information on business registration and permit requirements. D4 suggested that *“A solution could be a course or detailed information on the Commerce Department’s website that provides a comprehensive database on all required permits and steps of the application before issuing permits or starting a business”*. Studies have found that structured guidance and simplified bureaucratic processes significantly enhance market entry and reduce the administrative burdens for new businesses (Azhar et al., 2024; Pan & Huang, 2022).

5.7.2 Policies and Regulations

The interviews revealed that Saudi government policies have significantly improved the business registration process, reduced administrative complexity, and encouraged entrepreneurship. Most participants reported that the introduction of online registration systems, reduced bureaucratic steps, and centralised platforms made starting a business easier. These findings align with broader research showing that simplified registration procedures and efficient government support improve business formation rates and entrepreneurial growth (Antoine, 2021; Salami et al., 2023).

The participants highlighted the benefits of these streamlined systems. D4 stated, *“I can now issue work permits, like those for fashion shows, directly from the Ministry of Culture's website, as the process has been streamlined and is available online”*. Similarly, S2 described the speed and efficiency of the online platform: *“I can now obtain a commercial register in just two minutes, significantly simplifying the setup process of obtaining official documents and starting a business”*. Research supports these observations, indicating that modernised business registration platforms, including digital systems and centralised platforms, have increased business formation rates and overall economic activity by reducing administrative burdens (Domenico et al., 2024). A comparative study of 20 countries found that simplified business registration, flexible labour regulations, and centralised platforms increase business formation rates and entrepreneurial success (Usmany, 2024).

However, challenges remain in this regard. D2 noted that *“While the process is simpler, acquiring official documentation from the Ministry of Commerce is still slow: Registering a brand with the Ministry of Commerce is not complicated, but it is time-consuming”*. This reflects broader evidence showing that while digital platforms improve registration efficiency, delays in documentation processing can undermine such gains. Research highlights that reducing the time and cost of official registration processes is critical for fostering business growth (Pratama, 2023). The findings demonstrate that Saudi Arabia's move toward streamlined business registration is consistent with the global pattern of improving the ease of doing business. Research confirms that simplified registration procedures increase business formation rates and entrepreneurial activity, particularly when they are integrated with broader economic reforms (Antoine, 2021; Salami et al., 2023).

5.7.2.1 Municipal Approvals

Municipal approvals are crucial for establishing physical stores in Saudi Arabia, with requirements covering signage, infrastructure, and safety. Participants highlighted significant regulatory challenges, noting that the complexity and inconsistency of these regulations create barriers to entry and operational inefficiencies in the fashion industry. For instance, D5 described the process of meeting municipal requirements for rental spaces and obtaining the necessary licences as *“lengthy and unpredictable”*. This sentiment was echoed by D8, who lamented the lack of clarity from the municipal authorities. D2 noted, *“There have been instances where municipal inspections for minor issues, including inoperative lighting or a slightly cracked tile, resulted in fines and led to the closure of businesses”*. D8 shared a personal experience that underscores these inconsistencies: *“After receiving initial approval to*

convert a shop in Mecca's residential area into a workshop, I faced a sudden regulation change that banned such conversions, forcing my shop to close”.

The challenges faced by small businesses due to regulatory burdens and their impact on compliance costs are well documented and mirrored globally (Arshed et al., 2014). For example, In the Indonesian fashion industry, complex bureaucratic licencing requirements significantly hinder entrepreneurs by increasing the costs and time required to start a business (Sapar, 2024). Research in Peru and Brazil has shown that excessive inspections and regulatory challenges affect business operations (Barrón et al., 2020; Prado et al., 2024). In contrast, some regions have successfully integrated municipal involvement to support the fashion industry's growth. Dunedin, New Zealand, utilised city council initiatives for rebranding and cluster development (Thompson-Fawcett, 2007), while Swedish municipalities focused on promoting sustainable fashion consumption (Thidell et al., 2019). The Swedish city of Gothenburg has experienced rapid organisational changes through the continuous adoption of management fashions over the past decade (Brorström & Norbäck, 2024). In South Africa, a local municipality intervened in the fashion industry as part of the creative industries sector, using a special-purpose vehicle model for industrial development (Pather, 2015). These examples highlight the potential for municipalities to play a constructive role in fashion industry development through initiatives ranging from economic development to sustainability.

The interviews underscored the need for more straightforward processes and transparency. Participants criticised the variability and inconsistency of regulations across cities, which complicated compliance. S1 pointed out, *‘There is no standardised platform or official resource to educate designers about all the laws and regulations related to starting a project’*. This reflects the findings of Alasiri et al (2025) and Sarabdeen (2023), who emphasise the need for standardised platforms for regulatory guidance because of the unpredictability of municipal regulations.

Furthermore, participants discussed diversifying fashion business licences to include options for workshops or studios, reflecting the industry's evolving needs. D2 suggested, *“Currently, the options are limited to opening a retail store or a tailor shop. However, in the fashion industry, we need workshop or studio licences, which allow us to showcase products and have a tailoring business in the same establishment”*. Research supports targeted licencing policies tailored to specific business models that can promote growth and innovation (Devas & Kelly, 2001; Kazungu et al., 2013).

Participants also stressed the importance of allowing fashion workshops and studios in residential areas to support designers with limited resources to create sustainable fashion. D7

noted, *“Being able to open licensed workshops in their homes would be a game-changer for many designers facing financial constraints”*. D9 observed during trips to Turkey and Egypt that home-based craft and fashion workshops could help preserve the cultural heritage and boost the economy. Studies show that home-based businesses can reduce startup costs and help informal entrepreneurs formalise and flexible zoning laws and mixed-use permits can foster local creativity and economic revitalisation (Alzamel, 2024; Rajih & Sufyani, 2024).

5.7.2.2 Labour and Workforce Policies

Participants identified labour policies, particularly the Saudization policy, as critical factors impacting their businesses. The Saudization policy, enforced by the Ministry of Commerce, mandates the employment of Saudis in MSMEs, and the Ministry of Labor and Human Resources funds a part of their salaries. Although the policy aims to increase the employment of Saudi citizens across various industries, it presents unique challenges for the fashion industry in particular.

Participants highlighted the difficulty in finding and training Saudi employees with the necessary skills in the fashion industry. S3 noted that *“The current labour and recruitment regulations limit designers’ ability to bring in the necessary skills to develop their projects”*. Similarly, D1 argued that *“Forcing fashion businesses to hire Saudis is unsuitable for the fashion sector because there’s a lack of needed skills”*. This aligns with a study on the hotel industry, which found that Saudization has been more successful in the public sector than in the private sector, where businesses face difficulties in finding qualified Saudi workers and adapting to new (Azhar et al., 2018). This study emphasises the need for industry-specific policies to address skills mismatches and recruitment challenges.

Participants stressed that training Saudi employees should be a collaborative effort with educational institutions, rather than being left to the discretion of business owners. As D1 explained, *“It is easier to find skilled Saudi workers in other industries. For the fashion industry, if we have to hire Saudi workers, it would be better if institutes trained them instead of us. I am just starting out and need to focus on production now. Training workers myself could slow down both my work and production”*. This sentiment is echoed in research on the banking sector, where successful Saudization has been linked to providing industry-specific training and revising working hours to attract and retain Saudi talent (Azhar et al., 2016).

Another key recommendation is to reduce the fees for hiring foreign workers. Participants argued that current labour importation policies impose significant financial

burdens on entrepreneurs. D2 suggested that *“If the fees and regulations were reduced, then it would be much easier for entrepreneurs”*. This concern is consistent with Abdulkarim and Khalid’s (2018) study Abdulkarim and Khalid (2018), which showed that strict Saudization quotas and high compliance costs reduced productivity and increased firms’ exit rates in some industries. This suggests that relaxing some financial and administrative barriers could improve business sustainability while still supporting Saudization goals.

5.7.2.3 Logistics and Infrastructure

The Saudi government plays a crucial role in developing transportation infrastructure, including roads, ports, and airports, which facilitates the movement of goods within and outside the country, reduces transportation costs, enhances supply chain efficiency, and improves the ability of fashion businesses to expand and grow (Lopez-Ruiz et al., 2020). Participants emphasised the significant impact of logistics policies on the fashion industry’s success, particularly the challenges related to import/export regulations, customs procedures, and transportation.

Complex import regulations hinder fashion entrepreneurs’ access to essential materials, causing delays, increasing costs, and reducing market competitiveness. D5 explained, *“I’ve ordered handmade embroidered pieces from India, but the delay in customs clearance delayed the collection’s production before Ramadan”*. D3 explained how taxes on imported sustainable products increase costs: *“We rely on sustainable fabrics, which I import from China. However, taxes are problematic. Even if we try to import from China to reduce the cost of the fabric, the taxes force us to increase the final product price”*. High taxes and customs duties increase the prices of raw materials and final products, further reducing their competitiveness.

Streamlining customs procedures, improving systems, and reducing clearance times can enhance operational efficiency and reduce costs (Almalki & Alkahtani, 2022). Inadequate transportation infrastructure, particularly in remote areas, is a barrier to business expansion. Participant D9 from northern Saudi Arabia noted, *“I face difficulties in delivering my products to my customers in major cities such as Riyadh or Jeddah because of the weak transportation infrastructure in my village. This can lead to delays in delivery and increased transportation costs”*. Saudi Arabia’s efforts to improve its logistics infrastructure, including regional logistics hubs and improved road networks, are aimed at addressing these (Aljabhan & Elnahrawy, 2024; Almalki & Alkahtani, 2022).

Fashion businesses aiming to expand internationally face difficulties in terms of shipping costs and tracking shipments. D4 stated, *“Many of our followers on social media request*

delivery to the Gulf countries. I know it is an opportunity for me, but unfortunately, there are difficulties tracking shipments and delays in international shipping". Partnering with established logistics firms has helped businesses overcome these challenges. D6 stated, *"I subscribed to an annual logistics contract with DHL, a global logistics company, and benefited from their advanced solutions in managing shipments, which reduced delays and improved the experience for my customers, which enabled me to ship to Gulf countries"*.

However, long registration processes and complex compliance requirements remain barriers for some businesses (Omran, 2024). Participants emphasised the importance of government policies that reduce logistics costs, such as subsidies and tax incentives for shipping. D4 noted, *"Customers always complain about the delivery cost or the delay in receiving shipments, but when I partnered with a better company, my sales increased. However, I had to raise the prices of the final products to cover the shipping costs"*. Government policies aimed at facilitating international shipping agreements and improving customs procedures can enhance the global reach of Saudi fashion brands. Improving Saudi Arabia's ranking in the World Bank's Logistics Performance Index (LPI) by investing in logistics infrastructure could increase the trade volume and reduce shipping costs (Almalki & Alkahtani, 2022). S4 highlighted the competitive importance of fast and reliable delivery services: *"Fashion designers should contract with reliable logistics companies because 70% of online customers refuse to purchase again after a bad delivery experience. Customers choose the designer who offers the fastest delivery, especially during peak seasons such as Eid"*. Saudi Arabia's Vision 2030 includes substantial investments in logistics infrastructure and global trade networks, which are expected to improve supply chain efficiency and market access for fashion companies (Jaziri et al., 2020).

5.7.3 *The impact of policy on EO*

Policies can significantly impact the EO levels of MSMEs. In Saudi Arabia, research has confirmed that effective policy frameworks can enhance the EO of MSMEs, leading to improved innovation, competitive advantage, and market performance (Aloulou, 2024). The interviews highlighted the initiatives introduced by the Saudi Fashion Commission to support innovation in the fashion sector. These include training programs for designers and enabling participation in local and international events, which have helped fashion MSMEs present their products and brands more innovatively. D6 stated, *"The Fashion Commission's support during the '100 Saudi Brands' exhibition allowed me to refine my brand and present*

it with innovation and design style. The training sessions helped me anticipate market trends and develop designs that met international standards”.

Fashion designers in Saudi Arabia can protect their work through IP rights, including industrial design, copyright, trademark, and patent rights. According to Saudi Authority for Intellectual Property (2024), originality is the key requirement for protecting fashion designs under copyright or industrial design laws, which helps designers maintain a competitive edge.

However, securing IP protection in the Saudi fashion industry presents several challenges for several reasons. Designers have raised concerns about the lack of specific IP laws tailored to fashion, leading to increased legal uncertainty. D7 noted, *“IP laws are easier to apply in other industries. We need clear laws that distinguish between inspiration and plagiarism”*. D2 also mentioned the difficulty in finding relevant information: *“I tried to understand IP laws on the authority’s website, but there’s not enough guidance for fashion designs”*. The complexity and cost of registering IP discourage many designers from seeking protection against IP infringement. Only two participants reported successfully securing trademarks or patents, while the others struggled with a lengthy registration process. Participant D8 explained: *“Protecting my brand identity took 4 months, and I wasn’t able to register the trademark or the designs”*.

Research shows that simplifying IP registration encourages MSMEs to invest more in innovation. Aloulou (2024) found that streamlined processes improve Saudi firms’ ability to introduce new products. Similarly, Rusadi et al. (2023) highlighted the importance of strong IP frameworks in the digital era, where designs can be easily copied. Without effective protection, fashion MSMEs face greater risks of intellectual property theft and market loss.

Wilson (2020) argues that broader market policies, such as fair competition laws and easier market entry, provide greater benefits than narrowly targeted policies. Strengthening IP protections would not only protect creative output but also boost industry-wide innovation. Naim (2021) suggests that although GCC countries have largely aligned their IP laws with international standards, effective enforcement is still lacking because of local cultural and legal challenges.

However, the fashion industry in GCC countries faces challenges in effectively enforcing IP laws. Issues such as the persistence of counterfeit markets, limited awareness among local designers and businesses of their IP rights, and the complexities of cross-border enforcement hinder the full potential of IP protection. Moreover, the cost of securing IP rights remains a barrier for small enterprises in the industry sector. León et al. (2016) found that high

fees and legal complexities discourage MSMEs from seeking protection, especially in developing regions.

Strengthening regional cooperation and enhancing legal frameworks, especially in trade dress and copyright protection, are crucial for securing a sustainable and innovative fashion ecosystem (Mills, 2009; Saardchom, 2017).

Table 5.5: Summary of Thematic Findings of Policy Domain within Saudi FEE

Themes	Findings
Government Support and Vision 2030	- Initiatives under Vision 2030 have significantly increased government support for businesses, especially in the fashion sector. - It is recommended that the fashion industry be recognised as a distinct sector to improve how policies are made and to enhance the accuracy of data about the market.
Policies and Regulations	- Online registering for new businesses has been improved, making it easier to start fashion businesses.
Municipal Approvals	- Fashion entrepreneurs face significant challenges due to complex and inconsistent municipal regulations, particularly when trying to obtain approvals needed to open and operate stores. - There is a need for establish new for business licenses that suit better different type of stories in fashion industry.
Labour and Workforce Policies	- The Saudization policy creates difficulties in finding locally skilled employees for the fashion industry. - More cooperation with educational institutions to train employees and simpler policies for hiring foreign workers are suggested to alleviate these challenges.
Logistics and Infrastructure	- Poor infrastructure and complicated regulations related to imports and exports severely limit the efficiency and competitiveness of fashion businesses. - Improved customs procedures and better transportation infrastructure are advocated to support the expansion of businesses and reduce operational costs.
Impact of Policy on EO	- Strong and effective policies are essential for enhancing the entrepreneurial orientation of fashion MSMEs, leading to superior innovation and improved market performance. - Clearer laws on intellectual property and more straightforward registration processes are needed to encourage investments in innovation within the fashion industry.

5.8 Market

This section explores the current state of Market in the FEE, focusing on two key themes: (1) Customer Preferences, and (2) Market Access.

5.8.1 Customer Preferences

The findings reveal both positive and negative consumer attitudes toward local fashion brands. There has been a noticeable shift in Saudi consumer preferences toward local designs, driven by a growing sense of cultural pride and an increased appreciation for local craftsmanship. Participant S4 stated, *“Fashion has become a way of expressing our history and cultural diversity. It’s more than just clothing; it’s a statement of identity”*. Similarly, D6 reinforced this perspective, noting, *“Saudi consumers have become more aware of Saudi traditional garments, which has increased their preference for Saudi-made fashion”*. Participant S5 noted, *“There’s growing appreciation for designs that reflect Saudi culture and identity; it’s becoming a point of pride”*. This suggests that local brands can leverage cultural authenticity as a key differentiator in the competitive fashion industry.

This strong and consistent demand for traditional and modest fashion presents significant opportunities for local fashion MSMEs to meet this growing need. While global brands offer contemporary fashion options, a clear gap remains in the availability of locally inspired and traditional clothing. Participant S6 emphasised this by stating, *“Saudi customers tend to prefer local brands because they specialise in traditional attire like abayas, jalabiyas, and men’s thobes garments that international brands typically don’t offer”*. This trend underscores the competitive advantage of local brands in catering to the distinct cultural and religious preferences of Saudi consumers. Saudi consumer behaviour exhibits a strong inclination toward fashion. Participant S6 noted, *“Saudis love to refresh their wardrobes and enjoy having special outfits for every occasion”*.

Regional variations also shape Saudi consumer preferences, indicating the need for localised product strategies in marketing. For instance, Participant S1 stated, *“In Qassim, people prefer traditional black abayas, so we only make subtle adjustments like adding minimal prints. Coloured abayas are less common here”*. This highlights the importance of understanding regional differences and tailoring product offerings to meet localised demand. Despite the growing demand for local brands, pricing is a significant challenge. Some participants indicated that local products are often perceived as expensive, which can deter potential buyers from purchasing them. Participant D1 explained, *“Honestly, there’s a*

perception issue; some people still see Saudi products as overpriced, which makes them hesitant to buy”.

While Saudi consumers are hesitant to pay premium prices for local brands, they are more accepting of high prices from established global brands, such as Louis Vuitton or Valentino. As Participant D7 observed, *“When customers see a dress or bag made in Saudi Arabia, they often feel the price is too high compared to global brands. But when it’s a Louis Vuitton or Valentino piece, they do not question the price and are willing to pay a premium”.* However, this perception appears to be shifting as the product quality improves. D1 added, *“Lately, we have seen significant improvements in quality and materials. Many customers now prefer Saudi brands over international ones when the quality surpasses that of well-known global brands”.* This reflects an increasing willingness among Saudi consumers to purchase local products when they perceive their value to be equivalent to or greater than that of global brands.

These findings align with the existing literature, highlighting a growing preference for Saudi-made fashion driven by cultural pride and an appreciation for local craftsmanship (Almatrudi et al., 2023). Studies suggest that local brands can enhance their competitiveness by improving product quality, increasing consumer satisfaction, and strengthening their brand image (Klabi, 2020; Mkedder et al., 2021). Additionally, leveraging national identity through targeted marketing strategies aligned with consumer ethnocentrism can significantly boost the success of local brands (Šapiæ, 2018). The dominance of foreign brands underscores the need for ongoing improvements in local product quality to maintain consumer trust and foster loyalty (Alsulami, 2018; Šapiæ, 2018). This focus on local identity reflects the unique dynamics of the Saudi market, where cultural conformity influences fashion choices more than individual expressions.

In contrast, Western markets, including Europe and North America, tend to emphasise personal expression over cultural conformity, with fashion trends shaped more by celebrity culture, media, and seasonal changes than by social or religious expectations (Okonkwo, 2016). However, international brands remain highly valued in Saudi Arabia for their perceived quality and prestige, with Saudi youth particularly favouring U.S. apparel brands (Abalkhail, 2018a; Almatrudi et al., 2023; Alsulami, 2018; Šapiæ, 2018). Luxury fashion continues to serve as a marker of social status, driven by emotional value, perceived quality, family influence, and cultural expectations (Algumzi, 2022; Aqeel, 2016).

5.8.2 Market Access

Participants identified access to distribution channels as one of the most significant strategic barriers to market entry, particularly in the competitive fashion industry. Social media, especially Instagram, has emerged as a dominant sales channel, providing a direct connection with customers and reducing operational costs. All participants used social media, particularly Instagram, as their primary channel to reach customers and sell their products online. Some participants also reported having an online website store, whereas only one participant mentioned owning a physical store. These findings are consistent with the survey results of this study, which indicate that 71% of participants use social media as their primary sales channel, while 36% have their own website. In contrast, only 19% of participants reported having a physical store, and 17.5% relied on multi-brand stores for sales. A smaller proportion, 3.5%, used an online multi-brand website (e.g. Amazon and Etsy) as sales channels.

Participants highlighted the financial and operational challenges associated with physical stores, including rent, interior design, municipal fines, and complexities involved in securing a suitable location and obtaining the necessary licences. These challenges require considerable time and effort, contributing to a preference for online shopping. Participant D1 explicitly stated that *“Online sales are preferred to avoid the expenses associated with rent and the complexities of obtaining licenses”*. Similarly, D3 elaborated, *“Opening a physical store required navigating numerous requirements. The financial burden of store decoration and operational challenges”*. High rental costs and complex licencing procedures create significant barriers to opening physical stores. Participant S1 shared that *“Many designers are forced to operate from home or rely on social media to reach consumers”*.

Despite the relatively low percentage of physical store ownership, participants emphasised the importance of maintaining a physical store in the fashion sector. Physical stores remain critical because customer interaction with the product, seeing, touching, and trying it, plays a significant role in purchasing decisions. This finding aligns with research indicates that Korean and Chinese senior consumers prefer offline channels for the emotional connection they provide with brands, whereas younger consumers exhibit a stronger affinity for online channels (Lee & Yu, 2023).

Participants highlighted their growing reliance on social media and direct-to-consumer models. S4 explained, *“Instagram is where our customers discover us. We post a new collection, and within minutes we see enquiries and direct messages”*. This insight reflects broader trends in the fashion industry, where e-commerce is viewed as a key driver of growth and efficiency (Ratchford et al., 2022; Silva et al., 2020). Studies have shown that improving

multi-brand channel distribution in fashion retailing increases efficiency. For instance, improvements in supply flow, layout changes, and the use of computer applications have increased productivity and reduced cycle time, space occupation, and labour costs (Silva et al., 2018).

Several also emphasised the importance of combining online and physical store presence to meet diverse customer preferences. Participant D9 explained, *“Some customers will follow us on Instagram, but they won’t buy until they’ve tried the dress on in the store”*. This hybrid approach illustrates the growing adoption of multi-channel and omni-channel models in the fashion industry, where brands seek to leverage the strengths of both online and offline environments to drive customer engagement and loyalty (Bai et al., 2024; Kumar Percherla, 2021). Similarly, small-scale fashion retailers have been shown to benefit from omnichannel strategies, which increase technical efficiency by 17% compared to purely online channels (Ryu et al., 2019). Despite the advantages of online sales, the participants also reported challenges associated with e-commerce. One recurring problem is product returns. D2, *“We had a case where a customer ordered a dress for a wedding, wore it, and then tried to return it, claiming it did not fit”*. Delivery challenges were also noted, reflecting broader logistical issues in the fashion industry. D5 commented, *“We have had orders lost in transit. Customers get frustrated”*. E-commerce adoption for ready-made clothing faces challenges due to concerns over sizing accuracy, material quality, and delivery reliability (Alzahrani & Al-Dabbagh, 2020).

Social media platforms such as Instagram and Facebook have had a significant impact on fashion shopping behaviour in Saudi Arabia by increasing consumer engagement and brand visibility (Trad & Al Dabbagh, 2020). Despite the growing influence of social media, perceived risks related to payment security, product quality, and return policies limit wider adoption of online shopping (Abood, 2018). Targeted social media campaigns have proven effective in enhancing customer loyalty and increasing brand value for fashion brands operating in the Saudi market (Alatawy, 2022).

While participants expressed a preference for relying on their own sales channels, existing research suggests that multibrand channels could enhance market reach and operational efficiency. Participants explained why some companies preferred to manage their own sales channels. D8 stated, *“Owning our own store gives our brand strength and independence, enhancing customer trust”*. D6 mentioned, *“By selling in our own store, we avoid direct comparisons with other brands, which can distract customers”*. Research suggests that while multichannel strategies increase customer attraction, brand awareness, and

profitability, they may also reduce profit growth compared to single-channel models (Nguyen, 2019).

The shift from physical stores to e-commerce and social commerce reflects broader changes in the global fashion sector. Luxury and fashion brands increasingly rely on B2B, B2C, and direct-to-consumer models (Kumar Percherla, 2021). Vertical integration increases brand loyalty, especially for weaker brands, by providing a better customer experience and product access (Schramm-Klein et al., 2008). Furthermore, distribution patterns vary by region. For example, in Colombia, smaller sizes are in higher demand in central areas, highlighting the importance of adapting distribution strategies to local market conditions (Contreras-Velásquez et al., 2024).

These findings highlight the strategic importance of integrating online and offline sales channels to balance cost efficiency, customer reach and brand positioning. Fashion brands that leverage multichannel and hybrid strategies are well-positioned to enhance customer engagement and achieve long-term growth. Effective distribution strategies are critical for expanding consumer reach, improving brand value, and achieving sustainable business performance. Multichannel distribution allows fashion brands to cater to diverse consumer preferences through physical stores, online platforms, and mobile applications, thereby enhancing convenience and accessibility (Ko et al., 2017; Ryu et al., 2019).

Table 5.6: Summary of Thematic Findings of Market Domain within Saudi FEE

Theme	Main Findings
Customer Preferences	- Growing preference for local designs driven by cultural pride and appreciation for Saudi craftsmanship. - Strong demand for traditional attire (e.g., abayas, jalabiyas, thobes), which creates a competitive advantage for local brands. - Preferences vary by region (e.g., black abayas preferred in Qassim), highlighting the need for localised product strategies. - Saudi products are perceived as expensive compared to international brands, but improving product quality is shifting perceptions.
Market Access	- Instagram is the primary sales channel for local fashion brands, providing direct customer engagement and reducing costs. - High rental costs, complex licensing processes, and operational difficulties discourage physical store ownership. - Combining online and offline channels helps increase customer engagement and trust. - Issues with product returns, delivery reliability, and payment security remain significant barriers to broader adoption. - Participants believe Selling through own stores enhances brand identity and customer loyalty, avoiding direct comparisons with competitors.

5.9 Application of RBV to EO of Saudi fashion MSMEs

RBV asserts that firms attain and sustain competitive advantages by strategically utilising resources that are Valuable, Rare, Inimitable, and Non-substitutable (Barney, 1991). These resources span both tangible assets, such as capital and infrastructure, and intangible assets, such as innovation, proactiveness, and risk-taking. Together, these traits constitute EO, which has been widely recognised as a core intangible resource within the RBV framework (Lumpkin & Dess, 2001; Lumpkin & Dess, 2015).

Evidence from this study supports the notion that differences in entrepreneurial performance among Saudi fashion MSMEs are shaped not only by access to FEE resources such as funding, training, and institutional programs but also by the entrepreneur's capacity to strategically convert these assets into competitive advantage. This perspective aligns with RBV's emphasis on resource deployment as a form of strategic behaviour (Wernerfelt, 1984), and is consistent with scholarship recognising EO as a critical mediator of resource effectiveness (Dionysus & Arifin, 2020; Ferreira et al., 2011).

The qualitative findings revealed distinct connections between specific FEE resources namely policy incentives, training support, and networking programs and the three dimensions of EO: risk-taking, innovativeness, and proactiveness. These interactions illustrate how external enablers are internalised as strategic capabilities within the RBV logic.

5.9.1 Policy Incentives and Risk-Taking

Although several public support mechanisms are available in the Saudi fashion ecosystem, such as financial grants and institutional partnerships, many entrepreneurs face constraints including bureaucratic inefficiencies, centralised service delivery, and limited transparency. Despite these obstacles, those with stronger EO particularly in risk-taking demonstrated a capacity to strategically navigate and utilise policy incentives. For instance, the survey in this study revealed that approximately 76% of fashion MSMEs in Saudi Arabia offered similar products, such as abayas and evening gowns. However, not many fashion MSMEs consider innovation. Entrepreneurs who demonstrate real innovation have received better support. For instance, D5 successfully aligned her sustainability-focused jewellery line with an environmental grant scheme: *"I knew my idea of using sustainable materials would catch the Fashion Commission's attention"*. Similarly, D3 leveraged institutional support to produce functional clothing for dialysis patients, resulting in a government-assisted manufacturing partnership. D3 stated, *"My project focuses on designing apparel for dialysis patients. I introduced this initiative to the fashion association, and they*

assisted in facilitating a partnership with a specialized manufacturer to produce the garments”.

These examples mirror prior findings that policy incentives play a crucial role in reducing the perceived risks of entrepreneurial ventures, thereby encouraging bold, opportunity-seeking behaviours (Rihayana et al., 2024). In line with RBV, such incentives are considered valuable and difficult to replicate particularly in nascent sectors thereby conferring an advantage when strategically harnessed (Sarsiti, 2024). This accords with research suggesting that risk-taking enables entrepreneurs to reconfigure and enhance their human capital and operational base, especially in resource-constrained contexts (Stone, 2016; Thai & Mai, 2023; Sidek et al., 2019). This example reflects the findings of Khan et al. (2020), who found that innovative SMEs are better positioned to capitalise on institutional resources such as funding and partnerships, further highlighting the RBV's assertion that intangible capabilities drive long-term advantage.

5.9.2 Training Support and Innovativeness

The findings underscore the importance of proactivity in navigating and capitalising on available support structures. For instance, D5 travelled abroad to attend an international training program organised by the Saudi Fashion Commission in partnership with global institutions. This initiative resulted in new design approaches, a new network, and an overhaul of her brand identity. D5 stated, *“It was a transformation for my brand. I learnt new design technology, and I came back with new partnerships and design directions that completely reshaped my project”*. Moreover, D6 faced the challenge of finding skilled local labour in jewellery-making, as such competencies were not taught in local universities. In response, she developed a hybrid solution involving in-house training programs and international talent acquisition to ensure quality and sustainability in production. D6 stated, *“It was hard to find someone who understood gilding techniques for jewellery, so I brought in an expert from abroad and trained two Saudis with him. It was an expensive move, but it helped me break into the jewellery market”*. In contrast, entrepreneurs such as participants D1 and D2 were forced to shut down their ventures due to their inability to recruit skilled tailors and embroiderers, highlighting the detrimental effect of weak entrepreneurial orientation and lack of proactive problem-solving.

This aligns with Stone (2016), who emphasised that calculated risk-taking enables entrepreneurs to restructure and reconfigure human capital resources, unlocking access to high-value competencies that are not readily available in the local market. This dichotomy supports RBV findings emphasising resourcefulness and dynamic capability as essential to performance

(Zahra, 2021). This aligns with research showing that calculated risk-taking enhances resource structuring and access to niche competencies (Thai & Mai, 2023), thereby boosting competitive performance (Sidek et al., 2019). These cases affirm the role of training support as a form of intangible resource acquisition. Training enhances not only technical proficiency but also a firm's absorptive capacity its ability to internalise, adapt, and apply new knowledge (Jalali, Jaafar & Thurasamy, 2013; Arifin & Dewi, 2020). From an RBV standpoint, such capabilities are considered rare and difficult to imitate, serving as critical drivers of innovation in fashion MSMEs.

5.9.3 *Networking Programs and Proactiveness*

Entrepreneurs with high proactiveness were particularly adept at using ecosystem-based networks to anticipate trends and initiate strategic collaborations. D6's proactive decision to import expertise in gilding a skill not locally available enabled her to enter a niche market. In contrast, D1 and D2, who lacked such foresight and did not engage in external networks, were ultimately forced to cease operations due to workforce constraints. These findings support prior research showing that relational and informational capital gained through incubators, accelerator programs, and trade events serves as a valuable asset within RBV logic (Mahmoud et al., 2023). Such networks amplify a firm's responsiveness to emerging opportunities and reinforce its strategic agility (Stone, 2016).

5.9.4 *Strategic Use of Market Intelligence*

One major constraint identified by participants was the lack of local market intelligence. Many designers defaulted to imitation due to limited data access. However, entrepreneurs who invested in trend forecasting tools reported enhanced originality and market alignment. As D4 observed: *"When we invest in subscriptions to global fashion forecasting websites, we always create unique products that people actually want."*

This finding aligns with De Nuccio (2025), who argued that strategic use of data enhances value proposition design and market differentiation. It is further supported by RBV research linking market-driving capabilities to long-term performance gains (Khan et al., 2022). The integration of RBV with the Saudi fashion entrepreneurship ecosystem affirms that EO is not merely a firm-level trait, but a strategic mechanism for transforming external supports into performance-enhancing resources. Entrepreneurs exhibiting higher levels of innovativeness, proactiveness, and risk-taking consistently demonstrated a stronger ability to mobilise FEE resources for competitive advantage. These findings reinforce the conceptual

coherence between EO and RBV in the context of a rapidly evolving, resource-scarce creative economy.

5.10 Chapter Summary

This chapter presents a thematic analysis of the FEE, aiming to understand how finance, support systems, culture, policy, market, and education impact EO and BP among Saudi fashion MSMEs. The study finds that although formal financial support exists, most fashion entrepreneurs depend on personal savings and family due to strict lending criteria and limited access to venture capital. Saudi culture promotes entrepreneurship as a mark of independence, and Vision 2030 has increased the acceptance of fashion entrepreneurship. However, deep-rooted stereotypes, such as fashion being a low-status, feminine field, persist. Limited infrastructure and sector-specific innovation hubs constrain EO, especially among rural and emerging designers. The Saudi fashion market is shaped by regional preferences and a growing demand for culturally rooted, modest, and contemporary designs. However, challenges include high service costs, limited access to quality subcontractors, and a lack of tailored marketing platforms for these services. Educational institutions rarely support the creation of startups or market entry. Curricula are often generic, lacking industry relevance and practical training components. Finally, the study found that the RBV is highly relevant to understanding the success of Saudi fashion entrepreneurs, as those with strong EO were significantly better at converting both tangible (e.g. funding and training) and intangible (e.g. knowledge, networks, and brand identity) resources into sustained competitive advantage, confirming that EO acts as a key strategic enabler within the RBV framework.

CHAPTER 6: CONCLUSION

6.1 Chapter overview

The overall aim of this research was to develop an effective FEE framework for Saudi Arabia. The proposed framework seeks to establish, promote, and reform FEE to enhance the EO and BP of fashion MSMEs. Study 1, employing a quantitative approach, tested broad hypotheses regarding the positive relationships between the components of FEE and both EO and BP:

H1: The Fashion Entrepreneurship Ecosystem positively influences Entrepreneurial Orientation.

H2: The Fashion Entrepreneurship Ecosystem positively impacts Business Performance.

H3: Entrepreneurial Orientation positively influences Business Performance.

H4: Entrepreneurial Orientation mediates the relationship between the Fashion Entrepreneurship Ecosystem and Business Performance.

Study 2 builds upon the results and addresses the limitations of Study 1. While the initial study confirmed the strength of the associations between the ecosystem components and performance outcomes, Study 2 utilised a qualitative approach to delve into the 'how' and 'why' behind these relationships. This phase explored the underlying mechanisms and contextual factors that either facilitate or hinder the theoretical linkages proposed in the conceptual model. A significant concern in Study 1 was that some findings were inconsistent with the existing literature, prompting further investigation in Study 2 to confirm, refine, or refute the hypothesised relationships. This is particularly pertinent in the context of the unique challenges and opportunities within the Saudi fashion industry, which may not align perfectly with broader global trends. Study 2 addresses the following two main research questions:

RQ4: What is the current landscape of the Saudi FEE?

RQ5: How does each domain of the Saudi FEE impact EO and BP?

6.2 Summary of Research Findings

6.2.1 *Quantitative study*

This study examines the impact of FEE on the EO and BP of fashion MSMEs in Saudi Arabia. This addresses a notable gap in empirical research on FEE, particularly in the Saudi context. The analysis for H1 revealed a significant positive effect of FEE on EO ($\beta = 0.308$, $t = 4.145$, $p < 0.001$), indicating that FEE plays a critical role in fostering the EO of Saudi fashion MSMEs. However, this effect size was comparatively lower than those reported for more developed EEs in the literature. For instance, the Dutch ecosystem strongly influenced entrepreneurial outcomes, with an R^2 value of 0.526 (Stam & Van De Ven, 2021). This suggests that the Saudi FEE, while progressing, still faces limitations in areas such as capital access, networking, and policy support, contributing to a less pronounced influence on EO.

In examining H2, the analysis confirmed a positive relationship between FEE and BP ($\beta = 0.149$), suggesting that a conducive EE facilitates business growth by enhancing access to resources, market opportunities and industry-specific support (Isenberg, 2011b; Wurth et al., 2023). Although the effect size was relatively small, this is consistent with findings in creative industries, where the influence of EEs on BP is often influenced by intangible, non-linear elements such as creativity, cultural values, and design identity (Chapain et al., 2010; Görgens & Stevens, 2013; McRobbie, 2018). This implies that factors such as creative autonomy, market competition, and strategic agility may play equally or more critical roles in determining business outcomes in fashion MSMEs (Brydges & Pugh, 2021; Clark et al., 2017; Loots et al., 2021; Radzi et al., 2017). Regarding H3, the findings revealed a notably strong relationship between EO and BP ($\beta = 0.469$), underscoring the critical role of entrepreneurial orientation in enhancing business performance. The EO dimensions—innovativeness, proactiveness, and risk-taking—contributed significantly to the competitive positioning and performance of Saudi fashion enterprises. This supports global findings that EO enhances BP by promoting innovation and strategic responsiveness, particularly in dynamic and design-driven sectors like fashion (Covin & Wales, 2019; Rauch et al., 2009; Yuefen et al., 2024; Zainol & Wan Daud, 2011)

Finally, the results for H4 indicate that EO partially mediates the relationship between FEE and BP. Partial mediation occurs when the mediator, EO, explains part but not all of the relationship between the independent variable, FEE, and the dependent variable, BP (Nitzl, Roldán, et al., 2016). In this study, FEE had a direct effect on BP and indirectly influenced BP through EO. This means that EO helps translate ecosystem support into tangible performance

outcomes. For instance, a Saudi fashion startup benefiting from ecosystem resources such as training (Kwartawaty et al., 2023), finance (Dhar et al., 2024), and networking (Lamidi & Rahadhini, 2024) may experience better BP both directly through access to those supports and indirectly by developing stronger EO traits such as innovativeness and proactiveness that further drive success (Dhewanto et al., 2023; Pramesti, 2024). Empirical evidence supports this mediation. For example, Wang (2016) found that EO partially mediates the relationship between environmental dynamism and SME performance. Similarly, Hossain and Azmi (2021) confirmed the partial mediation of EO between EE factors and SME success in the creative industry. These findings that while FEE provides essential environmental conditions, EO is pivotal in translating these conditions into tangible business success. This result also aligns with the RBV theory, which posits that firm performance depends not only on the availability of external resources but also on how strategically these are leveraged internally through capabilities such as EO (Agrawal et al., 2024; Ferreira et al., 2011; Lubis, 2022; Wernerfelt, 1984). Thus, ecosystem support must be complemented by a strong EO to fully realise performance benefits.

6.2.2 Qualitative study

This study provided insights into the Saudi FEE, including finance, culture, support, education, and policy, highlighting the challenges, opportunities, and the broader structural and cultural factors shaping the entrepreneurial landscape for Saudi fashion designers.

First, the finance domain revealed that Saudi fashion entrepreneurs rely heavily on informal funding sources despite the availability of formal financial support. Most interviewees indicated that personal savings and family contributions were the primary sources of funding, with bank loans and venture capital underutilised due to procedural complexities and high collateral requirements. For example, six out of nine participants noted that they funded their ventures through personal savings or family loans rather than seeking formal financial support.

Participants highlighted that securing bank loans was difficult due to stringent requirements, including the need for a government-employed guarantor. This observation aligns with global research indicating that fashion businesses often struggle to secure funding due to the perceived high-risk nature of the industry and limited understanding of creative business models by financial institutions. Formal funding mechanisms, such as the Social Development Bank, were noted as valuable sources of support due to favourable terms, including interest-free loans and extended repayment periods. However, despite the availability of these options, many entrepreneurs cited barriers such as limited financial literacy and

awareness of funding opportunities. Participants also highlighted the challenges faced by female entrepreneurs, who encountered additional institutional and cultural biases when seeking funding. The limited availability of tailored financial products for the fashion sector and the perception of fashion as a high-risk industry further constrained access to capital. This suggests a need for targeted financial products and more accessible funding channels to support innovation and growth within the Saudi fashion sector.

Second, the cultural domain demonstrated that Saudi society values entrepreneurship as a symbol of independence and success. The interviews revealed that entrepreneurial activity is encouraged, with religious and cultural values playing a significant role in shaping business practices. Many participants noted that the teachings of the Prophet Muhammad, who was himself a trader, inspire business activity and ethical conduct in commerce.

However, the perception of fashion as a low-status profession persists, particularly concerning tailoring, which is often seen as menial work. This perception creates barriers for individuals seeking to establish careers in fashion design or tailoring. Participants highlighted the influence of regional differences on fashion entrepreneurship, with certain areas such as Qassim and the Eastern Province having strong commercial traditions that support business activity.

Participants also noted a lack of local media coverage of fashion entrepreneurs' success stories, which limits the visibility and recognition of local talent. This lack of representation in the media was seen as a missed opportunity to inspire future entrepreneurs and attract investment into the sector. The interviews also identified a strong culture of independence among Saudi fashion designers, with many preferring to work alone rather than engage in partnerships. Concerns about protecting intellectual property and creative control were cited as primary reasons for this reluctance to collaborate. Despite this, some participants acknowledged the potential benefits of strategic partnerships, particularly in accessing funding and sharing market risks.

Third, the support domain highlighted critical gaps in the availability and accessibility of specialised support services for fashion entrepreneurs. Participants expressed frustration with the lack of dedicated fashion incubators and the misalignment of existing incubator services with the needs of the fashion industry. Most incubators focused on technology startups, providing little relevant guidance for fashion businesses. The high cost of consultancy services and limited access to experienced industry mentors further compounded the problem. Some participants reported concerns about conflicts of interest among consultants, noting that many consultants also operated their own fashion brands, raising doubts about impartiality.

Geographical disparities in access to support services were also evident, with most resources concentrated in major urban centres such as Riyadh and Jeddah. Entrepreneurs in peripheral regions faced additional barriers due to the limited availability of local incubators, mentors, and networking opportunities. Language barriers also restricted access to international grants and support programs, as many opportunities required proficiency in English. Government institutions, including the Ministry of Commerce, Monsha'at, and the Saudi Fashion Commission, were recognised for their role in facilitating business registration and supporting entrepreneurial activity. However, participants emphasised the need for more targeted support structures, such as fashion-specific chambers of commerce and dedicated fashion R&D centres. The lack of industry-specific support structures limits the ability of fashion businesses to innovate and scale effectively.

Fourth, the educational institution domain revealed significant progress in the development of fashion education in Saudi Arabia, with several universities establishing fashion design programs in recent years. However, participants noted that the quality and relevance of fashion education remain inconsistent. While government-funded education has expanded access to fashion programs, the curriculum often lacks alignment with international standards and industry needs. Participants highlighted challenges such as slow accreditation processes, limited access to experienced faculty, and a lack of practical training opportunities.

Participants also reported a gap in the availability of entrepreneurship education tailored to the fashion industry. Existing business courses were often too generic and failed to address the specific needs of fashion entrepreneurs. The absence of dedicated fashion incubators within universities further limited the ability of students to translate their creative ideas into viable business models. This disconnect between academic training and market demands suggests the need for more integrated and industry-focused fashion education programs.

The findings also highlighted a lack of effective collaboration between academic institutions and the fashion industry. Participants noted that while fashion students produce promising research, there are few mechanisms for translating these insights into commercial products or business opportunities. The absence of dedicated fashion research centres and intellectual property support further restricts the commercialisation of student innovations.

Finally, the policy domain revealed that while government initiatives such as Vision 2030 have created a more supportive environment for entrepreneurship, participants noted gaps in policy implementation and regulatory alignment. For example, the registration and licensing process for fashion businesses was described as complex and time-consuming. Participants also cited a lack of tailored tax incentives and trade policies to support the growth of fashion

businesses. Some participants expressed frustration with the absence of clear guidelines for importing raw materials and exporting finished products, highlighting the need for more streamlined trade regulations.

6.3 Evaluative Assessment of the Saudi Fashion Entrepreneurship Ecosystem

The empirical findings of this study suggest that the Saudi Fashion Entrepreneurship Ecosystem (FEE) can best be described as moderately developed. This classification is based on both quantitative results and qualitative insights, which together indicate that while the ecosystem has made notable progress particularly under the influence of Vision 2030 it continues to face structural and functional limitations that constrain its overall effectiveness.

Quantitative analysis revealed that the FEE has a significant but limited influence on EO. This suggests that while FEE domains such as finance, education, and policy contribute to shaping innovativeness, risk-taking, and proactiveness among fashion MSMEs, the ecosystem does not yet provide a sufficiently enabling environment to substantially foster entrepreneurial orientation at scale (Chapter 4, Section 4.5.2). In contrast, EO demonstrated a stronger direct impact on BP, highlighting that internal capabilities, rather than ecosystem support alone, are primary drivers of business success (Chapter 4, Section 4.6.3).

The direct effect of the FEE on BP was positive but relatively modest, indicating that while the ecosystem contributes to business outcomes, its role is supportive rather than transformative (Chapter 4, Section 4.6.2). EO partially mediated the relationship between FEE and BP, reinforcing its role as a strategic capability through which external resources must be effectively leveraged to generate improved performance (Chapter 4, Section 4.6.4).

Qualitative findings provided depth to these patterns. Entrepreneurs described ongoing difficulties in accessing sector-specific financial and institutional support, particularly among female founders and those located outside major urban centres (Chapter 5, Sections 5.3.2, 5.5.2). While institutions such as the Social Development Bank and the Fashion Commission have expanded programming in recent years, many participants reported that available services were either misaligned with their sectoral needs or too difficult to access due to administrative complexity and limited localisation (Chapter 5, Section 5.5.3). Entrepreneurial education was also identified as lacking in practical, fashion-focused content, with curricula in local institutions often disconnected from real-world industry demands (Chapter 5, Section 5.6.2).

Although these efforts mark clear progress, interview data pointed to a misalignment between high-level policy initiatives and the specific operational needs of fashion

entrepreneurs. While programs such as Saudi 100 Brands have positioned local designers on international platforms, several participants viewed these as replicated models from European fashion ecosystems, lacking adaptation to Saudi Arabia's local context (Chapter 5, Section 5.6.3). This concern is reflected in ecosystem development literature, which strongly emphasises that effective entrepreneurship ecosystems must be contextually grounded and developed in response to local institutional, cultural, and sectoral conditions (Isenberg, 2011; Stam & Spigel, 2016). Transplanting programs without adjusting them to domestic realities can result in structural misfits between support offerings and entrepreneurial capacity (Spigel, 2017; Autio et al., 2014).

For example, while global exposure was widely valued, many participants expressed a need for more accessible, locally relevant programs, especially for early-stage or home-based designers (Chapter 5, Sections 5.3.2, 5.5.1). They called for inclusive funding mechanisms, mentorship opportunities, and practical training in e-commerce and supply chain management. One entrepreneur shared that although they attended branding workshops, they still lacked assistance in navigating product registration and scaling challenges. These accounts reveal that the current top-down approach to ecosystem building may overlook bottom-up needs, especially among marginalised or underserved subgroups within the fashion sector.

Recognising the ecosystem as moderately developed provides a constructive lens through which to interpret both the progress achieved and the remaining structural challenges. It also reinforces the need for more sector-specific and inclusive interventions particularly in education, funding accessibility, and programme design to unlock the ecosystem's full potential and better align it with the objectives of Vision 2030 (Chapter 6, Section 6.5; Almuzel & Anderson, 2020).

6.4 Research Limitations and Recommendations for Future Research

This section outlines the limitations of this study and provides recommendations for future research aimed at extending the findings and addressing the identified gaps. Although this study has made several significant contributions, it has three main limitations that warrant critical reflection: scope, methodological, and theoretical.

6.4.1 *Scope Limitations*

The first limitation is the narrow contextual scope, as the proposed FEE model was tested exclusively within the Saudi fashion industry. Although this offers deep contextual insights, it restricts the model's applicability to broader geographic or industrial contexts. Saudi Arabia's

unique institutional, economic, gender, and cultural dynamics significantly influence entrepreneurial activity, and these factors may not translate to other settings (Juho & Kauppinen, 2017). Research confirms that EEs are highly context-dependent and influenced by localised institutional structures and social norms (Kapturkiewicz, 2022; Sipakoly, 2024). Even within Saudi Arabia, regional disparities exist; for example, entrepreneurial experiences in Riyadh and Jeddah benefit from stronger infrastructure and government support than those in smaller cities (Almuzel, 2022; Roomi et al., 2023). These variations mirror broader international findings that EE performance is shaped by contextual factors such as access to finance, infrastructure, and human capital, which differ widely between advanced and emerging economies (Cao & Shi, 2021; Sipakoly, 2024; Xie et al., 2019). Empirical research suggests that the impact of EO on BP varies across national and sectoral ecosystems. For instance, studies in Malaysia, Indonesia, and China show that dimensions such as innovativeness or risk-taking affect firm performance differently depending on the ecosystem's regulatory and infrastructural context (Alam et al., 2022; Barinta et al., 2022). Similar findings have been reported in China and Malaysia, where EO effectiveness is mediated by infrastructure, regulatory frameworks, and financial access (Mustafa et al., 2019; Olutuase et al., 2018; Torabi et al., 2020).

This study's exclusive focus on the fashion sector presents further limitations. Future research should investigate other creative industries (e.g. design, crafts, and music) and non-creative sectors (e.g. tech startups and green enterprises) to determine whether challenges such as funding, supply chain issues, or talent availability are unique to fashion or common across sectors. Numerous studies in other creative sectors have revealed distinct ecosystem dynamics, underscoring the need for sector-specific investigations (Brydges & Pugh, 2021; Loots et al., 2021; Purbasari & Wijaya, 2019; Stam, 2018). Further studies should test the FEE model in different geographical regions, both within and beyond Saudi Arabia, to enhance its generalisability and predictive validity. Cross-national comparative studies, such as Acs et al. (2017), demonstrate that the interactions between institutions and entrepreneurial agencies significantly affect entrepreneurial outcomes across diverse contexts. Replication across sectors and countries is essential. For example, Nicotra et al. (2018) found that the configuration and strength of ecosystem elements differ significantly between Italy's creative industries and technology clusters, highlighting the importance of cross-context validation for ecosystem models.

6.4.2 Methodological Limitations

This study primarily relies on self-reported data to assess EO and BP, a method widely used in entrepreneurship research but often critiqued for its vulnerability to response biases, such as social desirability or self-enhancement (Donaldson & Grant-Vallone, 2002; Podsakoff et al., 2003). These biases may lead to over- or underestimations of entrepreneurial abilities and business outcomes, thus affecting the reliability and validity of the findings. Moreover, the sampling strategy, which utilised online surveys and social media platforms, may have inadvertently excluded entrepreneurs from traditional or rural areas who lack digital access or engagement. While the achieved response rate of 34% aligns with norms in online survey research (Baruch & Holtom, 2008), the potential for non-response bias remains, as non-participants may differ significantly from respondents.

A notable demographic finding from the first study was that over 90% of the sample was female. This underscores the importance of conducting targeted research on gender-specific dynamics in the Saudi fashion industry. Although the sample composition may limit the study's broader representativeness, it also provides a unique opportunity to explore female entrepreneurs' nuanced experiences in a traditionally male-dominated business environment (Henry et al., 2016). Several studies conducted in Nigeria (Abdulraheem et al., 2024), Pakistan (Malik et al., 2023), Europe (Aakko & Niinimäki, 2018), and Indonesia (Mashithoh & Ginting, 2016) highlight the unique entrepreneurial challenges and opportunities faced by female fashion designers, which shape their BP. Therefore, this emphasises the need for future studies to investigate female-led fashion MSMEs within the Saudi FEE. Furthermore, comparative studies between male- and female-led MSMEs in the Saudi fashion sector can help identify disparities in entrepreneurial resources, performance metrics and strategic decision-making. These insights are essential for developing relevant entrepreneurial policies and support mechanisms for the Saudi fashion sector's FEE.

This study employed a cross-sectional design that captured data at a single point in time. Although useful for identifying relationships, this approach cannot capture the dynamic and evolving nature of EEs. Given Saudi Arabia's ongoing transformation under Vision 2030, entrepreneurial ecosystems are rapidly evolving, influenced by new policies, changing market demands, and cultural changes. Since data collection in 2023, the Saudi FEE has become more inclusive and sustainable, especially for women entrepreneurs in the apparel sector (Alshibani et al., 2024). Research has identified cultural, institutional, and financial improvements in Saudi Arabia's ecosystem (Aljarodi et al., 2024; Osama & Adam, 2024). Understanding these changes requires a longitudinal ethnographic approach to capture the complex social

relationships and governance factors underpinning EE development (Cho et al., 2021; Scott et al., 2022). This approach is essential to ascertain whether improvements in the EE directly enhance the BP or if other dynamics, such as market trends, digital adoption, or evolving policy landscapes, play a role (Yuan et al., 2021). Therefore, Future research should focus on these areas by employing longitudinal studies to explore the dynamic nature of ecosystems and their impact on entrepreneurship (Cho et al., 2021). This will help policymakers and stakeholders design more targeted support structures and better navigate the complexities of the evolving entrepreneurial landscape in Saudi Arabia (Elnadi & Gheith, 2021; Ribeiro et al., 2023).

6.4.3 Theoretical Limitations

Although this study adopts a robust theoretical framework, it is constrained by its focus on the established constructs of EO and BP. Emerging dimensions such as sustainability and digital transformation, which are increasingly critical in EEs, have not been explored in depth. The current study reported a low R^2 value for EO (0.095), suggesting that the model may benefit from additional explanatory variables in future studies. Numerous studies have demonstrated the value of incorporating mediating and moderating factors to better explain the relationship between EO and BP. For example, network innovation agility (Mulyana & Hendar, 2020) and dynamic marketing capabilities (Nyoman et al., 2025) have been found to mediate this relationship in fashion SMEs. Other influential factors include digital infrastructure (Li & Luo, 2020), marketing orientation (Kozlowski et al., 2018), innovation orientation (Bianchi et al., 2017), and network capabilities (Choi, 2018), all of which have been shown to significantly shape the EO–BP link. Additionally, energised professional networks enhance the impact of EO on marketing performance (Riyoko & Widodo, 2022).

The effectiveness of the EO is also context-dependent and shaped by cultural norms and institutional frameworks. In Pakistan, for instance, entrepreneurial culture significantly moderates the EO–performance relationship, whereas government support shows limited influence (Mustafa et al., 2019). Therefore, to enhance the model's robustness and contextual sensitivity, future research should consider integrating relevant mediating and moderating variables, including technological, institutional, and cultural dimensions of the digital divide. In conclusion, this study provides important insights into the Saudi FEE, particularly in the fashion industry. However, several limitations constrain the broader applicability of these findings. Future research should address these limitations by expanding to other geographic and industrial contexts, adopting mixed and longitudinal methods, exploring gender dynamics,

and incorporating emerging theoretical dimensions into the research model. These steps will enhance the robustness and relevance of research on entrepreneurial ecosystems, offering valuable guidance to scholars, practitioners, and policymakers.

A further theoretical limitation of this study concerns the scope of internal firm-level constructs examined. Specifically, the research deliberately focused on EO as the sole micro-level internal resource mediating the relationship between the FEE and BP. While EO characterised by innovativeness, proactiveness, and risk-taking has been widely recognised as a core strategic intangible resource (Lumpkin & Dess, 1996; Covin & Slevin, 1989), this narrow focus excludes other salient internal capabilities that may offer complementary or alternative explanatory value. Constructs such as strategic capabilities, entrepreneurial self-efficacy, and innovation capacity have been extensively identified in the literature as critical drivers of entrepreneurial effectiveness, particularly in volatile or resource-constrained environments (Teece, 2007; Newman et al., 2019; Lichtenstein & Brush, 2001). The omission of these constructs, although intended to preserve model parsimony, may limit the theoretical depth of the study and obscure more nuanced firm-level dynamics.

Future research is therefore encouraged to adopt a broader theoretical framework that incorporates these additional constructs. In particular, extending the RBV by integrating SCT offers a compelling direction. SCT emphasises the agentic capabilities of entrepreneurs, including self-efficacy, self-regulation, and learning behaviours (Bandura, 1997; Zhao et al., 2005), which are essential for understanding how individuals engage with and convert ecosystem resources into competitive advantage. Such a combined RBV–SCT approach would facilitate a more holistic understanding of how internal competencies interact with ecosystem-level conditions to shape entrepreneurial performance. This extension would not only contribute to theoretical development in entrepreneurship scholarship but also provide practical insights for designing more comprehensive and context-sensitive ecosystem support strategies, particularly in emerging creative sectors such as fashion.

6.5 Theoretical contributions

This thesis makes several theoretical contributions by advancing the understanding of the Saudi FEE and its impact on the EO and BP of fashion MSMEs. Through a multilevel analysis, this study offers insights into the dynamics of EEs in emerging economies and addresses specific gaps in the literature.

A central contribution of this research is the expansion of knowledge on FEE in emerging markets, particularly in the Middle East. Previous studies have predominantly examined markets in developed countries, such as Italy (Lazzeretti & Capone, 2020; Vacca & Vandi, 2025), Sweden, Finland (Colliander, 2015; Salonoja, 2013), and Canada (Brydges & Pugh, 2021). By investigating the Saudi context, this study provides a nuanced understanding of how local FEE operates within its unique sociocultural, economic, and institutional landscape. It builds on prior work in the creative industries and entrepreneurship ecosystems (Potts, 2011), offering empirical evidence from a non-Western context. By mapping the Saudi FEE, this study deepens the understanding of how local conditions, such as cultural norms and regulatory frameworks, influence ecosystem development, distinguishing it from ecosystems in more established markets. This study contributes to the literature on entrepreneurship in the Gulf region (Ali et al., 2019; Almuzel & Anderson, 2020; Dixit et al., 2024).

This study further contributes to the literature on EO and BP, especially in the creative industries. EO, comprising innovativeness, proactiveness, and risk-taking (Covin (Covin & Slevin, 1989; Lumpkin & Dess, 2001), has been widely studied across industries and regions. However, its role in the fashion sector in emerging markets remains underexplored. By applying EO concepts to the Saudi fashion industry, this study investigates how entrepreneurial traits influence firm outcomes in a trend-driven and competitive environment. The findings offer contextualised insights into how EO can enhance BP, demonstrating its utility in enabling firms to identify opportunities and navigate challenges in a rapidly evolving sector (Putria & Gunawan, 2019; Sarsiti, 2024). This study also extends EO research by examining its interactions with ecosystem-level variables in an emerging market setting.

In examining the structure of EEs, this thesis underscores the importance of sociocultural, economic, and institutional factors in shaping entrepreneurial activity (Isenberg, 2011a; Stam, 2015). This study investigates how global frameworks for entrepreneurial ecosystems can be meaningfully adapted to local conditions in Saudi Arabia. For example, societal attitudes towards entrepreneurship, gender-specific barriers, and access to finance play critical roles in influencing entrepreneurial engagement and outcomes. The research thereby offers a localised perspective that enhances the broader literature on entrepreneurship ecosystems in non-Western contexts (Bruton et al., 2010; Tolstykh et al., 2020).

This thesis also presents a conceptual framework that integrates the domains of the Saudi FEE, EO, and BP. Drawing on Spigel's (2017) definition of entrepreneurial ecosystems as networks of social, political, economic, and cultural elements supporting innovative startups, the framework positions EO as a mediating factor. By directly linking ecosystem domains to

business outcomes, the framework builds on Isenberg's (2011) and Stam and Van de Ven's (2019) structured models. It illustrates how key ecosystem components, such as finance, education, and market access, interact to shape entrepreneurial behaviour and influence firm performance. This approach offers a focused and integrative perspective on how interconnected ecosystem elements collectively support entrepreneurial success in industries such as the fashion industry.

Applying the entrepreneurial ecosystem framework in the Saudi context represents another important contribution of this study. While previous studies have demonstrated the utility of this framework in identifying challenges and enablers across various sectors (Brydges & Pugh, 2021; Colliander, 2015), this study extends its application by mapping the specific features of Saudi FEE. It identifies the unique challenges faced by Saudi MSMEs, including limited access to domestic manufacturing, gaps in entrepreneurship education, and bureaucratic inefficiencies (Morant-Martínez et al., 2019). Simultaneously, it highlights enabling factors, such as government initiatives under Vision 2030, which have facilitated the growth of the fashion sector. By applying the ecosystem framework in this context, this study offers a detailed understanding of how entrepreneurial environments can be structured to support MSME development in emerging markets.

Finally, this thesis connects theoretical insights with practical recommendations aimed at policymakers and industry stakeholders. This underscores the importance of integrating entrepreneurship training within fashion education, drawing on international models from Australia and the United Kingdom, where experiential learning and industry collaboration are emphasised (Lazzeretti & Capone, 2020; Shi et al., 2012). The study also calls for investment in local manufacturing capabilities and streamlined regulatory procedures to strengthen MSME growth in India. These recommendations are grounded in the conceptual framework developed by this study, offering a balanced and evidence-based approach to address structural constraints and enhance ecosystem effectiveness.

6.6 Practical Implications for Stakeholders

Finance

This research implies that fashion entrepreneurs continue to rely heavily on personal and familial capital due to limited access to formal finance and institutional barriers. This raises concerns about the inclusiveness and effectiveness of the financial ecosystem for creative

industries. Therefore, advancing financial literacy, funding strategies and investment accessibility through policy making will assist the growth of Saudi FEE. Policymakers should prioritise reforms that enhance the accessibility and transparency of financial programs, particularly by streamlining loan conditions (e.g., guarantor requirements) and addressing gaps in venture capital and incubator funding (Alsaiani, 2022; Altokhais, 2018). Educational institutions must integrate targeted financial literacy training into design and creative programs to equip graduates with essential budgeting, pitching, and investment-readiness competencies (Akinyede, 2023; Polisetty et al., 2021). Entrepreneurs are encouraged to pursue collaborative financing strategies and partnerships that mitigate risk while maintaining brand autonomy (Kariv & Coleman, 2015).

Culture

This research suggests, while Saudi society increasingly encourages entrepreneurship, fashion remains culturally undervalued in some regions. Risk aversion and fear of failure are prevalent and undermine entrepreneurial orientation, particularly for women. Therefore, this research implies change on the policy-making front to promote cultural narratives that normalise failure as part of the entrepreneurial process and elevate fashion entrepreneurship through inclusive media representation and national success stories (Global Entrepreneurship Monitor, 2024a; Fridhi, 2020; Alshebami & Seraj, 2022). Educators should frame fashion entrepreneurship as both economically and culturally significant, embedding social entrepreneurship and cultural heritage within course design (Hasanin, 2021; Fahrati, 2023). Entrepreneurs may strategically embrace cultural narratives to position their ventures as both modern and rooted in Saudi identity, thereby challenging social perceptions (Alzahrani et al., 2015).

Support Systems

This research also discovered that the fashion entrepreneurs face limited access to industry-aligned incubators, high service costs, and geographic inequities in support availability. Many consulting services are also unregulated and lack industry expertise. Therefore, presenting policymakers with an action point to create fashion-specific incubators beyond metropolitan centres, and enforce professional standards within the consultancy sector to prevent conflicts of interest (Cech & Leonard, 2001; Dipaola et al., 2014). Educational institutions can bridge support gaps by establishing university-linked incubators and co-curricular mentorship programs aligned with the creative sector (Dlamini et al., 2022).

Entrepreneurs are encouraged to critically evaluate consultancy services, build peer learning communities, and advocate for support structures tailored to fashion-specific needs (Braun & Suoranta, 2024).

Educational Institutions

The research additionally suggests, although fashion education has expanded in Saudi Arabia, programs remain disconnected from international standards and lack embedded entrepreneurship training and commercialisation pathways. Therefore, the policymakers should support international collaborations, accelerate accreditation processes, and incentivise industry engagement within higher education (Gonaim, 2021; Aljendan, 2024). Educators must prioritise curriculum reform to integrate fashion-specific business education, entrepreneurship theory, and applied learning experiences (Sousa et al., 2019; Jamaluddin et al., 2019). Entrepreneurs should engage with universities to inform research agendas, provide mentorship, and co-develop commercially viable research outputs (Lehmann et al., 2022; Audretsch et al., 2021).

Policy and Market

Regulatory improvements have facilitated start-up processes, but a lack of industry-specific chambers, innovation policy, and strategic investment limits sectoral competitiveness. Although policymakers should establish fashion-focused chambers of commerce, regional policy forums, and funding mechanisms that incentivise sectoral R&D, particularly in sustainable and heritage-based fashion (Seraj et al., 2022; Gamba, 2019; Todeschini et al., 2020). Educators should actively include policy literacy and design governance in fashion curricula to prepare graduates to navigate institutional environments holistically. Therefore, entrepreneurs should also make active efforts to participate in regulatory consultations and position their ventures within broader national goals, such as Vision 2030, to attract policy support. Promote cross-sector collaborations between fashion and industries like petrochemicals, military, and AI for material and product innovation (Todeschini et al., 2020; Nayak et al., 2022).

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APPENDICES

Appendix A: Ratings for individual NECI pillars for 16 high-income economies

Country	Financing for entrepreneurs	Governmental support and policies	Taxes and bureaucracy	Governmental programs	Basic school entrepreneurial education and training	Post school entrepreneurial education and training	R&D transfer	Commercial and professional infrastructure	Internal market dynamics	Internal market openness	Physical and services infrastructure	Cultural and social norms
Canada	4.89	4.54	5.01	4.86	3.64	4.66	4.25	5.46	5.24	4.33	6.22	5.08
France	5.37	5.29	4.87	5.88	2.44	5.07	4.49	5.76	3.93	4.66	6.74	4.38
Germany	4.84	4.1	4.18	6.43	2.57	4.98	4.59	5.77	5.53	4.94	5.73	4.09
Italy	4.47	4.43	3.79	4.39	3.46	4.62	4.28	4.91	4.96	4.25	5.73	4.44
Luxembourg	4.62	4.26	4.84	4.98	4.14	5.18	4.69	4.58	3.85	4.38	5.6	4.53
Netherlands	5.86	5.09	6.01	5.95	5.47	6.28	5.44	6.38	5.19	6.22	7.03	6.35
Norway	4.67	3.47	4.69	5.41	3.38	4.8	4.45	5.95	4.11	4.44	7.38	4.96
Qatar	5.14	5.75	6.3	5.91	6.31	6.49	5.35	6	6.07	5.13	7.23	6.69
Republic of Korea	5.14	6.35	6.1	6.36	4.49	5.35	5.26	5.3	7.48	5.03	7.32	6.02
Saudi Arabia	6.47	6.57	6.38	6.39	4.44	5.54	5.14	6.34	7.17	6.01	7.96	7.4
Slovenia	4.61	3.85	4.31	5.3	3.21	4.8	4.16	5.68	6.56	4.87	6.53	4.11
Sweden	4.95	3.53	4.75	4.67	3.94	4.44	4	5.14	4.5	4.32	6.92	5.57
Switzerland	5.84	5.1	5.61	6.07	3.54	5.59	5.96	6.2	3.82	4.87	7.64	6
United Arab Emirates	7.28	7.92	7.46	7.46	7.65	7.81	7.75	7.76	8.01	7.34	7.6	8.12
United Kingdom	4.83	3.66	5.15	4.1	3.01	4.55	3.93	5.26	5.53	4.69	5.4	5.54
United States	4.86	3.8	4.98	3.79	3.63	4.82	3.85	5.62	4.93	4.17	6.75	6.66
Green: Highest Scores for each indicator Red: Lowest Scores for each indicator												

Appendix B: Questionnaire

Dear Participant,

My name is Marwah Alharbi, a PhD candidate who is studying Fashion Entrepreneurship at RMIT University. I am currently conducting extensive research on the Saudi Arabian fashion industry entrepreneurial ecosystem. The purpose of the research is to assess the current state of the Saudi fashion industry, in order to design a suitable model that provides meaningful support and growth for fashion-related projects in Saudi Arabia. To achieve this goal, the researcher created the following questionnaire to explore your perspective as a Saudi fashion entrepreneur about any obstacles experienced, and support required to create a fashion brand. Therefore, your participation in this research would be greatly appreciated, as you may provide some interesting comments and perspectives on this topic. The results of this research will have both theoretical and practical ramifications for fashion entrepreneurship. The findings from the survey will inform how future services are developed to ensure they reflect, and meet, your expectations and needs.

After completing the questionnaire, you will receive two free copies (Arabic - English) of the Fashion Trends Forecasting Report for S/S 2022 and A/W 22-23.

If you need any further information regarding this research, please click the following link <https://docs.google.com/document/d/1o5Bi38s-CyRrIfk5ErxEpt5Gzq3Sd87Hh4tohtzWBzQ/edit#heading=h.jc8lfndrkaby>

Many thanks for your anticipated cooperation.

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Tick the box to confirm that you have read the Participant Information sheets and agree to take part in this study.

- ☐ Yes, I Consent
- ☐ No, I don't Consent

Do you have a fashion brand in Saudi Arabia?

- ☐ Yes
- ☐ No

Section 1: Demographic Information

Please tell me about you by answering the following questions as accurately as possible.

Your gender:

- ☐ Male
- ☐ Female

Your age group:

- ☐ <18
- ☐ 19-25
- ☐ 26-30
- ☐ 31-40
- ☐ 41-50
- ☐ 51-60
- ☐ Above 60

What is your highest level of education?

- ☐ High school or less
- ☐ Diploma
- ☐ Associate degree
- ☐ Bachelor
- ☐ Master Doctorate

What is your field of study?

- ☐ Fashion
- ☐ Design Business
- ☐ Fashion Business
- ☐ Other (please specify)

Which university did you graduate from?

Have you attended or completed any business courses/training?

- ☐ Yes Please (please specify)
- ☐ No

Your main occupation:

- ☐ Employed in full-time work
- ☐ Employed in part-time work
- ☐ I am a student
- ☐ Self-employed Unemployment

Do you have business experience before starting this business?

- ☐ Yes Please (please explain)
- ☐ No

What is your position?

- ☐ Designer of own brand
- ☐ Owner
- ☐ CEO (professional manager)
- ☐ Employee

- Other (please specify)

Section 2: Firm Characteristics

Please tell me about your business by answering the following Questions.
In which year was the enterprise established?

How many equivalent full-time employees do you have currently?

- 1 to 5
- 6 to 49
- 50 to 249

Where is your business located?

- Northern Province
- Southern Province
- Central Province
- Eastern Province
- Western Province

Where do you currently sell your products? (Select all that apply)

- Social media (e.g., Facebook, Instagram)
- Company owned website
- Company owned physical store or boutique
- Multi-brand stores
- Online multibrand website (e.g., Amazon, Esty)
- Other (please specify)

What is the legal structure of your business?

- Partnership
- Family trust
- Sole Trader
- Public company
- Other (please specify)

In which of these categories would you classify your business? (Select all that apply)

- Womenswear Menswear
- Childrenswear
- Accessories (footwear, bags, leather goods)
- Jewellery
- Other (please specify)

What type of product do you Sell? (Select all that apply)

- Activewear
- Bridalwear or evening wear
- Outdoors
- Casualwear
- Formalwear
- Abayas Athleisure

- Other (please specify)

How do you manufacture your products?

- Import and wholesale
- Domestic clothing manufacturers
- Overseas clothing manufacturers
- At home
- Made-to-measure
- Other (please specify)

What was your main motivation to start your fashion business?

- Innovative new business idea or products
- Independence
- Take advantage of a market opportunity
- Increase income
- To obtain prestige or to be socially recognized
- Escape unemployment
- Other (Please specify)

Which type of finance did you use when you started the business? (Select all that apply)

- Bank loan
- Angel investor
- Family and/or friends
- Venture capital
- Savings
- Equity Incubator
- Crowdfunding

What obstacles did you face when accessing financial support?

- Application procedures were complex
- Interest rates are high
- Collateral requirements were too high
- Size of loan and maturity were insufficient
- Difficulty preparing a feasibility study
- I am unable to create relationships with prospective investors.
- I am unaware of financing options and finance institutions in Saudi Arabia.
- I do not have capacity or skills to persuade an investor to invest in their project.
- Other (Please specify)

Section 2: The Saudi Fashion Entrepreneurial Ecosystem

As an entrepreneur starting a new business, please indicate the degree to which you agree or disagree with each of the statement below by ticking (✓) the most appropriate option

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Finance					
1- In Saudi Arabia, there is sufficient equity funding available for fashion businesses.	i	i	i	i	i
2- In Saudi Arabia, there is sufficient debt funding available for fashion businesses.	i	i	i	i	i
3- In Saudi Arabia, there is sufficient government subsidies (support) available for fashion businesses.	i	i	i	i	i
4- In Saudi Arabia, there is sufficient funding available from private individuals (other than founders) for fashion businesses.	i	i	i	i	i
5- In Saudi Arabia, there is sufficient venture capitalist funding available for fashion businesses.	i	i	i	i	i
6- In Saudi Arabia, there is sufficient funding available through initial public offerings (IPOs) for fashion businesses.	i	i	i	i	i
Government programs					
7- In Saudi Arabia, a wide range of government assistance for fashion businesses can be obtained through contact with a single government entity	i	i	i	i	i
8- In Saudi Arabia, there are adequate government programs to help fashion businesses	i	i	i	i	i
9- In Saudi Arabia, the people working for government institutions are competent and effective in supporting fashion businesses	i	i	i	i	i
10- In Saudi Arabia, almost anyone who needs help from a government program for fashion businesses can find what they need	i	i	i	i	i
11- In Saudi Arabia, government programs aimed at supporting fashion businesses are effective	i	i	i	i	i
12- In Saudi Arabia, fashion hubs and business incubators provide effective support for fashion businesses	i	i	i	i	i

Market	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
13-In Saudi Arabia, fashion businesses can easily enter new markets	i	i	i	i	i
14-In Saudi Arabia, the fashion businesses can afford the cost of market entry	i	i	i	i	i
15-In Saudi Arabia, fashion businesses can enter markets without being unfairly blocked by established fashion firms.	i	i	i	i	i
16-In Saudi Arabia, the anti-trust legislation is effective and well-enforced.	i	i	i	i	i
17-In Saudi Arabia, the availability of market information such as fashion trends forecasting is an obstacle for fashion businesses.	i	i	i	i	i
18-In Saudi Arabia, access to distribution channels (retailers) is an obstacle for fashion businesses.	i	i	i	i	i
Culture					
19-In Saudi Arabia, the national culture is highly supportive of individual (fashion entrepreneurs) success achieved through own personal efforts	i	i	i	i	i
20-In Saudi Arabia, the national culture emphasises self-sufficiency, autonomy, and personal initiative.	i	i	i	i	i
21-In Saudi Arabia, the national culture encourages entrepreneurial risk-taking.	i	i	i	i	i
22-In Saudi Arabia, the national culture encourages creativity and innovativeness.	i	i	i	i	i
23-In Saudi Arabia, the national culture emphasizes the responsibility that the individual (rather than the collective) has in managing his or her own life.	i	i	i	i	i
24-In Saudi Arabia, the creation of fashion businesses is considered an appropriate way to become rich.	i	i	i	i	i
25-In Saudi Arabia, most people consider becoming a (fashion) entrepreneur as a desirable career choice.	i	i	i	i	i
26-In Saudi Arabia, successful (fashion) entrepreneurs have a high level of status and respect.	i	i	i	i	i
27-In Saudi Arabia, you will often see stories in the public media about successful (fashion) entrepreneurs.	i	i	i	i	i
28-In Saudi Arabia, most people think of (fashion) entrepreneurs as competent, and resourceful individuals	i	i	i	i	i

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Entrepreneurial education					
29-In Saudi Arabia, colleges and universities provide good and adequate preparation for starting up and growing fashion businesses	i	i	i	i	i
30-In Saudi Arabia, the vocational education system provides good and adequate preparation for starting up and growing fashion businesses	i	i	i	i	i
Talent					
31-In Saudi Arabia, there is no shortage of the types of employees you need for your fashion businesses.	i	i	i	i	i
32-In Saudi Arabia, you can afford to hire the employees you need for your fashion business.	i	i	i	i	i
33-In Saudi Arabia, you are satisfied that the skill levels of people are sufficient for your fashion business's needs.	i	i	i	i	i
Support					
34-In Saudi Arabia, there are enough subcontractors, mentors, qualified experts, suppliers, and consultants to support fashion businesses.	i	i	i	i	i
35-In Saudi Arabia, fashion businesses can afford the cost of using subcontractors, suppliers, mentors, qualified experts and consultants.	i	i	i	i	i
36-In Saudi Arabia, it is easy for fashion businesses to get good subcontractors, suppliers, and consultants.	i	i	i	i	i
37-In Saudi Arabia, it is easy for fashion businesses to get good, professional legal and accounting services.	i	i	i	i	i
38-In Saudi Arabia, it is easy for fashion business to get good banking services (checking accounts, foreign exchange transactions, letters of credit and the like).	i	i	i	i	i
39-In Saudi Arabia, it is easy for fashion businesses to access to business incubator accelerator.	i	i	i	i	i

Entrepreneurial Orientation.

Please indicate the degree to which you agree or disagree with each of the statement below by ticking (✓) the most appropriate option.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. Our company favours a strong emphasis on R&D, technological leadership, and innovations	i	i	i	i	i
2. Many new lines of products were marketed in the past 3 years	i	i	i	i	i
3. Changes in product lines have usually been quite dramatic	i	i	i	i	i
4. Typically, we initiate actions to which competitors then respond	i	i	i	i	i
5. We are very often the first business to introduce new products, administrative techniques, processing technologies	i	i	i	i	i
6. Our company has a strong tendency to be ahead of others in introducing novel ideas or products	i	i	i	i	i
7. The term “risk taker” is considered a positive attribute for people in our business.	i	i	i	i	i
8. People in our business are encouraged to take calculated risks with new ideas.	i	i	i	i	i
9. Our business emphasizes both exploration and experimentation for opportunities.	i	i	i	i	i

Business Performance

Please evaluate the performance of your business over the previous three years relative to your major competitors (1=much worse than competitors, 7=much better than competitors)

	1	2	3	4	5
In relation to the firm's objectives, the level of ROI was	i			i	i
In relation to the firm's objectives, the level of sales increase was	i	i	i		
In relation to the firm's objectives, the degree of customer satisfaction was	i	i	i	i	i
In relation to the firm's objectives, the satisfaction of your employees was	i	i	i	i	i
The overall results of the firm in the past year were	i	i	i	i	i

If you interested, please put your contact info for further interview.

If you have any comments and concerns on overall survey, please provide it here.

--- Thank you very much for your cooperation ---

Appendix C: The interview protocol and questions

I would like to thank you for participating in this interview as a part of my PhD research project about the Saudi fashion entrepreneurship ecosystem. You have been selected because of your position in (The institution name) which plays an essential role in the development of the entrepreneurship in the Saudi Arabia. Your contact details were obtained from the institution's website. As I have mentioned to you before, the purpose of this study is to design a fashion supportive model that is appropriate for the Saudi fashion entrepreneurship ecosystem. We have planned this interview to last no longer than one and half hours. I would like to remind you that you have signed the consent form that give me your permission to audio record our conversation. For your information, only researchers on the project will be privy to the tapes which will be eventually destroyed after the project is over. Since your participation is completely voluntary, in case you feel distressed, and you want to discontinue the interview or the use of the recorder, please let me know. Before we begin the interview, do you have any concerns or questions? If any questions arise at any point in this study, you can feel free to ask them at any time. I would be more than happy to answer your questions. Then with your permission we will begin the interview.

Before we conclude this interview, would you add anything else?

Section 1: Role of the stakeholder
What type of support do you provide to Saudi fashion entrepreneurs? In your opinion, what is the impact of this support on the Saudi fashion entrepreneurial ecosystem?
Section 2: Domains of the fashion entrepreneurial ecosystem
From your perspective, in general, what are the main challenges for fashion entrepreneurs in Saudi Arabia? In general, how would you rate the Saudi fashion entrepreneurial ecosystem?
In your experience, who is the most important person/organisation to contact if somebody (entrepreneur, support organisation, etc) wants to gain access to the entrepreneurial ecosystem and relevant networks? From your perspective, which are the most relevant actors in the local entrepreneurial ecosystem in terms of influencing entrepreneurial success in a significant way? What do they do? Do you think there are important players or services missing in the fashion entrepreneurial ecosystem?
Human Capital Are entrepreneurial education programs available in fashion schools and institutes? What types of educational and training support are available for Saudi fashion entrepreneurs? How do you view the current educational role in promoting fashion entrepreneurship? Do you think that fashion education in Saudi Arabia encourage fashion graduates to start their enterprises? What skills do you think are necessary for a fashion entrepreneur to succeed?
Finance What type of financial services is available for fashion entrepreneurs? What financial business services providers focus their services on the needs of fashion entrepreneurs?
Support How do you view the adequacy of support for fashion entrepreneurs at the present time? How do you view formal institutions sponsoring fashion designers starting their own business in Saudi Arabia? To what extent do you think the Saudi fashion stakeholders would support fashion entrepreneurs? Are there suitable incubation/accelerator programs for fashion? Are specialised business services, such as legal, tax or accounting services, offered?

Do entrepreneurs have a body representing their interests vis à vis the public sector and support organisations? Is physical infrastructure, such as office space or maker spaces, available for entrepreneurs? Are large corporations present that support entrepreneurs as part of their business activity or corporate social responsibility programs? Do the media report on fashion entrepreneur success stories?
Culture What do you think is society's perception of fashion entrepreneurship? How is fashion entrepreneurship understood and valued by society?
Market What market information is necessary for fashion entrepreneurs to facilitate the development of their businesses? How/where can an entrepreneur access this information? Network and cooperation Which formal/informal mechanisms are in place for stakeholder coordination? How would you assess the intensity of cooperation between the stakeholders, ranging from information exchange to coordination and co-production? How would it be ideal? In your experience, which are the main barriers to effective interaction between stakeholders in Saudi FEE?
Policy To what extent does the government contribute to an enabling environment for fashion entrepreneurs (policy framework in place)? What regulatory barriers are particularly burdensome for fashion entrepreneurs? From your perspective, how does the political situation in the country affect the fashion entrepreneurial ecosystem? How would it be ideal? Do you know if there is a strategy in place to promote fashion entrepreneurship?

Thank you for your participation! I really appreciate your time.

Appendix D: Participant Information Sheet/Consent Form for Research-Survey

RMIT Classification: Trusted

RMIT Classification: Trusted



Participant Information Sheet/ Consent Form

Title: Mapping the Saudi Fashion entrepreneurial Ecosystem.

Senior Supervisor	Caroline Tan
Associate Supervisors	Carolina Quintero
Principal Research	Marwah Alharbi

Dear Participant,

You are invited to take part in this research project, which is called *Mapping the Saudi Fashion entrepreneurial ecosystem*. You have been invited because of your position at [name of institution], which plays an essential role in the development of fashion entrepreneurship in Saudi Arabia. Your contact details were obtained from the institution's website.

This Participant Information Sheet tells you about the research project. It explains the processes involved in taking part. Knowing what is involved will help you decide if you want to take part in the research.

Please read this information carefully. Ask questions about anything that you don't understand or want to know more about.

If you decide you want to take part in the research project, you may continue to the survey after reading this Participant Information Sheet. By completing the survey, you are telling us that you:

- Understand what you have read.
- Consent to take part in the research project.

You may download a copy of this Participant Information Sheet to keep.

1. What is the purpose of this research?

The purpose of the research is to design a fashion supportive model that is appropriate for the Saudi fashion entrepreneurship ecosystem. This research will contribute to critical knowledge of the understanding the fashion entrepreneurship ecosystem.

This research will be used by the researcher Marwah Alharbi to obtain a PhD in Fashion and Textiles from RMIT University. This research has been initiated by the researcher, Marwah Alharbi, supervised by Dr. Caroline Tan, and Dr. Carolina Quintero.

2. What does participation in this research involve?

If you decide to take part in the research project, you will be required fill an online survey which will take approximately 30 minutes to complete.

3. Do I have to take part in this research project?

Participation in any research project is voluntary. If you do not wish to take part, you do not have to. If you decide to take part and later change your mind, you are free to withdraw from the project at any stage. Submitting your completed questionnaire is an indication of your consent to participate in the study. You can withdraw your responses any time if you change your mind about having them included in the study before we have analysed and published the results.

4. What are the possible benefits of taking part?

There are no direct benefits for participating in the study; however, you may appreciate contributing to knowledge in understanding the current state of the Saudi fashion entrepreneurship ecosystem.

5. What are the risks and disadvantages of taking part?

This project will use an external site to create, collect and analyse data collected in a survey format. The site we are using is Qualtrics. No personal information will be collected in the survey so none will be stored as data. Once we have completed our data collection and analysis, we will export the data to the RMIT server where it will be stored securely for five years. The data on the host server will then be deleted and expunged.

6. What if I withdraw from this research project?

If you do consent to participate, you may withdraw any time before the completion of the survey, as the survey is anonymous. As you anonymise during the study, it will not be possible for us to withdraw your data.

7. What happens when the research project ends?

When the research project ends in October 2023, you will be provided with a summary of the results.

8. What will happen to information about me?

It is anticipated that the results of this research project will be published and/or presented in a variety of forums. In any publication and/or presentation of the research, information will be provided in such a way that you cannot be identified, except with your express permission. All data will be stored in a password protected computer of the principal research student and on the University Network. This system provides a high level of manageable security and data integrity, secure remote access, and is backed up regularly.

In accordance with relevant Australian and/or Victorian privacy and other relevant laws, you have the right to request access to the information about you that is collected and stored by the research team. You also have the right to request that any information with which you disagree be corrected. Please inform the research team member named at the end of this document if you would like to access your information. Any information that you provide can be disclosed only if (1) it is protect you or others from harm, (2) if specifically allowed by law, (3) you provide the researchers with written permission. Any information obtained for the purpose of this research project that can identify you will be treated as confidential and securely stored.

9. Who has reviewed the research project?

All research in Australia involving humans is reviewed by an independent group of people called a Human Research Ethics Committee (HREC). This research project has been approved by the RMIT University HREC. This project will be carried out according to the *National Statement on Ethical Conduct in Human Research* (2007). This statement has been developed to protect the interests of people who agree to participate in human research studies.

10. Further information and who to contact.

If you want any further information concerning this project, you can contact the researcher on +(613) 9925 1302 or any of the following people:

Research contact person

Name	Dr Caroline Tan
Position	Senior supervisor
Telephone	+(613)9925 1302
Email	Caroline.tan@rmit.edu.au

11. Complaints

Should you have any concerns or questions about this research project, which you do not wish to discuss with the researchers listed in this document, then you may contact:

Reviewing HREC name	RMIT University
HREC Secretary	Vivienne Moyle
Telephone	03 9925 5037
Email	humanethics@rmit.edu.au
Mailing address	Manager, Research Governance and Ethics RMIT University GPO Box 2476 MELBOURNE VIC 3001

Consent Form

Acknowledgement by Participant – by continuing to the survey:

- ☐ I have read and understood the Participant Information Sheet.
- ☐ I understand the purposes, procedures and risks of the research described in the project.
- ☐ I have had an opportunity to ask questions, and I am satisfied with the answers I have received.
- ☐ I freely agree to participate in this research project as described and understand that I am free to withdraw at any time during the project without affecting my relationship with RMIT.
- ☐ I understand that I can print or download a copy of this document to keep.

Appendix E: Letter of Ethical Approval

RMIT Classification: Trusted



College Human Ethics Advisory Network (CHEAN)
College of Design and Social Context (DSC)
NHMRC Code: EC00237

Notice of Approval

Date: **21 May 2021**

Project number: **2021-24306-14393**

Project title: **Mapping the Saudi fashion entrepreneurial ecosystem**

Risk classification: **Low risk**

Chief investigator: **Associate Professor Rebecca Yang**

Status: **Approved**

Approval period: From: **20 May 2021** To: **20 May 2024**

The following documents have been reviewed and approved:

Title	Version	Date
Risk Assessment and Application Form	1	21/05/2021
Data Management protocol	1	21/05/2021
Interview Schedule	1	21/05/2021
Participant information sheet	1	21/05/2021
Recruitment material	1	21/05/2021
Survey	1	21/05/2021

The above application has been approved by the RMIT University CHEAN as it meets the requirements of the *National Statement on Ethical Conduct in Human Research* (NHMRC, 2007).

Terms of approval:

1. Responsibilities of chief investigator

It is the responsibility of the above chief investigator to ensure that all other investigators and staff on a project are aware of the terms of approval and to ensure that the project is conducted as approved by CHEAN. Approval is valid only whilst the chief investigator holds a position at RMIT University.

2. Amendments

Approval must be sought from CHEAN to amend any aspect of a project. To apply for an amendment, use the request for amendment form, which is available on the HREC website and submitted to the CHEAN secretary. Amendments must not be implemented without first gaining approval from CHEAN.

3. Adverse events

You should notify the CHEAN immediately (within 24 hours) of any serious or unanticipated adverse effects of their research on participants, and unforeseen events that might affect the ethical acceptability of the project.

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4. Annual reports

Continued approval of this project is dependent on the submission of an annual report. Annual reports must be submitted by the anniversary of approval of the project for each full year of the project. If the project is of less than 12 months duration, then a final report only is required.

5. Final report

A final report must be provided within six months of the end of the project. CHEAN must be notified if the project is discontinued before the expected date of completion.

6. Monitoring

Projects may be subject to an audit or any other form of monitoring by the CHEAN at any time.

7. Retention and storage of data

The investigator is responsible for the storage and retention of original data according to the requirements of the *Australian Code for the Responsible Conduct of Research* (R22) and relevant RMIT policies.

8. Special conditions of approval

Nil.

In any future correspondence please quote the project number and project title above.

Associate Professor Marsha Berry
Chairperson, College Human Ethics Advisory Network (CHEAN B)
RMIT University

cc: Dr Fotini Toso (CHEAN secretary)
Dr Carolina Quintero Rodriguez (Co-investigator)
Mrs Marwah Dakhilallah D Alharbi (Research student)
Dr Stephen Wigley (Co-investigator)